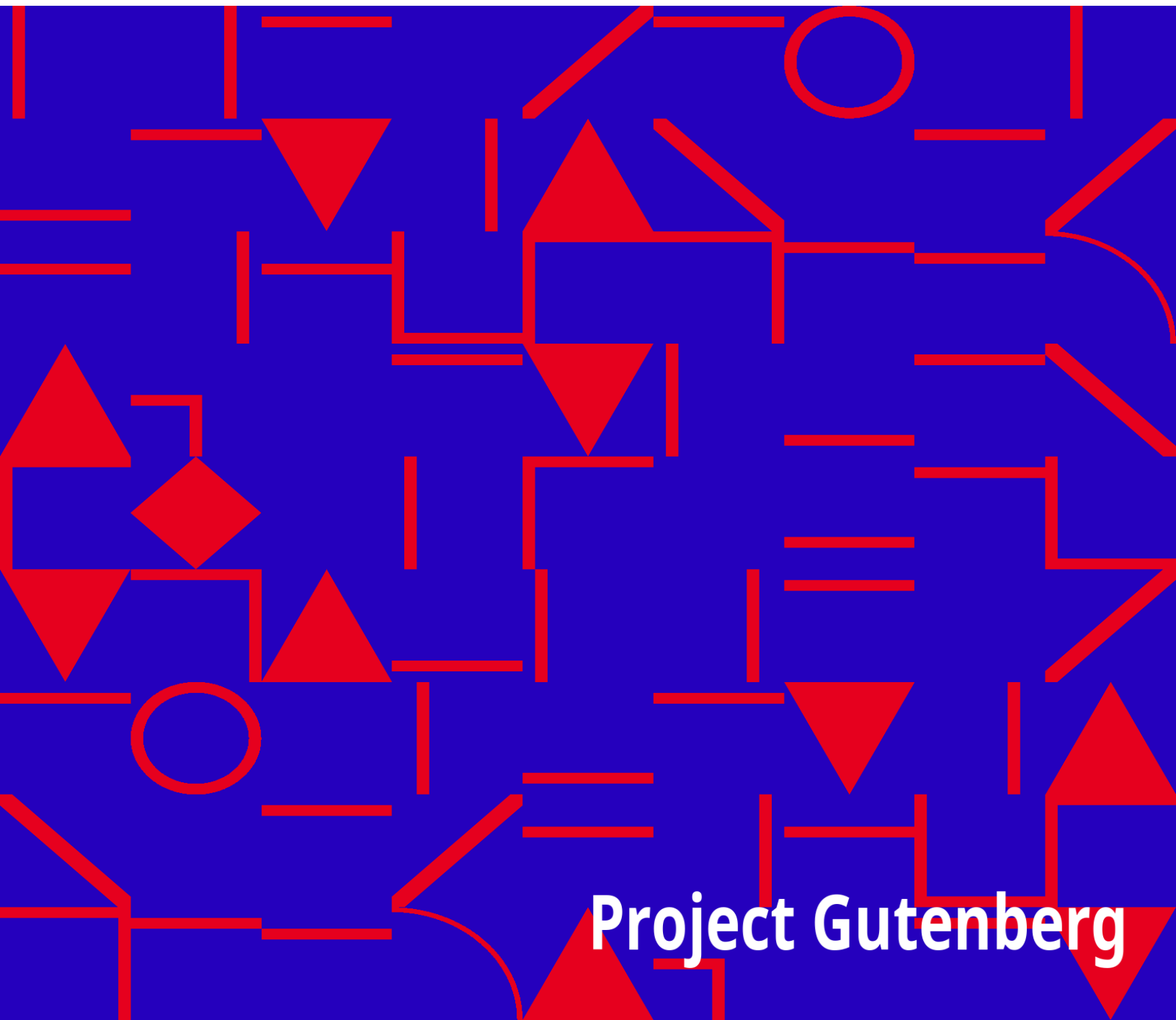


The New Gresham Encyclopedia. Estremoz to Felspar

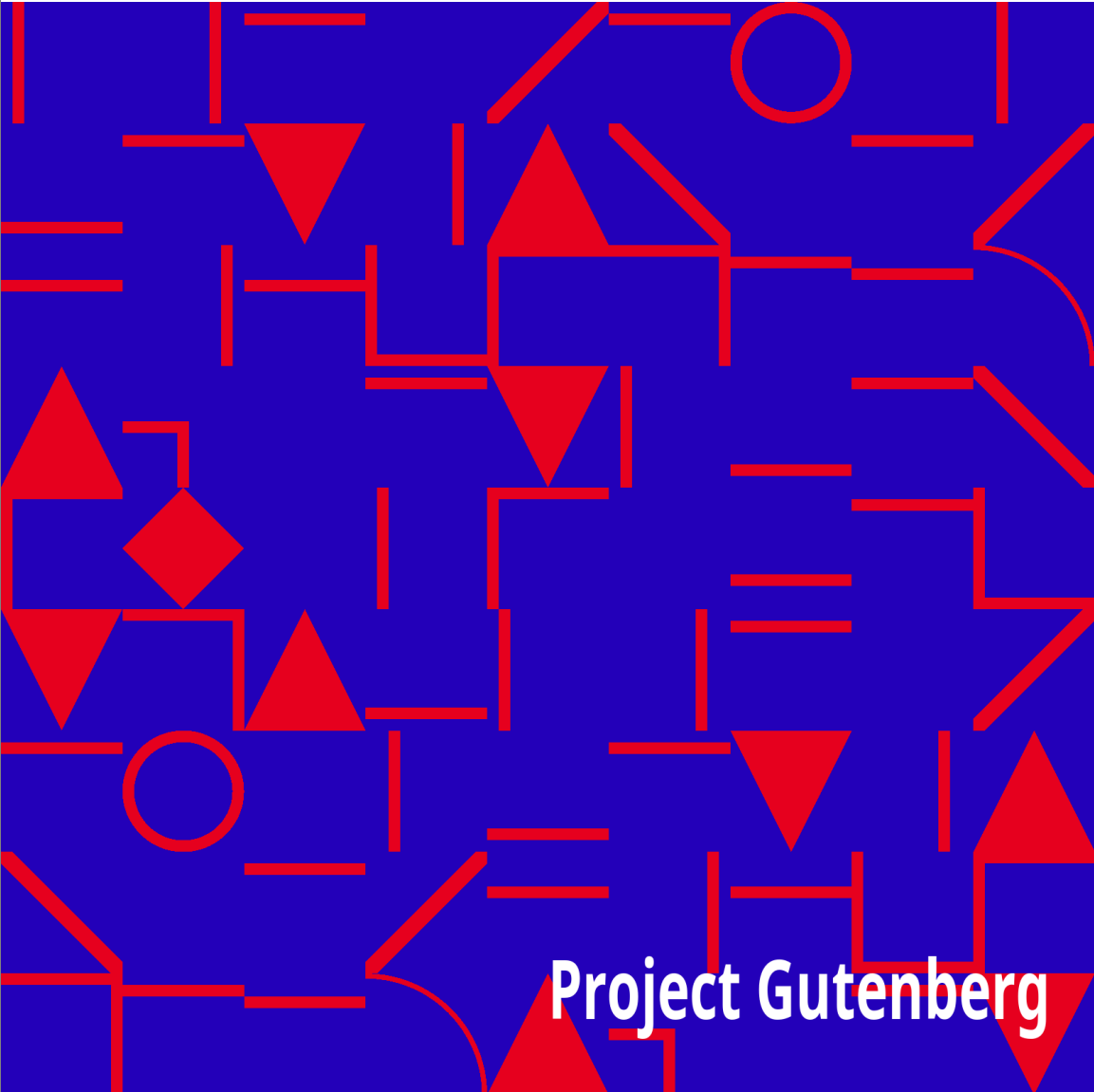
Various



Project Gutenberg

The New Gresham Encyclopedia. Estremoz to Felspar

Various



Project Gutenberg

The Project Gutenberg eBook of The New Gresham Encyclopedia. Estremoz to Felspar

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The New Gresham Encyclopedia. Estremoz to Felspar

Author: Various

Release date: April 12, 2011 [eBook #35845]

Most recently updated: January 7, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/35845

Credits: Produced by Jonathan Ingram, Keith Edkins and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE NEW
GRESHAM ENCYCLOPEDIA. ESTREMOZ TO FELSPAR ***

Transcriber's note: A few typographical errors have been corrected. They appear in the text like this, and the explanation will appear when the mouse pointer is moved over the marked passage.

**THE
NEW • GRESHAM
ENCYCLOPEDIA**

VOLUME · IV · PART · 3



***The* GRESHAM · PUBLISHING
COMPANY · *Limited***

**66 CHANDOS STREET · STRAND
LONDON W.C.2.
1922**

LIST OF PLATES AND MAPS

VOLUME IV PART 3

ESTREMOZ to FELSPAR

PLATES

	Page
ETHNOLOGY	304

MAPS IN COLOUR

EUROPE	322
--------	-----

KEY TO PRONUNCIATION

The method of marking pronunciations here employed is either (1) by marking the syllable on which the accent falls, or (2) by a simple system of transliteration, to which the following is the Key:—

VOWELS

ā, as in *fate*, or in *bare*.

ä, as in *alms*, Fr. *âme*, Ger. *Bahn* = á of Indian names.

a', the same sound short or medium, as in Fr. *bal*, Ger. *Mann*.

a, as in *fat*.

a", as in *fall*.

a, obscure, as in *rural*, similar to *u* in *but*, è in *her*: common in Indian names.

ē, as in *me* = *i* in *machine*.

e, as in *met*.

è, as in *her*.

ī, as in *pine*, or as *ei* in Ger. *mein*.

i, as in *pin*, also used for the short sound corresponding to ē, as in French and Italian words.

eu, a long sound as in Fr. *jeûne* = Ger. long ö, as in *Söhne*, *Goethe* (Goethe).

eu, corresponding sound short or medium, as in Fr. *peu* = Ger. *ö* short.

ō, as in *note*, *moan*.

o, as in *not*, soft—that is, short or medium.

ö, as in *move*, *two*.

ū as in *tube*.

u, as in *tub*: similar to *è* and also to *a*.

u", as in *bull*.

ü, as in Sc. *abune* = Fr. *û* as in *dû*, Ger. *ü* long as in *grün*, *Bühne*.

u', the corresponding short or medium sound, as in Fr. *but*, Ger. *Müller*.

oi, as in *oil*.

ou, as in *pound*; or as *au* in Ger. *Haus*.

CONSONANTS

Of the *consonants*, **b, d, f, h, j, k, l, m, n, ng, p, sh, t, v, z**, always have their common English sounds, when used to transliterate foreign words. The letter **c** is not used by itself in re-writing for pronunciation, **s** or **k** being used instead. The only consonantal symbols, therefore, that require explanation are the following:—

ch is always as in *rich*.

d, nearly as *th* in *this* = Sp. *d* in *Madrid*, &c.

g is always hard, as in *go*.

h represents the guttural in Scotch *loch*, Ger. *nach*, also other similar gutturals.

n', Fr. nasal *n* as in *bon*.

r represents both English *r*, and *r* in foreign words, which is generally much more strongly trilled.

s, always as in *so*.

th, as *th* in *thin*.

th, as *th* in *this*.

w always consonantal, as in *we*.

x = ks, which are used instead.

y always consonantal, as in *yea* (Fr. *ligne* would be re-written *lēny*).

zh, as *s* in *pleasure* = Fr. *j*.

Estremoz´, a town of Portugal, in the province of Alemtejo, 22 miles west of Elvas. Pop. about 8000.

Eszek (es-sek´), or **Esseg**, a town of Yugo-Slavia, formerly in Hungary, on the Drave, 13 miles from its confluence with the Danube. It has barracks, town house, normal school, an important trade, and several fairs. Pop. 31,000.

Étampes (ā-tän´p), a town of France, department of Seine-et-Oise, 30 miles s. by w. of Paris. Pop. 9450.

Étaples, a town of Northern France, department of Pas-de-Calais, on the right bank of the estuary of the Canche, 17 miles south of Boulogne. During the European War, Étaples became a place of great importance. It was a huge British encampment, including many hospitals, and a cemetery with over 11,000 graves. It was also of importance as a training-centre, and the famous 'Bull-ring' was there. Pop. 6000.

Etap´pen (Ger.), a department in Continental armies the business of which is to relieve the commanders of the field army of all responsibility for their communications in the rear. The officers of this department supervise all arrangements for loading and unloading at stations, forwarding, feeding, and billeting.

Eta´wah, a town, Hindustan, United Provinces, capital of district of the same name, on left bank of the Jumna, picturesquely situated among ravines, and richly planted with trees. It has some good buildings, and a considerable trade. Pop. 45,350.—The district has an area of 1694 sq. miles, and a pop. of 760,120.

Etching, a method of engraving lines upon a metal plate by means of acid, whence the term has come to denote an impression taken on paper or similar material from the etched plate. Sometimes, though incorrectly, applied to a line-drawing in pen and ink. The usual process is to cover the plate (generally of copper) with an *etching-ground* of waxes and resins, on which the lines are opened up by means of a sharp-pointed *etching-needle*, either from a design transferred to the ground, or by the artist working directly. The lines are then *bitten* by putting the plate into dilute nitric or hydrochloric acid, the back and edges being protected by *stopping-out varnish*. The plate is removed when the lightest lines are sufficiently bitten. If some lines need deeper biting, the rest may be covered with stopping-out varnish, and the plate replaced in the acid; or acid may be applied locally. If a plate is removed before biting is complete, in order to take a trial impression, it is recovered with a transparent ground, additional lines opened up if necessary, and *rebitten*. In *soft-ground etching*, the ground is mixed with tallow, thin paper laid upon it, and the design firmly drawn thereon with a pencil. When the paper is removed, the ground adheres to it where the lines were drawn. The plate is bitten in the usual way. This produces the effect of a chalk or pencil drawing. *Dry-point* is a method of working direct on to the copper with a sharp point, which raises a *burr* on each side of the line, giving it a characteristic quality. Dry-point, etching proper, and engraving proper are often combined in one plate; and a mixture of etching with mezzotint or aquatint is not uncommon. In *printing*, a matter of first-rate importance, the ink is rubbed into the lines and superfluous ink *wiped* from the surface of the plate, ink being left in any place where a tint is required. Impressions may then be taken by hand; but a press is generally used, being more expeditious and yielding more even results. The papers used are various, but those of Japanese make are most popular. The number of good impressions possible from one plate is limited by the wearing of the plate; in particular, the burr of dry-point soon disappears. A *state* is the name given to each stage in the progress of a print, which is the result of new work on the plate. Differences due to variations in the amount of ink used, or to wiping, do not constitute states; but the addition of a title, artist's signature, &c., will make a state. As distinct from the engraved line, the etched line has a freedom and spontaneity resembling that made by pen or pencil.

The process was apparently used as a means of decorating metal some time before prints were taken. The earliest-known etching is dated 1513. Among the first to use the process was Albrecht Dürer (1471-1528), who between 1515 and 1518 produced six plates on iron, showing great power and precision, but

hardly realizing all the qualities of the medium. Among his followers, the German Little Masters, Hans Sebald Beham and Albrecht Altdorfer were responsible for some interesting plates, as was Lucas van Leyden (1494-1533), the Dutchman. Among the earliest Italian etchers were Francesco Mazzuoli (1503-40) and Andrea Schiavone (1522-82), who show more freedom and delicacy than the Germans. But at this time etching was mainly the by-product of artists whose chief work was painting or engraving. Its great period opened in the seventeenth century. Jacques Callot (1592-1635), born at Nancy, who worked there and at Rome, produced about one thousand plates of small size, the most important being two series of the *Miseries of War*. He is remarkable for his fine sense of design, the fantastic, grotesque quality of his figures, and the delicacy and variety of his line, obtained by rebiting and by combining engraving with etching. Claude Lorrain (1600-82), the landscape painter, possibly under the influence of Callot, produced some fifty plates, very delicately etched, and suggestive of atmosphere. Sir Anthony Van Dyck (1599-1641), beside being court painter to Charles I of England, was the greatest Flemish etcher of his day. His eighteen etched portraits of famous contemporaries (fifteen of which were included in his *Iconography*, published 1645), in their direct handling and vivid characterization, are among the finest work of the kind ever done. But the central figure in etching, not only of the seventeenth century but of all time, is Rembrandt van Ryn (1606-69), whose work is unrivalled both in quality and influence. His etchings show the same realism, understanding of humanity, and creative imagination which mark all his work. Roughly, they fall into three groups, according to the time at which they were produced. In the first period, the ordinary etched line is mainly used, and the artist is evidently feeling his way; in the second, chiaroscuro is more marked, and dry-point used freely; in the third, the handling is very free and vigorous, chiaroscuro becomes the dominant feature, and dry-point the usual method. Development on these lines marks all his plates, which consist of (1) portraits, e.g. *Jan Six* (c. 1646) and *Clement de Jonghe* (1651); (2) figure compositions, many of scriptural subjects, which include the masterpiece *Christ receiving Little Children* (c. 1650), commonly known as 'The Hundred Guilder Print'; (3) landscapes, e.g. *The Goldweaver's Field*, the least numerous class, but one which has inspired the greatest mass of work. The Dutch painters contemporary with or following Rembrandt were in some cases prolific etchers, notably Ferdinand Bol and Adrian van Ostade, and reproduce in that medium the characteristics of their painting. In the eighteenth century etching fell somewhat into disuse, save in Italy, where G. B. Tiepolo (1696-1770), the decorative painter, produced some fifty plates, and Antonio Canale

(Canaletto, 1697-1768) showed in his few etchings the same power to express structure and aerial perspective as in his painting. More prolific was G. B. Piranesi (1720-88), who published a series of views of the Classical and Renaissance architecture of Rome, professedly with an archæological aim, but of great artistic interest. His imaginative power, bold design, and vigorous handling are best seen in the fantastic plates of his *Carceri*. In England, William Hogarth (1697-1764) produced a few etchings; Thomas Rowlandson (1756-1827), the caricaturist, used etching as the basis of his aquatints; and John Crome (1768-1821), the landscape painter, etched some characteristic plates. But it was Francisco Goya (1746-1828) whose work ushered in a new era. His bitterly satirical *Caprichos* (1793-6, 72 plates), *Proverbios* (1810-5, 18 plates), *Desastres de la Guerra* (c. 1810, 82 plates), and the more popular but no less remarkable *Tauromaquia* (c. 1815, 33 plates illustrating bull-fighting), in all of which the bitten line is allied with aquatint, show a powerful and fantastic imagination, brilliant design and draughtsmanship, and superb technique. Widely different in character are the 71 plates of the *Liber Studiorum*, one of the most remarkable works of J. M. W. Turner (1775-1851). In these etching merely provides the ground plan for the use of mezzotint, or, more rarely, aquatint. In the nineteenth century the revival inaugurated by Goya was carried on in France by several of the Barbizon group of landscape painters, notably by J. F. Millet (1814-75), responsible for some simple but impressive plates. A more important figure as an etcher is Alphonse Legros (1837-1911), whose admirable portraits recall those of Van Dyck, though elsewhere he shows something of Goya's taste for the grotesque. This last also appears in the work of Charles Méryon (1821-68), one of the greatest of French etchers, whose feeling for decorative design and decisive handling are best seen in his views of Paris. Apart from other etchers of the period are Jules Jacquemart and Félix Braquemond, remarkable for their exquisite delicacy in the reproduction of surface texture. Of the Impressionist painters, Camille Pissarro (1830-1903) produced some very individual plates, marked by the use of broken lines and much rebiting, in the effort to secure atmospheric effect. Similarly, by means of open shading and absence of outline, Anders Zorn (1860-1918), the Swede, has aimed at reproducing the play of light round objects; but his portraits are his best work. The chief figure in nineteenth-century etching, however, is J. A. M'N. Whistler (1834-1903), whose *French Set* (1858), *Thames Set* (1871), *Venice Set* (1880), and *Twenty-six Etchings* (1886) show his delicate yet decisive handling, his economy of means, his feeling for design, and his power of securing luminosity and atmosphere. Part of his success was due to insistence upon printing his own plates. His brother-

in-law, Sir Francis Seymour Haden (1818-1910), the distinguished doctor, also took a prominent part in the revival of etching, and in his plates showed remarkable skill. The same accomplishment marks the work of William Strang, A.R.A. (died 1921), who has produced many notable portraits of contemporary celebrities, including R. L. Stevenson and Thomas Hardy. Another artist of great technical skill, excelling in the use of dry-point, is Félicien Rops (1833-98), a Belgian, whose work is remarkable for its union of satire and licentiousness. The most notable living etchers are chiefly found in England, and include Sir Frank Short, famous also for his mezzotints; D. Y. Cameron, a disciple of Whistler, though of marked individuality in his treatment of architecture; Muirhead Bone, whose architectural work is unrivalled and has inspired many followers; James M'Bey; and Augustus John, who stands apart from his contemporaries in his preference for figure subjects. In France, Jean-Louis Forain has produced some remarkable work, notably series dealing with the life of Christ, and with Lourdes, which show his satiric power and a very distinctive technique. See *Engraving*.—BIBLIOGRAPHY: A. M. Hind, *A Short History of Engraving and Etching* (very complete and authoritative). For technical details, M. Lalanne, *Etching*; Paton, *Etching and Mezzotint Engraving*.

Ete'ocles and **Polyni'ces**, two heroes of ancient Greek legend, sons of Œdipus, King of Thebes. After their father's banishment from Thebes, Eteocles usurped the throne to the exclusion of his brother, an act which led to an expedition of Polynices and six others against Thebes. This war is known as the Seven against Thebes, and forms the basis of Æschylus's *The Seven against Thebes*. The two brothers fell by each other's hand. See *Antigone*.

Ete'sian Winds (Gr. *etos*, year), winds which, blowing over the Mediterranean regions from a general northerly direction during some weeks of the summer, replace the heated air that rises from the Sahara and other parts of Africa. By carrying with them moisture from the sea, they add greatly to the fertility of Egypt.

Ethane, (C₂H₆), a hydrocarbon belonging to the paraffin series. It is a colourless inflammable gas, and is found amongst the gaseous constituents of the Pennsylvanian oil-wells.

Eth'elbert, King of Kent, born about A.D. 560, died 616. He succeeded his father, Hermenric, and reduced all the English states, except Northumberland, to the condition of his dependents. Ethelbert married Bertha, the daughter of

Caribert, King of Paris, and a Christian princess, an event which led indirectly to the introduction of Christianity into England by St. Augustine. Ethelbert was the first English king to draw up a code of laws.

Ethelbert, King of England, son of Ethelwulf, succeeded to the government of the eastern side of the kingdom in A.D. 857, and in 860, on the death of his brother Ethelbald, became sole king. His reign was much disturbed by the inroads of the Danes. He died in 866.

Eth'elred I, King of England, son of Ethelwulf, succeeded his brother Ethelbert in A.D. 866. The Danes became so formidable in his reign as to threaten the conquest of the whole kingdom. Ethelred died in consequence of a wound received in an action with the Danes in 871, and was succeeded by his brother Alfred.

Ethelred II, King of England, son of Edgar, born A.D. 968, succeeded his brother, Edward the Martyr, in 978, and, for his want of vigour and capacity, was surnamed *the Unready*. In his reign began the practice of buying off the Danes by presents of money. After repeated payments of tribute, he effected, in 1002, a massacre of the Danes; but this led to Sweyn gathering a large force together and carrying fire and sword through the country. They were again bribed to depart; but, upon a new invasion, Sweyn obliged the nobles to swear allegiance to him as King of England; while Ethelred, in 1013, fled to Normandy. On the death of Sweyn he was invited to resume the government, and died at London in the midst of his struggle with Canute (1016).

Eth'elwulf, King of England, succeeded his father, Egbert, about A.D. 837, died 857. His reign was in great measure occupied in repelling Danish incursions; but he is best remembered for his donation to the clergy, which is often quoted as the origin of the system of tithes. Alfred the Great was the youngest of his five children.

Eth'endun, Battle of, the victory which Alfred the Great gained over the Danes (878), and which led to the treaty with Guthrum, the Danish king of East England. The locality is doubtful.

Ether, or **Æther**, sometimes called *luminiferous ether* to prevent confusion with the well-known volatile liquid of the same name, a hypothetical medium filling the whole of what seems to be empty space, and even the interstices between the atoms of material bodies. Most thinkers believe that such a medium must be postulated if we are to explain the transmission of physical

actions between bodies at a distance from one another. With the exception of ordinary mechanical pressures and tensions, the simplest examples of influences that can pass across space are sound and light. Sound, we know, is carried by the air, a medium more subtle than solid or liquid bodies, but still easily recognizable by its effects on our senses, and by its mechanical, physical, and chemical properties. We know a good deal about air, and about the process that goes on when sound is passing through it. But the ether is incomparably more elusive than air. It affects the sense of sight, indeed, as the air affects the sense of hearing; but, so far as we know, it has no weight, no specific heat, no chemical affinity. Except that it is the medium which conveys light, electric and magnetic actions, and possibly gravitation, we know extremely little about it. An extreme school of modern physicists is even inclined to deny, or at least to ignore, its existence altogether.

Early speculators regarded the ether as a species of fluid, which could be displaced by ordinary matter, so that upholders of the wave theory of light necessarily thought of waves like those of sound, in which the direction of vibration is in the line of transmission, for no other kind of wave can occur in a fluid. Young and Fresnel, however, insisted on the view that the movements of the medium are at right angles to the direction of propagation, and pointed out that this might be explained by supposing the medium to possess elasticity of shape. The obvious objection to the conception of a solid which permits the planets to move through it with apparently perfect freedom was met long afterwards by Stokes and Kelvin, who instanced such substances as shoemaker's wax and jelly, which are rigid enough to be capable of elastic vibration, and yet permit bodies to pass through them with more or less ease. Fresnel's work called attention to the subject of the elasticity of bodies, and led to the discovery of the general equations of vibration of an elastic solid by Navier in 1821. Navier's equations, slightly generalized, were used by Cauchy with a certain amount of success to explain reflection, refraction, and the phenomena of crystal-optics. In 1837 George Green published a variety of elastic solid theory which was a decided improvement on Cauchy's, but many difficulties remained, and it is now almost universally agreed that the vibrations of an ordinary elastic solid do not furnish an exact parallel to the vibrations which constitute light. One of the chief difficulties is that in an ordinary elastic solid two types of waves can occur, one distortional, with the displacement of a particle perpendicular to the direction of transmission, and the other dilatational, with the displacement along the line of transmission, as in sound. Waves of light must be of the distortional kind, and the velocity of

the other kind of wave may be quite different from the velocity of light. A kind of ether in which this difficulty of the longitudinal wave does not occur was imagined by Cauchy and afterwards discussed by Lord Kelvin, who called it the contractile, or labile, ether. This is an elastic body with negative compressibility, like homogeneous foam which is prevented from collapsing by attachment to the sides of a containing vessel. Another type of quasi-elastic solid was brought forward by James MacCullagh in 1839. MacCullagh's solid possesses what may be called elasticity of rotation, but offers no resistance to deformations in which elementary parts of the solid preserve their orientation. The equations of motion of this ether devised by MacCullagh are very similar to those obtained much later from a very different physical point of view by Clerk Maxwell. Elastic solid theories, however, have fallen into the background before the advancing popularity of the electromagnetic theory of James Clerk Maxwell. Maxwell's equations of the electromagnetic field are deduced from easily demonstrable experimental facts, supplemented by the characteristic hypothesis that the electric current always travels in a closed circuit, even in cases where, as in the discharge of a condenser, the material circuit is open, so that the path of the current has to be completed through the ether. Other essential features of Maxwell's view are that electric, magnetic, and electromagnetic action is transmitted by means of stresses in a medium which possesses some sort of elasticity and inertia not exactly of an ordinary mechanical kind, and that the energy of all such action resides in the medium. 'Maxwell's equations', especially as modified by H. A. Lorentz so as to take account of the atomic structure of electricity, are fundamental in modern electrodynamics and the electron theory of matter. The form of Maxwell's equations shows that electromagnetic action can be propagated in waves with a definite velocity, which depends on the specific inductive capacity and the magnetic permeability of the medium. Maxwell had no difficulty in showing from experimental data that the velocity given by his theory, which turns out to depend on the ratio of the electrostatic and electromagnetic units of charge, is identical with the known velocity of light. He concludes that waves of light are electric waves. The actual production of waves by electrical means was experimentally demonstrated by Sir Oliver Lodge, and more completely by Heinrich Hertz, and is now a commonplace of wireless telegraphy and telephony. The question of the nature of the mechanical process by which physical actions are carried on in the ether weighed heavily on Maxwell, as on other nineteenth-century physicists. Mechanical models of many kinds have been devised to represent ethereal action. Were it sufficient for the purpose, certainly nothing could be simpler than the elastic solid model. Other models

of much interest are the gyrostatic ether and the vortex sponge ether of Lord Kelvin, and the molecular vortex ether of Maxwell. It is recorded that the celebrated mathematician Gauss had made out a theory of electrodynamics, but always declined to publish it because he was unable to devise a mental picture of the physical action represented by his mathematics; and it was probably a similar reason that led Lord Kelvin to declare, so late as 1904, that "the electromagnetic theory has not helped us hitherto". Sir J. J. Thomson has developed a theory of moving tubes of electric force, which produce magnetic fields by their motion. Possibly light may consist of tremors in these tubes, and if the tubes are discrete, it may become practicable to reconcile the modern quantum theory (q.v.) with the phenomena of interference of light, with which at present it seems to be utterly inconsistent.

The extraordinary developments in both theoretical and experimental physics during recent years have diverted attention to some extent from the question of the constitution of the ether, and the problem of its mode of working is more frequently considered from a mathematical and pseudo-metaphysical point of view than from the old standpoint of Newtonian dynamics. It was from a question about the ether, however, that the theory of relativity, the most important of recent speculations, took its origin. Is the ether fixed, or does it move? Is it carried along with the earth in its motion round the sun, or does the ether pass through the atoms of material bodies as the sea passes through the meshes of a net? The elastic solid analogy, and the simplicity of the classical explanation of the aberration of light, are evidence in favour of a fixed ether. But the celebrated interference experiment of Michelson and Morley, which was capable of detecting a comparatively small relative velocity of earth and ether, gave a null result. Various electrical experiments also point to the conclusion that the medium in which optical and electrical effects take place is carried along with the earth in its motion. We are thus placed in a dilemma. We must either reconcile the idea of a fixed ether with the Michelson-Morley and kindred experiments, or we must explain aberration on the supposition that earth and ether move together. Both alternatives have had their supporters. Those who, like Sir Joseph Larmor and Sir Oliver Lodge, believe in a fixed ether rely on the hypothesis of the 'Fitzgerald contraction', according to which bodies moving through the ether with velocity v are contracted in the direction of their motion by the fraction $\sqrt{1 - v^2/c^2}$ of their length, c being the velocity of light. This contraction is in ordinary cases very small, amounting only to a few inches for the diameter of the earth when moving round the sun. The hypothesis follows naturally enough from the accepted theory of the motion of

electrons, and leads to a perfectly simple explanation of the Michelson-Morley result. The most prominent champion of a moving ether was Sir George Stokes. He assumed that, so far as the earth's motion through it is concerned, the ether behaves as a perfect liquid, so that it moves along with the earth, and he proved that aberration would be unaffected by this motion, provided it is everywhere irrotational, or free from spin. Stokes's theory has been extended by Larmor so as to cover a very important set of phenomena found by Arago and Airy, and explained in a general way by Fresnel. These phenomena relate to the velocity of light in material media which are in motion relative to the earth, running water for example. Fresnel proved that all the experimental results are explained if the velocity of light in the water, with respect to the earth, is given by the formula $c' + v(1 - 1/m^2)$, where c' is the velocity of light in still water, v is the velocity of the water relative to the earth, and m is the index of refraction of water. At present the fashionable view of all the phenomena is that taken in Einstein's theory of relativity (q.v.), which makes revolutionary suppositions with respect to the measurement of space and time, and assumes that the velocity of light is a universal constant, independent of the motion either of the source of light or of the observer. Once its initial assumptions are granted, the theory undoubtedly gives simple and natural explanations of the chief optical and electrical phenomena, and in particular leads at once to Fresnel's formula given above. Most English writers on the subject, among whom A. S. Eddington, E. Cunningham, and A. N. Whitehead are prominent, continue to believe that an ether exists, in spite of the fact that as relativists they hold that no experiment can ever enable us to determine our motion through it.—BIBLIOGRAPHY: E. T. Whittaker, *History of the Theories of Æther and Electricity*; Sir Joseph Larmor, *Æther and Matter*; A. S. Eddington, *Space, Time, and Gravitation*; O. W. Richardson, *Electron Theory of Matter*; R. W. Wood, *Physical Optics*.

Ether, or **Ethyl Ether**, $(C_2H_5)_2O$, a colourless, inflammable liquid produced by distillation of alcohol with concentrated sulphuric acid. It is almost immiscible with water, lighter than alcohol, has a sweet taste, and evaporates rapidly in air, producing extreme cold. The vapour of ether mixed with air forms an explosive mixture. Ether is a valuable solvent for many organic substances, fats, oils, &c., and is also used in surgery as an anæsthetic.

Etherege (eth'-è-rej), Sir George, English writer of comedy, born about 1635, died about 1691. He studied at Cambridge, travelled afterwards on the Continent, and then returned to enter himself at one of the Inns of Court.

Devoting himself less to legal studies than to literature and society, he wrote several plays. In 1664 he had his first comedy represented, *The Comical Revenge, or, Love in a Tub*, which was well received. Four years later his *She Would if She Could* appeared, a brilliant play, though frivolous and immoral. Eight years afterwards (1676) he produced his best comedy, *The Man of Mode, or Sir Fopling Flutter*. Etherege's plays are witty and sparkling, and the characters, genuine portraits of the men and women he saw, are vividly if lightly drawn.

Ethical Culture. It has been said by a prominent leader of the movement for ethical culture that the one dogma admitted is the doctrine of personality. The movement eliminates from its teaching all hitherto received religions, admits no Christian symbolism, and acknowledges neither a personal Creator nor a personal Saviour. Christ is, however, highly revered as a man. The world as it *should be* is regarded not as an unattainable though beautiful ideal to be admired and longed for, but as a possible reality to be achieved by strenuous concerted action. The means by which it is hoped to bring about this much-to-be-desired result is the reaction on each other of carefully selected and highly cultivated personalities. Such virtues, therefore, as kindness, pity, justice, charity, temperance, and chastity are deemed less necessary as a personal moral duty in each human being than as a means by which a perfect world may be attained. Man's duty is towards no divine being, but to his fellow-man. In place of that help from above which theologians deem needful to attain even a short step in the direction of perfection, the power of conscience is considered as sufficient for all needs, and disciples are counselled that they should

More strictly, then, the inward judge obey,

since they no longer believe in divine anger or approval.

Germes of the movement may be found in many writers, and Emerson seems to have foreseen it when he said: "The mind of this age has fallen away from theology to morals. I conceive it to be an advance." But the obvious founder of ethical societies was Felix Adler (born 1851), who, in 1876, established in New York a Society of Ethical Culture. He also set in motion such useful work as training-schools, kindergartens, and nursing. In 1885 his associate, W. Salter, established the Chicago Ethical Society. Both have written extensively on the subject; while English supporters of the movement include Sir Leslie Stephen, Sir J. Seeley, Professor Sidgwick, and others. Several ethical societies

exist in Britain, carrying out much educational and philanthropic work. There are both Sunday services and Sunday schools, and in many cases the branches are more or less closely affiliated with labour and its associations. With regard to this community of work and aim, it may be noted that while ethical culturists look forward to a time when no man shall exploit his fellow human beings for personal ends, absolute equality for all is not promised, being recognized as impossible.—BIBLIOGRAPHY: Felix Adler, *Creed and Deed*; W. M. Salter, *Ethical Religion*.

Eth'ics, otherwise called *Moral Philosophy* or *Morals*, is the science which treats of the nature and laws of the actions of intelligent beings, considered as to whether they are *right* or *wrong*, *good* or *bad*. Its subject-matter is human conduct and character in view of a standard or ideal. It refers to constant elements in human nature, and, like æsthetics and logic, is of universal application. The science is more or less closely connected with theology, psychology, politics, political economy, and jurisprudence, but what most strictly belongs to it is the investigation of the principles and basis of duty or the moral law, and an inquiry into the nature and origin of the faculty by which duty is recognized. Various answers have been given to the question why we call an action good or bad, such as that it is consistent or not with the will of God, or with the nature of things, or with the greatest happiness of the greatest number, or that an inward faculty decides it to be such or such; and a great variety of *ethical* systems has been proposed. The foundations of the leading systems were laid in antiquity, the names of Socrates, Plato, Aristotle, Epicurus, the Cynics, and the Stoics being especially prominent. All the Greek philosophers, however, considered ethics from an individualistic standpoint, and paid little attention either to politics or to sociology. The introduction of Christianity brought a new element into ethical speculation, and among Christians ethics were intimately associated with theology, and morality was regarded as based on and regulated by a definite code contained in the sacred writings. The speculations of the Greeks were not, however, disregarded, and some of the ablest Christian moralists (as Augustine, Peter Lombard, Erigena, Anselm, and Aquinas) endeavoured to harmonize the Greek theories with the Christian dogmatics. Most modern ethical systems consider the subject as apart from theology and as based on independent philosophical principles, and they fall into one of two great classes—the utilitarian systems, which recognize as the chief good, happiness, or the greatest possible satisfaction of the tendencies of our nature; and the rationalistic systems, which recognize that ideas of law and obligation can have their source only in reason.

Utilitarianism has been rightly called universal hedonism, as distinguished from the hedonism of Epicurus, which was egoistic. The first of the modern Utilitarian school in England was Hobbes (1588-1679). Among subsequent names are those of Cudworth, Locke, Clarke, Shaftesbury, Butler, Hutcheson, Hume, Adam Smith, Reid, Paley, Whewell, Bentham, J. S. Mill, &c. Paley held that men ought to act so as to further the greatest possible happiness of the race, because God wills the happiness of men, and rewards and punishes them according to their actions, the divine commands being ascertained from Scripture and the light of nature. Bentham's utilitarianism is considerably different from Paley's. It was entirely dissociated from theology or Scripture, and maintained that increase of happiness ought to be the sole object of the moralist and legislator, pleasure and pain being the sole test of actions. To utilitarianism as a special development belong the later 'evolution ethics' represented by Herbert Spencer, in which biological conceptions, such as 'the preservation of the human race', take the place of the Benthamite criterion for determining what is good and bad in actions. Another theory of ethics places the moral principle in the sentimental part of our nature, that is, in the direct sympathetic pleasure or sympathetic indignation we have with the impulses which prompt to action or expression. By means of this theory, which he treats as an original and inexplicable fact in human nature, Adam Smith explains all the phenomena of the moral consciousness. In considering the ethical systems of the Rationalistic school, systems which recognize that the ideas of law and obligation can have their source only in reason, the question, what is the source of the laws by which reason governs, gives rise to a number of psychological theories, amongst which we may notice Clarke's view of the moral principles as rational intuitions or axioms analogous to those of mathematics; Butler's theory of the natural authority of conscience; the position of Reid, Stewart, and other members of the later Intuitional school, who conceive a moral faculty implanted in man which not only perceives the 'rightness' or 'moral obligation' of actions, but also impels the will to perform what is seen to be right. Very similar, as far as classification goes, is the position of Kant, who holds that reason recognizes the immediate obligation of certain kinds of conduct, and that an action is only good when done from a good motive, and that this motive must be essentially different from a natural inclination of any kind.—BIBLIOGRAPHY: H. Sidgwick, *The Method of Ethics*; A. C. Bradley, *Ethical Studies*; H. Spencer, *Principles of Ethics*; L. Stephen, *The Science of Ethics*; *The English Utilitarians*; W. Wundt, *Ethics*; J. Martineau, *Types of Ethical Theory*; A. Sutherland, *The Origin and Growth of*

the Moral Instinct; E. Westermarck, *The Origin and Development of Moral Ideas*; W. R. Sorley, *The Moral Life*; C. Read, *Natural and Social Morality*.

Ethiopia, or **Æthiopia** (Heb. *Cush*), in ancient geography, the country lying to the south of Egypt, and comprehending the modern Nubia, Kordofan, Abyssinia, and other adjacent districts; but its limits were not clearly defined. It was vaguely spoken of in Greek and Roman accounts as the land of the *Ichthyophagi* or fish-eaters, the *Macrobii* or long-livers, the *Troglodytes* or dwellers in caves, and of the *Pygmies* or dwarf races. In ancient times its history was closely connected with that of Egypt, and about the eighth century B.C. it imposed a dynasty on Lower Egypt, and acquired a predominant influence in the valley of the Nile. In sacred history Ethiopia is repeatedly mentioned as a powerful military kingdom (see particularly *Is.* xx, 5). In the sixth century B.C. the Persian Cambyses invaded Ethiopia; but the country maintained its independence till it became tributary to the Romans in the reign of Augustus. Subsequently Ethiopia came to be the designation of the country now known as Abyssinia (q.v.), and the Abyssinian monarchs still call themselves rulers of Ethiopia.

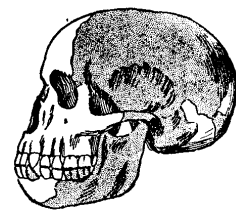
Ethiopian Language, The, or more accurately the *Geez* language, is the old official and ecclesiastical language of Abyssinia, introduced into that kingdom by settlers from South Arabia. In the fourteenth century it was supplanted as the language of the Christian Church of Abyssinia by the Amharic. It is a Semitic language resembling Aramaic and Hebrew as well as Arabic. It has a Christian literature of some importance. The principal work is a translation of the Bible, including the Old and New Testaments and *Apocrypha*, to which are appended some non-canonical writings, such as the *Shepherd of Hermas* and the *Book of Enoch*. The language is to some extent represented by the modern dialects of Tigre, and by that spoken by some nomadic tribes of the Sudan. For the Ethiopian or Abyssinian Church, see *Abyssinia* and *Copt*.

Ethmoid Bone (Gr. *ēthmoeides*, like a sieve), a light spongy bone situated in the upper part of the cavity of the nose. The olfactory nerves pass upward through its numerous perforations to reach the brain.

Ethnology and Ethnography, sciences dealing with man, the aim of the former being to analyse and interpret the meaning of the social phenomena of mankind, as shown in their customs, languages, institutions, &c., the latter being more concerned with descriptive details and the orderly collection of facts relating to particular tribes and localities. Both terms, however, are used

very loosely and in a variety of ways, often being confused with *anthropology*, the general science or natural history of mankind, of which the other two are parts. Anthropology, again, is sometimes used in the narrower sense implied in the word *somatology*, the study of the physical structure and distinctive characteristics of the various races of mankind. When the term ethnology is used by the politician or journalist, in most cases it is intended to refer to the racial components in a given territory; in other words, it is used in the sense in which the scientific writer would employ the word anthropology. For instance, when the endeavour was made in the earlier part of the nineteenth century to liberate the Greeks from Turkish dominion, the plea was put forward that they differed in race; and the delimitation of the territory of the Greek state was claimed on what was called 'the ethnological basis', the geographical distribution of people of Greek nationality. Even since then, and especially during the European War and the subsequent attempt at a settlement, claims have been put forward to fix the boundaries of Italy, Yugo-Slavia, Czecho-Slovakia, Poland, &c., on the basis of race and nationality. But further confusion arises from the attempt to apply this anthropological or ethnological test in deciding whether physical type, language, religion, or social traditions and usages are to be the test of nationality. In this article it will be convenient to give the term ethnology its widest meaning, and to consider not merely the customs, beliefs, and institutions of various peoples, but also the early history of the human family, its differentiation into races and the significance of their geographical distribution, and the different phases of culture which are found in the various communities even of the same race.

During the last eighty years the discovery of a series of fossilized remains of extinct genera and species of the human family and of apes has given us a glimpse of the origin and early history of mankind. Man's ancestors probably parted company with those of the anthropoid apes somewhere in the neighbourhood of Northern India early in the Miocene period; and before the close of the Pliocene period their descendants had gradually acquired the highly developed brain and the intelligence which imply the emergence of the distinctively human characteristics. The most significant token of the attainment of the status of men was the acquisition of the power of speech, which enabled its possessors to hand on the accumulated knowledge and the fruits of experience, and so enormously to increase their powers. The earliest-known



The Piltown Skull.
As restored by Dr.
Smith Woodward
and Mr. Frank
O'Barlow. The dark
portions are those
actually recovered.

representative of the human family was the Ape-man, *Pithecanthropus*, who at the end of the Pliocene period wandered east as far as Java, where the fossilized remains of a skull were found thirty years ago by Professor Eugen Dubois. At a later date a much more highly developed type, one, moreover, that was much closer to the ancestry of modern men than the aberrant Ape-man of Java, wandered as far west as England, where a representative of this extinct genus was discovered by the late Mr. Charles Dawson in 1912 at Piltdown, in Sussex. This very primitive member of the human family has been called the 'Dawn-man' or *Eoanthropus* by Dr. Smith Woodward. He has a brain which, though poorly developed, is definitely human, but his face (and especially the jaws) retains considerable resemblance to that of an ape. Of the other fossilized remains of extinct varieties of the human family, the most important are those known respectively as Heidelberg man and Neanderthal man. The former is almost as old as the Piltdown man, and its former existence was revealed by the discovery in the Mauer Sands, near Heidelberg, in 1908, of a very massive and chinless jaw. At a much later date Europe was inhabited by a brutal species of mankind, Neanderthal man, which became extinct when in the Neolithic Age men of our own species made their way into Europe and completely superseded the less efficient Neanderthal species. The latter were men of vast strength, with short, clumsy, thick-set limbs, a stooping gait, thick neck, and a great flattened head with a coarse face. These people inhabited Europe in the days when the elephant and the woolly rhinoceros lived there; they made the rough stone implements known as Mousterian. But, in spite of their enormous strength, these people were not able to hold their own in competition with the nimbler wits and the more skilled hands of *Homo sapiens*, who introduced into Europe a more finished technique in making implements, and revealed his genius and manual dexterity in the remarkable pictures which he painted on the walls of caves, especially in Southern France and Northern Spain. We have no information concerning the place of origin or the course of the wanderings of these earliest members of our own species.

But an extremely primitive race has survived until the present time to demonstrate the original type of *Homo sapiens*. The aboriginal Australian, like all existing races of men, belongs to the same species as ourselves, but it represents with singularly little modification the original type and colouring of *Homo sapiens*. Fossilized remains of the proto-Australian race have been found in Queensland (at Talgai) and in Java (at Wadjak); but the wandering of the race from its original Asiatic centre of characterization is indicated by the survival of remnants of this people in the pre-Dravidian jungle tribes of India

(mainly in the Deccan), the Vedda of Ceylon, the Sakai of the Malay Peninsula, the Toala of Celebes, and other peoples of the Malay Archipelago, whose existence blazes the track from India to Australia.

The Australian race is on the average about 5 feet 2 inches in height; their skin is dark-brown or black; hair black and wavy or curly; skull typically long (dolichocephalic), with a relatively small brain-case; the nose is flat and broad, and the jaws large and prominent. What lowly culture these people now possess has been mainly acquired within relatively recent times by contact with more civilized peoples.

Long after the proto-Australians separated from the rest of mankind and wandered east, another group wandered west, and, probably in tropical Africa, became specialized in structure to become the Negro race. The negro, like the Australian, retains many primitive characters, such as the black skin, and the small brain, but in other respects, such, for example, as the extremely flattened and curved hair ('pepper-corns'), he has become highly specialized and sharply differentiated from all other varieties of mankind. At an early period in the history of the race the negro divided into two groups—a pygmy variety or Negrillo, and the ordinary tall negro. One of the branches of the pygmy stock became further specialized in structure (in the course of which the black colour of the skin was lost), and became the Bushman race which has gradually been pushed into the deserts of South Africa (see *Hottentot*).



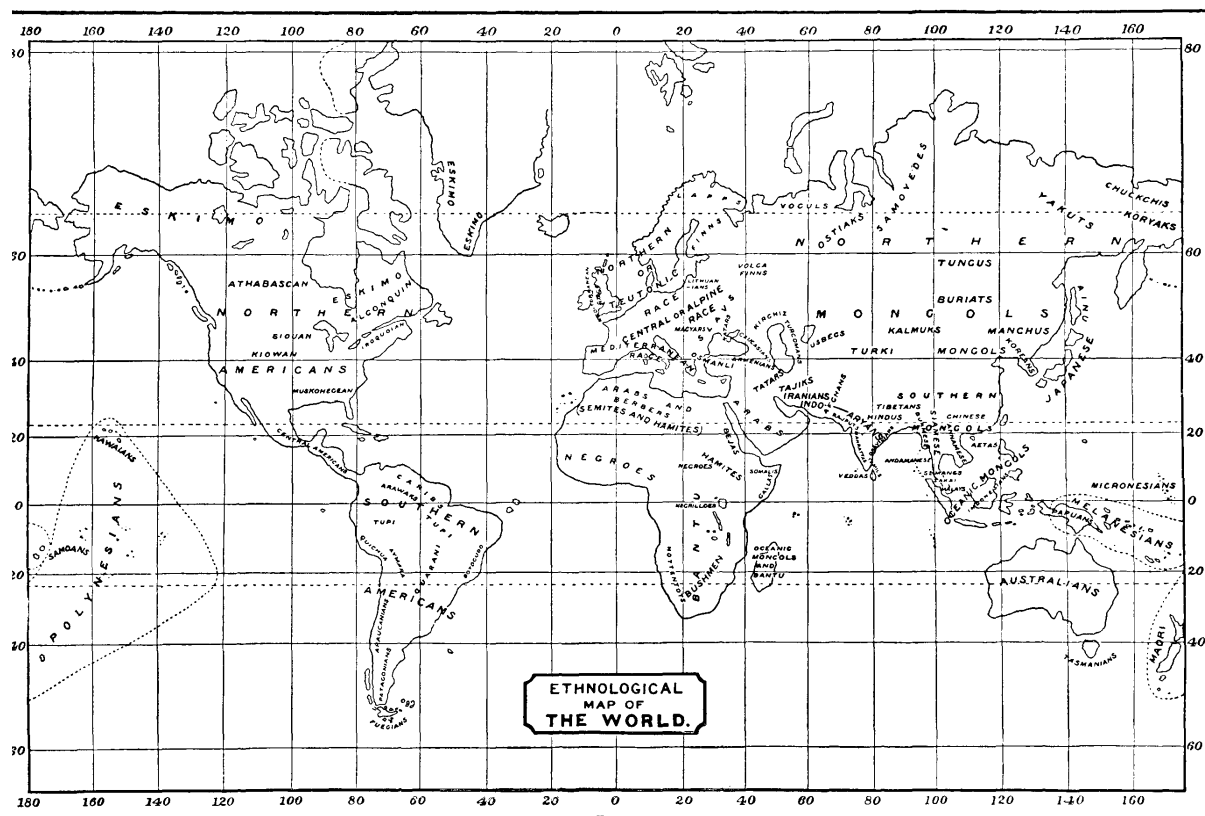
Australian Aborigine, from east side of Lake Eyre.

After the differentiation of the Negro race into pygmy and tall varieties, representatives of both divisions spread along the southern coast of Asia, the former, known in the East as Negritos, reaching the Andaman Islands, the Malay Peninsula (Semangs), the Philippines (Aetas), and New Guinea (Pygmies), and the taller Negroids to Melanesia, New Guinea, and the neighbouring islands of the Malay Archipelago. Many authorities regard the extinct Tasmanian people as a branch of this race.

But the vast majority of the Negro race is found in Africa south of the Sahara Desert and the populations sprung from them in the American continent and the West Indies.

The African negro is subdivided into two main groups known respectively as Nilotic and Bantu; but in addition there are the pygmies of the equatorial and Congo regions, and the Bushmen and Hottentots of the Kalahari Desert, Namaqualand, Lake Ngami, and the Orange River. The Nilotic negro ranges across the continent from Somaliland to Nigeria, and is differentiated from the Bantu chiefly from the fact that along this belt there has been a constant passing to and fro of Hamitic and Semitic peoples for many centuries, leading not only to very considerable recent racial admixture, but also to cultural and especially linguistic influences, which have brought about the breaking up of the population into a series of nations of varied speech and customs. Among the Bantus, on the other hand, although their culture is lower than that of the Nilotic negroes, there is more uniformity both of race and customs. In race they are negro mixed in early times with the proto-Hamitic peoples of East Africa, whence the mongrel population moved south, driving the Bushmen and Hottentots before them.





After the ancestors of the Australian and Negro races had separated from the rest of mankind, which had spread throughout a great part of Asia, North-Eastern Africa, and Europe, the coming of the Glacial epoch created barriers of ice which shut up the various groups each within its own domain. Somewhere in Eastern Asia, possibly in the basin of the Yellow River, the proto-Mongolian race gradually assumed its characteristic traits. In East Africa and the neighbouring tract of Asia the ancestors of the Brown or so-called Mediterranean race were free to roam east and west from India to the African and European coasts of the Atlantic. Farther north, probably in Europe, the Nordic or Blond race (in the map labelled Northern or Teutonic race) assumed its distinctive features; and somewhere in the region between its area of characterization and that of the Yellow race—probably in the region to the north-east of the Caspian—the so-called Alpine (Armenoid) or proto-Slav race developed. It is distinguished from the Brown and Blond races by the broad skull and heavy jaw, no less than by the robustness of build and the great tendency to hairiness; from the Mongolian people the Armenoids are distinguished by the prominence of the nose, the character of the hair, and the colour of the skin. The term Alpine, which is usually applied to this race, is

singularly inappropriate; for, although in Europe and Asia Minor the members of this race show a partiality for high mountains, the vast majority of the members of the race dwell in the plains of Russia, which also may have been the original home of the race. In view of the topographical relationship of the area of characterization of this race to the homes of the other races, Nordic, Brown, Negro, Australian, and Mongol, arranged in a great arc around it, it might not be inappropriate to call the so-called Alpine (Armenoid or proto-Slav) race by the non-committal title 'Central'. At the close of the Glacial epoch, when the melting of the ice unlocked the domains of these races, members of the Central race poured into Asia Minor and Syria, and down to the head of the Persian Gulf; they also made their way north of the Caspian and Black Sea into Europe, mingling there with the Nordic people. But they also moved east in Siberia and mingled with the proto-Mongolian race. It was soon after this event that members of the proto-Mongolian stock, possibly with some admixture of people of the Central race, wandered to North-Eastern Asia and crossed the Behring Strait to colonize America for the first time. From the north-west coast of America these immigrants in course of time made their way south and east, until eventually the whole of the New World from Hudson Bay to Cape Horn was inhabited. Many centuries afterward (especially between 300 B.C. and A.D. 1000) there was a great influx of a variety of other peoples from Polynesia and the Old World on to the Pacific coast of the Americas, which profoundly altered the physical type of the population of Central America and the Andean coast.

The members of the proto-Mongolian race who remained in Asia spread over a large area from the Arctic Ocean south to Tibet, China, Indo-China and the Malay Archipelago, the Philippines, Formosa, Nicobar Islands, and Madagascar. Their domain became divided geographically into three minor areas of characterization, of the Northern, Southern, and Oceanic Mongols respectively. In the Northern Mongols are included the Koreans, the Japanese and the people of Liu-Kiu, the Tungus (including the Manchu, Gold, &c.), the Kalmuks, Buriats, Koryaks, Chuckchis, Kamchadales, Gilyaks.

The Central race became differentiated into a considerable number of varieties. Apart from the Slavs, there were several groups which became isolated the one from the other in Asia Minor and Syria. One of these developed in an extreme degree the characteristic features of the race—the brachycephaly, the prominence of the nose and the high-ramus of the jaw. These are the Armenian and kindred people. Another branch gave origin to the Northern Semites, who

made their way into Palestine and Mesopotamia. (The Southern Semites belong to another race—the Brown.) Another branch of the Central race preceded these two in making their way to the sea-coasts of the Levant and the Persian Gulf—this may be called the Maritime branch of the Central race.

The Slav branch of the Central race was making its way into Europe long before the Neolithic phase of culture there. It passed north of the Black Sea via Poland. But at the end of the Neolithic phase there were two streams of other branches of the race—the true Alpine subdivision passing from Anatolia into the Carpathians and the Alps, to Switzerland, Bavaria, Savoy, and Brittany, and the Maritime division passing round the coasts to the Iberian Peninsula, the British Isles, and Western Europe.

The Brown race spread in East Africa from Somaliland to the Mediterranean and all its coasts, to Western Europe, and the British Isles; in the other direction to Arabia, the shores of the Persian Gulf, and eastward along the coast to India, where it mingled with the pre-Dravidian (proto-Australian) population to give rise to the mongrel Dravidian people. The spread of these Brown people farther east into the Malay Archipelago explains the origin of the Indonesians, who occupied the islands before the coming south of the Mongols, but after the proto-Australians and the proto-Negroes had passed through towards Australia and Melanesia respectively.

From very early times there has been an intermingling of the different races. In East Africa every degree of intermingling of the Hamitic branch of the Brown race has been taking place for more than sixty centuries with negroes, both of the Sudanese and the Bantu stocks. At a later time Arabs poured into Africa and added their quota to the mixture. In India the original pre-Dravidian (proto-Australian) aborigines became diluted with a large influx of the Brown race to form the Dravidian people, who acquired a high civilization from the west. At a later date people of the Central race speaking an Aryan language swarmed through the north-western frontier and introduced their language and culture into India.

Before this happened the Brown race had extended farther east and provided the basis for the population of Indonesia, supplanting to a great extent the earlier proto-Australian and Negroid peoples there. Then the Malays came down from the north and added to the Indonesian mixture a strong Mongolian element. Colonists from the Malay Archipelago settled in Madagascar and added to its mixture of Brown (Semites and Hamites) and Black (Bantu)

elements representatives of the Mongolian (Malay) race. In the course of their maritime expeditions the Malay Archipelago gave to Japan a not inconsiderable contribution both of people and culture.

But the area of the most complex admixture of races in ancient times was Siberia. With the melting of the ice barriers at the close of the Glacial epoch the proto-Mongolian and proto-Central peoples came into intimate contact; and to this mixture was added a proto-Nordic element, as well as a not inconsiderable influx of members of the Brown race, who came from the south through Turkestan to exploit the gold and copper of the Yenesei region. The presence of their dolichocephalic skulls in a region where brachycephaly is the rule has been a perpetual puzzle to anthropologists, who at the present time attempt to solve the problem by assuming the presence of an aboriginal race of long-headed people, who were exterminated by the Mongols and the Turks. The greed for the riches of the head-waters of the Yenesei has made Siberia the home of strife for fifty centuries. This has led not only to a puzzling admixture of races in the affected area, but has started raids of Mongols and Turks, which at various times extended as far as Europe (Huns and Avars), India, and China. So mixed are the races in Siberia that it is not easy to determine whether some of them should be classed as mainly Turki or mainly Mongol; and this applies also to the colonies (Bulgars, Magyars, Finns, Lapps, &c.) which at various times the Asiatic invaders left behind them in Europe, each of which has been profoundly altered by admixture since then.

In the great Mongolian domain that occupies so great a part of Northern and Eastern Asia there are certain definitely alien elements. The Yakuts (of the region near the Lena River) are definitely Turki in race, and the curious hairy Ainus (of Yezo, Sakhalin, and some of the Kurile Islands) are certainly members of the Central race.

A peculiar branch of the northern Mongols is clearly differentiated from the rest to form the Eskimo people who occupy Greenland and Arctic America. They present a marked contrast to the American Indians. The American Indian may be regarded essentially as a branch of the proto-Mongolian race mixed to some extent with a proto-Central element. But on the Pacific littoral there has been considerable admixture with a variety of peoples from Eastern Asia and Oceania for several centuries (c. 300 B.C. to A.D. 1000). Although the peoples conform on the whole to a definite type as regards the characters of their hair and features, there is a considerable range of variation as regards height, skull-form, and other racial features. The people of the states where a high

civilization prevailed ten centuries ago—Mexico, Central America, Peru, and Chile—are clearly differentiated from the rest of the American population by the more obtrusive evidence of admixture with Polynesian and Asiatic peoples. In addition to the peoples of the north-west coast (Haidas and Salish) and of the ancient civilizations (Mayas and Aztecs) of Central America and Mexico, the population of North America can be divided (see map) into the following tribes: (1) Athabaskan, (2) Algonquin, (3) Iroquoian, (4) Siouan, (5) Shoshonean (in the map called 'Kiowan'), (6) Muskhogean, and (7) Pueblo (not indicated in the map, but in Arizona, north of Mexico).

In South America the centre of the ancient civilization was in the region of the Quichua (Inca) and Aymara peoples. The semi-civilized Chibcha people occupied the table-land of Bogota. To the south of Peru the coastal people (Araucanians) were to some extent influenced by the more highly civilized Incas to their north. The presence of gold in the Matto Grosso region of Brazil attracted men from Peru, and set in motion migrations of people towards the Rio de la Plata in the south and towards Venezuela in the north. Among the linguistically distinct peoples found in the latter area are the Tupi, Arawaks, and Caribs. A very primitive people, the Botocudo, occupy the eastern coast of Brazil south of the River San Francisco.

From the beginning man was a maker of implements of stone and bone: but for a vast number of centuries he was merely a hunter who did not attempt anything more in the way of industry. Civilization probably originated in the Nile Valley when men found barley growing there naturally, and discovered that it provided them with a supply of food which could maintain them throughout the year. When the population in the valley increased, so that the natural supply of barley became inadequate, men learned to imitate the inundation, and by scraping channels in the sand to render the desert fertile. Thus was agriculture and irrigation invented, and thus were men led to organize the labour of the community under the direction of a leader who was primarily an irrigation engineer, but eventually became a king and the god Osiris, the dead king, whose reputation as the bestower of life-giving water became apotheosized as the giver of life and immortality.

Pottery was probably invented as an outcome of the mode of life and the needs of these early agriculturists, and the domestication of cattle and the use of their milk for food helped to neutralize the ill-effects of a too exclusively cereal diet. Other events followed in the train of this first adoption of a settled mode of life. The disposal of the dead in the sands that fringed the area of

cultivation, and the natural preservation of the corpse that often resulted, shaped the beliefs of the people with reference to the fate of the dead. Incidentally it led to the invention of the arts of the carpenter, the stone-mason, and the embalmer; and as an outcome of these practices architecture, as well as the ritual of the temple, had its origin.

Long before these events primitive man had begun to ponder over the meaning of death. At first he associated it with such injuries as he had learned by experience killed animals that he hunted; and as the escape of blood caused unconsciousness and death, he framed the belief that blood was the substance of consciousness and of life. To exchange blood was to share knowledge; to give blood was to confer fresh vital substance, i.e. to minimize the risk of extinction or prolong the existence of living or dead. This is the fundamental idea underlying all religious belief and ritual—the giving of life and immortality.

But the act of birth is also a process of life-giving. The cowrie-shell (and subsequently other shells and the pearls contained in them) came to be regarded as a symbol of this life-giving power, and an amulet which could protect both the living and the dead from the risk of extinction. The demand for these precious elixirs of life became so intense that they acquired a fictitious value as currency, and models of them were made to serve as amulets in their stead. The beauty and the lightness of the models of such shells made of the soft useless plastic metal found in the Egyptian and Nubian deserts was probably the means by which gold first acquired any value, and afterwards by confusion came to be credited with the same life-giving attributes as were at first bestowed merely upon the form of the amulets made from it. Thus gold came to be regarded as an elixir of life, and men began to search for the precious substance far and wide, incidentally spreading abroad the germs of the arts and crafts, the beliefs and practices of our common civilization. The use of malachite as a cosmetic provided the circumstances that eventually led men to discover how a gold-like substance, copper, could be obtained from the green ore; and in course of time it came to be realized that the metal was useful for other purposes than the mere making of amulets and jewellery. When the full value of copper as a material for making tools and weapons was fully appreciated, the ore became of tremendous economic importance, and men sought for it far and wide, as they had previously prospected for flint and gold.

The people who introduced the Neolithic culture into Europe brought with them from Egypt a knowledge of agriculture, of pottery-making, of

domestication of animals, of linen, and of the characteristic burial customs and religious beliefs. But these rudiments of civilization were also diffused to Crete and Cyprus, to Syria and Asia Minor, to Elam and Sumer by prospectors searching for the things which the growth of civilization was making valuable, the incense and the timber, the gold and precious stones, the copper and other metals. It is probable that the germs of Egyptian civilization were first planted in Elam by men prospecting for copper, and that Sumerian and Babylonian civilization received their initial inspiration in this way. Crete was inoculated with the germs of civilization by Egypt directly, as well as indirectly, from Asia Minor, which was subjected to the double influence of Egyptian and Sumerian culture. In the Age of Copper, Elamite culture was diffused abroad by miners to Turkestan and Baluchistan, thence respectively to Siberia and China (Shensi province), and to India. In the neighbourhood of the south-eastern corner of the Caspian the alloy bronze was probably invented soon after 3000 B.C. by mixing tin and copper; and the influence of this epoch-making event rapidly spread to Babylonia, to Crete, and to Europe, where it inaugurated the Age of Bronze. It also spread to China, to India, and many centuries later across the Pacific to Central America.

The needs of the early Egyptians compelled them to devise sea-going ships, which in turn became the models of the Cretans, the people of East Africa, the Babylonians, the Phœnicians, and the Greeks. These ships trafficked in the Eastern Mediterranean and the Red Sea, then farther west and east, to the Atlantic seaboard of Western Europe and the shores of the Indian Ocean. The search for gold and pearls led early mariners to Southern India and Ceylon, to Burmah and Indonesia, to the whole coast-line of Eastern Asia, New Guinea, and Melanesia, and in course of time to Polynesia and the coasts of Central America and Peru. Wherever these adventurers found gold or copper, pearls or precious stones, they settled to exploit these sources of wealth, and incidentally planted the germs of their methods of cultivation, their stonework, their burial customs and beliefs. Such expeditions were probably responsible for introducing into Polynesia its first colonists, a mixture of people of Brown and Maritime Central races, mingled with other elements in the course of their easterly wanderings. The earliest movement into Polynesia apparently took with it a considerable element of Melanesian blood, which eventually was carried to New Zealand and the Moriori Islands in the south, and to Easter Island and the American coast in the east. The germs of the ancient civilizations of Central America and Peru were carried across the Pacific from

Cambodia and Indonesia between the years 300 B.C. and A.D. 1000, the periods of greatest activity being probably the third and fourth centuries A.D.

The elements of this imported culture were planted in Honduras and Guatemala and the Isthmus region (Costa Rica, Panama, and Colombia), and from there spread in the fifth century A.D. to Yucatan and then to Mexico. It also spread from the isthmus down the Pacific littoral of South America, the earliest centre of civilization being the region around Lake Titicaca. From Mexico the culture spread in a degraded form up the Mississippi to the Great Lakes, as well as north-west into Arizona.—BIBLIOGRAPHY: A. H. Keane, *Man, Past and Present*, revised edition by A. Hingston Quiggin and A. C. Haddon, is a useful guide to the literature of anthropology and ethnology; see also Robert Munro, *Prehistoric Britain*; W. J. Sollas, *Ancient Hunters and their Modern Representatives*; M. Boule, *Les Hommes fossiles*; G. Elliot Smith, *The Migrations of Early Culture*; such periodicals as the *Journal of the Royal Anthropological Institute*, *Man*, and especially *L'Anthropologie*, give the current literature.

Eth'yl, the name given to the radicle C_2H_5 , contained in ether, $(C_2H_5)_2O$, alcohol, C_2H_5OH , &c. *Ethyl* has not been isolated, as it immediately combines with another *ethyl* group forming diethyl or butane, $C_2H_5-C_2H_5$. *Ethyl chloride*, C_2H_5Cl , formed by the action of hydrochloric acid on alcohol, is much used for the production of low temperature—as a local anæsthetic. *Ethyl nitrite*, $C_2H_5NO_2$, constitutes *sweet spirits of nitre* when dissolved in alcohol.

Eth'ylamine, $C_2H_5NH_2$, an organic base formed by the substitution of 1 atom

of hydrogen in ammonia by ethyl group. Thus
$$\begin{array}{c} \text{H} \\ \diagup \text{N} - \text{H} \\ \diagdown \text{H} \end{array} \text{ ammonia} \rightarrow$$

$$\begin{array}{c} \text{C}_2\text{H}_5 \\ \diagup \text{N} - \text{H} \\ \diagdown \text{H} \end{array} \text{ ethylamine.}$$
 It has the odour and many of the characteristic

reactions of ammonia, but unlike ammonia it is inflammable and liquid at ordinary temperature.

Eth'ylene, or **Olefiant Gas**, C_2H_4 , an unsaturated hydrocarbon, the first member of the olefine series. It is a colourless gas with a faint odour, and burns with a bright luminous flame. It is a constituent of ordinary coal-gas, and

may be obtained from alcohol by heating it with twice its volume of concentrated sulphuric acid.

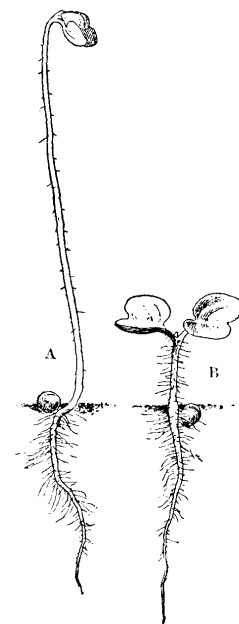
Étienne (ā-ti-ān), **St.**, a town of Southern France, department of Loire, on the Furens, 32 miles s.w. of Lyons. It has spacious streets with substantial houses, but, owing to the number of public works, presents a dingy appearance. The principal buildings and institutions are the cathedral, an ancient Romanesque structure; the town house, court-house, exchange, communal college, mining school, gallery of arts, library, and museum. The town stands in the centre of one of the most valuable mineral fields of France; and in addition to the extensive collieries, blast-furnaces, and other ironworks in the vicinity, has manufactures of ribbons, silks, cutlery, and fire-arms. Pop. 148,656.

Etiolation (Fr. *étioier*, to blanch), or **Blanching**, of plants, is a state produced by the absence of light, by which the green colour is prevented from appearing. It is effected artificially, as in the case of celery, by raising up the earth about the stalks of the plants; by tying the leaves together to keep the inner ones from the light; by covering with pots, boxes, or the like, or by setting in a dark place. The green colour of etiolated plants may be restored by exposure to light. Etiolated plants are also abnormal in other respects; the stems, or in some cases the leaves, become extraordinarily elongated, and the internal structure undergoes modification in various ways.

Etiology (Gr. *aitia*, cause, and *logos*, discourse, account), a biological term introduced by Huxley, and denoting that branch of biology which deals with the origin and mode of development of organic beings. In medicine the word *etiology*, signifies the study of the causes and origin of disease. The term is also applied in philosophy to the science of Cause and Effect.

Etive (et'iv), **Loch**, an inlet of the sea on the west coast of Scotland, Argyleshire, nearly 20 miles long, of very unequal breadth, but at the broadest part about 1½ miles.

The scenery of its shores is very beautiful. About 3 miles from the sea, at Connel Ferry, a ridge of sunken rocks crossing it causes a turbulent rapid, which at half-tide forms a sort of waterfall.



Two Seedlings of
Sinapis alba of
Equal Age
A, Grown in the
dark, etiolated.
B, Grown in
ordinary daylight,
normal. The roots
bear root-hairs.

Etna, or **Ætna, Mount**, the greatest volcano in Europe, a mountain in the province of Catania in Sicily; height, 10,758 feet. It rises immediately from the sea, has a circumference of more than 100 miles, and dominates the whole north-east part of Sicily, having a number of towns and villages on its lower slopes. The top is covered with perpetual snow; midway down is the woody or forest region; at the foot is a region of orchards, vineyards, olive groves, &c. Etna thus presents the variety of climates common to high mountains in lower latitudes, oranges and lemons and other fruits growing at the foot, the vine rather higher up, then oaks, chestnuts, beeches, and pines, while on the loftiest or desert region vegetation is of quite a stunted character. A more or less distinct margin of cliff separates the mountain proper from the surrounding plain; and the whole mass seems formed of a series of superimposed mountains, the terminal volcano being surrounded by a number of cones, all of volcanic origin, and nearly 100 of which are of considerable size. The different aspects of the mountain present an astonishing variety of features—woods, forests, pastures, cultivated fields, bare rocky precipices, streams of lava, masses of ashes and scoriæ, as also picturesque towns and villages. From the summit the view presents a splendid panorama, embracing the whole of Sicily, the Lipari Islands, Malta, and Calabria. The eruptions of Etna have been numerous, and many of them destructive. That of 1169 overwhelmed Catania and buried 15,000 persons in the ruins. In 1669 the lava spread over the country for forty days, and 10,000 persons are estimated to have perished. In 1693 there was an earthquake during the eruption, when over 60,000 lives were lost. One eruption was in 1755, the year of the Lisbon earthquake. There were also eruptions in 1832, 1865, 1874, 1879, and 1886. Among more recent eruptions are those of 1892, 1899, 1911, and 1914. An eruption is ordinarily preceded by premonitory symptoms of longer or shorter duration. The population of the district of Etna is about 300,000.

E'ton, a town of England, in Buckinghamshire, on the left bank of the Thames, 22 miles west of London. An iron bridge connects it with Windsor, on the opposite side of the river. Eton derives its celebrity wholly from its college, called the King's College of Our Lady of Eton beside Windsor, one of the great public schools of England, founded by Henry VI in 1440. The building, which was commenced in 1441 and finished in 1523, has received important additions in recent times in the shape of mathematical and science schools, and a museum. The college foundation now consists of provost, headmaster, lower master, seventy scholars, and two conducts (or chaplains). The oppidans, or

boys not on the foundation, number about 1100. They are mostly lodged and boarded in the masters' houses. Pop. 3366.

Etrépilly, (1) a small town of France, department of Aisne, is situated near Château-Thierry. Millstones are obtained from quarries in the neighbourhood. (2) A small town of France, department of Seine-et-Marne, stands on the left bank of an affluent of the Marne. Agricultural implements are manufactured.

Etru'ria (Gr. *Tyrrhenia*), the name anciently given to that part of Italy which corresponded partly with the modern Tuscany, and was bounded by the Mediterranean, the Apennines, the River Magra, and the Tiber. The name of Tusci or Etrusci was used by the Romans to designate the race of people anciently inhabiting this country, but the name by which they called themselves was Rasena (or perhaps more correctly Ta-rasena). These Rasena entered Italy at a very early period from the north, and, besides occupying Etruria proper, extended their influence to Campania, Elba, and Corsica. Etruria proper was in a flourishing condition before the foundation of Rome, 753 B.C. It was known very early as a confederation of twelve great cities, each of which formed a republic by itself. Amongst the chief were Veii, Clusium, Volsinii, Arretium, Cortona, Falerii, and Faesulae; but the list may have varied at different epochs. The chiefs of these republics were styled *lucumōnes*, and united the office of priest and general. They were elected for life. After a long struggle with Rome, the Etruscan power was completely broken by the Romans in a series of victories, from the fall of Veii in 396 B.C. to the battle at the Vadimonian Lake (283 B.C.). The Etruscans had attained a high state of civilization. They carried on a flourishing commerce, and at one time were powerful at sea. They were less warlike than most of the nations around them, and had the custom of hiring mercenaries for their armies. Of the Etruscan language little is known, although about 6000 inscriptions have been preserved. It was written in characters essentially the same as the ancient Greek. The Etruscans were specially distinguished by their religious institutions and ceremonies, which reveal tendencies gloomy and mystical. Their gods were of two orders, the first nameless, mysterious deities, exercising a controlling influence in the background on the lower order of gods, who manage the affairs of the world. At the head of these is a deity resembling the Roman Jupiter (in Etruscan *Tinia*). But it is characteristic of the Etruscan religion that there is also a Vejovis or evil Jupiter. The Etruscan name of Venus was *Turan*, of Vulcan *Sethlans*, of Bacchus *Phuphluns*, of Mercury *Turms*. Etruscan art was in the main borrowed from Greece. For articles in

terra-cotta, a material which they used mainly for ornamental tiles, sarcophagi, and statues, Etruscans were especially celebrated. In the manufacture of pottery they had made great advances; but most of the painted vases popularly known as Etruscan are undoubtedly productions of Greek workmen. The skill of the Etruscans in works of metal is attested by ancient writers, and also by numerous extant specimens, such as necklaces, ear-rings, and bracelets. The bronze candelabra, of which many examples have been preserved, were eagerly sought after both in Greece and Rome. A peculiar manufacture was that of engraved bronze mirrors. These were polished on one side, and have on the other an engraved design, taken in most cases from Greek legend or mythology. The Etruscans showed great constructive and engineering skill. They were acquainted with the principle of the arch, and the massive ruins of the walls of their ancient cities still testify to the solidity of their constructions. Various arts and inventions were derived by the Romans from the Etruscans.—
BIBLIOGRAPHY: G. Dennis, *Cities and Cemeteries in Etruria*; Seymour, *Up Hill and Down Dale in Ancient Etruria*.

Etruria, a village of England, in Staffordshire, between Hanley and Burslem, famous as the place where Josiah Wedgwood established his pottery works in 1769. Pop. 8056.

Etruria, Kingdom of, in Italy, founded by Napoleon I in 1801. Its capital was Florence. In 1807 Napoleon incorporated it with the French Empire.

Etruscan Vases, a class of beautiful ancient painted vases made in Etruria, but not strictly speaking a product of Etruscan art, since they were really the productions of a ripe age of Greek art, the workmanship, subjects, style, and inscriptions being all Greek. They are elegant in form and enriched with bands of beautiful foliage and other ornaments, figures and similar subjects of a highly artistic character. One class has black figures and ornaments on a red ground—the natural colour of the clay; another has the figures left of the natural colour and the ground painted black. The former class belong to a date about 600 B.C., the latter date about a century later, and extend over a period of about 300 or 350 years, when the manufacture seems to have ceased. During this period there was much variety in the form and ornamentation, gold and other colours besides the primitive ones of black and



Etruscan Pottery (from Veii)

red being frequently made use of. The subjects represented upon these vases frequently relate to heroic personages of the Greek mythology, but many scenes of an ordinary and even of a domestic character are depicted. The figures are usually in profile: temples are occasionally introduced; and many curious particulars may be learned from these vase pictures regarding the Hellenic ritual, games, festivities, and domestic life.

Ett'rick, a pastoral district of Scotland, in Selkirkshire, watered by the Ettrick, and anciently part of Ettrick Forest, which included Selkirk with parts of Peebles and Edinburgh. The Ettrick receives the Yarrow 2 miles above Selkirk, and enters the Tweed 3 miles below. The *Ettrick Shepherd*, the Scottish poet James Hogg, was a native of this district.

Etty, William, an English painter, born in 1787, died in 1849. He studied at the Royal Academy, worked long without much recognition, but at length in 1820 he won public notice by his *Coral Finders*. In 1828 he was elected an academician. Among his works, which were greatly admired, are a series of three pictures (1827-31) illustrating the *Deliverance of Bethulia by Judith, Benaiah* (one of David's mighty men), and *Women Interceding for the Vanquished*. All these are very large pictures, and are now in the National Gallery of Scotland (Edinburgh). Others of note are: *The Judgment of Paris*; *The Rape of Proserpine*; and *Youth at the Prow, and Pleasure at the Helm*. Etty especially excelled at painting undraped figures.

Etymol'ogy (Gr. *etymos*, true, and *logos*, account), a term applied (1) to that part of grammar which treats of the various inflections and modifications of words and shows how they are formed from simple roots; (2) to that branch of philology which traces the history of words from their origin to their latest form and meaning. Etymology in this latter sense, or the investigation of the origin and growth of words, is amongst the oldest of studies. Plato and other Greek philosophers, the Alexandrian grammarians, the scholiasts, the Roman Varro, and others wrote much on this subject. Their work, however, is made up of conjectures at best ingenious rather than sound, and very often wild and fantastic. It was not till recent times, and particularly since the study of Sanskrit, that etymology has been scientifically studied. Languages then began to be properly classed in groups and families, and words were studied by a comparison of their growth and relationship in different languages. It was recognized that the development of language is not an arbitrary or accidental matter, but proceeds according to general laws. The result was a great advance

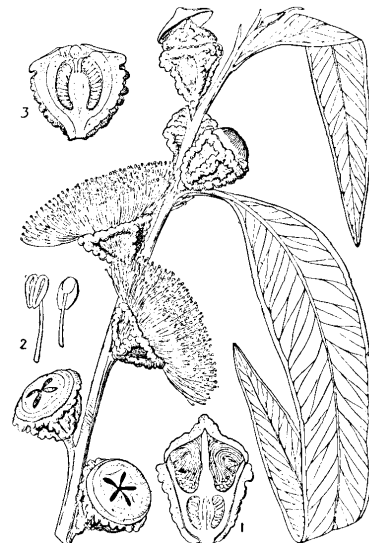
in etymological knowledge and the formation of a new science of *philology*.— Cf. W. W. Skeat, *The Science of Etymology*.

Eu (*eu*), a town in Northern France, department of Seine-Inférieure, about 17 miles north-east of Dieppe. It is notable for its old twelfth-century church and the celebrated Château d'Eu, part of which was destroyed in 1902. Pop. 4900.

Eubœ'a, formerly called Negropont, a Greek island, the second largest island of the Ægean Sea. It is 90 miles in length; 30 in greatest breadth, reduced at one point to 4 miles. It is separated from the mainland of Greece by the narrow channels of Egripo and Talanta. It is connected with the Bœotian shore by a bridge. There are several mountain peaks over 2000 feet in height, and one over 7000 feet. The island is well-wooded and remarkably fertile. Wine is a staple product, and cotton, wool, pitch, and turpentine are exported. The chief towns are Chalcis and Karysto. The island was anciently divided among seven independent cities, the most important of which were Chalcis and Eretria, and its history is for the most part identical with that of those two cities. With some small islands it forms a modern nomarchy, with a pop. of 116,900.

Eubu'lus, a Greek comic poet, who flourished at Athens about 375 B.C. His subjects were chiefly mythological, and he delighted in ridiculing the tragic poets, especially Euripides.

Eucalyp'tus, a genus of trees, nat. ord. Myrtaceæ, mostly natives of Australia, and remarkable for their gigantic size, some of them attaining the height of 480 or 500 feet. In the Australian colonies they are known by the name of gum trees, from the gum which exudes from their trunks; individual species are known as 'stringy bark', 'iron bark', *karri*, or *jarrah*. The wood of some is excellent for building and many purposes. The *E. globūlus*, or blue gum, yields an essential oil which is valuable as a febrifuge, antasthmatic, and antispasmodic. The medicinal properties of this tree also make it useful as a disinfectant, and as an astringent in affections of the respiratory passages, being employed in the form of an infusion, a decoction, or an extract, and cigarettes made of the leaves being also



Eucalyptus globulus
1, Section of unopened flower.
2, Anthers. 3, Section of fruit.

smoked. The *E. globulus* and the *E. amygdalina* are found to have an excellent sanitary effect when planted in malarious districts such as the Roman Campagna, parts of which have already been reclaimed by their use. This result is partly brought about by the drainage of the soil (the trees absorbing great quantities of moisture), partly perhaps by the balsamic odour given out. *E. mannifera* and others yield a sweet secretion resembling manna. Some, especially *E. rostrata*, yield a kind of gum kino. The *Eucalyptus* has been introduced with success into India, Palestine, Algiers, and Southern France.

Eucharist (ū'ka-ris; Gr. *eucharistia*, from *eu*, well, and *charis*, grace), a name for the sacrament of the Lord's Supper, in reference to the blessing and thanksgiving which accompany it.

Eucharistic Congresses, gatherings of the Roman Catholic clergy and laity, held with the object of glorifying the Sacrament of the Eucharist, were inaugurated by Bishop de Sègur, of Lille. The first congress, held in that city (1881) excited little but local interest; but the movement rapidly developed, succeeding congresses being held at Avignon (1882), Liège (1883), Paris (1888), Jerusalem (1893), Lourdes (1899), Rome (1905), and elsewhere. In 1908 the congress held in London was attended by Cardinal Vannutelli, the first Papal legate to visit England for three centuries, by six other cardinals, fourteen archbishops, and seventy bishops. A proposal to carry the Sacrament through London in procession aroused much opposition, and the project was abandoned on the personal intervention of Mr. Asquith, then Premier.

Euchre (ū'kèr), a card-game very popular in America, is usually played by two or four persons. After the cut for deal five cards are dealt (either by twos and threes or by threes and twos) to each player, and the uppermost card of those undealt is turned up for trump. The first player has the option either to 'order up' (namely to make this card trump) or to pass. In the latter case it is left to the next player to decide if he will play first or pass, and so on till the turn of the dealer comes. He must either play on this trump or turn it down, when all the players have again in turn their choice of making a new trump or passing. If a trump is 'ordered up' or taken in the first round, the dealer may take it into his cards, discarding in its place his poorest card. If the player who elects to play wins five tricks, he counts two; if he wins three tricks, he counts one; if he wins fewer than three tricks, he is *euchred*, and each independent opponent counts two. The cards rank as at whist, except that the knave of the trump suit, called the *right bower* (from the Ger. *bauer*, a peasant), is the

highest card, the knave of the other suit of the same colour being the second highest.

Eucken, Rudolf Christoph, German philosopher and theologian, born in East Friesland in 1846. Educated at the Universities of Göttingen and Berlin, he was professor of philosophy at Basel from 1871 to 1874, when he obtained a similar appointment at Jena. Opposed both to utilitarianism and positivism. Eucken is one of the leaders of those German philosophers who maintain that the spiritual interests of man should be taken into consideration, and oppose the philosophic systems which treat life only from the physical and biological points of view. His spiritualistic philosophy has found many adherents, and his works are very popular. In 1908 he won the Nobel prize for literature, and in 1910 he was made a D.D. of the University of Glasgow. His works include: *The Life of the Spirit* (1909), *The Problem of Human Life as viewed by the Great Thinkers* (1909), *The Meaning and Value of Life* (1909), *Main Currents of Modern Thought* (1911), *Life's Basis and Life's Ideal* (1911), *Can we still be Christians?* (1913).

Euclid (*Eucleidēs*), of Alexandria, a distinguished Greek mathematician, who flourished about 300 B.C. His *Stoicheia* (Elements of Geometry), in thirteen books, are still extant, and form the most usual introduction to the study of geometry. The work was known to the Arabs, translations of it having appeared in the time of Harun-al-Rashid and of Al-Mamun. It was translated from the Arabic into Latin by Adelard of Bath, and an English translation from the Latin appeared in 1570. The severity and accuracy of Euclid's methods of demonstration have as a whole never been surpassed. Besides the *Elements*, some other works are attributed to Euclid.—Cf. R. S. Heath, *The Thirteen Books of Euclid's Elements*.

Euclid (*Eucleidēs*), of Megara, an ancient Greek philosopher, the founder of the Megaric school of philosophy, and a pupil of Socrates.

Eudiometer (Gr. *eudios*, serene), an instrument originally designed for ascertaining the purity of the air or the quantity of oxygen it contains, but now employed generally in the analysis of gaseous mixtures. It consists of a graduated glass tube, either straight or bent in the shape of the letter **U**, hermetically sealed at one end and open at the other. Two platinum wires, intended for the conveyance of electric sparks through any mixture of gases, are inserted through the glass near the closed end of the tube, and approach but do not touch each other. To determine the proportion of oxygen in a given

specimen of air, hydrogen is introduced into the tube with a measured volume of the air, and the mixture is fired by an electric spark. Water is formed, and the quantity of oxygen can be estimated from the diminution of volume. In a mixture of gases, chemical absorbents may be used to remove the gases one by one, the amounts present being determined by the successive changes of volume.

Eugene (ū-jēn'), or François Eugène, Prince of Savoy, fifth son of Eugène Maurice, Duke of Savoy-Carignan, and Olympia Mancini, a niece of Cardinal Mazarin. He was born at Paris 18th Oct., 1663, and died in Vienna 21st April, 1736. Offended with Louis XIV, he entered the Austrian service in 1683, serving his first campaign as a volunteer against the Turks. Here he distinguished himself so much that he received a regiment of dragoons. Later, at the sieges of Belgrade and Mayence, he increased his reputation, and on the outbreak of war between France and Austria he received the command of the Imperial forces sent to Piedmont to act in conjunction with the troops of the Duke of Savoy. At the end of the war he was sent as commander-in-chief to Hungary, where he defeated the Turks at the battle of Zenta (11th Sept., 1697). The War of the Spanish Succession brought Eugene again into the field. In Northern Italy he outmanœuvred Catinat and Villeroi, defeating the latter at Cremona (1702). In 1703 he commanded the Imperial army in Germany, and in co-operation with Marlborough frustrated the plans of France and her allies. In the battle of Blenheim, Eugene and Marlborough defeated the French and Bavarians under Marshal Tallard, 13th Aug., 1704. Next year Eugene, returning to Italy, forced the French to raise the siege of Turin, and in one month drove them out of Italy. During the following years he fought on the Rhine, took Lille, and, in conjunction with Marlborough, defeated the French at Oudenarde (1708), and Malplaquet (1709), where he himself was dangerously wounded. After the recall of Marlborough, which Eugene opposed in person at London, without success, and the defection of England from the alliance against France, his further progress was in a great measure checked. In the war with Turkey, in 1716, Eugene defeated two superior armies at Peterswardein and Temesvar, and, in 1717, took Belgrade, after having gained a decisive victory over a third army that came to its relief. During fifteen years of peace which followed, Eugene served Austria as faithfully in the Cabinet as he had done in the field. He was one of the great generals of modern times.—Cf. G. B. Malleson, *Prince Eugene of Savoy*.

Eugenia (so named in honour of Prince *Eugene*), a genus of Myrtaceæ, nearly related to the myrtle. It contains numerous species, some of which produce delicious fruits. Cloves are the dried flower-buds of *E. caryophyllata*.

Eugenics has been defined as "the study of agencies under social control that may improve or impair the racial qualities of future generations, either physically or mentally". It is concerned with the investigation of the physical, mental, and moral traits of mankind, and especially with the factors of inheritance of desirable and undesirable qualities. The interest in the subject is largely due to the untiring zeal of the late Sir Francis Galton, who devoted most of his life to the study of the manifold problems that came within the scope of 'eugenics', and, in accordance with the terms of his will (1908), founded the Galton Chair of Eugenics in the University of London. The library and laboratory of the Galton benefaction form part of the Department of Applied Statistics, under the direction of Professor Karl Pearson, F.R.S., at University College, London, who is also the editor of the journal *Biometrika*, which is devoted to the statistical side of the problems of anthropology and heredity. The aim of the Galton laboratory is to collect material relating to human heredity, and to investigate its significance; and also to extend the knowledge of eugenics by professional instruction, lectures, publications, and experimental work. The scope of its activities will best be appreciated by the study of such works as the late Sir Francis Galton's *Natural Inheritance* (1889) and *Essays in Eugenics* (1909), and Professor Karl Pearson's *Groundwork of Eugenics* (1909), *Practical Problems of Eugenics* (1909), and *State of National Eugenics* (1909). *The Treasury of Human Inheritance*, issued in parts from the Galton laboratory, is a monumental record of facts relating to the hereditary transmission of human qualities. The Eugenics Education Society, under the presidency of Major Leonard Darwin, has for its aim the stimulation of public interest in the subject, and the discussion of the problems of heredity. It issues a journal, *The Eugenics Review*, now in its twelfth year.

It has long been known that by means of careful selection of parents it was possible to breed horses, cattle, dogs, &c., and a great variety of food- and flowering-plants, with desirable qualities highly developed. But it is obvious that such direct methods cannot be applied to human beings for the purpose of breeding men and women with special traits. What the eugenic societies aim at doing is to educate the people to realize the far-reaching effects of the inheritance of good or bad qualities, in the hope that such knowledge may exert some influence in the choice of partners in matrimony. But their efforts

are especially directed to the exposure of the disastrous results that may ensue from the contamination of a family by the intermarriage of one of its members with an individual subject to some hereditary defect of a physical, mental, or moral nature.

The study of eugenics is intimately related to a wide range of subjects: to genetics, which explains the laws that govern the heredity of specific traits in man, and suggests certain practical applications of the rules of breeding to race improvement by cutting off undesirable strains and by selecting mates desirable from the eugenic standpoint; to the study of biographies of individuals and the genealogies of families, for the purpose of obtaining data for the investigation of the working of inheritance; to anthropology, history, and archæology, law and politics, economics and sociology, medicine and psychology, and statistical science.

The tremendous stimulus which the rapid development of eugenics has given to the wider recognition of the significance of heredity in human affairs has tended to obscure the importance of social environment and individual experience, especially in children of tender age, in shaping the attitude of the individual. Education is a vastly more important factor—the manner and attitude of the teacher, rather than the subject-matter of his or her lessons—than the eugenic enthusiasts, with their over-emphasis on the dominance of hereditary influences, are willing to admit. In the causation of many diseases, commonly reputed to be hereditary, such as tuberculosis and certain forms of insanity, the social and physical circumstances probably play a more important part than heredity in determining the onset of the illness, even when some undoubted hereditary aptitude to fall a victim to one or other of these affections is admitted. In no branch of medicine or sociology is this fallacy more fruitful of error than in the domain of mental disease. Apart from certain physical defects of the nervous system and specific infections, such as syphilis, the causes of mental alienation are to be sought rather in some maladjustment to the individual's social circumstances, often the result of some emotional disturbance, even in early childhood, which created the attitude of mind that eventually determined the mental conflict expressed by the insanity. The study of the effects of the strain of war has shown that anxiety, if sufficiently intense and prolonged, can produce mental disturbance in anyone, whatever his heredity and antecedents. By over-emphasizing the importance of inheritance in the causation of such conditions as insanity and epilepsy, and ignoring the effects of the profound social disturbance an insane parent may inflict upon

any home, and especially upon the impressionable minds of young children in it, the eugenic societies have been responsible for raising up a growing body of opposition to their views. Not only in the domains of medicine and psychology, but also in those of ethnology and sociology, there is a feeling that the eugenic claims have been pushed too far. But when the subject of eugenics has been pruned of these extravagances, it will exert a far-reaching influence upon social and political organization and events by compelling respect for the vast importance of heredity as a factor that plays some part in determining the physical, mental, and moral qualities of mankind. References to the voluminous literature will be found in *The Eugenics Review* (published by the Eugenics Education Society, Kingsway, London).

Eugénie (eu-zhā-nē), Marie de Guzman, ex-Empress of the French, born at Granada, in Spain, 5th May, 1826, died at Seville 11th July, 1920. Her father, the Count de Montijo, was of a noble Spanish family; her mother was of Scotch extraction, maiden name Kirkpatrick. On 29th Jan., 1853, she became the wife of Napoleon III and Empress of the French. On 16th March, 1856, a son was born of the marriage. When the war broke out with Germany, she was appointed regent (15th July, 1870) during the absence of the emperor, but on the 4th Sept. the Revolution forced her to flee from France. She went to England, where she was joined by the Prince Imperial and afterwards by the emperor. Camden House, Chislehurst, became the residence of the imperial exiles. On 9th Jan., 1873, the emperor died, and six years later the Prince Imperial was slain while with the British army in the Zulu War. In 1881 the empress transferred her residence to Farnborough, in Hampshire. During the European War she established a hospital at Farnborough. In 1918 she handed over to Clemenceau the letters which she had received from William I in 1870. The letters shed a striking light upon the ambitions of Prussia. She was buried in the mausoleum at Farnborough.—BIBLIOGRAPHY: De Lano, *The Empress Eugénie*; Tschuddi, *Eugéne, Empress of the French*; Stoddart, *The Life of Empress Eugénie*; E. Legge, *The Empress Eugénie and her Son*.

Euge'nius, the name of four Popes.—1. **Eugenius I**, elected 8th Sept., 654, while his predecessor, Martin I, was still living; died in 657 without having exerted any material influence on his times.—2. **Eugenius II** held the see from 824-827.—3. **Eugenius III**, born at Pisa, was a disciple of St. Bernard of Clairvaux. He was raised to the popedom in 1145; was obliged to quit Rome in 1146 in consequence of the commotions caused by Arnold of Brescia; returned with the help of King Roger of Sicily in 1150, and died in 1153.—4. **Eugenius**

IV, from Venice, originally called Gabriel Condolmero, was raised to the popedom in 1431. In consequence of his opposition to the Council of Basel he was deposed. He died in 1447.

Eugenol, or **Allylguaiacol**, is found in cloves, the leaves of cinnamon, and other plants. About 90 per cent of *clove oil* is composed of eugenol.

Eu'gubine Tables, the name given to seven bronze tablets or tables found in 1444 at the town of Gubbio, the ancient Iguvium or Eugubium, now in the Italian province of Perugia, bearing inscriptions in the language of the ancient Umbrians, which seems to have somewhat resembled the ancient Latin as well as the Oscan. They seem to have been inscribed three or four centuries B.C., and refer to sacrificial usages and ritual.

Euhem'erism, a method or system (so named from its founder Euhemerus, a Greek philosopher) of interpreting myths and mythological deities, by which they are regarded as deifications of dead heroes and poetical exaggerations of real histories.

Eulenspiegel (oi'len-spē-gl), Till, a name which has become associated in Germany with all sorts of wild, whimsical frolics, and with many amusing stories. Some such popular hero of tradition and folk-lore seems to have really existed in Germany, probably in the first half of the fourteenth century, and a collection of popular tales of a frolicsome character, originally written in Low German, purports to contain his adventures. The earliest edition of such is a Strasbourg one of the year 1515 in the British Museum. Better known, however, is that of 1519, published also at Strasbourg by Thomas Mürner (under the title *Howle-glass*). The work was early translated into English and almost all European tongues. A modern English translation appeared in 1890.

Euler (oi'lér or ü'lér), Leonard, a distinguished mathematician, born at Basel in 1707, died at St. Petersburg (Petrograd) in 1783. He was educated at the University of Basel under the Bernouillis, through whose influence he procured a place in the Academy of St. Petersburg. In 1741 he accepted an invitation from Frederick the Great to become professor of mathematics in the Berlin Academy, but in 1766 returned to St. Petersburg, where he became director of the mathematical class of the academy. Euler's profound and inventive mind gave a new form to the science. He applied the analytic method to mechanics, and greatly improved the integral and differential calculus. He also wrote on physics, and employed himself in metaphysical and

philosophical speculations. Amongst his numerous writings are: the *Theoria Motuum Planetarum et Cometarum*, *Introductio in Analysin Infinitorum*, and *Opuscula Analytica*.

Eu'menes (-nēz), the name of two kings of Pergamus.—1. **Eumenes I** succeeded his uncle Philetærus 263 B.C. He reigned for twenty-two years, and then died in a fit of drunkenness.—2. **Eumenes II** succeeded his father Attalus 197 B.C., and, like him, attached himself to the Romans, who, as a reward for his services in the war against Antiochus of Syria, bestowed upon him the Thracian Chersonesus and almost all Asia on this side of the Taurus. He died in 159 B.C.

Eumenides (ū-men'i-dēz). See *Furies*.

Eumycetes, or **Higher Fungi**, a common name for those Fungi which possess a septate mycelium. They also have a well-marked type of 'principal' spore—either the *ascospore* (Ascomycetes) or the *basidiospore* (Basidiomycetes)—and rarely produce definite sexual organs. Opposed to Phycomycetes.

Eunomians, the followers of Eunomius, Bishop of Cyzicum, in the fourth century A.D., who held that Christ was a created being of a nature unlike that of the Father.

Eu'nuch, an emasculated male. The term is of Greek origin (*eunouchos*, from *eunē*, a couch or bed, *echein*, to hold or guard); but eunuchs became known to the Greeks no doubt from the practice among Eastern nations of having them as guardians of their women's apartments. Eunuchs were employed in somewhat similar duties among the Romans in the luxurious times of the empire, and under the Byzantine monarchs they were common. The Mohammedans still have them about their harems. Emasculation, when effected in early life, produces singular changes in males and assimilates them in some respects to women, causing them in particular to have the voice of a female. Hence it was not uncommon in Italy to castrate boys in order to fit them for soprano singers when adults.

Euon'yimus, the spindle trees or prickwoods, a genus of shrubs or trees, nat. ord. Celastrineæ, containing about fifty species, natives of the temperate regions of the northern hemisphere. The root-bark of *E. atropurpureus* is the source of euonymin, a bitter principle with a powerful stimulating effect on the liver.

Eupato´ria, formerly Koslov, a seaport on the western coast of the Crimea, government of Taurida. It was here that the allied forces landed at the commencement of the Crimean War (14th to 18th Sept., 1854). Pop. 30,432.

Eupato´rium, a genus of plants, chiefly natives of America, belonging to the nat. ord. Compositæ. Their roots are perennial, possessing a rough, bitter, or aromatic taste; the flowers are small, white, reddish, or bluish, in corymbs. Amongst the many species are *E. cannabīnum*, or hemp-agrimony, a British plant.

Eupen (oi´pen), a town and district of Belgium, formerly part of Rhenish Prussia, 7 miles s.s.w. of Aix-la-Chapelle. It has manufactures of woollen and linen cloth, hats, soap, leather, and chemicals; paper, flax, and worsted mills; and an important trade. The town was ceded to Prussia at the Peace of Paris in 1814. On 26th May, 1919, Eupen was occupied by Belgian troops, and by the Treaty of Versailles Eupen and Malmédy were handed over to Belgium. Pop. 13,540.

Eupho´nium, a brass bass instrument, generally introduced into military bands, and frequently met with in the orchestra as a substitute for the superseded ophicleide. It is one of the saxhorn family of instruments. It is tuned in C or in B flat, and is furnished with three or four valves or pistons.

Euphorbia. See *Spurge*.

Euphorbia´ceæ, the spurgevents, a nat. ord. of herbaceous plants, shrubs, or very large trees, which occur in all regions of the globe. Most of them have an acrid milky juice, and diclinous or monœcious flowers. The fruit is dry or slightly fleshy, and three-lobed. Among the genera are: *Euphorbia*, which yields an oil used as a powerful cathartic; *Croton*, affording croton-oil; the *Ricīnus commūnis*, or castor-oil plant; the *Buxus sempervirens*, or box-wood plant; the *Manihot utilissima*, which yields the food known as tapioca or cassava. In most members of the genera the milky juice contains caoutchouc.

Euphor´bium, a yellowish-white body, which is the solidified juice of certain plants of the genus *Euphorbia*, either exuding naturally or from incisions made in the bark. It is a powerfully acrid substance, virulently purgative and emetic.

Euphra´tes, or **El Frat**, a celebrated river of Western Asia, Mesopotamia, having a double source in two streams rising in the Anti-Taurus range. Its total length is about 1750 miles, and the area of its basin 260,000 sq. miles. It flows

mainly in a south-easterly course through the great alluvial plains of Babylonia and Chaldæa till it falls into the Persian Gulf by several mouths, of which only one in Persian territory is navigable. About 100 miles from its mouth it is joined by the Tigris, when the united streams take the name of Shatt-el-Arab. It is navigable for about 1200 miles, but navigation is somewhat impeded by rapids and shallows. The melting of snow in the Taurus and Anti-Taurus causes a flooding in spring. The water is highest in May and June, when the current, which rarely exceeds 3 miles an hour, rises to 5. In the Bible (*Gen. xv, 18*) the Euphrates is *The River*, or *The Great River*.

Eu'phuism (Gr. *euphues*, well endowed by nature), an affected style of speech which distinguished the conversation and writings of many of the wits of the court of Queen Elizabeth. The name and the style were derived from *Euphues, the Anatomy of Wit* (about 1580), and *Euphues and his England* (about 1582), both written by John Lyly (1554-1606). A well-known euphuist in fiction is Sir Piercie Shafton in Scott's *Monastery*. Scott, however, had not studied Lyly sufficiently, and Sir Piercie raves bombastically rather than talks euphuistically. The chief characteristics of genuine euphuism were extreme artificiality and numerous allusions to natural history embellished by imagination.

Eu'polis, an Athenian comic poet, who flourished about 429 B.C. Neither the date of his birth nor that of his death is known with certainty. He belongs, like Aristophanes and Cratinus, to the Old Comedy. His works are all lost except small fragments. According to Suidas, he produced seventeen plays, seven of which won the first prize. His best-known plays are the *Kolakes* (Flatterers), in which he attacked the prodigal Callias, and the *Baptæ* (Dippers), in which he attacked Alcibiades and the exotic ritual practised at his clubs.

Eura'sians (syncopated from European-Asians), a name euphemistically given to the 'half-castes' of India, the offspring of European fathers and Indian mothers. They are particularly common in the three presidential capitals—Calcutta, Madras, and Bombay. Belonging strictly to neither race, Eurasians are not infrequently ostracized by both; and their anomalous position often exerts a baneful influence upon their character. They generally receive a European education, and the young men are often engaged in Government or mercantile offices. The girls, in spite of their dark tint, are generally very pretty and often marry Europeans.

Eure (*eur*), a river of North-West France, which rises in the department of the Orne, and falls into the Seine after a course of 124 miles, being navigable for about half the distance. It gives its name to a department in the north-west of France, forming part of Normandy; area, 2330 sq. miles. The surface consists of an extensive plain, intersected by rivers, chief of which is the Seine. It is extensively cultivated; apples, pears, plums, and cherries form important crops, and a little wine is produced. The mining and manufacturing industries are extensive, and the department has a considerable trade in woollen cloth, linen and cotton fabrics, carpets, leather, paper, glass. Evreux is the capital. Pop. 303,092.

Eure-et-Loir (*eur-è-lwär*), a department in the north-west of France, forming part of the old provinces of Orléannais and Île-de-France; area, 2293 sq. miles. A ridge of no great height divides the department into a north and a south basin, traversed respectively by the Eure and the Loire. The soil is extremely fertile, and there is scarcely any waste land. A considerable portion is occupied by orchards and vineyards, but the greater part is devoted to cereal crops. The department is essentially agricultural, and has few manufactures. The capital is Chartres. Pop. 251,259.

Eure'ka (Gr. *heurēka*, I have found it), the exclamation of Archimedes when, after long study, he discovered a method of detecting the amount of alloy in King Hiero's crown. Hence the word is used as an expression of triumph at a discovery or supposed discovery.

Eurhythmics, a general term, but usually used to denote a system of education evolved by Émile Jaques-Dalcroze of Geneva. This form of training bears on all art, but especially on the art of music. Eurhythmics is essentially an original contribution to education. It aims at training musical sense on the broadest lines, using the body as an instrument of expression. Breaking away from preconceived ideas of music as a phenomenon of sound only, M. Dalcroze claims that music is innate. From this standpoint it follows that musicality as such is capable of cultivation apart from instrumental performance. Rhythm, not being a quality confined to music, but found common to all art, and fundamental to life, can, therefore, be developed from within the human being. This the Dalcroze system claims to do. Rhythm of sound plays a leading part in that it is allied to movement. Exercises at the piano are played to which the pupil listens, and to which he responds in movement—movement so closely allied to the music that it is a form of musical imagery. The technique is developed on simple lines to serve this end only. The system is progressive,

starting from elementary rhythmic structure, and ending with complete musical form. It is far-reaching in educative purpose. It claims to free innate rhythm, to develop it for individual self-expression; to bring mind and body into closer unity, and in their interaction to give poise to both; to train accurate musical listening, ready assimilation of musical language and its spontaneous translation into terms of movement; to give musical experiences which shall be heard and felt; to cultivate musical expression and creation (in movement); to blend self-discipline with emotion.

Euripides, the last of the three great Greek writers of tragedies, was born about 480 B.C., and died 406 B.C. Tradition declares that he was born at Salamis, on the very day of the Greek naval victory there. He was, as far as we can tell, of good birth; at any rate, he was well educated, and was able to live a life of ease and leisure, and to collect one of the largest libraries of the time. The comic poets, especially Aristophanes, delighted to say that his mother, Cleito, was a cabbage-woman, but there is probably little or no truth in this statement. Euripides was originally trained as an athlete, but conceived an intense dislike for that occupation. Greatly daring, he expressed his view openly (Fragment 284). Like a popular modern dramatist, his recreation was probably 'anything except sport'. He then took to painting, but abandoned it in favour of writing tragedies. His first play (not preserved), the *Peliades*, was produced when he was twenty-five years of age. He is said to have written ninety-two dramas, eight of which were satyr-plays. Ancient critics allow seventy-five of these to have been genuine. During his long career he only won the first prize five times. Euripides did not take any part in public life, but devoted himself entirely to a life of speculation and to writing plays. There is a tradition, not, however, on a very firm basis, that he was twice married, and that both marriages were failures. He is represented by Aristophanes as a woman-hater, but indeed he portrays women more sympathetically than Æschylus or Sophocles. The women had little cause to congratulate themselves on securing Aristophanes as a champion, for his scorpions are far more stinging than Euripides' whips. Euripides left Athens about 409 B.C., and went to the court of King Archelaus in Macedonia. There he died in 406 B.C.; according to some accounts, he was killed by savage dogs which were set on him by some of his rivals at the king's court.

Seventeen tragedies and one satyr-play have been preserved to us. The latter (*The Cyclops*) is interesting as being the only example of a satyr-play which we possess. In itself it is not amusing. It has been admirably translated by

Shelley. The seventeen tragedies in the order of their production are: *Alcestis*, *Medea*, *Hippolytus*, *Hecuba*, *Andromache*, *Ion*, *Suppliants*, *Heracleidæ*, *Hercules Furens*, *Iphigenia among the Tauri*, *Trojan Women*, *Helena*, *Phænissæ*, *Electra*, *Orestes*, *Iphigenia at Aulis*, and *The Bacchæ*. The *Rhesus*, a feeble production long attributed to Euripides, is almost certainly not his work.

The work of Euripides still retains the power of arousing strong likes and dislikes. He has had sturdy supporters and fanatical detractors. The truth is that if the tragedies of Æschylus and Sophocles are looked upon as models for all Greek tragedy, Euripides falls far short of his models. Euripides, however, though he died shortly before Sophocles, belonged to a younger and quite different generation, and held different views about art, morality, religion, and almost everything of importance. His aim was rather different from that of the earlier poets, and he must be judged, not by their standards, but on his own merits. His own merits are amply sufficient to justify the high opinion held of him in the ancient world, and supported by many of the greatest of the moderns. The dethroning of Euripides was the result of a German conspiracy, carried out with much energy by Niebuhr, and with even more by Schlegel. They enjoyed themselves while pulling Euripides to pieces much as schoolboys who have detected a flaw in the armour of their master. Many proofs can be adduced that Euripides was not a sophistical trifler; but one glance at his bust is enough to assure anyone of unbiased judgment that he was a man of remarkable breadth of mind and intellectual gifts. The fact remains, however, that the extant plays of Euripides are of very unequal merit. The *Helena* is not a good play; it was ridiculed by Aristophanes, but he did not succeed in making it much more absurd than it was already. The *Hecuba* and the *Heracleidæ* are not well constructed, and the *Electra* and *Orestes* challenge too directly the masterpieces of the earlier tragedians. In his greatest plays, however, Euripides can bear comparison with any poet. The *Medea* is a play which still never fails to please; the *Hippolytus* and the *Ion* are admirable dramas and admirably constructed; above all, the *Bacchæ* is a masterpiece, more picturesque than any other Greek tragedy, a play not unworthy to be set near *The Tempest* and *Cymbeline*.

Euripides has been accused by his detractors of degrading his art, because he opened his plays with a prologue and ended them with the intervention of a god. Both devices, if not desirable, are quite pardonable. Possible plots were becoming more and more scarce; Euripides did not wish to adopt trite themes,

and so went into the by-ways of mythology, or adopted a less well-known alternative version of a well-known legend. He could not count on his audience already possessing enough knowledge of the story to enable them to understand his plays without a prologue. The *deus ex machina*, as the god who ends some of the plays is called, was often warranted or required by the plot which called for a conventional ending. Euripides has also been accused, by Aristophanes and by many less entertaining writers, of taking away all the dignity of tragedy. It is quite true that he is a realist. Sophocles represented men as they ought to be, Euripides represented them as they were. This was an unforgiveable offence in the eyes of the 'men of Marathon' at Athens. The tragic heroes were not mere stage characters, they considered; they were often ancestors or national heroes, and it was impious to represent them as speaking ordinary language, or sharing the weaknesses of ordinary men. Euripides did do this, did it intentionally, and did it excellently. He came at an awkward transition period, and the lack of success of some of his work is owing to the impossibility of pouring new wine into old bottles. The old tragedy was too tightly bound by convention to suit Euripides, who wished to portray living men and women, and to have an exciting plot. The new comedy—the romantic comedy of Menander—had not yet been invented. Had it been, Euripides would surely have written comedies. The comic poets of the next century turned to him for a model, and it was one of them, Philemon, who said that if he were quite sure that dead men retained their perception he would hang himself to see Euripides. Euripides is, in fact, the earliest writer of romantic plays, a fact well illustrated by his *Alcestis*, which is one of his best plays. In it tragedy and comedy are harmoniously blended, and it has a happy ending.

For better and for worse Euripides is a very modern poet, and makes a special appeal to the present generation. But his pathos, his wide sympathies, and his wonderful poetry have appealed to the best judges in all ages. Theocritus, Virgil, Ovid, Horace, Milton, and Browning have been among his admirers; his detractors include a few Teutonic professors, and a few who honour the memory of Æschylus and Sophocles on the other side idolatry.—BIBLIOGRAPHY: A. W. Verrall, *Euripides the Rationalist*; G. G. A. Murray, *Euripides and his Age* (Home University Library); W. B. Donne, *Euripides* (Ancient Classics for English Readers); Sir J. P. Mahaffy, *Euripides: an Account of his Life and Works*; N. J. Patin, *Étude sur Euripide*; P. Masqueray, *Euripide et ses idées*. There is a complete verse translation by A. S. Way, and verse translations of several plays by G. G. A. Murray. There is a 'transcript' of the *Alcestis* in

Browning's *Balaustion's Adventure*, and of the *Hercules Furens* in his *Aristophanes' Apology*.

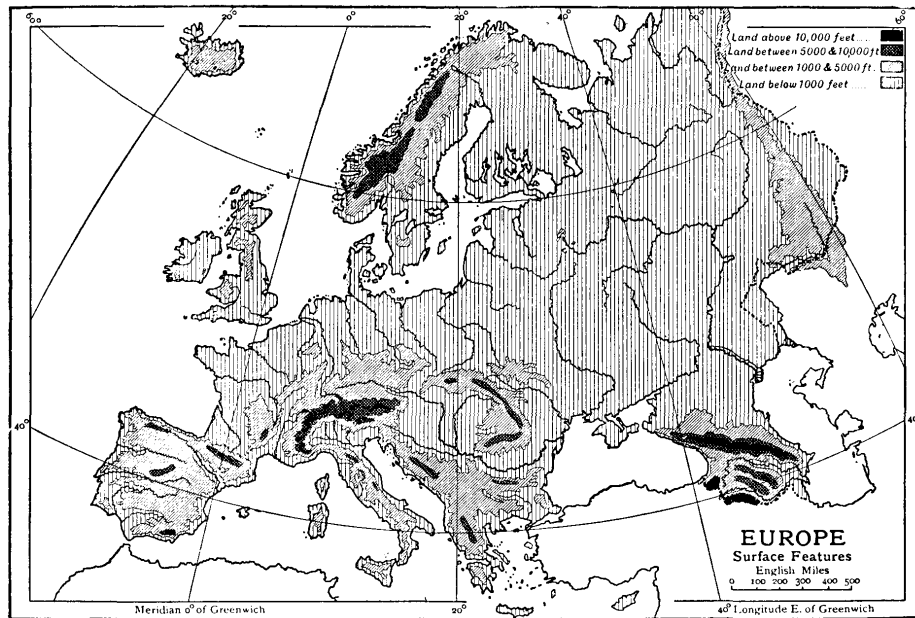
Euripus (ū-rī'pus), in ancient geography, the strait between the Island of Eubœa and Bœotia in Greece.

Euroc'lydon, a tempestuous wind of the Levant, which was the occasion of the shipwreck of the vessel in which St. Paul sailed, as narrated in *Acts*, xxvii, 14-44. The north-east wind is the wind evidently meant in the narrative; and an alternative reading adopted in the revised version is *euraculon* (euraquilo) or north-easter.

Euro'pa, in Greek mythology, the daughter of Agēnor, King of the Phœnicians, and the sister of Cadmus. The fable relates that she was abducted by Jupiter, who for that occasion had assumed the form of a white bull, and swam with his prize to the Island of Crete. Here Europa bore to him Minos, Sarpēdon, and Rhadamanthus.

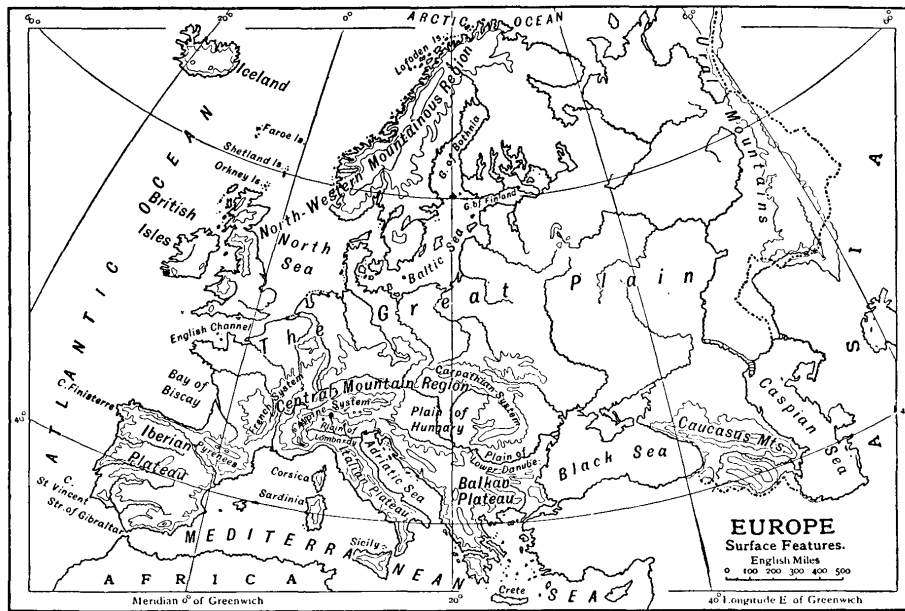
Europe, the smallest of the great continents, but the most important in the history of civilization for the last two thousand years. It forms a huge peninsula projecting from Asia, and is bounded on the north by the Arctic Ocean; on the west by the Atlantic Ocean; on the south by the Mediterranean, the Black Sea, and the Caucasus Range; on the east by the Caspian Sea, the Ural River, and the Ural Mountains. The most northerly point on the mainland is Cape Nordkyn, in Lapland, in lat. 71° 6'; the most southerly points are Punta da Tarifa, lat. 36° N., in the Strait of Gibraltar, and Cape Matapan, lat. 36° 17', which terminates Greece. The most westerly point is Cape Roca in Portugal, in long. 9° 28' w., while Ekaterinburg is in long. 60° 36' E. From Cape Matapan to North Cape is a direct distance of 2400 miles, from Cape St. Vincent to Ekaterinburg, north-east by east, 3400 miles; area of the continent, about 3,865,000 sq. miles. Great Britain and Ireland, Iceland, Novaya Zemlya, Corsica, Sardinia, Sicily, Malta, Crete, the Ionian and the Balearic Islands are the chief islands of Europe. The shores are very much indented, giving Europe an immense length of coast-line (estimated at nearly 50,000 miles). The chief seas or arms of the sea are: the White Sea on the north; the North Sea on the west, from which branches off the great gulf or inland sea known as the Baltic; the English Channel, between England and France; the Mediterranean, communicating with the Atlantic by the Strait of Gibraltar (at one point only 19 miles wide); the Adriatic and Archipelago, branching off from the

Mediterranean: and the Black Sea, connected with the Archipelago through the Hellespont, Sea of Marmora, and Bosphorus.



Surface.—The mountains form several distinct groups or systems of very different geological dates, the loftiest mountain masses being in the south central region. The Scandinavian mountains in the north-west, to which the great northern peninsula owes its form, extend above 900 miles from the Polar Sea to the south point of Norway. The highest summits are about 8000 feet. The Alps, the highest mountains in Europe (unless Mount Elbruz in the Caucasus is claimed as European), extend from the Mediterranean first in a northerly and then in an easterly direction, and attain their greatest elevation in Mont Blanc (15,780 feet), Monte Rosa, and other summits. Branching off from the Alps, though not geologically connected with them, are the Apennines, which run south-east through Italy, constituting the central ridge of the peninsula. The highest summit is Monte Corno (9541 feet). Mount Vesuvius, the celebrated volcano in the south of the peninsula, is quite distinct from the Apennines. By south-eastern extensions the Alps are connected with the Balkan and the Despoto-Dagh of the south-eastern peninsula of Europe. Among the mountains of South-Western Europe are several massive chains, the loftiest summits being in the Pyrenees, and in the Sierra Nevada in the south of the Iberian Peninsula. The highest point in the former, La Maladetta or Mount Maudit, has an elevation of 11,165 feet; Mulahacen, in the latter, is

11,703 feet, and capped by perpetual snow. West and north-west of the Alps are the Cevennes, Jura, and Vosges; north and north-east, the Harz, the Thüringerwald Mountains, the Fichtelgebirge, the Erzgebirge and Böhmerwaldgebirge. Farther to the east the Carpathian chain encloses the great plain of Hungary, attaining an elevation of 8000 or 8500 feet. The Ural Mountains between Europe and Asia reach the height of 5540 feet. Besides Vesuvius, other two volcanoes are Etna in Sicily, and Hecla in Iceland. A great part of Northern and Eastern Europe is level. The *great plain* of North Europe occupies part of France, Western and Northern Belgium, Holland, the northern provinces of Germany, and the greater part of Russia. A large portion of this plain, extending through Holland and North Germany, is a low sandy level not infrequently protected from inroads of the sea only by means of strong dykes. The other great plains of Europe are the Plain of Lombardy (the most fertile district in Europe) and the Plain of Hungary. Part of Southern and South-Eastern Russia consists of steppes.



Rivers and Lakes.—The main European watershed runs in a winding direction from south-west to north-east, at its north-eastern extremity being of very slight elevation. From the Alps descend some of the largest of the European rivers, the Rhine, the Rhône, and the Po, while the Danube, a still greater stream, rises in the Black Forest north of the Alps. The Volga, which enters the Caspian Sea, an inland sheet without outlet, is the longest of European rivers, having a direct length of nearly 1700 miles, including windings 2400 miles. Into the Mediterranean flow the Ebro, the Rhône, and the Po; into the Black Sea, the Danube, Dnieper, Dniester, and Don (through the Sea of Azov); into the Atlantic, the Guadalquivir, the Guadiana, the Tagus, and Loire: into the English Channel, the Seine; into the North Sea, the Rhine, Elbe; into the Baltic, the Oder, the Vistula, and the Duna; into the Arctic Ocean, the Dvina. The lakes of Europe may be divided into two groups, the southern and the northern. The former run along both sides of the Alps, and among them, on the north side, are the lakes of Geneva, Neuchâtel, Thun, Lucerne, Zürich, and Constance; on the south side, Lago Maggiore, and the lakes of Como, Lugano, Iseo, and Garda. The northern lakes extend across Sweden from west to east, and on the east side of the Baltic a number of lakes, stretching in the same direction across Finland on the borders of Russia, mark the continuation of the line of depression. It is in Russia that the largest European lakes are found—Lakes Ladoga and Onega.

Geology.—The geological features of Europe are exceedingly varied. The older formations prevail in the northern part as compared with the southern half and the middle region. North of the latitude of Edinburgh and Moscow there is very little of the surface of more recent origin than the strata of the Upper Jura belonging to the Mesozoic period, and there are vast tracts occupied either by eruptive rocks or one or other of the older sedimentary formations. Denmark belongs to the Cretaceous period, as does also a large part of Russia between the Volga and the basin of the Dnieper. Middle and Eastern Germany, with Poland and the valley of the Dnieper, present on the surface Eocene formations of the Tertiary period. The remainder of Europe is remarkable for the great diversity of its superficial structure, rocks and deposits belonging to all periods being found within it, and having for the most part no great superficial extent. Europe possesses abundant stores of those minerals which are of most importance to man, such as coal and iron, Britain being particularly favoured in this respect. Coal and iron are also obtained in France, Belgium, and Germany. Gold is found to an unimportant extent, and silver is widely spread in small quantities. The richest silver ores are in Norway, Spain, the Erzgebirge, and the Harz Mountains. Spain is also rich in quicksilver. Copper ores are abundant in the Ural Mountains, Thuringia, Cornwall, and Spain. Tin ores are found in Cornwall, the Erzgebirge, and Brittany.



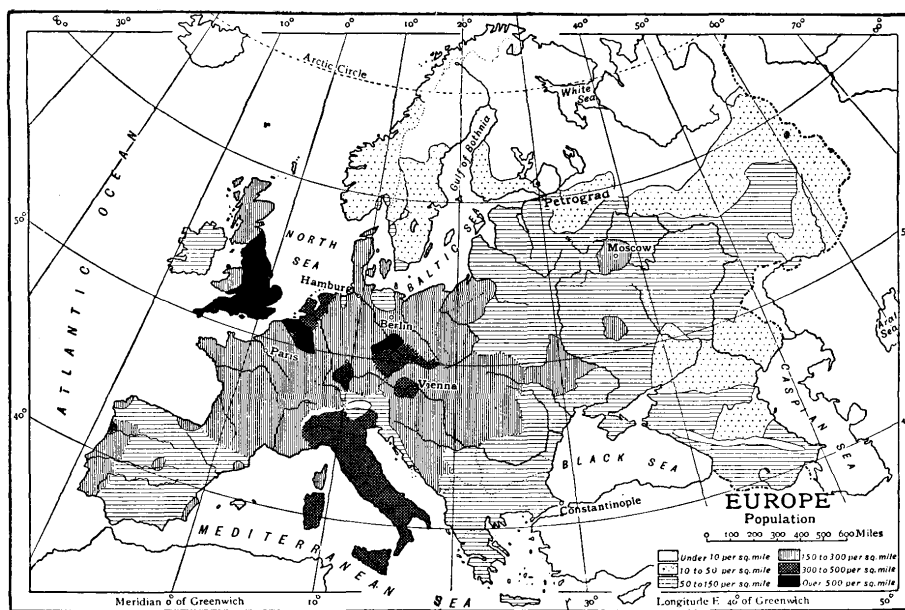
Climate.—Several circumstances concur to give Europe a climate peculiarly genial, such as its position almost wholly within the temperate zone, and the great extent of its maritime boundaries. Much benefit is also derived from the fact that its shores are exposed to the warm marine currents and warm winds from the south-west, which prevent the formation of ice on most of its northern shores. The eastern portion has a less favourable climate than the western. The extremes of temperature are greater, the summer being hotter and the winter colder, while the lines of equal mean temperature decline south as we go east. The same advantages of mild and genial temperature which western has over eastern Europe, the continent collectively has over the rest of the Old World. The diminution of mean temperature, as well as the intensity of the opposite seasons, increases as we go east. Peking, in lat. 40° N., has as severe a winter as Petrograd in lat. 60° N.

Vegetable Productions.—With respect to the vegetable kingdom, Europe may be divided into four zones. The first, or most northern, is that of fir and birch. The birch reaches almost to North Cape; the fir ceases a degree farther south. The cultivation of grain extends farther north than might be supposed. Barley ripens even under the seventieth parallel of north latitude; wheat ceases at 64° in Norway, 62° in Sweden. Within this zone, the southern limit of which extends from lat. 64° in Norway to lat. 62° in Russia, agriculture has little importance, its inhabitants being chiefly occupied with the care of reindeer or cattle, and in fishing. The next zone, which may be called that of the oak and beech, and cereal produce, extends from the limit above mentioned to the forty-eighth parallel. The Alps, though beyond the limit, by reason of their elevation belong to this zone, in the moister parts of which cattle husbandry has been brought to perfection. Next we find the zone of the chestnut and vine, occupying the space between the forty-eighth parallel and the mountain chains of Southern Europe. Here the oak still flourishes, but the pine species become rarer. Rye, which characterizes the preceding zone on the continent, gives way to wheat, and in the southern portion of it to maize also. The fourth zone, comprehending the southern peninsulas, is that of the olive and evergreen woods. The orange flourishes in the southern portion of it, and rice and even cotton are cultivated in some places in Italy and Spain.

Animals.—As regards animals, the reindeer and polar-bears are peculiar to the north. In the forests of Poland and Lithuania the urus, a species of wild ox, is still occasionally met with. Bears and wolves still inhabit the forests and mountains; but, in general, cultivation and population have expelled wild animals. The domesticated animals are nearly the same throughout. The ass and mule lose their size and beauty north of the Pyrenees and Alps. The Mediterranean Sea has many species of fish, but no great fishery; the northern seas, on the other hand, are annually filled with countless shoals of a few species, chiefly the herring, mackerel, cod, and salmon.

Inhabitants.—Europe is occupied by several different peoples or races, in many parts now greatly intermingled. The Celts once possessed the west of Europe from the Alps to the British Islands. But the Celtic nationalities were broken by the wave of Roman conquest, and the succeeding invasions of the Germanic tribes completed their political ruin. At the present day the Celtic language is spoken only in the Scottish Highlands (Gaelic), in some parts of Ireland (Irish), in Wales (Cymric), and in Brittany (Armorican). Next to the Celtic comes the Teutonic race, comprehending the Germanic and

Scandinavian branches. The former includes the Germans, the Dutch, and the English. The Scandinavians are divided into Danes, Swedes, and Norwegians. To the east, in general, of the Teutonic race, though sometimes mixed with it, come the Slavonians, that is, the Russians, the Poles, the Czechs or Bohemians, the Serbians, Croatians, &c. In the south and south-east of Europe are the Greek and Latin peoples, the latter comprising the Italians, French, Spanish, and Portuguese. All the above peoples are regarded as belonging to the Indo-European or Aryan stock. To the Mongolian stock belong the Turks, Finns, Lapps, and Magyars or Hungarians, all immigrants into Europe in comparatively recent times. The Basques at the western extremity of the Pyrenees are a people whose affinities have not yet been determined. The total population of Europe is about 400 millions; nine-tenths speak the languages of the Indo-European family, the Teutonic group, the Slavonic, and the Latin. The prevailing religion is the Christian, embracing the Roman Catholic Church, which is the most numerous, the various sects of Protestants (Lutheran, Calvinistic, Anglican, Baptists, Methodists, &c.), and the Greek Church. A part of the inhabitants profess the Jewish, a part the Mohammedan religion.



Political Divisions.—In 1921 Europe consisted of the following independent states, kingdoms, or republics: Albania, Armenia, Austria, Azerbaijan, Belgium, Bulgaria, Czecho-Slovakia, Denmark, Esthonia, Finland, France, Georgia, Germany, Great Britain and Ireland, Greece, Hungary, Italy, Latvia, Lithuania, Luxemburg, Monaco, Montenegro, Netherlands (Holland), Norway, Poland,

Portugal, Roumania, Russia, Serbia (Yugo-Slavia), Spain, Sweden, Switzerland, Turkey, Ukraine. Of these the following were kingdoms: Belgium, Bulgaria, Denmark, Great Britain, Greece, Italy, Montenegro, Netherlands (Holland), Norway, Serbia (Yugo-Slavia), Spain, and Sweden. Turkey (whose possessions in Europe were limited to Constantinople) was an empire, Luxemburg was a grand-duchy, Liechtenstein and Monaco principalities, whilst all the other European states were republics.

History.—Europe was probably first peopled from Asia, but at what date we know not. The first authentic history begins in Greece at about 776 B.C. Greek civilization was at its most flourishing period about 430 B.C. After Greece came Rome, which by the early part of the Christian era had conquered Spain, Greece, Gaul, Helvetia, Germany between the Danube and the Alps, Illyria, and Dacia. Improved laws and superior arts of life spread with the Roman Empire throughout Europe, and the unity of government was also extremely favourable to the extension of Christianity. With the decline of the Roman Empire a great change in the political constitution of Europe was produced by the universal migration of the northern nations. The Ostrogoths and Lombards settled in Italy, the Franks in France, the Visigoths in Spain, and the Anglo-Saxons in South Britain, reducing the inhabitants to subjection, or becoming incorporated with them. Under Charlemagne (771-814) a great Germanic empire was established, so extensive that the kingdoms of France, Germany, Italy, Burgundy, Lorraine, and Navarre were afterwards formed out of it. About this time the northern and eastern nations of Europe began to exert an influence in the affairs of Europe. The Slavs, or Slavonians, founded kingdoms in Bohemia, Poland, Russia, and the north of Germany; the Magyars appeared in Hungary; and the Normans agitated all Europe, founding kingdoms and principalities in England, France, Sicily, and the East. The Crusades and the growth of the Ottoman power are amongst the principal events which influenced Europe from the twelfth to the fifteenth century. The conquest of Constantinople by the Turks (1453), by driving the learned Greeks from this city, gave a new impulse to letters in Western Europe, which was carried onwards by the invention of printing and the Reformation. The discovery of America was followed by the temporary preponderance of Spain in Europe, and next of France. Subsequently Prussia and Russia gradually increased in territory and strength. The French revolution (1789) and the Napoleonic wars had a profound effect on Europe, the dissolution of the old German Empire being one of the results. The most important events in European history from the revolution of 1789 to 1914, the beginning of the European War, were: the

establishment of the independence of Greece; the disappearance of Poland as a separate state; the unification of Italy under Victor Emmanuel; the Franco-German War, resulting in the consolidation of Germany into an empire under the leadership of Prussia; and the partial dismemberment of the Turkish Empire. The European War, 1914-8 (q.v.), revolutionized the continent and altered the map of Europe. The chief results were the disintegration of the Dual Monarchy and of Russia, the abolition of the German Empire, and the deposition of hereditary rulers in the smaller German states, which instituted republican Governments. The following new states were formed from the constituent parts of Russia and Austria-Hungary: Albania, Armenia, Azerbaijan, Czecho-Slovakia, Esthonia, Finland, Georgia, Latvia, Lithuania, Yugo-Slavia, and Ukraine. Poland, dissolved in the eighteenth century, was again reconstituted. France regained Alsace and Lorraine, Turkey lost almost all her possessions in Europe, whilst Belgium, Denmark, Greece, Italy, and Roumania were greatly enlarged, acquiring new territories. All these alterations of boundaries and additions of territories were based on ethnological grounds, the new states being inhabited by peoples belonging to the same ethnical group and speaking the same language. See articles on the various countries.—BIBLIOGRAPHY: E. A. Freeman, *General Sketch of European History*; A. Hassall (editor), *Periods of European History*; *European History Chronologically Arranged*; A. S. Rappoport, *History of European Nations*; O. Browning, *General History of the World*; H. S. Williams, *The Historian's History of the World*.

European War, 1914-8. The European War, which began in Aug., 1914, and involved the greater part of the globe before the last shot was fired in Nov., 1918, had its ostensible origin in the assassination of the Austrian heir-apparent, the Archduke Franz Ferdinand, and his wife, at Serajevo, capital of Bosnia, once part of the ancient kingdom of Serbia. This crime was committed by a Bosnian student, but Austria-Hungary held Serbia responsible, and, inspired by Germany, sent an ultimatum on 23rd July, amounting to a demand that Serbia should surrender her independence. Two days later, notwithstanding that Serbia conceded every demand, with two reservations which she offered to submit to the Hague Tribunal, Austria-Hungary declared war on her. Germany, who had seen in the Serajevo tragedy a pretext for making her long-premeditated bid for world dominion, "knew very well what she was about in backing up Austria-Hungary in this matter", as the German Ambassador in Vienna frankly told the British representative at the time; and when Russia, as the traditional protector of the Slavs, mobilized her southern

armies to save Serbian independence if necessary, she threatened instant mobilization on her own part unless Russia stopped these military measures within twelve hours. It was technically impossible for Russia to do anything of the kind, but her protest to this effect was unavailing. Germany declared war on Russia on 3rd Aug., and as this inevitably involved war at the same time with Russia's ally France, she sent a note on the following day to Belgium demanding safe passage for German troops through Belgian territory, though Prussia as well as Great Britain, France, Russia, and Austria had guaranteed the neutrality and independence of Belgium by the treaty of 1839, repeatedly confirming this on subsequent occasions. When the British Ambassador in Berlin protested against the threatened violation of treaty rights, the German Chancellor, Bethmann-Hollweg, repudiated the treaty as a mere "scrap of paper".

On 3rd Aug., when Germany formally declared war on France—though her troops had already invaded French territory at various points—Belgium refused Germany's demands, and called on Great Britain and France for assistance. It was this call, and Germany's refusal on the following day to accede to the British demands that Belgian neutrality should be respected—declaring war on Belgium instead and violating her territory early that morning—which decided Great Britain to range herself wholly on the Franco-Russian side. The German Ambassador in London had already been warned (on 31st July) that we should be drawn into the struggle if Germany persisted in her threatened attack on France. Two days previously Germany had made the 'infamous bid' to Great Britain that if she would remain neutral no territory would be taken from France herself, though no undertaking could be given with regard to the French colonies. British mobilization orders were issued on 4th Aug., and at 11 p.m. on that date Great Britain declared war on Germany.

Fortunately the British navy was ready for any emergency, with the Grand Fleet—the command of which was given to Admiral Sir John Jellicoe—still assembled in full strength at Portland, after the manœuvres, the order for its dispersal having been countermanded on 27th July. Lord Kitchener, home on leave from Egypt, had also been stopped by a telegram from Mr. Asquith, then Prime Minister, as he was stepping on the Channel boat at Dover on his return journey (3rd Aug.), and two days later was appointed Secretary of State for War. Meantime the Austrians had already bombarded Belgrade (29th July); Italy had declined (1st Aug.) to be drawn into the conflict with her Austro-German partners of the Triple Alliance on the grounds that their war was an

aggressive one; and German troops, as already mentioned, had invaded France at several points on 2nd Aug., before formally declaring war on that country.

Western Front, 1914

The struggle on the Western front began in earnest on the following day, when war was declared on France and the Germans captured Trieux, near Briey, and Lunéville was bombarded by German aeroplanes. The German system of mobilization had been quicker than the French and Russian, but the opening moves filled the Allied commanders with too-confident hopes. Although slower to mobilize than the Germans, a Russian army under Rennenkampf succeeded in invading East Prussia in force; the Belgians made a magnificent stand for their frontier fortresses when the Germans, denied the right of way which they had demanded, endeavoured to force the great highway of Western Europe which passes through Liége; and the French, besides checking the enemy at Dinant, had already recovered part of the lost provinces of Alsace-Lorraine.

On 16th Aug. the First British Expeditionary Force, under General Sir John French, completed its landing at Boulogne, and four days later had arrived and concentrated on the line Avesnes-Le Cateau, on the left or exposed flank of the French Fifth Army under General de Lanzerac. It consisted of 50,000 infantry with its artillery, and five brigades of cavalry—some 70,000 troops altogether, a mere drop in the ocean compared with the millions of men who were marching to battle for the great military powers, but destined to play a part in the forthcoming struggle out of all proportion to its size.

The position at this juncture was, briefly, as follows: the Germans having at length captured the last forts of Liége, with its gallant commander General Leman, were overrunning Belgium. Brussels had just been evacuated (20th Aug.), and the main Belgian army, menaced by greatly superior forces of the enemy, and disappointed in its hope of effective support from the Franco-British troops, was retiring to seek the protection of the forts of Antwerp. Having occupied Brussels on the 20th, the German Higher Command appointed Baron von der Goltz as Governor. A reign of terror in Belgium had already been inaugurated as part of Germany's deliberate policy of 'frightfulness', including the ruthless execution of civilians on unsubstantiated charges of shooting at the invaders.

The French armies, under the supreme command of General Joffre, who, like Lord Kitchener, had been an engineer student when the Franco-Prussian War broke out, and had been Chief of the General Staff since 1911, were now disposed for the double purpose of meeting the threatened German onslaught and preparing the counter-offensive on which French doctrines of strategy had been based. Starting from the Swiss frontier there were nine divisions forming the Alsace force, the main offensive group, consisting of the French First and Second Armies, being extended along the Lorraine frontier, and the Third Army about Verdun. The Fourth Army formed the mass of manœuvre held in reserve behind the centre, while the Fifth, whose left wing was now extended by the British Expeditionary Force, faced the Ardennes as far as the Belgian frontier.

Germany was not seriously alarmed by the spectacular advance of the French into their lost provinces. It suited the strategy of her War Staff to keep the French mass of manœuvre as far as possible from the point at which it would soon be sorely needed; and their feint attacks in the direction of Longwy, Lunéville, and Belfort were designed to strengthen the belief that their real offensive would come in the frontal assault which the French dispositions had assumed. Germany, however, had always intended to strike through Belgium when the time came to deliver the knock-out blow to France before Russia had time to mobilize her millions.

The German advance was proceeding according to the plan which had been worked out in detail as far back as 1904 by the soldier-scholar of the Garde-Ulanen, Count von Schlieffen, who died two years before his great scheme was put into execution. Based on the assumption that Germany and Austria-Hungary would have to fight France, Russia, Great Britain, and Belgium without the aid of Italy, it provided for an immediate attack by the right wing of the German army of such weight and ferocity as to destroy the French left by a single blow, and then roll up the main French armies one after the other. The South and Russian fronts were meantime to be lightly held, everything being staked on the sudden, overwhelming blow in the north through Belgium. One of the bitter controversies in Germany, after the war, raged round the responsibility for the failure of this plan, the execution of which devolved on General von Moltke, nephew of the great strategist of the Franco-Prussian War. The Kaiser believed that the name of Moltke would strike terror into the hearts of Germany's enemies, but the second Moltke lacked the genius of his

predecessor, and the course of events proved that he was not equal to the task of carrying out so prodigious a plan.

It was doubly necessary to strike at once with an immediate maximum of strength now that Britain had already ranged herself alongside the Allies. This maximum of strength was attained long before France had completed her mobilization, and enabled Germany to launch her unexpected blow with crushing effect. She had reckoned, however, without the stubborn defence of the Belgians in the opening moves of the game, a defence which clogged the wheels of her mighty war machine at the critical moment; and was wholly unprepared for Britain's great achievement in transporting her 'insignificant' but indomitable army, without a hitch, complete in every detail, and establishing it in its place in the line of battle, hundreds of miles from its base, in less than three weeks from the declaration of war. Clearly there was no time to be lost in solving the military problem on the Western front before the Russians could throw their full weight into the scales.

The secret of Germany's sudden attempt to overwhelm the Allied left by an outflanking movement was well kept. The position in Belgium was obviously grave; but Joffre still clung to the belief that if the Germans attacked the Allied left in force, they would leave their own position in front of the French Fifth Army so exposed as to give him an opening for a successful counter-stroke with de Lanzerac's troops in co-operation with the British. Up to the 22nd General French's preparations were all in the direction of offensive action on these lines; his two corps had taken up their positions through Binche and Mons and along the canal to Condé.

The German tide which now swept through the plains of Belgium entirely upset the Allied calculations. General French woke on 22nd Aug. to find the troops of the French Fifth Army on his right in unmistakable retreat. The full force of the German blow, delivered by von Buelow's Second Army, had been felt by de Lanzerac's troops on the Sambre at daybreak, and had pressed them back from the river. The British position held by the 1st Corps (1st and 2nd Divisions) under General Sir Douglas Haig, the 2nd Corps (3rd and 5th Divisions) under General Sir H. Smith-Dorrien, and the Cavalry Division under General Allenby, became isolated by the retreat of de Lanzerac—"the most complete example", as Lord French long afterwards described him, "of the Staff College pedant whose 'superior education' had given him little idea of how to conduct war". De Lanzerac asked General French if he would attack the flank of the German columns which were pressing him back from the

Sambre, but the British Commander, who had received definite instructions from Lord Kitchener that his command was to be an entirely independent one, "and that you will in no case come in any sense under the orders of any Allied general", replied that with his own position so seriously threatened by the retreat of de Lanzerac's troops such an operation was impracticable, but he agreed to retain his present position for the next twenty-four hours.

The British army fulfilled this pledge, and the barrier thus held and maintained during the subsequent retreat, though shattered in parts, saved the French left from being outflanked by the invading right wing of the Germans under von Kluck. The whole situation became extremely critical on the following day (23rd Aug.). Namur, the forts of which had been regarded as impregnable, fell before the crushing attack of the heavy Austrian howitzers brought up by the advancing Germans; the French thrust into Alsace-Lorraine had just been countered by the German Fifth Army under the Crown Prince Rupprecht of Bavaria, which compelled the French to retreat from all but a corner of Alsace; and the main German attack, launched at the other end of the line, forced the French back both from the Sambre and the Meuse. The French Fifth Army, the position of which was considerably weakened by the fall of Namur, was attacked both by von Buelow's army in front, and by a Saxon army under von Hausen on its right. It was forced back until von Hausen found a gap on its right flank, through which he proceeded to pour his Saxons with the object of rolling up the French Third and Fourth Armies under Ruffey and Langle de Cary. These retreated in turn, to recover alignment with de Lanzerac's Fifth Army, which had retreated from the British right. The British army was thus left 'in the air', outflanked not only on the right, where von Buelow was now advancing on it from Charleroi, but also on the left, where von Kluck's right wing was sweeping down in full force from the north-west.

The onslaught on the British front began shortly after noon with a bombardment of some 600 guns along the whole line of 25 miles; followed by a great frontal attack in mass formation. The British troops, all experts at musketry, used their rifles with such deadly effect that the frontal attack crumpled up. The line held; but with the German tide surging round on either flank the position became increasingly critical. Under the threatened turning movement General Smith-Dorrien withdrew from the Mons salient, and before nightfall took up a fresh line some 3 miles south of the canal. The advanced troops of the 1st Corps had not been seriously engaged, and held their ground. It was not until late that night that the desperate situation on his right was fully

revealed to General French; and when news also arrived from Joffre that the British army would probably be attacked the next day by at least three German corps and two cavalry divisions, it became clear that a general and immediate retirement was inevitable. What actually happened was that the enemy attacked with no fewer than four corps, and at least two cavalry divisions.

The Retreat from Mons

The great retreat began shortly after dawn on the 24th with a feint attack by the 1st Division, under cover of which the 2nd Corps moved back 5 miles, and then stood in turn to protect the retirement of the 1st Corps. Further withdrawals were effected that day by alternate corps, covered by heavy rear-guard actions, until the 1st Corps had reached the line between Maubeuge and Bavai, with the 2nd Corps extending the line from Bavai to Bry. Von Kluck's army, though kept in check by the retreating troops, followed closely on their heels and round their left flank, their design apparently being to turn the British left and press them back on Maubeuge, the fortress close on their right rear, which, well fortified and provisioned as it was, offered, as General French afterwards pointed out (in 1914), a terrible temptation to an army seeking shelter against overwhelming odds. Bazaine's example at Metz in 1870, and a shrewd suspicion that the German move was deliberately planned with that end in view, proved sufficient reasons for avoiding the trap. A further retreat was accordingly ordered to the line Le Cateau-Cambrai, some miles farther back.

Tournai, which was held by a French Territorial brigade, fell that day. There was nothing apparently to prevent the German host at this juncture from continuing its course to the coast and seizing the Channel ports as far as the Seine. That, doubtless, would have been included in the programme had the Germans anticipated a campaign of any considerable duration. The Kaiser, however, had promised his troops that they should be home again "before the leaves fall"; and to bring this about it was necessary to settle with the Allied army once and for all. Where von Moltke failed, according to Ludendorff and other critics after the war, was in not striking farther to the north or north-west, and in not throwing still more weight into the scale from his left wing.

On the 25th the French were still retreating all along the line save at Maubeuge, the garrison of which held out until 7th Sept., and at Longwy, north of Verdun, which fell on 28th Aug. The British army, battle-worn and suffering severely from the heat, but resisting all the German efforts to turn its western

flank, marched stubbornly back, gallantly assisted by Allenby's cavalry. The French were a day's march ahead of them when the British reached the Le Cateau position. General French decided, therefore, that, sorely as the troops needed rest, there was nothing for it but to resume the retreat at daybreak, and issued orders to that effect. The hardest fighting on the 25th had fallen to the 1st Corps at Landrecies, where Haig's weary troops were violently attacked at nightfall, before they could snatch any rest, by fresh enemy troops sent forward in pursuit in motors and lorries. The German infantry paid dearly for their temerity in advancing through the narrow streets of the town in close order, two or three British machine-guns mowing them down in hundreds. The attack was a disastrous failure.

The 2nd Corps did not reach Le Cateau until ten or eleven o'clock that night, thoroughly exhausted after a hard day's fighting and marching. Smith-Dorrien had lost heavily in the operations, and was so convinced that his troops were unfit to resume the march at daybreak that he elected to stand and abide by the result. The magnificent fight put up by his troops on the following day, assisted by Allenby and Sordet's cavalry, and two divisions of French Territorial troops under d'Amade, which had been detailed to guard the British left flank, saved the situation, and averted, in the considered opinion expressed by General French five years later, "a stupendous repetition of Sedan". The actual result was a total loss of some 14,000 officers and men, about 80 guns, and numbers of machine-guns, as well as quantities of ammunition and material. According to General French, these losses heavily handicapped the British army in the subsequent stages of the retreat, and were felt throughout the first battle of the Marne and the early operations on the Aisne. In his dispatch of Sept., 1914, the British Commander-in-Chief had written of this battle in eulogistic terms. It was not till some time later, he explains, that he came to know the full details of the battle and to appreciate it in all its details. For General Smith-Dorrien it is urged that his stand at Le Cateau broke the full force of the German pursuit, and checked its course in time.

On the 27th the shattered 2nd Corps, having broken off the action, continued the retreat with the 1st Corps. On the 28th Gough, with the 3rd Cavalry Brigade at St. Quentin, and Chetwode, with the 5th at Cérizy, turned on the leading German cavalry at both these places and threw them back on their main bodies in confusion. For the first time since the retreat began the worn-out British infantry, having reached the line of the Oise between Noyon and La Fère, were able to rest and sleep in peace.

On the 29th the British troops reached the line Compiègne-Soissons, the Germans on the same day occupying La Fère and Amiens, as well as Reims and other towns along the French front. Bapaume held out until the rolling-stock had been removed from Amiens, but the flood-tide of invasion now seemed to be carrying everything before it. Uhlans threatened to cut Sir John French's communications with his base at Boulogne and Dieppe. The base was accordingly transferred to St. Nazaire, at the mouth of the Loire. Timely help came to the retiring British troops on the 29th by a brilliant counter-stroke near Guise on the part of the French Fifth Army on their right; but neither the British nor the French troops on de Lanzerac's right were in a position to make a stand in support of that reaction. The Aisne was forced by the invaders on the 28-29th, and Rheims, Châlons, and Laon abandoned to them within the ensuing forty-eight hours. Falling back doggedly from the Aisne and the Oise, the British troops withdrew on 2nd Sept. to Chantilly-Nanteuil, the German advance having been checked on the previous day by the 4th (Guards) Brigade in a stiff rear-guard action at Villers-Cotterets.

The great retreat was coming to an end. Victory and Paris seemed within the enemy's grasp. He had—as he thought—so shattered the British army that it was now entirely negligible as a fighting force. He was ignorant of the real strength of the force that was gathering on the British left north of Paris—the new French Sixth Army under General Maunoury. It seemed both to von Kluck and the German Higher Command that they had only the shaken French Fifth Army seriously to reckon with on the Allied left, and, as von Kluck was considered more than strong enough for the task, von Moltke took the Garde Reserve Corps and 11th Army Corps from his right wing to East Prussia, where the Russians were now carrying the war well into the Fatherland.

The help rendered by the Russians at this critical phase of the war was invaluable, and played no small part in the approaching struggle on the Marne. In his fears for the safety of Paris, Joffre was naturally anxious to profit by this relief, and discussed with Sir John French the possibility of taking the offensive at the earliest possible moment. There appears to have been some misunderstanding as to Sir John's plans at this point. The British Commander-in-Chief declares that he had every intention of remaining in the line and filling the gap between the French Fifth and Sixth Armies, but the French Higher Command was apparently under the impression that he was determined not to fight any more until his troops had been given a week to reorganize and refit. Lord Kitchener himself hurried to Paris to clear the matter up, but "full

accord", according to President Poincaré, long afterwards, "was not re-established without trouble". As soon, however, as the offensive was ordered, continued the same authority, the British Commander-in-Chief gave his assistance without reserve. "His army fought with magnificent courage, and Great Britain played a brilliant part in the common victory."

In the meantime the retreat continued, the British, on 2nd Sept., reaching the line of the Marne towards Lagny and Meaux, with the French Fifth Army, now under the command of Franchet d'Esperey, on their right, retiring on Château-Thierry, and Maunoury's new Sixth Army, on their left, retiring towards Paris. It was at this point that von Kluck made the fatal mistake of dismissing the British army as practically crushed and out of action. Diverting the advance of the German First Army, he left Paris on his right in order to deal what he hoped would be a decisive blow at the French Fifth Army south of the Marne. By 5th Sept. the British army had fallen back to the Forest of Crécy to bring it in line with the French Fifth Army.

Not only was the British army at length receiving sorely needed reinforcements, but the French army was every moment increasing in strength and numbers as it fell back on its reserves. Besides the French Sixth Army on the British left, another new French army had sprung into being behind the marshes of St. Gond—the Ninth, under Foch, who filled the gap between Franchet d'Esperey's Fifth and Langle de Cary's Fourth Army—behind Vitry. Eastward the line was continued by the French Third Army, now commanded by Sarraill in place of Ruffey; and Castlenau's Second Army, now fighting the battle of the Grand Couronné de Nancy which stemmed the German invasion at this point, and prevented the threatened envelopment on the Allies' right, where the Kaiser himself had gone to inspire the troops of Prince Rupprecht of Bavaria.

First Battles of Marne and Aisne

Secure on the right, Joffre was at last able to deliver the great counter-stroke on the left which the Germans had invited by their tremendous bid for swift and decisive victory. The retreat came to an end on 5th Sept., when Joffre gave Sir John French his final plans for the coming offensive, and von Kluck, ignorant of the recuperative powers of the British, as well as of the strength of the French Sixth Army on their left, marched across their front in pursuit of d'Esperey's Fifth. That night the Germans crossed the Marne, and the Grand and Petit Morin—two streams which branch off roughly parallel to one another

south of the Marne—while some of their patrols reached the Seine, there catching a fleeting glimpse of the capital where they confidently hoped the French would soon be brought to terms.

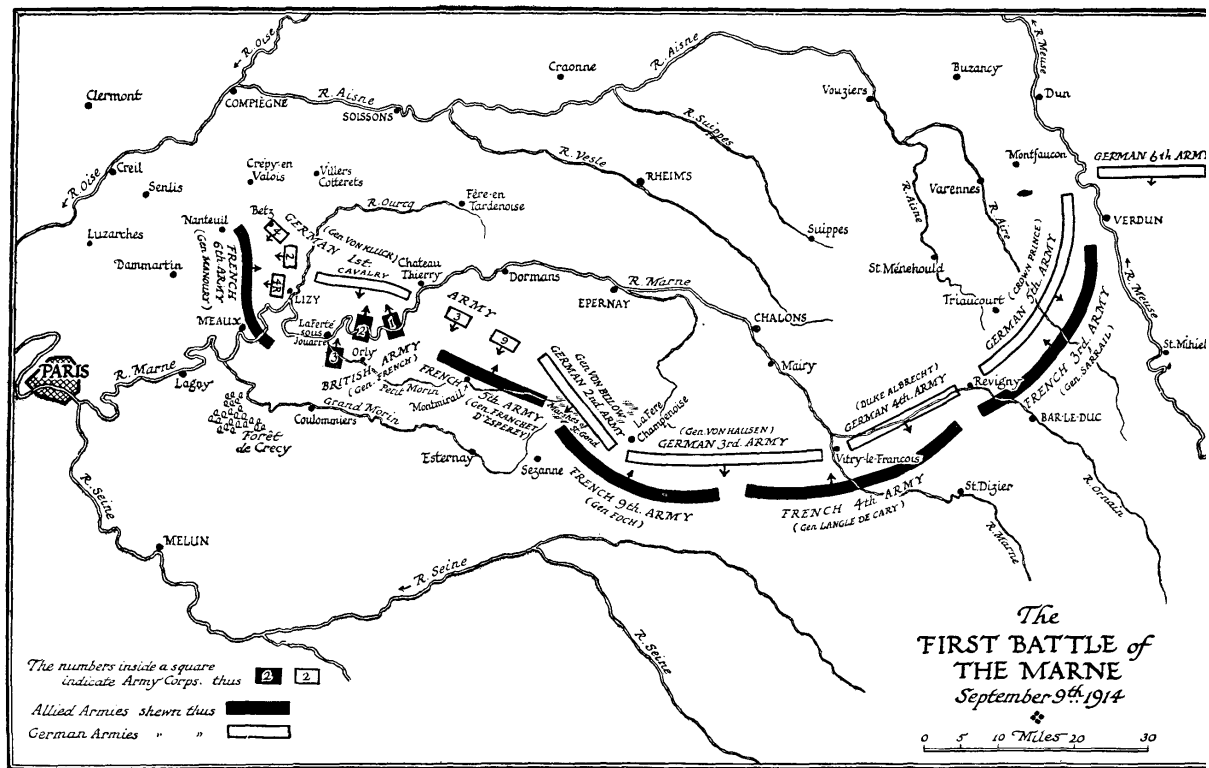
When at last the retreat came to an end, the British army had been reinforced by the 4th Division, which, with the 19th Infantry Brigade—and subsequently the 6th Division—became the Third Army Corps under General Pulteney, who arrived in France to take command of it on 30th Aug. Deficiencies in armament and material had also been partially made good, but, most important of all, Sir John French bore witness, "the promise of an immediate advance against the enemy had sent a thrill of exultation and enthusiasm throughout the whole force".

The first battle of the Marne had scarcely opened on 6th Sept., 1914, when von Kluck, realizing that Maunoury's force on his extreme right was becoming dangerous, sent two army corps northwards to deal with it. Maunoury had already crossed the Marne and fought the first battle of the Ourcq on the 5th. The dispatch of the two German corps to keep him in check made a way now for the British troops, when, according to plan, they turned on the invaders with the object of assailing their flank with the French Sixth Army on their left; while the French Fifth Army, and the French armies to its right, made a simultaneous frontal attack.

Both Joffre and French were under the impression that the German thrust was still in full career when their counter-stroke was delivered. Already, however, the tide had begun to turn. Von Kluck, realizing too late—what should have been obvious from the first—that his communications were being seriously threatened on the Ourcq, saw that retreat was inevitable unless he could crush the forces gathering so ominously against his right flank. The opening of the battle of the Marne thus became on von Kluck's part an effort to overwhelm Maunoury on his right, while he kept the British army and French Fifth Army at bay with strong rear-guards and cavalry. The surprise of the day to the Germans was probably the remarkable part played by the British, who, instead of being practically wiped out, as the enemy fondly believed, attacked with an energy and dash which carried everything before them, and, but for filling their allotted rôle of maintaining alignment with the French armies on each flank, would doubtless have advanced farther than they did. As it was, the progress made was considerable. The Germans were driven back to the Grand Morin, and the line of that stream made good on the following day.

Meantime the French Fifth Army on their right, materially helped by this success, had also recovered a good deal of ground, while Foch and Langle de Cary, farther east, held their own against the fierce assaults of the German centre. A last desperate effort was being made to hack a way through at this point, and Sarraill, on Langle de Cary's right, had to give way a little along the Meuse. That day the Germans reached the most southerly point of their advance, at Provins. The deciding phase of the battle, however, was developing with dramatic swiftness on von Kluck's right wing. Maunoury was hard pressed by the repeated onslaughts of the enemy, whose heavy reinforcements at this point held the issue in the balance for several days. General Gallieni, the Governor of Paris, hurried up fresh troops to Maunoury in motor-buses and taxis, and the French line held.

The British army helped matters considerably by driving the Germans across the Grand Morin at Coulommiers on the 7th, and on the following day from the Petit Morin, thus also helping d'Esperey with the French Fifth Army, on its right, to continue his advance farther east as far as Montmirail. On the 9th came the decisive blows which removed all doubts as to the issue of the battle. Von Kluck's retreat on his left flank exposed the right of von Buelow's Second Army, which was further jeopardized by a gap which appeared on its left, where it should have linked up with von Hausen's Third Army. This double opening gave Foch, facing von Buelow in the marshes of St. Gond, the opportunity which he sought of smashing the enemy's centre. He seized it by a series of lightning blows which drove the German centre back on the morning of the 10th in complete disorder, pursued by Foch's victorious infantry.



All the reinforcements sent to von Kluck were now of no avail against the French Sixth Army, which had been fighting against odds since 6th Sept., helped not a little by Pulteney's 3rd Corps on its right flank. Maunoury carried the Ourcq on the 9th, and Pulteney's corps was able to cross the Marne, after stiff fighting at La Ferté-sous-Jouarre, at dawn on the following day, when the German retreat became general. The left of the British 2nd Corps had crossed the Marne at Nanteuil, where the bridges were found unbroken and the enemy gone, on the morning of the 9th, but was ordered not to advance too far north until the 1st and 2nd Corps were firmly established on the northern bank. The 1st crossed later in the day at Charly-sur-Marne and Saulchéry, clearing the ground of the enemy and making many captures; but the 3rd Corps had a harder task at La Ferté-sous-Jouarre, and, as already mentioned, was not completely established on the other side until the following morning. The first battle of the Marne ended on the night of the 10th with the enemy in full retreat to the north and north-east and the Allies in hot pursuit. By the 12th he had been driven back from the Seine a distance of 65 miles, and the great German plan of a sudden crushing defeat of the Allies in the West had collapsed like a house of cards.

The hopes thus raised among the Allies of a speedy termination of the war in their favour were, on the other hand, equally illusory. Though the Germans lost heavily in officers and men, as well as in guns and other war material, their retreat was no disorderly flight. Many desperate rear-guard actions were fought all along the line, but the enemy retired steadily to prepared positions on the Aisne, where the eyes of all the commanders were to be opened to the possibilities of trench warfare under modern conditions. It needed many hard lessons before the truth was driven home.

When the first battle of the Aisne opened on 13th Sept., the British army already had its outposts on that river, the main body being in positions somewhat to the south, between Soissons and Bourg. Throwing bridges across during the night, the advance was continued on the opposite bank on the following day, though not without heavy British casualties, amounting to between 1500 and 2000, including 3 commanding officers. The 6th Division arrived from England at this stage, and joined its own 3rd Corps on the left. Further advance was stayed by the strength of the enemy's entrenched positions, in which he now determined to make his stand. Here he was backed by an overpowering superiority in artillery which, with fierce counter-attacks on the part of his infantry, gallantly repulsed though they were, caused such severe losses that the British Commander-in-Chief was forced to assume a defensive rôle, while Maunoury, de Castlenau, and Foch each made stupendous efforts to break the enemy's line and renew the war of movement and manœuvre on which their military principles had been based. All, however, ended in the same dreary deadlock of entrenchments.

Failing to shift the enemy from these impregnable positions, Joffre endeavoured to outflank the German right wing, already threatened by Maunoury's advance along the Oise. Two new French armies were formed from the reserves to extend the Allied left—the Seventh, entrusted to de Castlenau, whose Second Army was transferred to Dubail, and the Tenth, the command of which was given to Maud'huy. De Castlenau's Seventh Army, though it failed to turn the enemy's flank—the movement having been anticipated by him—succeeded in extending the pressure of Maunoury's left, which had swung round by the 20th until it ran north from Compiègne to west of Lassigny, and in building the first section of Joffre's great besieging wall which, gradually extending from the Alps to the sea, became the impenetrable barrier between the enemy and his main objectives. The Allies' line was

continued by de Castlenau through Roye to Albert, and thence, by Maud'huy's Tenth Army, through Arras to Lens.

Von Moltke had now been superseded in the German Higher Command by Falkenhayn, who promptly countered Joffre's new strategy by similar extensions of the German front, thus beginning the outflanking race destined only to end in stalemate on the coast. While extending their right the Germans made a strenuous effort to regain the initiative by a blow with the army group nominally commanded by the German Crown Prince on Sarraill's flank on the Meuse. It was a blow aimed at Verdun and the whole of the Allied line, which it hoped to break through at this point and so take in the rear. Verdun, however, had been rendered impenetrable by miles of powerfully protected outer defences, and practically the only success which fell to the Crown Prince on this occasion was the capture of the Camp-des-Romains and St. Mihiel on the Meuse, thus creating the remarkable salient east of Verdun which was destined to remain until the Franco-American force flattened it out in the victorious advance of the Allies four years later. The German Crown Prince fared even worse a week later, when he attacked along the main road through the Argonne towards Verdun, only to be flung back. It was after this double failure that the Germans bombarded Rheims and shattered her noble cathedral.

The crucial phase of the struggle in the West had shifted towards the coast as the first battle of the Aisne died down on 28th Sept., and the campaigns began in Artois which led in due course to the fierce struggles for the Labyrinth, the Vimy Ridge, Lens, and Loos. The extension of the French left placed the British army in an anomalous position. Even before Joffre had begun to build his barrier in this direction the British Commander-in-Chief had felt strongly that his proper sphere of action was on the Belgian frontier on the left flank of the French armies, for the two-fold purpose of defending the Channel ports and being in position to concert combined action with the British navy. He suggested this move to the north to Joffre on 29th Sept., pointing out its strategical advantages and the possibility of doing so now that the position of his force on the right bank of the Aisne had been thoroughly well entrenched. Joffre agreed in principle to General French's proposal, but postponed the movement until 3rd Oct.

Retreat from Antwerp

By this time the critical situation of the Belgian army at Antwerp had become hopeless, and the danger of a German descent on the Channel ports suddenly

became acute. Since their retreat towards Antwerp after their evacuation of Brussels on 20th Aug., the Belgians had kept the Germans at bay by vigorous counter-attacks, and threatened their communications by sundry sorties from the fortress. These sorties and counter-attacks, calling for reinforcements at a time when every soldier was needed on the main fighting fronts, infuriated the Germans and led to the reign of terror which included the deliberate destruction of Louvain and similar outrages at Malines, Termonde, and elsewhere.

Having made his position secure on the Aisne, and brought up his heavy guns, the enemy began his bombardment of the outer forts of Antwerp on 28th Sept. By the 3rd of Oct. the Belgians were endangered not only by the besieging army, but also by the ever-lengthening German line which, having now been extended from Lassigny to Lille—only 38 miles from the sea—threatened to isolate the Belgian forces from the Franco-British armies. They accordingly decided that plans must be made at once to withdraw from Antwerp in the direction of Ghent, both to protect the coast-line and gain touch with the Allies. The British troops hurriedly sent to reinforce the Belgians—a brigade of Royal Marines and part of the recently formed Naval Division—had no influence on the fate of the fortress, but helped in protecting the flight of the citizens and in the final retreat of the Belgian army. Some 1500 of the Royal Naval Division were forced across the Dutch frontier and interned, and about 800 were made prisoners. The remnants of the British force, and the bulk of the Belgian army, escaped westward, leaving the Germans on 9th Oct. in possession of the deserted city. A little more and the German commander (von Beseler) might have closed the gap beyond the Scheldt through which this retreat had been made. Luckily for the Allies, too, the German Higher Command failed, as Lord French long afterwards expressed it, to gather the richer harvest which had been put within its grasp by the capture of Antwerp. There was then apparently no insuperable obstacle to an immediate German advance on Dunkirk, Calais, and Boulogne, before the Allied troops could arrive from the main theatres to prevent it. "As on the Marne, so at Antwerp, they were not prepared to seize the psychological moment and to play boldly for the great stake".

General French, who had been given no voice in the Antwerp dispositions, was now in the midst of the British move from the Aisne front to the north, where he was to be reinforced by the 7th Infantry and 3rd Cavalry Divisions, which had been landed on the Belgian coast to defend Zeebrugge and Ostend under

Rawlinson's command, and the Indian contingent, which had just arrived at Orleans under Willcocks.

First Battle of Ypres

The two corps under de Castlenau and de Maud'huy were now under the supreme command of General Foch, who had orders to control all the French armies operating in the northern theatre, and was confident that it was still possible to outflank the Germans and bend them back behind the Scheldt. At the end of a fortnight the British army had been successfully transported to the north from the Aisne—after successfully holding the line of that river for twenty-five days against many desperate efforts of the enemy to break through—and had taken up its position on the left of Maud'huy's corps, the Allies' line being extended thence into Flanders by the French 8th Corps (under d'Urbal), which had been called up by Joffre to this end, as well as to help the sorely tried Belgians. Meantime the Germans, besides pressing Maud'huy hard in front of Arras, and forcing his Territorials out of Lille, had driven the retreating Belgian army out of Houthulst Forest to the line of the Yser north of Ypres, whence it took refuge behind the Yser and completed the Allied line to the coast. Rawlinson's 4th Corps, covering the retreat of the Belgian army, had hard fighting most of the way before it succeeded in joining the main British army, the German forces from Antwerp concentrating westwards in ever increasing numbers. By the 15th Capper's 7th Division was east of Ypres, while Byng's 3rd Cavalry Division, a day earlier, had connected up with Gough's 2nd Cavalry Division in front of Kemmel and assisted in the capture of that position, a gain which proved of the utmost importance in the subsequent struggle for Ypres. Allenby's Cavalry Corps had greatly distinguished itself during the two previous days, driving the enemy back all the way in a magnificent sweep to the north and north-east.

A brilliant series of advances by Pulteney's 3rd Corps on 12-15th Oct., leading to the capture of Bailleul and Meteren, and the line Sailly-Nieppe, confirmed French in the belief which he shared with Foch that the enemy had exhausted his strength, and that the time was ripe for a strong offensive eastwards. The advance had scarcely been started, however, before reports arrived of a powerful offensive on the part of the Germans towards Ypres and the Yser. The power and weight of the enemy's opposition on the British front increased each day. Armentières and the Bois Grenier were won by 15th Oct., but the failure of the 4th Corps in its advance on Menin was one of numerous indications that

the Germans were being heavily reinforced. By the 21st all General French's worst forebodings were realized by the certainty established that the small German force which had been operating between Ostend and Menin on the 18th had been increased by no fewer than four corps. This discovery, in the British Commander-in-Chief's own words, "came like a veritable bolt from the blue". There was no longer any hope of continuing the offensive on the part of the comparatively weak British line, extended as it was on too long a front. It was a case of holding on now until relief arrived.

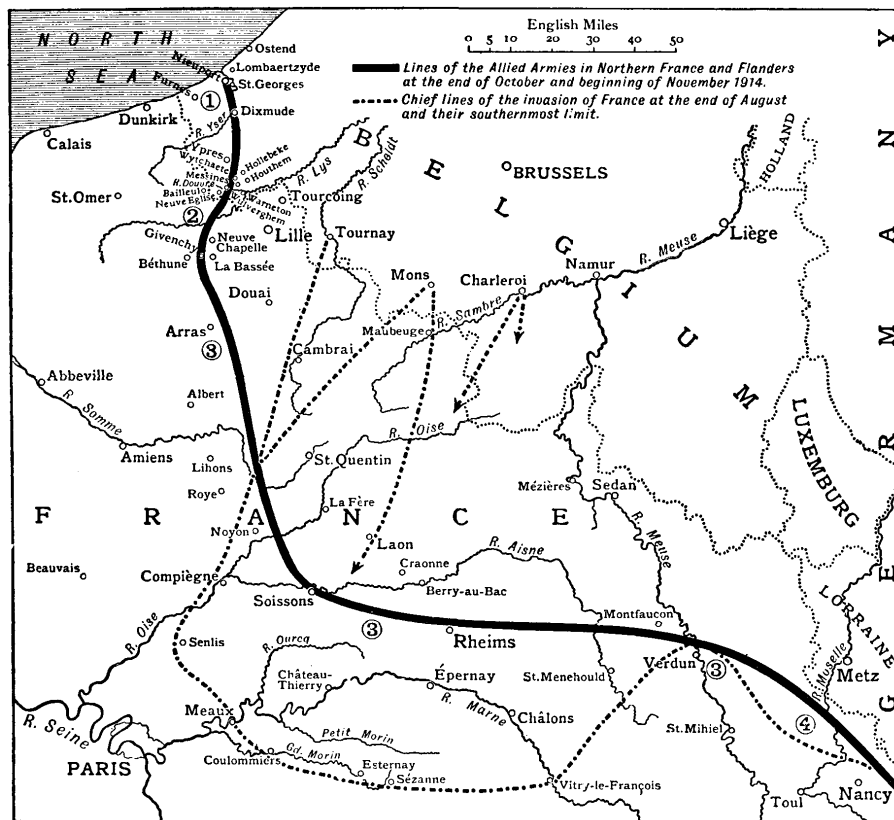
The threat was twofold. Not only were the Germans massing in tremendous strength in the north; they were also seriously threatening the British right. Maud'huy, round Arras, was fighting a battle which, like that which the British were waging round Ypres, was one of the landmarks of the war. He sent word on the 16th that the enemy was intent on driving in a wedge between Ypres and La Bassée—a threat which, had it matured, would have finally separated the Allies, and compelled the British either to surrender or be driven into the sea. Faced by this double threat, General French decided to risk the possible disaster on his right, and concentrate against the German tide in the north, which otherwise must gain the seaboard, with possibly fatal consequences to the whole British campaign.

Fortunately the First Army under Haig had already been ordered in the direction best calculated to meet the new situation, though there was no longer any hope of its original orders—to turn the enemy's flank and drive him back to Ghent—being carried out. In this first phase of the battle of Ypres, which lasted until the night of 26th Oct., the northern portion of the British line, notwithstanding the enemy's immense reinforcements, progressed slowly but surely, and took heavy toll in killed, wounded, and prisoners. A certain amount of ground was lost to the south, between Zonnebeke and La Bassée—the commanders of both the 2nd and 3rd Corps being anxious about their positions on more than one occasion—but at no point was any serious break made in the line. Maud'huy at the same time was gallantly keeping the enemy in check in the Arras region, though he could not drive him from Lens and the Vimy Ridge; while on the left the French and Belgians were withstanding repeated assaults on the swaying front between Dixmude and the sea. The line held, and was now more or less firmly established.

The second phase of the battle of Ypres consisted in the repeated attempts of the Germans to break through this line at all costs. It began on 27th Oct., and lasted through five of the most momentous days in the history of the British

army. On this day the French 9th Corps, which had been sent by Joffre to the assistance of the sorely tried British troops, took over the trenches in the northern part of the British salient. Capper's 7th Division, exhausted by incessant fighting and fearful losses, was temporarily attached to Haig's 1st Corps in the centre, and took over the ground south of the Ypres-Menin road. Byng's 3rd Cavalry Division was at the same time placed under Allenby. The 4th Corps was thus temporarily broken up, Rawlinson being sent home to supervise the preparation of the 8th Division for France.

The British line was further reduced in numbers during the opening days of this new phase of the battle by the repeated but unavailing attempt of the enemy to advance, the Germans meantime mounting up reserves of reinforcements for a decisive blow until, by the 30th, they outnumbered the battle-worn British by two to one. Then came the onslaught in full, almost overpowering strength. "October 31st and November 1st", afterwards wrote the British Commander-in-Chief in his story of *1914*, "will remain for ever memorable in the history of our country, for during those two days, no more than one thin and straggling line of tired-out British soldiers stood between the Empire and its practical ruin as an independent first-class Power." The storm centre of the British battle-line was the Wytschaete-Messines ridge, where Allenby's Cavalry Corps and Shaw's 9th Brigade of the 3rd Division withstood for forty-eight hours the supreme efforts of two and a half German army corps to dislodge them. The honours of those heroic days were shared by the French 9th Corps and the British 1st Corps (with the 7th Division attached) in their continued defence of the Ypres salient; and the British 2nd Corps, which held a long line on the right in difficult country, and, though forced to give up Neuve Chapelle on the 28th, withstood repeated attacks by superior numbers until the Indian Corps took over their positions.



Map showing lines of the Allied Armies in Northern France at the end of October and beginning of November, 1914

- 1, Yser line, defended by Franco-Belgian forces.
- 2, Ypres to La Bassée line, guarded chiefly by British troops.
- 3, French lines to a point above Verdun.
- 4, French lines adjacent to the Alsace-Lorraine frontier.

The culminating phase of the battle began on the 29th, when overwhelming masses of the enemy stormed the centre of the Ypres salient, held by Haig's 1st Corps and the 7th Division, and forced our troops back on Gheluvelt. The ground was recovered before nightfall, but fighting of the fiercest character continued all round and beyond the salient, the critical sector of the 31st extending from Gheluvelt to Messines, on the south, where the 1st Cavalry Division was heavily pressed, and the London Scottish received their costly baptism of fire. The Germans got into Messines that day, but were hurled out again, and the line in this sector was completely restored by nightfall. The climax of the crisis had been reached shortly after midday in the Gheluvelt area, when the 1st Corps, after doing more than could be expected of any men in their prolonged stand against the heaviest odds, was at last broken, part of the 1st Division falling back rapidly along the Ypres-Menin road, with the

Germans at their heels. "I felt", afterwards wrote General French, who was not more than a mile or so away at the time, "as if the last barrier between the Germans and the Channel seaboard was broken down", and he spent the worst half-hour his life had ever known. The situation was saved by Brigadier-General FitzClarence, V.C., commanding the 1st Guards Brigade of the 1st Division, who, on his own initiative, and in the nick of time, ordered the 2nd Worcesters to counter-attack. The Worcesters, who were in reserve to the 2nd Division, rushed up to fill the gap, and, saving the South Wales Borderers, drove the Germans out of Gheluvelt and re-established the line, which was completely restored before dark. FitzClarence was killed only a week or two later in the same part of the field.

The third phase of the battle of Ypres lasted from 1st Nov. to the 10th. Its most dangerous hours were at the very beginning, when both Messines and Wytschaete were lost, and only the timely arrival of the French 16th Corps, which partially restored the situation, and the devoted bravery of Allenby's Cavalry Corps, staved off this new threat of disaster. It is impossible here to follow all the confused operations in the remainder of this phase, in which fighting continued with varying intensity, and mingled success, all along the line from La Bassée to the sea.

The outstanding feature of the fourth and final phase, which extended from 11th Nov. to the 21st, was the succession of heavy assaults by the pick of the Prussian Guard, ordered by the Kaiser personally to carry the Ypres salient at all hazards. It failed, but not before the Germans had pierced the front along the Menin road in the first clash of arms on the morning of the 11th, a battalion of Royal Fusiliers being practically wiped out in gallantly disputing their passage. Haig met the situation "with the same grim determination, steadfast courage, and skilful forethought"—the words are those of Lord French—"which had characterized his handling of the operations throughout". The line was re-established, but only after fearful losses on both sides. The 1st (Guards) Brigade mustered at night only 4 officers and 300 men.

The French and Belgians were also attacked all along their line between Ypres and the sea—where British monitors swept the coast with shells for 6 miles inland—but the enemy was held off, save at Dixmude, which he captured and held. Between Dixmude, which had been stubbornly defended by Admiral Ronarc'h and his French marines, and Nieuport the sluices of the Yser had been opened by the Belgians, and the low-lying country across which the

Germans were striving to force a way so flooded as to render all their efforts futile.

It was during this final effort of the Germans to reach the Channel ports in 1914 that Lord Roberts arrived at the front to visit the Indian Corps, who had withstood some heavy assaults on the old line of the Second Army, between Armentières and La Bassée. Lord Roberts had scarcely fulfilled his mission, inspiring the troops with his presence at a critical time, when he was taken suddenly ill on 13th Oct. and died on the following day.

North of Armentières the British 3rd Corps under Pulteney, which held the line thence towards Messines, had its share of fighting on the left bank of the Lys, and though its deeds in maintaining its positions were overshadowed by the epic struggles in the salient, its minor battles played their part in the victory of First Ypres. The great battle died down with the failure of the supreme effort of the Prussian Guard. Rains and floods and mud combined to call a halt in the struggle for the Channel ports, and the Western front was now established for the winter. There were occasionally attacks by the Germans at Ypres and Festubert, and more than one attempt on their part to cross the flooded Yser on rafts; but all to no purpose; and before the end of the year (20th Dec.) a five days' battle between the Indian troops and the Germans round Givenchy left matters much as they were before, the British positions, with the aid of British troops, being held. The French also broke the monotony of trench warfare with encouraging advances in Alsace, towards Noyon, in the Argonne, and elsewhere, but no vital changes took place in the general situation.

Eastern Front, 1914

Foiled in their grandiose plans in the West, the Germans were forced to rest content with their valuable territorial gains in France and Belgium, and remain on the defensive there while they turned to the more threatening situation on the Eastern front. As already noted, the Germans had under-estimated the rapidity of the Russian mobilization. They had not anticipated an offensive on that front until they could spare as many reserves as necessary from the West, and the forces left to guard their vulnerable frontier of East Prussia were as inadequate to stay the unexpected advance which the Grand Duke Nicholas, who was in supreme command of the Russian armies, ordered under Generals Rennenkampf and Samsonoff, as were the Belgians to prevent the march of the Germans across their territory. By 25th Aug. the Russian armies, whose advance had begun as early as the 7th, had pushed so far ahead that all East

Prussia seemed in danger of falling into their hands. General von François, commanding the German troops, had been driven into Königsberg, the cradle of the Prussian monarchy; Gumbinnen, Justerberg, Allenstein, Soldau, had all been captured; and the hopes of a flight to Berlin before the Russian 'steam-roller'—too often raised in the early stages of the war—seemed not unlikely to be realized.

Germany's hour of danger, however, produced the man who was destined to play a ruling part in the remaining phases of the struggle—Paul von Hindenburg, a veteran of the Franco-Prussian War, then on the retired list. Hindenburg knew the topography of East Prussia by heart, and had commanded army corps in manœuvres along that frontier for many years. Appointed at this critical moment to supersede General von François, he collected 160,000 men from every available source, and by means of Germany's unequalled strategic railway system had concentrated them in a favourable position between Allenstein and Soldau for delivering the blow which would cut the communications of the southern army under Samsonoff, and smash it piecemeal in the treacherous marshes of the Masurian Lakes, the tracks through which, though well known to Hindenburg, were a veritable tangle to the Russians.

Tempted by their initial triumphs, the Russians had themselves courted disaster by placing themselves in precarious positions. Samsonoff's southern army had not only lost touch with the northern force under Rennenkampf, which had occupied Insterberg on the 23rd on its march on Königsberg, but had also failed to secure either Allenstein or Soldau. Hindenburg was quick to seize every advantage, and his lethal thrust on the 26th, when he retook Soldau and outflanked the Russian left, was followed by a similar enveloping movement on their right before they realized what had happened. Masses of German guns came up and completed the move. Too late the Russians fled along the only road left open to them—by way of Ortelsberg and Johannisberg, across a narrow slip of land between the marshes. Save for little more than one corps, which succeeded in escaping along this route before it was closed, practically the whole of Samsonoff's army was wiped out in this decisive battle of Tannenberg, as the victors named it. Samsonoff himself was killed by a shell on 31st Aug. Altogether the Russians lost in killed and wounded some 30,000; no fewer than 90,000 were taken prisoners.

Hindenburg, whose Chief of Staff was General von Ludendorff—already distinguished in the war as the leader in the assault on Liége—at once became

a national hero throughout Germany. The Central Powers, however, had little further cause for rejoicing on the Eastern front in 1914, once Hindenburg had been enticed to the Niemen by the rapid retreat of Rennenkampf after the Tannenberg *débâcle*. The Grand Duke Nicholas had sent General Ruzsky from Galicia to retrieve the situation, and the new leader made as good use of the Niemen River—a formidable obstacle to cross with its width of some 200 yards—as Hindenburg had done of the Masurian Lakes. The operations, which ended in the failure of the Germans to cross the river, and their heavy defeat at Augustovo after a sanguinary nineteen days' battle beginning on 1st Oct., restored confidence to the Russian army.

General Ruzsky had already made his mark in the opening campaign in Galicia, where the Austro-Hungarian armies, after invading Russian Poland at the opening of the war, were driven back in a series of mighty battles which left the Russians in possession of Lemberg and all Eastern Galicia. Brussiloff was meantime sweeping on towards the Carpathian passes, while Ivanoff, commanding the Russians in Poland, forced back the invading armies under Dankl and the Archduke Joseph Ferdinand beyond the Vistula and Cracow. Przemysl alone held out in Galicia, and this was invested by the Russians towards the end of September, when the Germans, far to the north on the Niemen, were rapidly losing the advantage and prestige they had won at Tannenberg. Hence Germany's increasing need for help from the Western front.

In order to check the Russian advance on the key position of Cracow, Hindenburg was now called from the East Prussian front to take over the supreme command of the unified German and Austro-Hungarian armies in a crushing blow at the Russian centre in Poland. Hindenburg's advance on Warsaw was planned by his Chief of Staff, Ludendorff, to keep pace with a parallel advance of the Austro-German armies in the south, intended to raise the siege of Przemysl and turn the Russians out of Lemberg. The combined forces of the Central Powers amounted to some two million men, outnumbering the Russians by at least half a million, and outgunning them completely. It speaks volumes not only for the fighting spirit of the Russian armies at this period of the war, but also for the strategy inspired by the Grand Duke Nicholas, that in both the first and second battles of Warsaw the combined Austro-German armies were both outfought and outmanœuvred. "The Grand Duke Nicholas", to quote from Lord French's tribute to his leadership some years later, "proved that he possessed that highest of military

gifts—the power of renunciation, of 'cutting down', of sacrificing the less essential for the more."

Foreseeing the danger of Hindenburg's march on Warsaw, the Grand Duke promptly recalled the first great Russian advance on Cracow, and the armies concerned were safely withdrawn behind the Vistula and the San before the enemy could cut the main line from Warsaw to Kiev. Ivanoff's army in Galicia conformed to the general movement. While thus suggesting the abandonment of Poland and a general retirement on Brest-Litovsk, the Grand Duke placed a field army in defence of Warsaw, assisted by Japanese heavy artillery, and prepared a great counter-offensive from the north-west, under cover of the guns of Novo Georgievsk.

Hindenburg's main blow was delivered at Josefov, higher up the Vistula, with the intention of taking Warsaw in the rear. Ruzsky had been brought down to take command at Josefov, and here repeated the disaster which he had recently inflicted on the Germans at Augustovo. This time he lured the enemy across the river before falling on him in difficult country, and cut him to pieces. Rennenkampf was equally successful with the counter-attack from Novo Georgievsk, striking so hard that the German left centre was forced back from the Vistula with heavy losses. With both flanks turned—for Ruzsky had followed up his victory by himself crossing the river at Novo Alexandriev, and driving the Germans back first to Radowa (15th Oct.) and then to Kielce (3rd Nov.)—Hindenburg had no alternative but to seek safety within the German frontier, and prepare another counter-stroke. The only success of the first advance on Warsaw was achieved by the Austrians under the Archduke Joseph Ferdinand and Generals Woyrsch and Dankl, who succeeded in temporarily relieving Przemyśl and recovering Jaroslav. Hindenburg's retreat, however, compelled them to withdraw in conformity to that movement. Przemyśl was again invested, the Austrians again fell back in Galicia, and the Russian advance on Cracow was renewed, the Grand Duke Nicholas having set his heart on reaching that convenient gateway both to Berlin and Vienna before the end of the year.

The second Russian advance on Cracow commenced with the second German advance on Warsaw, and both operations naturally reacted on each other. The Russians were at the very gates of Cracow by 5th Dec., but were then checked by an Austro-Hungarian counter-offensive which threatened both their flanks. The loss of the Dukla Pass (12th Dec.) forced a general retreat on this front, and was followed by the loss in turn of the Lupkow and Uszok Passes. By 20th

Dec., however, the Russians, who had fallen back behind the River Nida, turned on their pursuers and drove them north again until they were once more in possession of the Carpathian passes.

The Germans were no more successful in their second advance on Warsaw than the Russians had been in their march on Cracow. Hindenburg's urgent call for reinforcements, which had helped to relieve the pressure on the Western front in those critical days round Ypres, enabled him to attack all along the line on 18th Nov. The main assault was delivered by von Mackensen on the line held by Ruzsky from Gombin, on the Vistula, to Uniejov, on the Warta. Ruzsky's retreat to the Bzura in face of Mackensen's terrific onrush was one of the brilliant episodes of the war. It continued, with an increasing bulge, until Mackensen, on 23rd Nov., burst right through this and split it in halves. The halves held, and began to close up, with two German army corps all but bottled up within. But for some blunder in timing, for which Rennenkampf was held responsible, their only way of escape would have been cut off. As it was, after three days of desperate fighting, the German corps fought their way out—but only in remnants.

The Russians themselves had suffered heavy losses and were running short of ammunition, and when Hindenburg retaliated by blow upon blow in every sector, the Grand Duke Nicholas shortened his front, evacuated Lodz, and defended Warsaw from behind the Rawka and the Bzura. Here he sustained repeated onslaughts for nearly three weeks, the fiercest fighting taking place between the 19th Dec. and Christmas Eve against Bolimov and Sochaczew. In the end the Germans failed to hack a way through to Warsaw, just as they had failed a few weeks previously in their efforts to break through to Ypres and the Channel ports on the Western front. They were still 35 miles from the Polish capital at the close of 1914.

Opening Campaign in Serbia

In the meantime the Austro-Hungarian armies, which had somewhat atoned for their earlier failures by their successes in the recent advance which had temporarily relieved Przemyśl, were making little headway in their campaign against Serbia. That heroic little nation, though sadly depleted in her manhood by two years of Balkan warfare, was more than a match for her mightier neighbours in the opening stages of the European War. Austria's first punitive expedition, heralded by the bombardment of Belgrade on 29th July, was easily checked, the main invading forces being withdrawn to meet the new situation

on the Russian front. Serbia and Montenegro thereupon invaded Bosnia, and an unproductive campaign of withdrawals and advances led, towards the end of October, when Turkey threw in her lot with the Central Powers, to an Austrian invasion of Serbia in earnest. At first the Austrians, an army corps strong, and advancing in three columns, carried all before them. Valjevo fell on 29th Nov., and Belgrade ten days later. In withdrawing, however, the Serbians, ably led by the Crown Prince, Marshal Putnik, and General Mishitch, were falling back towards their bases and new supplies of ammunition; and in the battle of Rudnik, or 'Battle of the Ridges', which followed, turned on their pursuers and practically destroyed them. The battle lasted three days, at the end of which the remnants of the Austrian army corps, with the victorious Serbians hard on their heels, were in full flight to the frontier, leaving behind them 15,000 prisoners and 19 guns. All told their casualties amounted to some 80,000 and the bulk of their guns before the survivors were back in their own country, across the Danube, the Drina, and the Save.

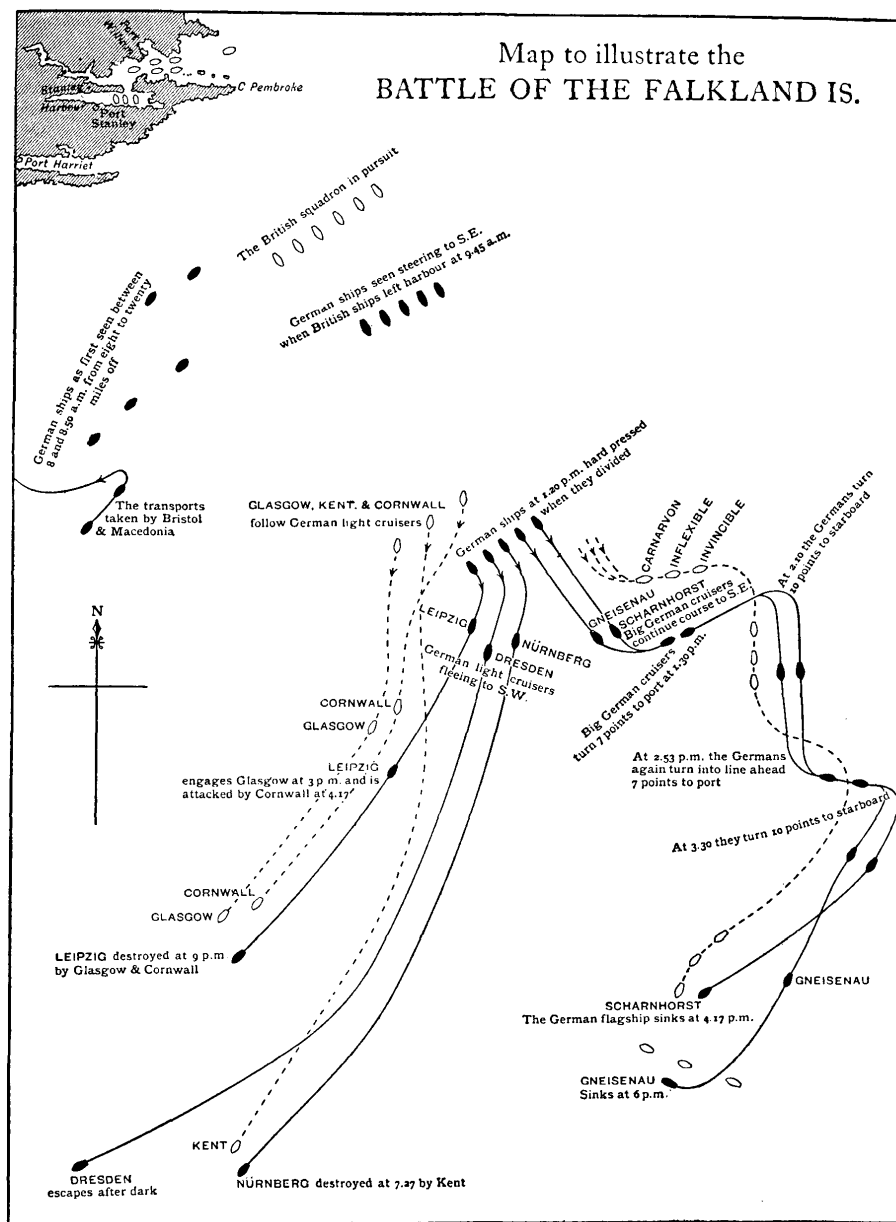
First Months of the War at Sea

At sea the first months of the war had seen the Germans very discreetly sheltering in Kiel and Wilhelmshaven, leaving it to submarines, mine-layers, raiders, and a few commerce-destroyers to do as much damage as possible and reduce the disparity between its own force and that of the Allied fleets.

The first naval action of any importance took place in the Bight of Heligoland, on 23rd Aug., when a force of destroyers, led by Commander Tyrwhitt in the new light cruiser *Arethusa*, with another light cruiser, the *Fearless*, and a flotilla of submarines under Commodore Keyes, served as a lure to draw out the Germans. They drew first a German force of destroyers and two cruisers (*Ariadne* and *Strasburg*) from Heligoland. The *Arethusa* greatly distinguished herself in the general engagement which followed, but, though she played the main part in driving the Germans back, she was so badly cut up that she had to be taken in tow at the end of the action. This was not until the Germans, deceived into the belief that the British ships were unsupported, sent out their heavier cruisers, the *Mainz*, *Köln*, and *Aurora*, just in time for Beatty's battle-cruisers, which arrived at the critical moment, led by the *Lion*, to settle the issue by sinking the three enemy cruisers above mentioned, besides a number of their destroyers. No British units were lost. Apart from this engagement the war in home waters for the next few months resolved itself into such isolated incidents as the sinking of the three old sister cruisers *Aboukir*, *Hogue*, and

Cressy by the German submarine commanded by Otto Weddigen (22nd Sept.); the destruction of four German destroyers by the British light cruiser *Undaunted*, supported by destroyers, on 17th Oct.; and raids by fast German cruisers on Yarmouth (3rd Nov.), on Scarborough, Whitby, and the Hartlepoons (16th Dec.), the first of which led to the loss of the German cruiser *Yorck* while crossing the minefield at Wilhelmshaven.

The scattered units of the German navy in other seas were gradually bottled up or destroyed. The Dreadnought *Goeben*, and the light cruiser *Breslau*, which were in the Mediterranean on the outbreak of hostilities, succeeded in evading the British fleet and reached the Dardanelles, where they played no small part in persuading the pro-German Turks, a few months later, openly to side with the Central Powers. Other German warships caught by the war in more distant waters included the light cruiser *Emden*, on the China station. The *Emden* disappeared for six weeks and then turned up in the Bay of Bengal, bombarding Madras on 22nd Sept., and capturing some twenty British steamers in the same month. On 28th Oct. she sank a Russian cruiser in Penang Roads, as well as a French destroyer, but twelve days later (9th Nov.) ended her eventful cruise off Cocos (Keeling) Islands, where she called to destroy the cable and wireless station. The telegraphists on the island sent out a warning message which reached the *Sydney*, a British cruiser of the Australian squadron then escorting Australian and New Zealand troops to the war. Within three hours the *Sydney* arrived on the scene, and having the range of the *Emden* with 6-inch guns to 4.1-inch guns, forced the German cruiser ashore. After losing 7 officers and 104 men the *Emden*, burning and half-sunk, surrendered.



Map to illustrate the BATTLE OF THE FALKLAND IS.
Map illustrating approximately—from Admiral Sturdee's Dispatch—the Battle of the Falkland Islands from start to finish. The map must not be regarded as showing the proper scale of distances, &c.

More dangerous than the *Emden* was the German cruiser squadron under Admiral Graf von Spee, which concentrated in the South Pacific from Kiao-Chau and elsewhere. The squadron consisted of the twin cruisers *Scharnhorst* and *Gneisenau*, each of 11,400 tons, 22 knots speed, and an armament which included eight 8.2-inch guns; and three smaller cruisers, *Dresden*, *Nürnberg*, and *Leipzig*, each of 3500 tons, and carrying ten 4.1-inch guns. On 1st Nov.

von Spee fell in with the weak British squadron under Admiral Cradock, who had been sent in August to protect the South Pacific trade, and was expecting reinforcements to cope with the German concentration. Cradock's squadron consisted of old ships like the cruisers *Good Hope* and *Monmouth*; the light cruiser *Glasgow*, and the armed liner *Otranto*; the pre-Dreadnought battleship *Canopus*, which had been left behind for repairs, being some twelve hours away. Outsteamed and outranged—though the *Good Hope* (14,000 tons) had two 9.2-inch guns on board—with the setting sun silhouetting their surfaces against the sky, they were no match for the Germans, who made the most of the added advantage which their inshore position gave them, obscuring their outlines when, as the light began to fail, they drew nearer to Cradock's ships and opened fire. The *Good Hope*'s two 9.2-inch guns could not find their target in the fading light, and were soon put out of action by von Spee's flagship, the *Scharnhorst*, whose eight 8.2-inch guns, like those of the *Gneisenau*, which meantime was engaged in a similar duel with the *Monmouth*, now had the British at their mercy. The 6-inch guns of Cradock's ships, almost awash in the rolling seas, were useless. At 7.50 p.m. the *Good Hope* blew up, but not before Cradock had ordered the *Glasgow* to get away with all speed and warn the *Canopus*. The useless *Otranto* had been ordered away before the battle opened. The *Monmouth*, after being silenced and set on fire by the 8.2-inch guns of the *Gneisenau*, was finally sunk by the *Nürnberg*. No survivors were picked up by the Germans, either from the *Good Hope* or the *Monmouth*. With Cradock perished in this naval disaster off Coronel some 1500 officers and men.

Meanwhile the *Glasgow*, making full use of her 25-knot speed, had warned the slow old *Canopus*, and together they made their way back to the Falkland Islands to await developments. Four days later Lord Fisher, who had just succeeded Prince Louis of Battenberg as First Lord of the Admiralty (20th Oct.), dispatched Admiral Sturdee with a squadron bent on avenging Cradock, and protecting the valuable base and coaling-station of the Falklands. Sturdee's squadron included the two first battle-cruisers built—the *Invincible* and *Inflexible*, each of 17,250 tonnage, with a speed of 27 knots, and eight 12-inch guns, besides sixteen 4-inch guns and five torpedo tubes. There were also four lighter cruisers—*Carnarvon*, *Kent*, *Cornwall*, and *Bristol*; and to these were added the *Canopus* when Sturdee reached the Falklands on 7th December. The *Glasgow* had already been picked up in the South Atlantic. The superiority both in number and weight of guns was now overwhelmingly on the side of the British.

Von Spee, who claimed to have suffered little loss in his victory off Coronel, had returned in the meanwhile to Valparaíso to refit, leaving again for the Falklands on 15th Nov. His programme apparently was to do as much damage as possible to the British base and coaling-station at Port Stanley; account for the *Canopus* and *Glasgow*, which he expected to find defending the port; and thence make for South Africa in support of the rebellion there. Only some twenty-four hours before he approached Port Stanley Admiral Sturdee had arrived, and the news sent by the signallers on the island at 8 a.m., that the unsuspecting enemy was approaching, found the crews grimy from coaling, but alert and ready. Von Spee sent the *Gneisenau* and *Nürnberg* ahead to shell the wireless station, but a salvo of 12-inch shells from the *Canopus* in the harbour at 9.20 a.m. caused them to change their course. It was not, however, until 9.45 a.m., when the *Invincible* and *Inflexible* put out to sea with the *Glasgow* and *Kent*, that the presence of the battle-cruisers was revealed to them. It was then too late to escape. The German ships were no match for the British battle-cruisers either in speed or gun-power. With the conditions of Coronel thus reversed, Von Spee, abandoning the attempt to run as hopeless, decided to die fighting, and met his death as gallantly as Cradock had done some five short weeks before. Both the *Scharnhorst* and her sister the *Gneisenau*, battered by the two British battle-cruisers, who were later joined by the *Carnarvon*, until they were mere helpless hulks, fought to the last before they capsized, the first at 4.15 p.m., the second just after six. Boats were ordered out to save survivors, and some 200 Germans were picked up from these and other ships that were sunk. The *Leipzig*, pursued by the *Glasgow* and *Cornwall*, kept up a running fight for three hours, and then, hammered to pieces, hauled her flag, but afterwards sank. The *Nürnberg*, after a longer chase, suffered a similar fate at the hands of the *Kent*, sinking an hour after surrendering. Only the *Dresden* escaped, seeking refuge at Juan Fernandez, but three months later she was called to account there by the *Kent* and *Glasgow* (14th March, 1915), when, after a five minutes' action, she was blown up by her commander.

With no enemy fleet at sea the task of the British navy was reduced to guarding British commerce from submarines and raiders, keeping watch and ward in the North Sea, and conveying troops to and from the widely scattered theatres of war. Every month added to Britain's commitments in various parts of the globe. All hope of an early peace had vanished by the end of 1914.

Mobilizing the Empire

Happily for the British Empire, Lord Kitchener had from the first anticipated a long-drawn struggle. His call for volunteers "for three years, or the duration of the war", showed how clearly he realized the gravity of the situation. One of his first demands had been for another half-million men to go on with, and history has recorded how nobly the young manhood of the nation responded to his call. All the Dominions and Overseas possessions rallied to the Motherland with equal enthusiasm. We have shown how Indian troops—fighting for the first time on European soil—had already stepped into the breach on the Western front. The Canadians, nearest at hand, were the first of the Dominions to follow suit, but Australia and New Zealand, before their campaigns in Egypt and Gallipoli—and subsequently in France and Flanders—had already occupied Samoa, the Bismarck Archipelago, and other German islands in the Pacific.

South Africa, 1914

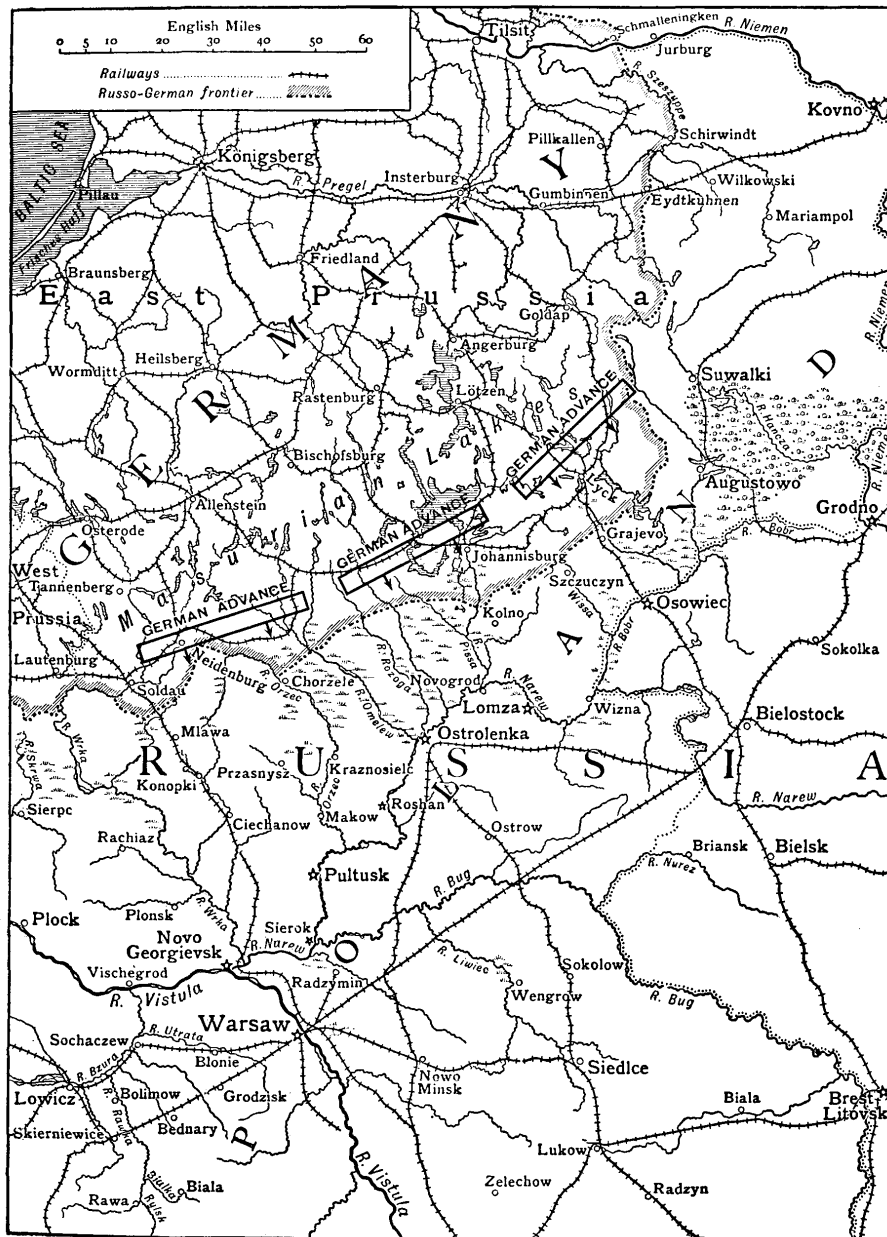
South Africa, on the outbreak of hostilities, had offered to carry the war into the German territory of South-West Africa, but General Botha had first to crush a revolt of Colonel Maritz's force; and this was succeeded by a more formidable rising in the Orange River Colony under De Wet, and in the Transvaal under Beyers. The response to Botha's call to arms proved the striking loyalty of the rest of the Union, and with the force thus mustered the South African Prime Minister completely defeated the rebels before the end of November. De Wet was captured, and Beyers was drowned in attempting to escape. The colonial campaign which followed will be dealt with subsequently.

1915 on the European Fronts

The Russian Campaign.—The heaviest burden of the war on the main fronts was now borne by Russia. Having failed to force a decision in the West, Germany looked to the Eastern front for compensating triumphs, confident that she could maintain her defensive positions against the Franco-British armies until such time as, with Austria-Hungary's help, she had brought Russia to her knees. It was also necessary to overawe Roumania and any other hesitating Balkan state that might be disposed to throw in its lot with Russia and her allies. Russia herself was provoking this reversal of German strategy by her renewed advances both on the Carpathian and East Prussian fronts. Hindenburg made two attacks on Warsaw early in the year (February and March), one by way of the Narew and the other by that of the Niemen, but

both failed, thanks chiefly to the indomitable spirit of the Russian infantrymen, ill-equipped though they were. It was not until Mackensen's great 'drive' began on the southern flank in Galicia that Germany's new strategy revealed itself. Russia had then reached the culminating point in her military career. Besides holding up the German offensive in Poland, she had made herself mistress of all East Galicia, Przemyśl having fallen to General Selivanoff on 22nd March after an investment of five months, thus releasing 100,000 men to reinforce the armies under Ivanoff, Dmitrieff, and Brussiloff—then battling for the passes which led through the Carpathians into the Plain of Hungary. Przemyśl alone yielded 126,000 Austrian prisoners, including the commander, General von Kusmanek, and 1000 guns. Between that period and the middle of April, when the Russians claimed possession of all the Carpathian heights along a front of 70 miles from south of the Dukla Pass to north of the Uszok Pass, another 70,000 Austrian prisoners were taken.

It was a dazzling, but an illusory triumph. The Russians had been deliberately led by the Austrians—under instructions from the German Higher Command—into their hazardous Carpathian adventure as part of the secret preparations for Mackensen's mighty blow elsewhere. Von Falkenhayn, then Chief of Staff, afterwards gave the credit for the plan to the German General Head-quarters. Germany, with all her factories turning out munitions of war far in excess of anything that the Allies could then muster, had been accumulating guns and ammunition for this purpose for months past, together with poison-gas and liquid fire, and a total force of some 2,000,000 well-armed men. Russia, on the other hand, though she might oppose this force with fully as many men, was coming to the end of her resources, and her troops were ill-equipped to meet the massed guns of the artillery brought against them when the German phalanx, after minor thrusts to left and right to cloak the real designs of the German Higher Command, began its overpowering advance on 1st May against the Dunajec lines, where Dmitrieff's Russian army believed itself securely entrenched. Mackensen's guns, opening up a way for the strongest army yet mustered under one general, blew the Dunajec lines to fragments. The Russian infantry clung to their positions to the last moment, but their rifles, often empty, were useless against high-explosive shells, or the waves of poison-gas which preceded the advance of Mackensen's shock troops.



Map to illustrate the German Attack north of Warsaw in February, 1915

On 5th May, with its front wholly turned, Dmitrieff's shattered army withdrew as best it could from the Dunajec lines to the San River. All Russia's gains in Galicia were destined to be sacrificed in similar fashion. Brussiloff's advance through the Carpathians was at once arrested; by 14th May, when Everts' army on the Nida had also fallen back, all the passes had been evacuated, though not without appalling losses. In the Bukovina, however, the Russian army under General Lechitsky maintained a stubborn resistance south of the Dniester until

27th June, when it fell back to the Gnila Lipa. It was high time to retreat. Przemyśl had again fallen into Austro-German hands (2nd June) as the first outstanding result of Mackensen's advance; Lemberg followed suit on 22nd June; and Halicz, abandoned by Brussiloff, fell on the day on which Lechitsky's army retreated from the Dniester to the Gnila Lipa. The end of June saw these positions abandoned and a further retreat in progress towards the line of the Lublin-Cholm railway.

Not cheaply were these spectacular triumphs won by the advancing armies of Mackensen and the Archduke Joseph Ferdinand. Their troops had been twice thrown back on the Dniester before that river had been finally won—a passage which cost them, all told, some 150,000 men; and in the successive retreats which followed, the Russian infantrymen turned repeatedly on their pursuers to prove that they were still capable of enforcing a price for every yard of ceded territory.

Mackensen's 'drive' was only part of the German Higher Command's plan for destroying the Russian armies in 1915. While the Austro-German phalanx was thus thrusting its way towards the Lublin-Cholm railway line, a simultaneous movement was in progress in the north, which had for its first objectives the great fortresses of the Polish salient, and Warsaw itself. Here Hindenburg, who was still in supreme command of the Austro-German forces on the Eastern front, had no longer General Ruzsky opposing him, Ruzsky having handed over the Russian northern command to Alexieff owing to ill-health. No matter how bravely the Russian infantrymen fought, or how ably they were led, they could not stand up against the hurricane of shot and shell which now blasted a path for the fully-equipped armies of von Below, von Eichhorn, von Gallwitz, von Scholtz, Leopold of Bavaria, and von Woyrsch. Prasnysz was won by von Gallwitz after a fierce battle in the middle of July, the Russians retiring to the shelter of the Narew fortresses guarding Warsaw from the north-east. Mackensen's advance to the south was resumed the next day, and the Grand Duke Nicholas, foreseeing the peril of this double threat, realized that his only hope lay in flattening the Warsaw salient and thus shortening his line. This sealed the fate of Warsaw, which was entered by Prince Leopold of Bavaria on 4th Aug. Ivangorod fell on the following day; Kovno on the 17th; Novo Georgievsk on the 19th. These losses, though deplorable, were not vital while the Russian armies still retained power to retaliate and recoup. Hindenburg strained every nerve to crush them once and for all. Ossowiec fell on 22nd Aug., the Russians resuming their retreat from the Niemen and Bobhr. Brest-

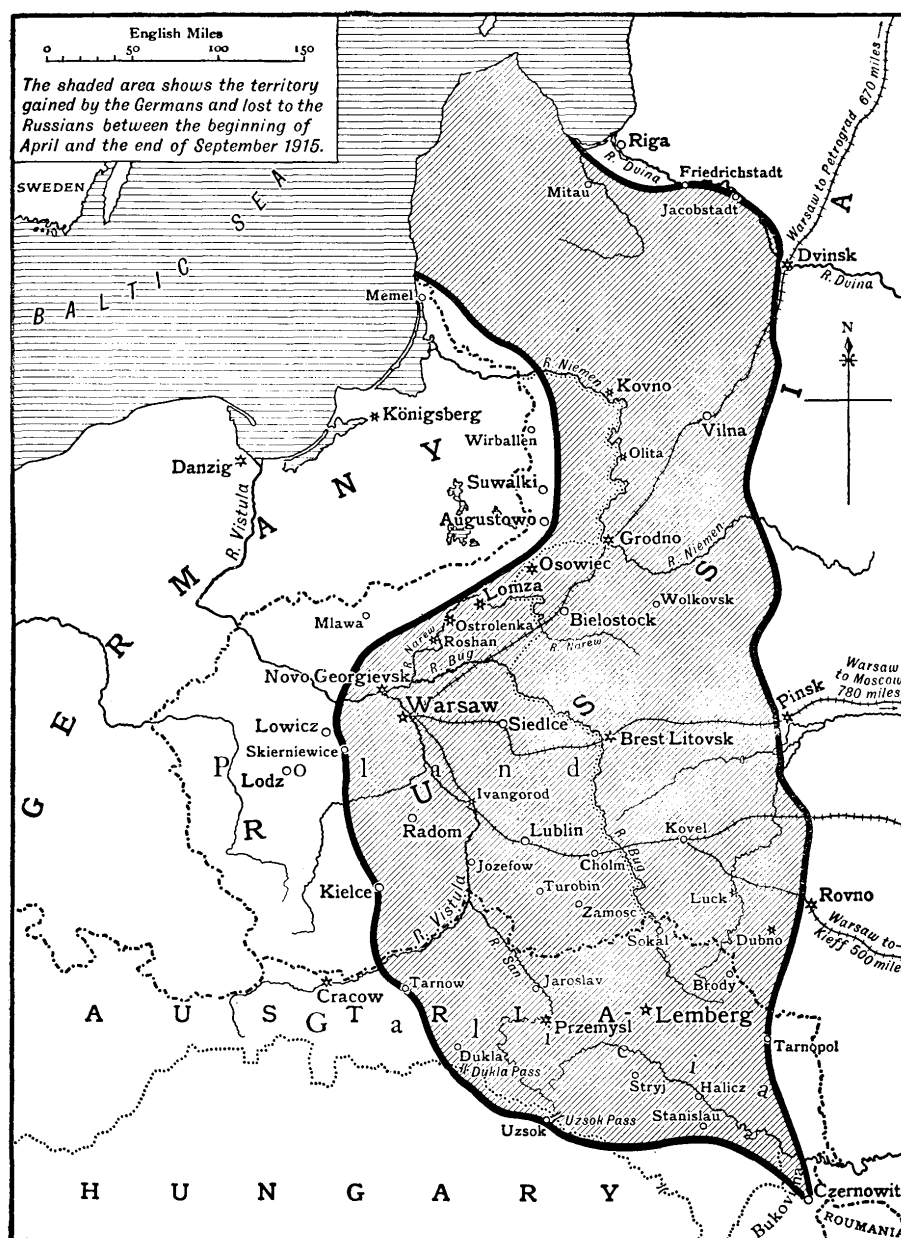
Litovsk had already been threatened by the converging movement of Mackensen and Prince Leopold of Bavaria. Seeing no hope of saving it in the face of the continued pressure, the Grand Duke Nicholas evacuated this most easterly of the Polish quadrilateral of forts on the 25th, having previously stripped it, as in the case of the other evacuated fortresses, of all war material.

Hindenburg strove to complete the discomfiture of the main Russian armies by a fresh advance on his extreme left, where von Below was ordered to push through Courland towards Riga, with Petrograd as the ultimate goal in the following year. On this front the Russians had already been forced to relinquish Memel, just across the German frontier, as well as Libau. German naval forces had shared in the operations on the Riga coast-line, and when von Below, after carrying Mitau, some 30 miles from Riga itself, met with prolonged resistance, they made an ill-fated attempt to capture the port from the sea. This was on 18th Aug., when the Russian fleet appeared on the scene while the German naval contingents were attempting to land in flat-bottomed barges at Pernau. The landing forces were annihilated, and the German ships beaten off with a loss of two cruisers and eight torpedo-boats. The Russians only lost an old gunboat in this one-sided action.

The naval operations, however, had little effect on the main issue. Russian fortresses continued for another month to fall like ninepins before the Austro-German armies. Grodno was evacuated at the beginning of September, and though General Everts escaped from Brest-Litovsk with his supplies and guns, he could not hold up Mackensen's irresistible march on Pinsk, even in the Pripet marshes, which were dry at that season of the year. Pinsk was occupied on 16th Sept. Nowhere was the pressure relaxed. In the south, where the flood-tide of the Teutonic advance had never set so strongly, the attack on the Volhynian fortresses had been vigorously opposed by Ivanoff; but Boehm-Ermolli entered Lutsk on 1st Sept., and the Austrians recaptured Brody on the same day.

The vital blow at this stage was being delivered in the north, where von Below, bent on reaching Riga for his winter quarters, was marching on the Dvina lines with the immediate object of crossing that river and turning the whole Russian front as far as Dvinsk. The extreme left flank of the Germans fought desperately for the Dvina crossing at Friedrichstadt, but failed to make it good, and the danger-point shifted towards Vilna, the ten days' battle for which was decided at Meiszagowla on 12th Sept. Though two Russian divisions of the Imperial Guard were brought up to defend this key position, they were

powerless to hold it against the great weight of German artillery. With its capture on 12th Sept., Vilna's fall became merely a matter of days. Before the Vilna armies could make good their escape, Hindenburg endeavoured to crown his triumph by outflanking them on both sides, von Eichhorn's cavalry sweeping round from Vilkomir in the north, and von Scholtz pressing forward, though less rapidly, on the southern side of the salient.



The Germanic Slice out of Russian Territory at the end of the Summer Campaign of 1915.

In this supreme crisis on the Eastern front, Ruzsky, recovered from his illness, returned to his command of the northern battlefields, and signaled his reappearance—not for the first time—by changing the whole complexion of affairs. Reinforcements enabled him in the first place methodically to evacuate the Vilna salient under their protection. Hindenburg endeavoured to counter this by rushing up cavalry reinforcements, with 140 guns, to support his outflanking thrust in the north, which, reaching Vidzy on the 16th and Vileika on the following day—this being well to the rear of the Vilna armies—threatened irremediable disaster to the retreating Russians. They were saved by the series of flank-guard battles securing their one avenue of retreat, and Ruzsky's counter-offensive from Dvinsk—a stroke so effective that the long German cavalry arm was in itself now in danger of being cut off. Vidzy was recaptured on the 20th; Smorgon, south of Novo Grodek, on the 21st; and Vileika before the end of the month.

On 15th Sept. Lord Kitchener had publicly declared that the Germans had "shot their bolt" on the Eastern front; and, so far as the immediate destruction of the Russian army as a force in being was concerned, this was true, though it was hard to believe while the wide sweep of the German advance was in full force. Ivanoff's reaction was equally marked in the south, where Brussiloff and Lechitsky took von Bothmer and Pflanzer-Baltin by surprise. Before the end of September the Austro-Germans had not only been pushed back to the Strypa, but had also lost both Dubno and Lutsk (23rd).

Germany's great summer offensive was over, but Hindenburg tried hard to secure good winter quarters in the north by a renewed advance on Dvinsk and Riga. A frontal attack was launched on Dvinsk on 3rd Oct. and was a costly failure. Ruzsky had defended Dvinsk with a semicircle of far-flung trenches on the Verdun model, against which the German shock-troops and guns could make practically no progress. After three weeks of vain endeavour Hindenburg shifted the attack to Riga, with no better success and heavy additional casualties. Thrust and counter-thrust succeeded one another with little change in the general situation until the end of November, when, after temporarily securing a crossing at Dahlend Island, south-east of Riga, in the River Dvina, Hindenburg was forced to abandon the attempt as futile. With the help of their fleet the Russians won their way back to Kemmern; and in their counter-offensive from Dvinsk in the same month recaptured Illutsk. All hope was then abandoned by the Germans of taking either Riga or Dvinsk that year. The German effort in the south, below the Pripet marshes, also slackened. Ivanoff

not only maintained the ground he had won, but scored several notable victories in the Strypa sector; but both here and along the Styr, where Lechitsky was opposing Bothmer, there was both give and take and nothing decisive—apart from the fact that Roumania was saved by this evidence of Russia's recuperative powers from choosing the wrong side.

In order further to influence the dubious attitude of Roumania, a fresh Russian offensive in the Bukovina was begun in the last days of the year, with Czernowitz as the objective; but as this rightly belongs to 1916 it will be dealt with in our summary of the operations for that year. Though Russia had not succumbed as a military power under the staggering blows she had received in 1915, she had lost 2,000,000 of her best fighting men, and the *moral* of her army was never so high again. Falkenhayn has hinted in his *Memoirs* that the Germans knew that the blind faith of the Russians in their rulers was already shaken before they started Mackensen's 'drive'. It could not be expected to endure in face of the criminal neglect and corruption which every day added to their hardships and losses at the front. The Russian court at that period has been described as a mixture of folly and intrigue, with 'dark forces' at work under pro-German influence, led by the impostor Rasputin. The Grand Duke Nicholas, who was above the treacherous influences now undermining all departments of the Russian system, had been transferred to the command in the Caucasus in the most critical hours of the Austro-Germanic advance, the supreme command of the Russian armies being taken over by the Tsar himself (5th Sept.), with Alexieff as Chief of Staff. The Tsar's motives were above suspicion; but he lacked the efficiency and generalship of the Grand Duke, and stood for a system which, under the searching test of war, was proving itself unworthy of the continued sacrifices of his subjects. The sacrifices were repeated in 1916, but the seeds were already sown of the red harvest which was to lead to Russia's downfall and the end of the Romanoffs.

The Balkans, 1915

The progress of German arms in 1915 had decided Bulgaria to throw in her lot with the Central Powers. Her price—fixed by secret treaty with Germany in July of that year—was the whole of Macedonia possessed by Serbia, and other valuable slices of territory. It was not until 12th Oct. that formal war was declared by Bulgaria against Serbia, five days after the fresh invasion of Serbia had begun under Mackensen's leadership, with two Austro-German armies, one under General Koevess, advancing west of Belgrade in a wide flanking

movement along the old roads over the Save and the Drina, and the other, under General von Gallwitz, advancing east of Belgrade against the main Serbian forces. Against this new Mackensen 'drive', with fully-equipped forces larger than the whole Serbian army, organized with all the Teutonic thoroughness which marked the same leader's Galician triumph, the Serbians had no chance, though they fought, as ever, with stoic resistance, and exacted a price for every inch of ceded territory. While they were thus stubbornly retreating, Bulgaria threw in two of her armies on the Eastern front, thus threatening, with the advancing Austro-German forces, to enclose them in a wide loop. The tragedy of it was that Serbia's allies were powerless to save her; and that Greece, who by the terms of her treaty with Serbia should have gone to her assistance as soon as Bulgaria attacked her, declined through King Constantine to do so, notwithstanding the insistent advice of his Prime Minister, M. Venizelos. Convinced, like King Ferdinand of Bulgaria, that Germany was winning the war, King Constantine maintained to the end an attitude which, though he chose to call it neutral, was never friendly towards the Allies. Russia had her hands too full to go to Serbia's aid, and though a Franco-British attempt was made as the net closed round the encircled Serbians, it was too late to save the situation.

The story of the Serbian disaster of 1915, when the fall of Monastir on 2nd Dec. robbed the Serbians of their last stronghold, is that of a desperate flight across the frontier and over the mountains of Albania and Montenegro to the Adriatic. Thanks to the Serbians' heroic efforts, the Austro-German armies had not been able to close the net tight, and though the Bulgarians followed hard on their heels, they could not quite complete their victory. All told, however, the Serbians lost some 50,000 men, killed, wounded, captured, or starved, in the retreat, together with their guns and equipment. Their aged monarch shared the retreat and succeeded in reaching Brindisi.

Meanwhile the Franco-British force, which, as already related, arrived too late to prevent this final act of the Serbian tragedy, had established a strong base at Salonika, notwithstanding Greek protests on the grounds of neutrality. It was not until 14th Oct. that the combined force, under the leadership of General Sarrail—the British column being commanded by General Mahon—began to move up the Vardar valley, the British advancing on the right towards Lake Doiran, and the French towards Strumnitza. Both forces were soon in touch with the Bulgarians, and fought a number of minor engagements in their forlorn hope of effecting a junction with the hard-pressed Serbians. Besides

being too late, however, the Franco-British forces were not strong enough to effect their purpose, and when the remnants of the Serbian army had been forced across the frontier towards the Adriatic, they were themselves attacked by powerful Bulgarian columns. The object of his expedition having been eliminated, General Sarrail prepared for retreat to his base. The Bulgarians did their best to harry his retirement. They launched a determined attack, which he repulsed with heavy loss; and then endeavoured to isolate the two columns by an assault on the British force at Lake Doiran. Though some 1300 casualties were sustained in defeating this attack, the British, acting as flank-guard to the French, enabled the retreat to be made jointly. By 13th Dec. the Allied troops, having administered a severe check to the pursuing Bulgarians two days previously, were across the Greek frontier in good order, and in due course had entrenched themselves about Salonika.

With the fall of Serbia came the collapse of Montenegro, in circumstances considerably less heroic than those which marked the Serbian retreat. The key position of Mont Lovtchen was abandoned to the Austrians with little if any show of resistance, and Cetinje, the capital, similarly entered by the invaders. King Nicholas of Montenegro sought refuge in Paris; Prince Mirko of Montenegro in Vienna.

The Serbian soldiers who survived the great retreat, numbering some 100,000 in all, were met on the Adriatic coast by units of the Italian fleet and transferred to Corfu—to recoup and refit for the later campaigns which were to lead to the recovery of their country.

Italy, 1915

Italy, whose warships were thus instrumental in salving the Serbian army, had thrown in her lot with the Allies by declaring war against Austria-Hungary on 23rd May, 1915. Austria had refused to offer adequate 'compensation' for her disturbance of the Balkans; and, moreover, the time had obviously arrived to complete Italian unity. A few weeks previously Italy had signed the Treaty of London, under which the Allies agreed to satisfy most of her territorial ambitions when the time came to share the spoils of victory an agreement which led to some of the most difficult problems in the final peace settlement. To Italy's honour be it added that she joined forces with the Allies when their prospects were none too bright, when they were able to report little or no progress either on the Western front or in Gallipoli, and Austro-German arms,

on the other hand, were beginning to carry all before them in Mackensen's great drive in Galicia.

Italy was in no position to throw her whole weight into the struggle in 1915. Though her war strength was reckoned at a million men, her army was ill-equipped with guns, especially with modern heavy artillery and machine-guns, and her industrial resources were wholly inadequate to make good the deficiency. The mountainous frontier which she had to defend, too, gave every advantage to the Austrians. She succeeded in seizing three of the passes, the Stelvio, Tonale, and Guidriari, on the east side of the Trentino, and in blocking others on the west side, as the opening moves of her campaign, the object being to secure her flank in the subsequent offensive operations which aimed at Trieste by an advance across the Isonzo. Though these operations succeeded in pinning to the Italian front considerable forces of Austro-Hungarian troops which might have been thrown into the Russian furnace, the Italian effort fell far short of its objectives. General Cadorna, the Italian Commander-in-Chief, won a number of small successes in deploying his Third Army on the right bank of the Isonzo during June and July, securing the bridge-heads at Caporetto—the scene of Italian disaster two years later—Plava, Gradisca, and Monfalcone, thus holding the western bank of the river from Tolmino down to the sea. But the Italians were now faced with powerful defences, buttressed by the Carso Plateau in the south, which could only be carried at that time at prohibitive cost. All attempts to capture these strongholds broke down, and though a footing was gained on the Carso, and slight gains were constantly reported from the Trentino, the operations along the Italian front settled down before the year was out to the give-and-take fighting which characterized the siege operations in the West.

Western Front in 1915

On the Western front neither France nor Great Britain was ready in 1915 to undertake any advance comparable with the great offensives of the Central Powers in the East. Russia in her agony complained that France was not doing enough, but all the Allies' efforts this year were crippled by their inability to supply the wholly unprecedented demands for munitions of war. Great Britain was still struggling months behind to catch up a foe who had been preparing for years. Mr. Lloyd George subsequently related how, in the month of May, 1915, when the Germans were turning out 250,000 shells a day, most of them high-explosives, Great Britain was producing a mere 2500 a day in high-

explosives, and 13,000 in shrapnel. The French, accustomed to supplying the demands of armies on a Continental scale, had naturally done considerably better than this, but even their most strenuous efforts were inadequate to cope with the enormous output of the German arsenals. Mr. Lloyd George retired from the Chancellorship of the Exchequer in order to assume control of the newly created Ministry of Munitions, which in due course more than made good all these defects. That, however, was not in 1915. Up to the end of that year, according to Lord French, "the scanty supply of munitions of war paralysed all our power of initiative, and at critical times menaced our defence with irretrievable disaster".

At the end of the first long winter of dreary trench warfare the British Commander-in-Chief deemed it necessary to undertake an offensive in order to prevent the *moral* of his army from deteriorating. Hence the battle of Neuve Chapelle, which, begun on 10th March, was fought with a small reserve of ammunition accumulated for the purpose, and had to be broken off after three days' struggle through lack of further supplies. The troops chosen for the main assault were Rawlinson's Fourth Army Corps, with the Indian Corps on the right. Following the preliminary bombardment, they quickly overran Neuve Chapelle itself and made 1000 yards progress on a 3-mile front. But to left and right the attacks were held up, and two further days' fighting failed to add to the gains—purchased at the excessive cost of 562 officers and 12,239 men. The total German losses, including 1680 officers and men as prisoners, were estimated as rather higher than this, but the net result, though ranking as a British victory, was admittedly disappointing.

Earlier in the year Lord French had endeavoured to convince Joffre that the proper rôle for the British army to fulfil was an advance on the extreme north in co-operation with the British navy. Joffre was unsympathetic, though he held out hopes of co-operating in such an advance with the French army at a later date. His plan for the 1915 campaign was to break through the German line from the south at Rheims, and from the west at Arras. To do this he must mass as many French corps as possible behind these points, meanwhile keeping the enemy busy elsewhere in order to prevent him from reinforcing the threatened positions. This general strategic idea, as Lord French has pointed out, was the foundation of all the Allied efforts in the West throughout 1915. It led to numerous local successes along many parts of the line, but no real advance was made towards the main objectives. These were not defined until the combined offensive was launched in September.

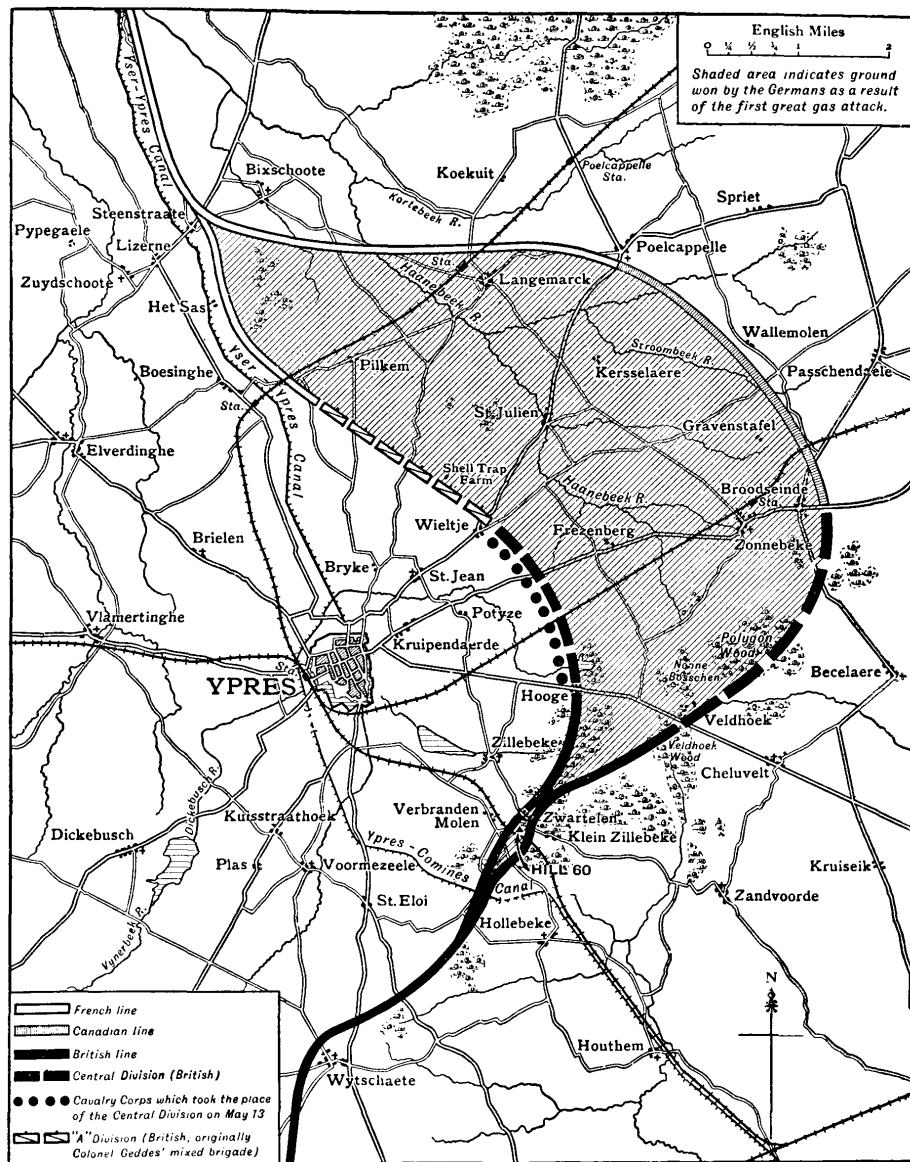
The Germans themselves, though content to leave to the Allies most of the attacking in the West in 1915, maintained a sufficiently active offensive-defensive. While the French in Alsace were making a fresh advance on Mulhouse at the beginning of the year, they counter-attacked at Soissons, after bombarding the cathedral on 9th Jan. It was only after a week's desperate fighting and heavy French losses—including a bridge-head on the Aisne—that they were checked. In Champagne the French managed to capture Perthes (8th Jan.), and strove valiantly but vainly to wipe out the St. Mihiel salient. The most ambitious effort of the opening months of the year was the British offensive at Neuve Chapelle, which, as already pointed out, failed largely through lack of ammunition. In his report on that battle the British Commander-in-Chief referred to the pressing need of "an almost unlimited supply of ammunition"; and the lack of it was the real explanation of the Allied failure in 1915.

Germany knew well enough how matters stood in this respect, and added ruthlessly to the handicap which their own superior supplies gave them by suddenly attacking with chlorine gas—the first use of poison-gas in the war. This was on 22nd April, following a grim struggle south-east of Ypres for Hill 60, the flattened remains of which, after five days' incessant fighting, remained in British possession. Having been careful beforehand to accuse the French of using poison-gas near Verdun on the 14th—a charge without justification—the Germans launched it in dense volumes from pipes previously laid down for the purpose north-east of Ypres. The attack was preceded by a heavy bombardment, the gas-clouds following at 5 p.m. on the 22nd. The Allied line was held at this point by French Colonial and Territorial troops, with the Canadian Division on their right. All unprotected as they were against this diabolical form of warfare, the French troops, gasping for breath, broke and fled. Many fell asphyxiated. With a gap in the Allied line 5 miles wide, the Canadians suddenly found their flank left in the air. Less affected by the gas than the French, they were chiefly instrumental in saving the situation by a valiant resistance until reinforcements could be sent to fill the gap.

The gap was evidently wider than the Germans either anticipated or realized; otherwise the disaster might have been irretrievable. As it was, the situation remained precarious until the 27th, when a counter-attack in conjunction with the French recovered some of the ground, and a large portion of the sorely tried Canadian Division was relieved by the Lahore Indian Division. Altogether seven British divisions were involved in this hard-fought battle, the

net result of which was to bring the Germans 2 miles nearer to Ypres on a 5-mile front, and to give the Allies a worse line to hold. Eight batteries of French field-guns were lost and four British guns of position. These last were recaptured by the Canadians, but the enemy had already destroyed them. In all the Allied casualties amounted to nearly 25,000. The Germans estimated theirs at 16,000. In his report on the gas-attack Sir John French declared that protest against this form of warfare would probably be useless, and Lord Kitchener intimated in the House of Lords on 18th May that retaliation might be inevitable. Respirators more or less effective were supplied to the troops, and the use of poison-gas, followed by liquid-fire—another German innovation—became permanent additions to the horrors of modern warfare.

Before the new battle round Ypres died down—it lasted, indeed, until the end of May—the storm centre shifted to the southern end of the British line, where it joined hands with the French left. Here General French began the battle of Festubert, undertaken to relieve the intense pressure on the troops at Ypres, but also serving as part of Joffre's general plan of attack in the direction of Lens and Lille. British and French alike were launched against the German lines on 9th May, the British taking the offensive between Rougebanc and Givenchy, and the French between Neuville St. Vaast and Notre Dame de Lorette. The renewed struggle for Ypres, however, had drawn heavily on the scanty British reserves of ammunition, and the preliminary bombardment of forty minutes proved wholly inadequate to crush the resistance offered by the enemy's numerous fortified posts when the First Army advanced to the attack. This disastrous engagement, in which the greatest bravery was displayed against overwhelming odds, cost over 12,000 casualties. It achieved nothing in the field, but the lessons which it taught led to the formation of the Coalition Government, with Mr. Lloyd George as Minister of Munitions. The second stage of the battle of Festubert, which began at midnight on 15th May, was more successful, the enemy's front-line trenches being captured on a front of 3000 yards; but the losses incurred in winning and holding the positions were disproportionately high.



The Ypres Salient before and after the Second Battle of Ypres, 22nd April-13th May, 1915

The French effort began more auspiciously as a result of the longer and more intense bombardment which preceded their attack on the 9th, but the series of minor successes which they won round Souchez, after weeks of incessant fighting, made little real impression on the defences of Lens. The truth was that Germany had so expanded her war-material factories that, with the aid of the Austrians, she could turn out sufficient shells and guns for her main offensive on the Eastern front, and at the same time overweight the Allies in the West.

Throughout the summer the line, though never quiescent, and often breaking out in furious bombardments, minor attacks and counter-attacks, and raids on both sides, remained little altered. The hardest fighting of all was round the war-scarred salient of Ypres, still held, as in the first gas-attack, by the Second Army, to which some of 'Kitchener's Men' were now attached. It fell to this advance-guard of the New Army to bear the brunt of the first attack on the British with liquid-fire, the Germans, who had already used this new device of the *flammenwerfer* against the French, employing it in another desperate assault on the British lines round Hooze. The New Army units fought with almost incredible gallantry, but were blinded by the unexpected, burning sheets of flame, and while they were still blind the enemy charged and took the first-line trenches on a front of some 500 yards. The losses were avenged on 9th Aug., when the 6th Division recovered all the captured positions, and 400 yards of German trench into the bargain.

With the arrival of the reinforcing British divisions of the New Army, General French was able to take over some 17 miles of additional front, the British line thus extending over about 50 miles, with the Belgians on the left holding the remaining 18 miles to the sea. This still left the French army 500 miles to hold, from the British right to the Alps.

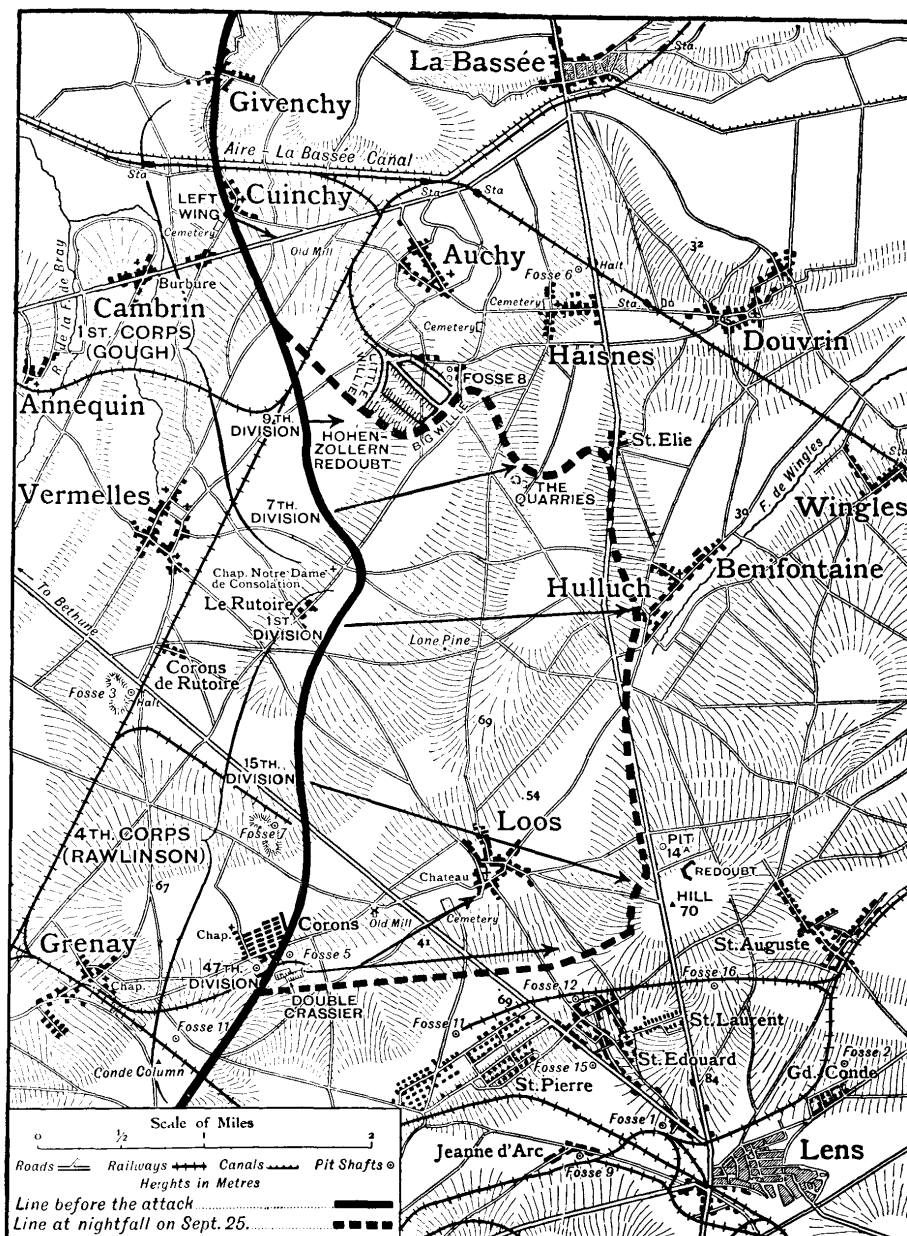
The summer of 1915 passed away without any great offensive on either side. The Germans, now at the flood-tide of their sweeping advance against Russia, were content to continue their vigorous offensive-defensive in the West. Besides the fighting already referred to, there was incessant warfare in the Argonne Forest, where the German Crown Prince was noisily active throughout the summer, threatening Verdun, but making no serious advance. The French continued the deadly trench-to-trench warfare in the Souchez area and the 'Labyrinth' region nearer Arras, and steadily tightened their hold on the reconquered corner of Alsace, consolidating their positions on the Hartmannsweilerkopf, which had been the scene of continuous fighting in the renewed advance towards Mulhouse. For the most part, however, the French, like the British, were now storing up reserves of ammunition and completing their dispositions for the joint offensive planned for the autumn, when the main objects of Joffre's general strategic idea for 1915 were for the first time clearly defined, though unattained.

A great Allied offensive in the West had become increasingly necessary in view of the prestige gained by the Central Powers, not only by their tremendous advance into Russia, but also by the Allies' disastrous campaign in

Gallipoli, where the failure of the Suvla Bay landing—subsequently dealt with in our account of the operations against Turkey—had just given the enemy additional cause for rejoicing. Joffre's plan of attack, designed "to drive the Germans out of France", was not ripe for execution until towards the end of September. Even then, though General French had for months been gradually accumulating troops and ammunition for the blow—the British army now numbering nearly a million men—full strength in both respects was still only half developed.

There were two main assaults in the combined offensive, launched on 25th Sept., 1915, the chief of which was in the French centre in Champagne, where de Castlenau, Joffre's right-hand man, attacked with Langle de Cary's Fourth Army on a 16-mile front between Auberive and Massiges, the object being to force the Germans back on the Aisne, and, if possible, cut off the army of the German Crown Prince in the Argonne. Second in importance was the Franco-British advance on the same day in Artois, General French's object being to push through between Lens and La Bassée on the north, while Generals Foch and d'Urbal, on his right, stormed the Vimy heights and attacked Lens from the south. Secondary operations were carried out at various other points in order to distract the enemy's attention, feint attacks being made by Sir Herbert Plumer with the 5th Corps—part of the Second Army—east of Ypres, where Bellewarde was temporarily taken; as well as by those units of the First Army occupying the line north of the Béthune-La Bassée Canal. Similar demonstrations were made at Bois Grenier, and along the slopes of the Aubers Ridge, where British and Indian troops alike fought heroically for ground which, though captured in the first onrush, could not be held under the powerful fire concentrated against them.

Battle of Loos



Map showing approximately the battle-lines of the First Army under Sir Douglas Haig at daybreak and at nightfall on 25th September, 1915

The main British attack was delivered by the First Army under Sir Douglas Haig along a front extending from La Bassée Canal in the north to the mining village of Grenay on the south, and is conveniently named the battle of Loos. The task of the 4th Corps (Rawlinson), with the 47th (London Territorial) Division on the right, the 15th (New Army) Division in the centre, and the 1st Division on the left, was to carry, as its first objective, Loos and the heights between Lens and Hulluch. The 1st Corps (Hubert Gough), with the 7th

Division on its right, the 9th (New Army) Division in the centre, and the 2nd Division (Horne) on the left extended to the canal, was to link up with the Fourth Army at Hulluch, taking in its stride the Hohenzollern Redoubt and the formidable German fortresses in the Quarries and Fosse 8. Had this general plan succeeded, and the Tenth French Army on the British right advanced in line with it, a great gap would have been made in the German positions in Artois, which, with a similar success from the mightier French blow in Champagne, would at least have brought relief to the hard-pressed Russians in the East, if it did not achieve all that Joffre had hoped for it. As it was, the Allied victory, both in Artois and Champagne, fell far short of its aims, and, save for insubstantial advances, was fruitful chiefly in bloody experience.

An intense bombardment for four days preceded the main British advance, and was followed by a gas-attack—used by the British on this occasion for the first time.^[1] Many parts of the German defences along the 7-mile front, however, were far from obliterated when the infantry advanced to the attack, especially on the left, where the 2nd Division was held up by impassable wire defences. The 9th Division had a similar experience on its left wing, where the undemolished defences of the Hohenzollern Redoubt became a veritable shambles. On its right, however, the leading Scottish battalions scored a fine success in reaching Fosse 8. On their right the 7th Division did equally well, capturing the Quarries and reaching Cité St. Elie, north of Hulluch, though subsequently forced to rest on the Quarries. In the meantime the 4th Corps, like the 1st, had been repulsed on the left and victorious on the right. Held up on the left by unbroken wire and undamaged German defences, the 1st Division could make no appreciable advance on the Lens-Hulluch road, thus exposing the left flank of the next division in line—the 15th (Scottish). The Scotsmen on their part had made a magnificent advance, first sharing with the Londoners of the 47th Division the honour of capturing Loos, and then rushing on impetuously to Hill 70, and beyond as far as Cité St. Auguste, on the outskirts of Lens. Had reserves been available at once to consolidate the gains thus won, Lens itself might have been taken, and a door flung open leading to Lille. But as it was impossible to say at which part of the long British line they would be needed, Sir John French had retained his reserves far in the rear. When at last they were forthcoming—the new 21st and 24th Divisions, hitherto untried—they were put into the battle-line after nightfall, tired and hungry after a heavy 8-mile march.

It was impossible to hold all the ground won during the day. Without reinforcements, and no sufficient artillery support, the 15th Division had already been forced to relinquish its hold on Cité St. Auguste, and fell back to the western slopes of Hill 70 on the Lens-Hulluch road. Misdirected and drenched with rain, the two tired reinforcing divisions could do little to help amid the confusion of the battlefield. Their supporting attacks at various points broke against the still intact German wire or simultaneous counter-attacks by the enemy. One brigade alone (the 72nd) lost 78 officers and 2000 men out of the 3600 with which it started. Altogether the losses of these two divisions amounted to 8000. Along the southern section they succeeded in maintaining contact with the 47th Division, and in the north helped the Scots of the 9th Division to maintain their hold on Fosse 8; but the Quarries, close by, were lost, and along most of this northern front the day's gains were gradually whittled away.

Another heavy handicap was the fact that the French Tenth Army, on the right of Rawlinson's Corps, had to postpone its advance until one o'clock in the afternoon—the British attack had been launched at 6.30 a.m.—and then was forced to direct the corps operating on its left in a south-easterly direction. This involved a considerable gap on the British right, when Rawlinson's men made their victorious advance. The Londoners of the 47th Division, however, not only held on to Loos, but formed a strong defensive flank which averted what might have been a complete disaster. Apart from the captured positions, 57 German officers and 3000 other prisoners had been taken during the day on the British front, together with 26 field-guns and 40 machine-guns.

On the following day the Germans counter-attacked in force and recovered Fosse 8, but on the 27th the Guards Division, under the Earl of Cavan, was sent forward and almost restored the earlier gains, including Chalk Pit Wood and the slopes of Hill 70. It cost the Guards 3000 casualties to make good the restored British line. They continued to hold it until the end of the month, when they were relieved. The 15th (Scottish) Division had also been withdrawn, after suffering no fewer than 6000 casualties. All told, the British losses in the battle of Loos amounted to 50,000 men and 2000 officers, including three divisional commanders—Major-General Sir Thompson Capper (7th Division), Major-General G. H. Thesiger (9th Division), and Major-General F. D. V. Wing (12th Division)—each of whom was killed. A series of costly counter-attacks on the enemy's part failed to make much impression on the new British line, and mounted up the German casualties until they were

estimated at many more than those of the British. The battle died down; the gains were consolidated; but the murderous struggle at close quarters for the Hohenzollern Redoubt and its adjoining entrenchments continued for weeks and months, an outstanding feature of which was the attempt of the 46th (North Midland Territorial) Division to carry the redoubt by storm on 13th Oct. The Midlanders' task was handicapped, like so many British operations at this period, by inadequate artillery preparation, and though they fought like veterans they could only win the western side of the stronghold at a cost of 4000 casualties.

The advance of the 10th French Army on the right of the British was held up in front of Souchez on the opening day of the combined offensive, but made better progress on 26th Sept., when d'Urbal's troops made themselves masters not only of long-contested Souchez, but also of Thelus, La Folie Farm, and most of the Givenchy Wood. But the Vimy heights, notwithstanding that some progress was made along their slopes, still barred the road to Lens from the south. On the 28th the French 9th Corps, at the British Commander-in-Chief's request, took over the defence of Loos, and the British line was rearranged.

The main French effort in 1915, as already pointed out, was in Champagne, where a solid week's bombardment paved the way for the great advance on 25th Sept. Inspired by Joffre's stirring Order of the Day, "Remember the Marne: Conquer or Die!", the French troops carried all before them on the greater part of the front. General Marchand's Colonial Division broke clean through 2 miles of the main German defences, Marchand himself falling severely wounded at the head of his men. The greatest advance was made on Marchand's right, from Navarin Farm to the Butte de Tahure, where an advance of 2½ miles was made before the day closed. But the troops in the centre were robbed of decisive victory by a double check on the wings. On the right the two German strongholds at the Butte de Mesnil and the Main de Massiges—comparable with the Hohenzollern Redoubt in their strength—held out stubbornly until, after days and nights of ceaseless combat, both fell into the attackers' hands. On the extreme left the assailants could make practically no headway.

Thenceforward the French advance made little progress towards the main objectives, though a breach was made in the enemy's second line in a fresh attack on the 29th; and a third advance (6th Oct.) won the village and Butte de Tahure. On 20th Oct. the Germans recovered the Butte de Tahure, and in other counter-attacks prevented the French from developing their first initial advance

into the greater victory which Joffre had hoped for it. The battle had yielded an impressive list of captures—the total number of German prisoners being over 23,000 before the end of September, and the captured guns 80—but the Allies' long line had not materially altered before the autumn offensive gave place to another winter of tedious siege warfare.

The year closed with the appointment of Joffre as Commander-in-Chief of all the French forces, General de Castlenau taking over the immediate command of the French troops in France; and the resignation of Sir John French—now created a Viscount of the United Kingdom, and appointed to the Home Command—after more than sixteen months of severe and incessant strain at the front. Lord French was succeeded by General Sir Douglas Haig, who had been singled out for promotion by his brilliant achievements since the British army first landed in France.

The Naval War in 1915

Throughout 1915 the operations at sea contained no movements so striking as some of those which marked the opening months of the war. The careers of all the scattered German cruisers were over, the last of them, the *Königsberg*, being finally destroyed in the Rufigi River, German East Africa, by the shallow-draught monitors *Severn* and *Mersey*, sent out for the purpose from Great Britain. The German High Seas Fleet remained in harbour, waiting for Lord Jellicoe to be tempted or goaded into some imprudent disposition of his forces. Hence the sudden raids on the British East Coast, begun in the closing months of 1914. They repeated them once too often, on 24th Jan., 1915, when the raiders, consisting of 4 battle-cruisers, 6 light cruisers, and a force of destroyers, were encountered off the Dogger Bank by the British battle-cruiser squadron under Admiral Beatty, consisting of the *Lion*, *Tiger*, and *Princess Royal*, as well as the *New Zealand* and *Indomitable*, and the light cruisers *Southampton*, *Nottingham*, *Birmingham*, and *Lowestoft*, together with the *Arethusa*, *Aurora*, and *Undaunted*. Outmatched by the 13.5-inch guns of Beatty's 'Cat' Squadron of battle-cruisers, the Germans made for home. In the hot chase which followed, the *Blücher*, last in line of the German battle-cruisers, was hit repeatedly, fell behind, and was eventually sunk by a torpedo from the destroyer *Meteor*. Her survivors were picked up by the *Arethusa*. Beatty's flagship, the *Lion*, which was leading the pursuit, was partly disabled by a chance shot and had to be towed home, Beatty himself following the chase at some distance in a destroyer. Before he could pick up his place in the

pursuit he met his three battle-cruisers returning, these having broken off the action owing to the increasing risk of straying into an enemy mine-field or of falling foul of the mines which the retreating Germans were strewing in their path. Two other German battle-cruisers had been set on fire by the British shells, the *Seydlitz* and the *Derfflinger*, but, with the rest of the German ships, they made good their escape. Taught by this experience, the Germans made no further naval raids on the East Coast.

In the following month (4th Feb.) Vice-Admiral von Pohl, Chief of the German Admiralty Staff, proclaimed a submarine blockade of the whole of the British Isles, declaring all the waters round Great Britain and Ireland a military area in which Allied merchant-ships were to be destroyed and neutral ships would incur danger of running the same risk. If the Germans thought they could scare British shipping away by these means, they were soon undeceived. They took heavy toll of peaceful shipping from the first, and shocked the rest of the world by the lengths to which they were prepared to go in developing this ruthless policy, but all their efforts failed to paralyse British trade as they anticipated. The crowning tragedy of this submarine campaign was the sinking of the Cunard liner *Lusitania* off the south coast of Ireland on 7th May, 1915, with its loss of upwards of 1000 non-combatants, including over 100 Americans. It was one of the German crimes against humanity in general and Americans in particular which brought the United States into the war on the side of the Allies in 1917.

1916 on the European Fronts

The Western Campaign.—During the year 1916 the unity of command which was postulated as an indispensable preliminary to the victory of the Allies so long before it became a fact was achieved neither as between Britain and France in the West and Russia in the East, nor even as between Britain and France along the Western front. The most that can be said of the co-ordination between the forces under the British and French Commanders-in-Chief is that liaison was established between them, and that as far as was possible each endeavoured to help the other by diverting to itself the energies of the German forces. The Western fighting embraced two main episodes: the powerful German attacks on Verdun, eventually unsuccessful; and the Allied offensive on the Somme, in which by far the greater share was borne by the British.

The year began with numerous German attacks widely separated in locality, and intended to mask the main offensive while keeping the French and British

occupied. Hartmannsweilerkopf, in Alsace (2nd to 8th Jan.); Champagne (9th Jan.); Givenchy, Arras, Neuville, Loos (14th Jan. to 6th Feb.); Vimy Ridge, Frise, Soissons, Ypres-Comines area, and Tahure (9th to 20th Feb.) were among them.

On 21st Feb. the new battle of Verdun began. This enterprise, though officially accredited to the German Crown Prince, was the design of General von Falkenhayn. Verdun was one of the four fortresses, of which the other three were Belfort, Toul, and Épinal, on which French armies defending the capital and the country from an invasion from the east would base themselves. The town lies sunk in the Meuse valley, and the German invasion in 1914 flowed past it along the heights of the Meuse down to St. Mihiel. At the beginning of the war Verdun was protected by an outer line of forts, with batteries pushed out in a circuit of 30 miles. The forts were not, however, strong enough in 1914, nor the perimeter of defence extended enough, to withstand the new artillery that had reduced Liège and Namur, and a fierce struggle went on during 1915 along the Meuse heights on the east of Verdun and beyond the low hills on the western side of the town and river, with a view to pushing out the defences all round. As previously pointed out, much had been done in this direction by General Sarrail, but not enough to deter von Falkenhayn and the German General Staff from selecting Verdun as a point for an attack which, if successful, would disorganize seriously the continuity of the French defences. In 1916, as for two years more, the problem of either combatant was to break through a line of trenches which extended continuously from the sea to Switzerland; and failing a complete break-through, comparable to that which had crippled Russia in 1915, to effect a fracture or a deep dent which would compel the loser to reconstruct his system of communications. At the best such a thrust might disclose a fatal weakness in the assailed; at the next best it would disastrously hamper his future activities.

Verdun as a fortress had strong and modern defences. West of the Meuse, north of the town, are low hills the chief of which is the Charny Ridge with dominating strategic points beyond known as Hill 304, Hill 295, Hill 265. The French lines were pushed beyond these into the woods of Avocourt and Forges, but below the heights of Montfaucon, which were the Crown Prince's headquarters. On the east of the Meuse the heights rise to a tableland severed by wooded ravines and overlooking the plain of the Woëvre. The line of French trenches embraced all this tableland and a good deal of the plain beyond. Its outer line ran in a bold convex curve from Forges and Consenvoye on the

Meuse to Fresnes on the Woivre, but it did not penetrate the woods of Forges or Spincourt, and it was below the gun positions on the hills of Ornes. Inside this outer circle was the inner line of Samogneux, Beaumont, Fosses Wood, and Bezonvaux. Inside that again the line of Bras, Douaumont Fort, Hardaumont Wood, Vaux Fort, and Eix.

The multiple defences were most elaborate between these two inner lines. A weak point was that though such defences would be very exacting of life and effort, yet the outer ones were not pushed out far enough to place the bridges of the Meuse out of reach of long-range gun-fire; and an overwhelming attack might have jammed a defending army on the east of the Meuse against the river. The French had provided against the possibility by the multiplication of transport, as well as of inner defences. The Germans hoped by the weight, volume, and suddenness of their attack to bring about the not impossible catastrophe. They massed an amount of artillery which, though surpassed afterwards in the war, was at that time the greatest assemblage that had ever been seen together, and accumulated a supply of ammunition exceeding the quantity which all previous experience prescribed. The heavier guns were placed at Ornes, Spincourt, and Forges. The woods below afforded cover for a concentration of men; and this concentration, amounting to fourteen divisions, with others in immediate reserve, was at first thrown at the 7-mile sector from Brabant-sur-Meuse to Herbebois, which was held by three French divisions under General Humbert.

The attack began on the morning of 21st Feb. with an artillery bombardment lasting four hours. The great weight of shell demolished the French first-line defences, so that the German troops had little to do but walk over them, while a remnant of the defenders fell back to their supporting positions. These were not sufficiently strong or well constructed to enable weak forces to hold them long against the force of three army corps (18th, 3rd, and 10th, with a Bavarian Division) which the Germans sent in after the guns had done their work. The effectiveness of the German artillery was due in part to its weight, and in part to the fact that French counter-battery work effected little, owing to the thick weather.

The trench systems in the Haumont and Caures Woods were carried, but the resistance of parts of the first line at Brabant, Herbebois, and elsewhere was even at this dangerous moment reducing the speed of the German advance, though the momentum was far from exhausted. It was not till next day that the first line was definitely abandoned by the French; and on 23rd Feb. the line

Samogneux-Herbebois was temporarily held. Before the morning of 24th Feb. the French contracted their line still further by drawing in their outposts from the Woevre. It seemed a matter for surprise at the time that no flank attack was made by the Germans in the Woevre; it had been perhaps thought an unnecessary extension of their general scheme, though the weather, which was bitter and snowy, was unfavourable for operations in that sodden plain.

But the German second wave of attack was now rising in fury, and General Pétain, who had undertaken the command of operations on the French side, was still awaiting reinforcements. The character of von Falkenhayn's attack had become clear, and while to the French the need for holding on was imperative, the Germans had a need no less urgent for hastening operations and exploiting their preliminary success to a point at which General Pétain could not repair the breach. They had, in fact, two days in which to achieve their aim—24th and 25th Feb. On the 24th they flowed round the Beaumont Woods and came close to the Talou Ridge, the Poivre Ridge, and the rest of the French line where it ran past Haudremont and Douaumont to Vaux. On the 25th they attacked the Poivre Ridge without much success, but pressed the more important sector of their attack close to Douaumont.

Next day, 26th Feb., brought the fateful hour of the struggle. Pétain's reinforcements were at hand. The Germans made their supreme effort on a 2-mile front at Douaumont, and the picked 24th Brandenburger Regiment was the spear-head of an assault which at one moment burst its way into the Fort Douaumont trenches between the village and redoubt—a fine feat of arms which evoked this telegram from the Crown Prince's head-quarters: "Douaumont, the eastern pillar of the Verdun defences, is solidly in German hands". The adverb alone was misplaced. The position was not held solidly, for Pétain's reserves, thrown in at the exact moment, flung back the Germans and prevented the leak in the defences from being widened by any further inrush.

This counter-attack, made by men of Balfourier's 20th Corps, marked, indeed, the turn of the struggle, for though Douaumont, and Vaux after it, were subsequently to be lost, together with many other historic redoubts and shell-battered points of vantage on either side of the Meuse, and though many thousands of lives were to be swept away in attack and counter-attack on the barren hills about Verdun, yet henceforward the assaults were no different, except in weight, from others which in 1916 and 1917 were projected by the Germans or the Allies on the amplifying complexities of the armoured defence lines. Von Falkenhayn's subsequent comment on the operations, which marked

the beginning of the creeping paralysis of his plan, was that violent French counter-attacks began, and the German forward movement on the heights was stayed.

Though the crisis was over there were many great moments of sleepless effort, of anxiety, and heroism in the months to come, for the last purposeful German assaults on the fortress were made on 15th June, and on 14th July the Germans were still occupying the French with assaults developing from the Thiaumont redoubts, which marked the farthest point southwards to which their long-sustained efforts had taken them. Following on these attacks was a considerable pause, during which the Germans were fully occupied elsewhere in dealing with the British attacks on the Somme. The last phase at Verdun in 1916 was that in which the French, inspired by the methods of General Nivelle, thrust the Germans out of all the positions so painfully won, and re-occupied by mid-December very much the same lines as those from which the great push of the last week in February had ejected them. The story of Verdun cannot here be told in detail; its principal events are dated as follows:

27th Feb.—Germans take Talou Ridge.

3rd March.—Germans enter Douaumont village.

7th March.—Germans, transferring their efforts to the west of the Meuse, capture Hills 360 and 265.

14th March.—Germans penetrate west of Verdun the line Béthincourt-Mort Homme.

20th March.—Germans enter Avocourt Wood.

29th March.—French recover Avocourt Redoubt.

1st April.—Germans, renewing their attacks east of Verdun, capture part of Vaux village.

10th April.—Germans make extended attack on both sides of the Meuse, failing at the Mort Homme, but gaining at Poivre Ridge.

5th May.—Germans, renewing westerly attacks, gain a footing on Hill 304.

20th May.—Germans in a great attack on Mort Homme capture summit of Hill 295. The attack next day enlarged the gains.

24th May.—Cumières and Fort Douaumont captured by Germans.

1st June.—Fresh German attack at Fort Vaux east of Verdun.

7th June.—Fort Vaux captured after six days' fighting.

17th June.—Attack renewed at Mort Homme. The attacks on both sides of the Meuse were prosecuted with increasing vigour till 28th, during which period the Germans took Hills 321 and 320, as well as Thiaumont Fort (23rd June) and Fleury (24th June). Fleury marked the point of their farthest advance towards the inner line of defences east of the Meuse at Forts Souville and Tavannes. The tide now paused, and on 30th June, a day before the British attack on the Somme, the French retook Thiaumont. The fighting went on in a restricted but incessant way through the rest of June and July, the French gradually improving their position. In August activity was renewed at Thiaumont and Fleury.

18th Aug.—French retake whole of Fleury.

9th Sept.—French retake trenches between Fleury and Douaumont.

24th Oct.—The French, after a long pause for readjustment, and now under the direction of General Nivelle, recapture village and fort of Douaumont, Haudremont quarries, and 4500 prisoners. They thus advanced to lines held in May.

3rd Nov.—Vaux recaptured. On 30th Nov. the German Crown Prince resigned the command of the Verdun front.

15th to 16th Dec.—General Nivelle (who succeeded General Joffre as French Commander-in-Chief on 12th Dec.) orders new attack at Verdun. Vacherauville, Poivre Ridge, Bezonvaux, Hardaumont recaptured with 11,000 prisoners.

In early 1916 the British army was still finding itself, and its new Commander-in-Chief, Sir Douglas Haig, regarded it as insufficiently trained and equipped for the great tasks which lay before it. As an instrument of war it was still not yet ready. It also remained, if not in a water-tight compartment in respect of the French armies at its side, yet with a separate command and in separate

control. This may have been merely the necessary consequence of its state of training; but it is certain that in 1916 there was no one with authority to compel that unity of action and command which in 1918, but not till then, directed the Franco-British armies as one force. Fortunately, perhaps, the German High Command elected to attack the French at Verdun instead of throwing their whole weight on the British, though there were numerous smaller actions along the worn and dangerous Ypres front and elsewhere in the first half of the year.

On 14th Feb. the enemy captured some 600 yards of 'International Trench', south-east of Ypres, but they were regained on 17th March, when a bitter and protracted struggle also began for the mine craters at St. Eloi. These were lost and recovered more than once, with heavy casualties on both sides. The Canadians, who had their full share of these costly operations, were again sorely tried at the beginning of June, when the Germans penetrated their front trenches in a surprise attack on the 2nd of that month. Major-General Mercer was killed in this assault, and General Williams captured. Eleven days later the Canadians atoned for this set-back by completely re-establishing their broken line. Throughout the first half of 1916 the enemy, not only round Ypres, but also round Loos, at Ploegsteert, Givenchy, and elsewhere, persevered in similar local attempts to keep the British occupied and upset their plans, while he was concentrating his chief efforts towards beating down the French defences at Verdun. The attack on Verdun, as already related, ultimately broke down, and the period in which it was at its height was utilized by Sir Douglas Haig to bring his forces as near as possible to the point at which they could undertake with success an attack of the first magnitude against the entrenched German lines. The date of the attack was premature, and was hurried on in order to take some weight off the harassed French armies at Verdun. It began on 1st July; the chosen *terrain* was the River Somme, and the great offensive, in which the French joined, was over a 28-mile front from Gommecourt, north of the Somme, to Dompierre, south of that river.

First Battle of the Somme

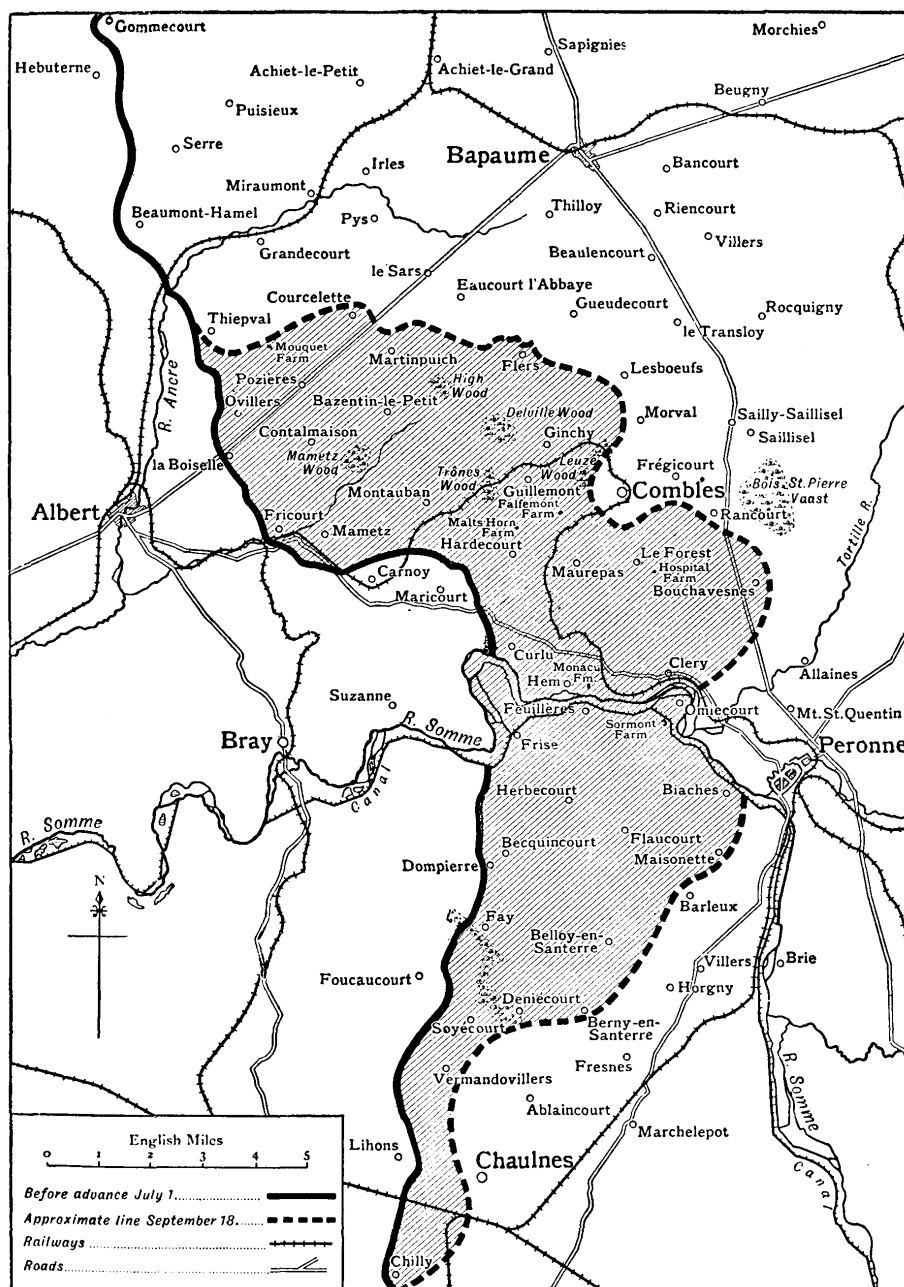
The German position in the Somme area was situated on the high ground which is the watershed between the Scheldt and the Somme. The ground runs east-south-east, and its hills fall into long irregular spurs divided by wide valleys. On the forward slopes of the hills the German first-line defences ran from the Somme at Curlu to Fricourt; at Fricourt the defence line turned north,

crossing the Ancre, thence passing over the summit of the watershed near Hébuterne and Gommecourt to Arras. Between the Somme and the Ancre a second line of defence had been constructed 2 miles behind the first, and on it had been lavished all the ingenuities of fortification which the German engineers afterwards developed in the so-called Hindenburg lines. South of the Somme, where the French were to co-operate with Sir Douglas Haig, the defences were not so elaborate; it was not here that the Germans, who were fully aware of the impending British attack, expected the blow to fall. They expected the greatest weight to be felt towards the Ancre.

The British preparatory bombardment, delivered by a force of artillery far greater than any British army had heretofore possessed, began on 24th June, and deluged the German positions with shells for a week. It was aided by the efforts of the Royal Flying Corps, which, at this time, had a decided superiority over that of the Germans. The British attack on 1st July began in broad daylight, and was delivered principally by Rawlinson's Fourth Army of five corps, with a subsidiary attack by the Third Army (Allenby) opposite Gommecourt, where one corps only was sent forward. The sectors attacked, beside that at Gommecourt, may be designated: Beaumont-Hamel; River Ancre, including Thiepval; La Boisselle and Contalmaison; Fricourt; River Somme at Montauban. The French attack, designed by Foch and delivered by the French Sixth Army (Fayolle), and Tenth Army (Micheler), was delivered along an 8-mile front, taking in a sector on either side of the Somme from Maricourt, through Frise and Dompierre, to Fay.

Severe as were the British preparatory and final bombardments, they did not succeed in demolishing the German systems of defences, and had left machine-gun nests intact. The efficacy of the machine-gun was one of the bitterest lessons to be learnt by the flower of the British armies of 1916, and the great losses of 1st July were largely due to the German handling of this weapon. Taking the British and French attack as a whole, it may be said to have failed towards the north and succeeded towards the south. The heaviest rebuff was inflicted on the corps of Allenby's Third Army which operated opposite Gommecourt. From Thiepval, across the Ancre, the Germans had massed their best fighting material and the greatest weight of their artillery. The 10th Corps (Morland), which included the famous 36th (Ulster) Division, Highlanders, and North Countrymen, did wonders, and actually penetrated the Thiepval Redoubt but could not hold on to its gains. Hunter-Weston's 8th Corps of picked troops, including the 29th Division from Gallipoli, found the

task of assaulting Beaumont-Hamel too strong for them. Farther south there were successes which increased in value towards the Somme. The 13th Corps (Congreve) carried Montauban and Mametz; the 15th Corps (Horne) surrounded Fricourt; the 3rd Corps (Pulteney) forced its way at great cost into La Boisselle.



The Allied Battlefield on the Somme: map showing approximately by the shaded area the Franco-British gains from 1st July to 18th September, 1916

The French armies, well handled, and aided by the advantage of finding the Germans less on their guard, made ground on either side of the Somme. Three corps participated in the assault—the 20th (Balfourier) from Maricourt to the Somme, where the hardest fighting was at Curlu and Hardicourt; the 1st Corps (Brandelat); and the 35th Corps (Allonier). The

last two walked through the Germans, and the French losses were light. As the result of the day's fighting along the whole extent of the front, the British captured 3500, and the French 6000 prisoners; but the casualties of the assailants were close on 50,000. The second day's fighting, though it emphasized the certainty that no great German defeat had been inflicted, enabled both the French and British commanders to enlarge the ground they had won, and the advantage was further exploited in the few days that followed. By the 5th of July over a front of 6 miles the Germans had been pushed back a mile; the British had captured 6000, the French 8000 prisoners.

This was the first blow in the Somme battle. Its results, compared with those of the German attack at Verdun, do not afford warrant for regarding it as a great victory. It became clear that there was no precedent to follow other than that set by the Germans at Verdun, namely that of systematically reducing the enemy's position. This heavy task was entered on by the British forces with unbroken determination; and the effort relaxed scarcely any of its vigour till 18th Sept., while the last big British attack on the Ancre began on 13th Nov. The principal events in this protracted struggle for positions and fortified strongholds after the opening phase already described were:

14th to 15th July.—British attack German second line, capturing Longueval, Trônes Wood, Delville Wood, and 2000 prisoners.

23rd July.—Second phase of Somme battle begun. Pozières captured 26th July.

16th Aug.—French take Belloy near the Somme; 1300 prisoners.

29th Aug.—Total British captures on Somme to date: 266 officers, 15,203 men, 86 guns.

15th Sept.—British advance (third phase of Somme battle) using tanks for the first time; Martinpuich and High Wood taken. Lesbœufs and Morval captured 25th Sept. Combles and Thiepval captured 26th Sept.

30th Sept.—Thiepval Ridge captured.

10th Oct.—French take Ablaincourt, south of Somme, and 1300 prisoners.

21st to 23rd Oct.—British take 1018 prisoners.

12th Nov.—French take Saillisel.

13th Nov.—Battle of the Ancre (fourth phase of Somme battle). British take 4000 prisoners.

29th Dec.—Sir D. Haig's dispatches relating to Somme battle. During the period 1st July to 18th Nov. were captured 38,000 prisoners, 125 guns, 514 machine-guns. The number of casualties inflicted on the Germans has not been made known. Those of the British amounted to 22,923 officers and 476,553 men. A number of these were, of course, not permanent casualties.

Russian Campaign, 1916

During the winter of 1915-6 the Russian armies were reorganized by General Alexieff under the nominal command of the Tsar. The Grand Duke Nicholas, as already stated, went to the Caucasus in 1915, and while Viceroy there the successful advance of General Yudenitch to Erzerum (captured 16th Feb.) was made. The Russian armies of the north were placed under General Kuropatkin (Riga to Dvinsk) and General Everts (Vilna to the Pripet), and the commands embraced respectively the Twelfth, Fifth, and First; and the Second, Tenth, Fourth, and Third Armies. In the southern group of armies, commanded by General Ivanoff till April, and by General Brussiloff afterwards, were included the Eighth Army (Kaledin) in the Rovno sector, Eleventh Army of Volhynia (Sakharoff), Seventh Army of Eastern Galicia (Scherbatcheff), and Ninth Army of the Dniester (Lechitsky). Facing the northern group of armies were German forces directed nominally by General Hindenburg, actually by General Ludendorff. The local commanders were von Below and von Scholtz (Riga to Dvinsk), von Eichhorn (Lake Narotch), von Fabeck and von Woyrsch with an Austrian army corps. A force under the nominal command of Prince Leopold of Bavaria connected these German armies with those which faced General Brussiloff (successor to Ivanoff). The Volhynian sector (Third Austro-Hungarian Army) was under von Brlog; Rovno sector (Fourth

Army) under the Archduke Joseph Ferdinand, stiffened by a reserve under von Linsingen. Farther south were General Boehm-Ermolli's army (Second), and the two armies of Bothmer and Pflanze-Baltin. The Russians, who had been recruiting far larger numbers than they could feed or employ, were much more numerous. The Germans were well entrenched and superior in artillery.

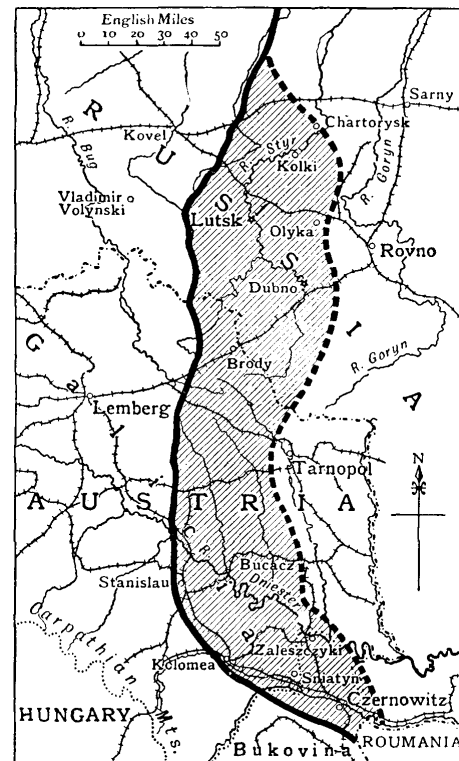
The German effort in 1916 was diverted to the West. The Russians, who were now better supplied in guns and ammunition than heretofore, seized the opportunity to take and keep the initiative in the East. They began before the end of 1915 with an offensive in Galicia on the Styr and Strypa, and continued their attacks through January; while in February there was severe fighting on the Dniester, in the Bukovina, and in Volhynia. The first full-dress attack was made, however, in the northern sector, where General Everts began the battle of Lake Narotch on 18th March. Fighting here was renewed eight times before 14th April, and the Russian gain on the Vilna road did not warrant the heavy losses (12,000), which were increased by a German counter-attack on 28th April.

The important part of the Russian campaign took place in the southern group of armies commanded by Brussiloff, who used his superiority of numbers against the Austrian generals and their very mixed troops with brilliant effect. On 4th June the Russian armies from the Pripet to the Bukovina were set in motion simultaneously against the long unequally guarded Austrian front, seeking the weak places. Generals Kaledin and Sakharoff, in the sectors nearest the Pripet, engaged the armies of von Brlog and the Archduke Joseph Ferdinand; and the Russian columns, though held up in the marshes supporting von Brlog near Kolki, swept through the archduke's defences like paper. They marched swiftly forward over rolling country to the Styr, driving the Austro-Hungarian levies before them. By 16th June the leading Russian columns were 12 miles from Vladimir Volhynok. North of this apex Kolki and Svidniki, on the Stokhod, were captured; south of the so-called Lutsk salient thus created Sakharoff captured Dubno, and was outside Brody on the 16th. In twelve days this most damaging attack captured 70,000 men, 83 guns, and created a salient which, at its greatest depth, was 50 miles from the 80-foot base from which it had been started. Von Linsingen's reserves were sent in, and Ludendorff took matters in hand.

General Scherbatcheff had simultaneously attacked von Bothmer from Kozlov to the Dniester. The Russian general reached Bucacz (8th June) and crossed the Strypa. He also captured 17,000 prisoners and 30 guns, but von Bothmer, athwart a good line of railway, could not be enveloped, and fell back sullenly and without disaster.

General Lechitsky, in the most southerly sector, struck with fury at Pflanzner-Baltin, and cut through his centre on the hills between the Dniester and the Pruth while turning his flank at the Dniester bridge-heads at Zaleszczyki and Biskupie. The net result was the wreck of Pflanzner-Baltin's army, which was forced to retreat across the Pruth to the Carpathians. Lechitsky captured 39,000 men; and Brussiloff's great attack had succeeded triumphantly on both wings. It had made less headway in the centre. There were two lines of subsequent pressure or advance open to him, one, the more northerly, towards Kovel; the other, with, as object, the further destruction of the southern Austro-Hungarian armies, towards Halicz.

Ludendorff, however, had by this time formed his plans for the restoration of equilibrium; and Linsingen, with his reserves, was employed to make amplification of the Russian success at the most northerly portion of the salient impracticable. Linsingen struck at the Stokhod River crossings. Brussiloff countered by bringing up a fresh army under General Lesch with the object of outflanking Linsingen in his turn; and another army, under General Rogoza, was ordered to occupy General Woyrsch's attention farther north. These manœuvres had considerable success, Lesch and Rogoza capturing 17,000 men. But though in these and subsequent engagements the largest numbers of captures fell to the Russians, and though in the extreme south they were again able to advance to the Carpathian passes, no disaster on the



Map illustrating the extent of Russian Recovery in the Summer Campaign of 1916

largest scale was inflicted. The Germans were able to withdraw their allies and to allow the Russian attack to wear itself out.

Nevertheless, the Russian victories were of immense service to the Allies, and by the autumn of 1916 it seemed that the prospects of the Central Empires were darker than at any period of the war. The Russian advance, in its resolution and generalship, need not shrink from a comparison with that with which Foch ended the war two years later. By the middle of September, Generals Kaledin, Lesch, Sakharoff, Scherbatcheff, with Bezobrazoff and Lechitsky in the south, had captured 370,000 prisoners, 450 guns, and an amount of supplies as great as that which fell into Ludendorff's hands at St. Quentin in 1918.

Balkan Campaign, 1916

After the conclusion of Brussiloff's triumphant dissipation of the Austro-Hungarian armies in the early autumn of 1916, the way was open for Roumanian co-operation with the Allies, and Roumania, though neither united nor completely ready, was urged to enter the war. This she did on 28th Aug., when Germany declared war on her, and Italy made a belated declaration of war on Germany. On 29th Aug. von Hindenburg was appointed Chief of the German General Staff in succession to von Falkenhayn, to whom was relegated the task of dealing with Roumania. The Russians during the rest of the year advanced towards the foot of the Carpathian passes and to the junctions of the knot of railways in South-Eastern Galicia, in order to gain complete contact with the Roumanians through the Bukovina. Meanwhile the Roumanians, instead of concentrating on their southern front, where a mixed force of Bulgarians, Turks, and Germans under the command of von Mackensen was preparing to take them in the flank, pressed forward through the easterly passes of the Carpathians into Transylvania. They advanced here some distance, practically striking a blow in the air, but neither raising the Transylvanian population nor capturing any strategic points. On 2nd Sept. Russian forces in aid of Roumania crossed the Danube into the Dobrudja, while on 3rd Sept. Brussiloff's troops won a considerable victory in South-East Galicia, and on 7th Sept. took Halicz. But this success was more than offset by the loss to the Roumanians on their southern front of Tutrakan, on the Danube, with 20,000 prisoners. Occurrences were symptomatic of what was to

come; and again, on the Eastern, as on the Western front, the Allies suffered from the lack of unity of command. The Russians and Roumanians joined hands on 10th Sept., but never concerted their strategy. Mackensen continued to advance along the Danube towards the vital Cernavoda Bridge, and so to threaten the whole of Southern Roumania, while the Russian forces which, on the east, had ventured into Roumania, found themselves by 18th and 19th Sept. faced with the new forces concentrated by the Germans against their eastern Transylvanian front. The rest of the Roumanian campaign is the history of the stages by which the two arms of these German-made 'nut-crackers' closed on the Roumanian armies, which had been placed in a false strategical position and were badly led. On neither front did the Roumanian soldiery, who fought well under very trying conditions, with inferior artillery and a poor medical service, give way without a struggle. Mackensen was stoutly held up on 20th Sept. in the Dobrudja, and on the Transylvanian side the Roumanians had a success on 27th Sept. But on 30th Sept. Falkenhayn developed his eastern attack near the Roter Turm Pass, and by 7th Oct. the whole Roumanian front in Transylvania was retiring by the way it had come. A week later it was out of Transylvania and defending the not very defensible passes.

On 20th Oct. Mackensen attacked on the whole line in the Dobrudja, and five days later he was on the vital Cernavoda Bridge. Constanza, the Roumanian Black Sea port, had fallen, and so far from ever being in a position to take Turkey or Bulgaria in the flank, the Roumanians were now themselves on the verge of being outflanked on the Danube. Meanwhile, on the other arm of the nut-crackers, von Falkenhayn, despite trifling setbacks, was pressing on. The Törzburg Pass (21st Oct.), Predeal Pass (23rd Oct.), Vulkan Pass (25th Oct.), Roter Turm Pass (31st Oct.) were all scenes of Roumanian reverses, and by 15th Nov. the bulletins were bringing the daily news that the Roumanian retreat continued. On 23rd Nov. Falkenhayn was advancing on Bucharest; Mackensen had crossed the Danube at Islatz and Simniza; and farther west Orsova and Turnu-Severin had fallen. All the German composite forces could now be deployed in Roumania, and the end followed swiftly. Mackensen and Falkenhayn were in touch on 26th Nov.; Campolung was captured 29th Nov.; Bucharest, Ploeshti, and Sinaia fell on 6th Dec.; and with them went the Roumanian oil-fields, the wells of which had, however, been very thoroughly damaged by Captain Norton Griffiths

and a small British party in order to prevent their use by the Germans. (They were restored in some eight months.)

On 8th Dec. the Germans estimated their Roumanian captures as 70,000 men and 184 guns; and it is true that only a portion, though a considerable one, of the Roumanian armies was able to effect a retreat with the Russians to the line of the Sereth defences. Fighting went on till the end of the year, and was continued into 1917 until the Roumanians were forced to sign the Treaty of Bucharest—revoked by the Allies at the end of the war.

Roumania's fate, following the tragedy of Serbia and the Allies' withdrawal from Gallipoli, strengthened the Greek military party round King Constantine, which was now openly pro-German. Against these influences M. Venizelos proved powerless, though Greek volunteers were at the same time joining the Venizelos party, and ready to fight with the Allies. This division of opinion in Greece was illustrated in the middle of August (1916), when two divisions of the 4th Greek Army Corps surrendered to the Bulgarians, who had advanced to the Greek port of Kavalla, while the 3rd Division of the same corps joined the Allies at Salonika. The pro-Germans had carried all before them at the last Greek elections (Dec., 1915), when the Venizelists declined to poll; and the danger of finding themselves suddenly attacked in the rear discouraged an Allied offensive against the Bulgarians until the autumn of 1916, when the newly equipped Serbian army arrived from Corfu, ready and eager to fight its way home, joining the force under General Sarrail, which already included Russian, Italian, and Portuguese contingents, besides French and British. Following pro-German riots against the Allied embassies in Athens, too, a 'pacific blockade' of the Greek coast had been enforced, and a firm Note presented to the Greek Government demanding the demobilization of the Greek army and a new general election, to be freely conducted. When these demands had been accepted and a new Government formed—though the king's pro-German sympathies remained as marked as ever—General Sarrail resumed, in September, the offensive against the Bulgarians. The main advance was undertaken by the French and Serbian divisions, with a Russian contingent, in the direction of Monastir, General Milne's British column meantime pushing the Bulgarians back from the Struma line. Two months' fighting saw the Serbians, who had borne the brunt of the attack at this point, marching back into Monastir in triumph, having turned the Bulgar-German

forces out of it on 18th-19th Nov. The British, at the same time, kept the enemy busy at the other end of the line, occupying a number of villages, and pushing the Bulgarians back beyond the railway between Seres and Demir-Hissar. With their heavy commitments elsewhere the Allies were for the time being unwilling to extend their military operations beyond Monastir.

Italian Campaign, 1916

Italy, who for political as well as military reasons had declined further assistance in the Balkans, had her share of hard fighting within her own frontiers in 1916. Before she could resume her advance on Gorizia and Trieste (held up in the early winter of 1915) the Austrians attacked in turn from the Trentino under General Conrad von Hoetzendorff, who, planning a drive on the Mackensen scale, aimed a blow at the tempting Venetian plains. The grand attack, supported by upwards of 2000 heavy guns on a 30-mile front between Val Sugana and Val Lagarina, and delivered on 14th May by some 350,000 first-class troops, smashed a way through in the centre. Though the flanks held firmly, and the Italians, roused to fury by the invasion, fought magnificently among the mountain heights, General Cadorna ordered the line to be withdrawn from its untenable positions until it was south of Asiago. Pressing their advantage with every means at their disposal, the Austrians announced in an Army Order on 1st June that only one mountain intervened between their troops and the Venetian plains. Cadorna, however, had now been reinforced, and two days later was able to reply that the Austrian offensive had been checked. For the rest of the month he was content, in this sector, to sustain the continued but unavailing assaults of the enemy, while he prepared his own great counter-attack on the Isonzo front, with Gorizia, the gateway to the plateau of the Carso which led to Trieste, as his immediate objective. This dramatic move, heralded by an intense bombardment on 6th Aug., was entrusted to the Duke of Aosta, whose Third Army, after three days' fighting of the fiercest description, carried the last heights defending the town and entered Gorizia in triumph. Following the retreating enemy across the Carso, the Italians, whose enthusiasm for the war had been greatly stimulated by this fine feat of arms—Italy's belated declaration of war on Germany followed upon the Gorizia victory—continued their advance across the northern end of that formidable plateau, winning a number of considerable battles, and capturing before the

end of the year between 30,000 and 40,000 prisoners, but never succeeding in mastering the Carso as a whole.

Naval War in 1916—Battle of Jutland

The battle of Jutland, which took place on 31st May, 1916, overshadows all other naval operations in that year; nevertheless, there were several other events of importance which preceded it, or were in some way related to its occurrence afterwards. For example, in the earlier months of the year the German raider *Moewe* was at large, and inflicted considerable damage on British shipping before returning safely to a German port; the mercantile submarine *Deutschland* left Germany for the United States and returned in safety; and another German submarine, U 53, also crossed the Atlantic with more belligerent intent, and sank several merchant vessels off Rhode Island on 8th Oct. The new development of the submarine war, in which Germany declared her intention of sinking merchant ships at sight, began on 1st March, and one of its most important consequences was the dispatch of a United States Note by President Wilson to Germany (18th April) in respect of the sinking without warning of the *Sussex* and other unarmed vessels. Another outcome of the German submarine warfare was the sinking of British hospital ships in the Mediterranean. Furthermore, following the battle of Jutland, and related in some respects to its only partially decisive character, the *Hampshire*, with Lord Kitchener and his Staff on board, was sunk off the north of Scotland (5th June) by striking a mine that is said by the Germans to have been laid by one of their submarines; and on 19th Aug. the German High Seas Fleet was able to come out again, though it sought no action, but avoided one. Two British light cruisers, the *Nottingham* and *Falmouth*, were sunk in the search for its whereabouts.

The 'partially decisive', or 'indecisive', character of the battle of Jutland are relative terms, and their exact implication has been, and must continue to be for a long time, a matter of controversy. On the one hand, the aim of Admiral von Scheer, the Commander-in-Chief of the German High Seas Fleet—to catch a portion of the British Grand Fleet and attack it while isolated and unsupported—was frustrated, and in that respect the German admiral failed. On the other hand, Admiral Sir John Jellicoe's purpose of destroying the German Fleet, if and when he succeeded in engaging it, also failed, as may be understood from its subsequent emergence from its

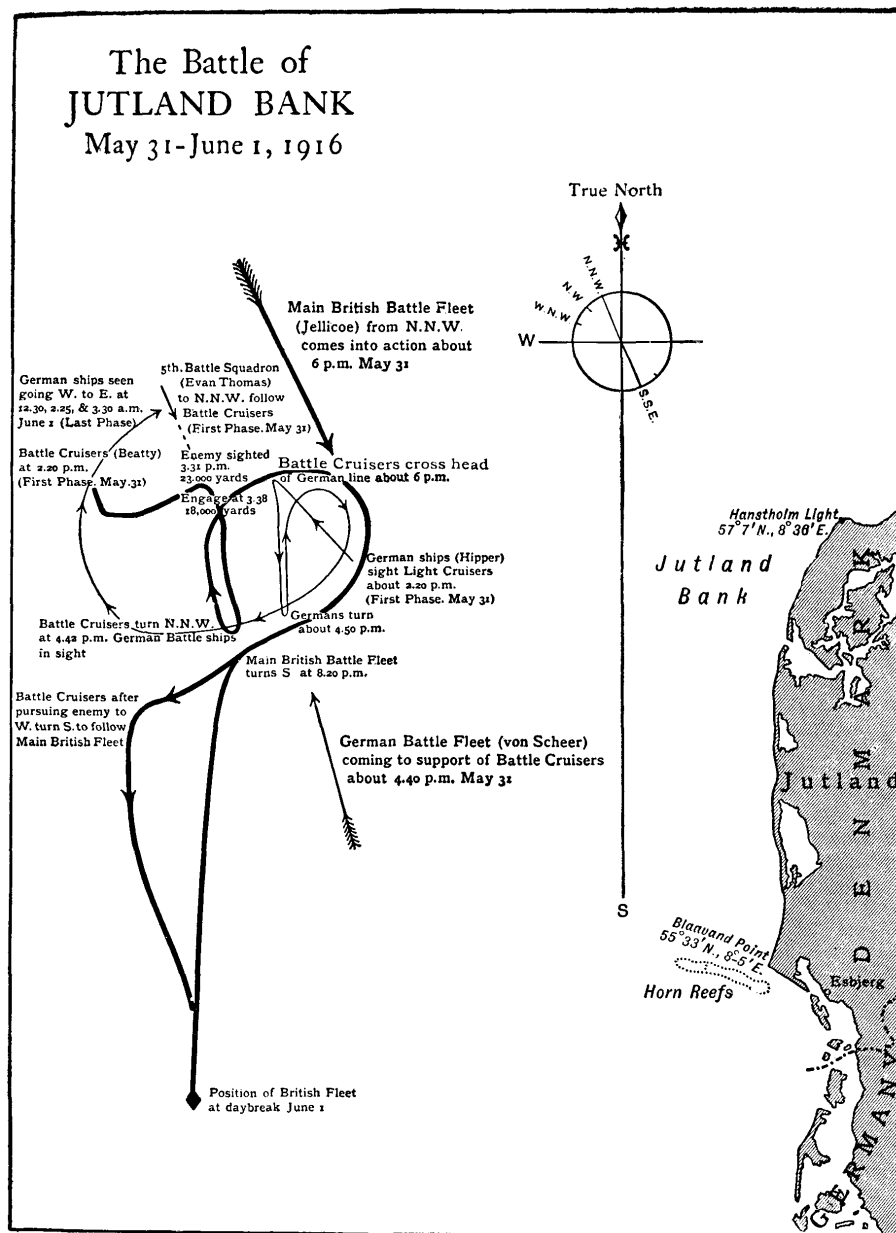
harbour in August, and the later development of the German submarine campaign, which could not have taken place had not the Germans possessed the framework of a fleet to support the under-water vessels. Sir John Jellicoe justifiably claimed that his action preserved intact the main forces of the British Grand Fleet, and left them as before in command of the outer seas, while demonstrating to the Germans that they could not again engage in a naval battle on a large scale with any hope of success. Admiral von Scheer was entitled to claim that he had engaged a superior British force, had inflicted on it more material damage than he had sustained, and had withdrawn the bulk of his forces to remain, as before, a menace, not to British safety, but to British unfettered control of the seas. The details, in outline, of the battle of Jutland are as follows.

On 30th May the Grand Fleet under Sir John Jellicoe left its three Scottish bases for a sweep of the North Sea, Sir David Beatty, with the battle-cruiser squadron, *Lion*, *Queen Mary*, *Princess Royal*, *Tiger*, *Indefatigable*, and *New Zealand*, and Sir Evan Thomas, with the four battleships of the *Queen Elizabeth* class, *Barham*, *Malaya*, *Warspite*, and *Valiant*, setting out from the most southerly of these bases, Rosyth. At 2 p.m. on 31st May Sir John Jellicoe, with the Battle Fleet in 6 divisions, was steaming line-ahead between Aberdeen and the north end of Jutland, in order to meet Sir David Beatty at an appointed rendezvous in the North Sea. Sir John Jellicoe's 6 divisions were, lined east to west, 1st Division (Jerram), *King George V*, *Ajax*, *Centurion*, *Erin*; 2nd Division (Leveson), *Orion*, *Monarch*, *Conqueror*, *Thunderer*; 3rd Division (Jellicoe), *Iron Duke*, *Royal Oak*, *Superb*, *Canada*; 4th Division (Sturdee), *Benbow*, *Bellerophon*, *Temeraire*, *Vanguard*; 5th Division (Gaunt) *Colossus*, *Collingwood*, *Neptune*, *St. Vincent*; 6th Division (Burney), *Marlborough*, *Revenge*, *Hercules*, *Agincourt*.

Both Jellicoe's and Beatty's forces had their attendant suites of destroyers, light cruisers, and other cruisers. The British Grand Fleet in all was constituted of 41 'capital ships', made up of 28 battleships, 9 battle-cruisers, and 4 armoured cruisers. It had also 103 'ancillary craft', made up of 25 light cruisers and 78 destroyers. The German fleet consisted of 20 battleships and 5 battle-cruisers, or 28 'capital ships'; there were also 11 light cruisers and 88 destroyers. In gun power and weight of projectile the Grand Fleet had a striking superiority over the German fleet, and Admiral

Jellicoe had apparently a valuable superiority in speed. In his own account of the battle he observes that the speed of some of the German ships had been underestimated.

There was no clear expectation on the British side of meeting the Germans when the Grand Fleet set out for its sweep on a line drawn from Wick to the opposite coast of Norway, with Beatty's 6 battle-cruisers and Evan Thomas's 4 battleships as advance-guard; and when von Scheer set out for the north from Heligoland Bight at daybreak, with an advance-guard of 5 cruisers, supported, 50 miles behind, by 16 Dreadnoughts and 6 slow pre-Dreadnoughts, he had no intention of seeking a general action.



The Battle of JUTLAND BANK May 31-June 1, 1916
Map showing the approximate positions of the British and German Fleets
at various stages of the battle.

The meeting of the advance squadrons began when both were on a level with the northern end of Jutland. Admiral Hipper, who commanded the German cruisers, turned round from north to south to rejoin his main fleet; he was then east of Admiral Beatty. Beatty followed him, at some disadvantage from smoke and haze. Evan Thomas's battleships were too far behind at this stage to join in the engagement. Hipper fended off Beatty

with destroyers as best he could in the hour before the German main fleet could come up, and in that hour *Queen Mary* and *Indefatigable* blew up, shells from the German ships, on which the system of fire control appeared to be more accurate than the British, reaching their magazines.

When the German main fleet was seen to be approaching in support, Beatty turned with his 4 remaining cruisers, and Evan Thomas's 4 battleships fell in behind. These 8 were stronger than the German advance 5, and swifter, so that Beatty did not execute a mere retreat but pressed on Hipper, making him turn east, and thereafter placing the British ships on the German line of retreat to Heligoland—'crossing the T', as the manœuvre is called.

Meanwhile Jellicoe's 6 battle divisions were coming on in an oblong of 6 lines of 4 ships each—the long sides of the oblong north and south, the short, east and west. Thus steaming, Jellicoe came into contact with Beatty and Evan Thomas engaged on the east side of the German line, whose head they had faced round and were themselves going south. Beatty and Thomas were thus between Jellicoe and the Germans, and it behoved the British Commander-in-Chief to see that his ships did not hurt one another with their fire. Jellicoe effected the necessary deployment, not in the manner that he had premeditated, but in that which circumstances forced him to employ. It was the less simple way, and the Grand Fleet was not in line till half-past six.

The Germans had no prudent course but to retreat, which they did in the haze and chemically-created smoke—both fleets going to the south-west, curving to west. The fleets were hammering each other as hard as they could; but when darkness came down the German fleet, badly damaged but not seriously diminished in numbers, was still fighting in retreat. Admiral Jellicoe, in his own account of the battle, remarks: "At 9 p.m. the enemy was entirely out of sight, and the threat of torpedo-boat destroyer attacks during the rapidly approaching darkness made it necessary for me to dispose the fleet for the night with a view to its safety from such attacks, whilst providing for a renewal of action at daylight".

The opportunity for renewal at daybreak did not come; nor was it likely to have come, since von Scheer's first preoccupation was naturally not to fight a superior force under conditions least favourable to himself. It is therefore

proper to state that the British Commander-in-Chief thought it wiser to break off action with his main fleet lest it should suffer too greatly in the turmoil and confusion of a night attack. The arguments in favour of this decision are several; the chief of them being that Admiral Jellicoe kept the British fleet and naval power intact, and another being those which the British admiral himself advanced, namely that he was not completely aware of how his own fleet and that of the enemy lay to one another, and that "the result of night actions between heavy ships must always be a matter of chance". Admiral Jellicoe did not feel justified in gambling on such a chance. He did what he told the Admiralty he should do in such circumstances, as recorded in a dispatch written on 30th Oct., 1914, and published at the end of the official *Battle of Jutland* in justification of his action: "If the enemy battle fleet were to turn away from an advancing fleet", he wrote on that occasion "I should assume that the intention was to lead us over mines and submarines, *and should decline to be so drawn*". The italics are Lord Jellicoe's own.

1917 on the European Fronts

The Russian Campaign.—On the Eastern front fighting on the grand scale had died down by the beginning of 1917, the Germans having exhausted the momentum of their advance under Mackensen, and accomplished their main purpose of putting Roumania out of account as a serious adversary. Along the line of the Sereth, and in the Bukovina, deadlock was reached in the spring of 1917. By that time the creeping paralysis which was seizing the Russian armies was making itself felt in this, their most distant tendon. Throughout April, May, and June the daily record of occurrences on the Eastern front is blank except for one attack by the Germans on the Stokhod (3rd April).

For the explanation of their quiescence the record of political events has to be scanned. It was well known on the Continent, though it was kept hidden from the British public during the winter of 1916-7, that the integrity of the Russian armies was crumbling, that soldiers were fraternizing with the enemy, and that a general revolution was being prepared by those forces of socialism and anarchy which had been thrust under, but had never lacked exponents, since the abortive revolution of 1905. The mismanagement, the corruption, and the bitter hardships of the war had given them their

opportunity, and these were the 'Dark Forces', more than the rogue Rasputin, the parasite of the Russian Court, which undermined the influence of the monarchy, and extinguished Tsar and Court, bureaucracy, aristocracy, and army in a common ruin. It was said in Europe during the winter of 1916-7 that the Allies would have to choose between the Russian monarchy and the Russian people; but neither the inertia of the Russian army nor the postponement of the re-opening of the Russian Duma acquainted the British public with the depth of the mischief that was working. The first inkling came on 12th March, 1917, when, following food riots in Petrograd, the Tsar ordered the suspension of the Duma. The Russian Revolution was the reply. Three Guard Regiments joined the people—the army had failed the monarchy. A Provisional Government was formed. Petrograd, Moscow, Kharkov, and Odessa joined it; and on 15th March the Tsar abdicated under compulsion. Several figures emerged from the crisis. M. Miliukoff, of the Constitutional Party, and Kerensky, a link with the Socialists, but the real forces at work did not at first show themselves.

On 24th March the army declared its loyalty to the Provisional Government, which two days before had been recognized by the Allies. For a time hopes were entertained that under this Provisional Government Russia would carry out her obligations to the Allies, and that her armies would fight; and every sort of device, including interchange of visits with representatives of British Labour and French Socialism, was employed to foster cordiality. The first sign of the essential futility of such hopes appeared on 4th May, when it was evident that the Russian Provisional Government was failing. A new coalition was formed, with Kerensky at its head, and loyalty to the Allies was urged and asserted. French and American missions visited Petrograd and Moscow, but no real consolidation was effected, though in the latter end of May and the beginning of June there was a remarkably deceptive appearance of it.

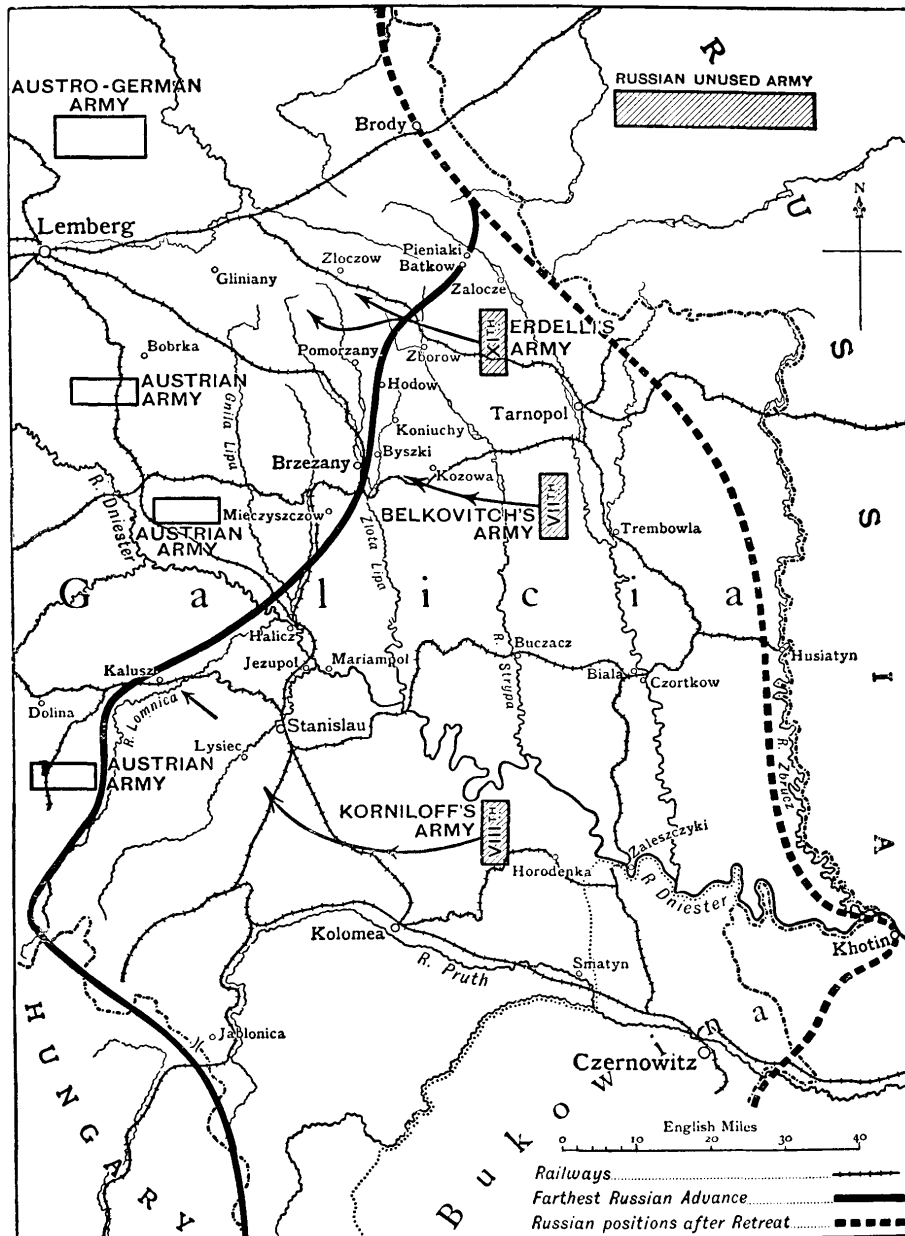
Under the spur of great efforts by M. Kerensky the army was stimulated into action once more, and a new offensive prepared. On such an offensive the Allies had placed high hopes, for the Russian armies in the spring of 1917 were better equipped and better provided with guns and ammunition than ever before. At first some of these hopes seemed destined to be realized. General Brussiloff, who had succeeded General Alexieff as

Commander-in-Chief of the Russian armies, consented rather reluctantly, under the insistence of Kerensky—at that time engaged in a life-and-death struggle with the forces of the Bolshevik party led by Lenin and Trotsky—to organize an attack on the Austro-Hungarian front south of the Pripet. It was believed that the soldiers of these southern armies were less tainted by Bolshevism than those in the northern armies, and that a success here might rally the country to its older standards of patriotism. Brussiloff entrusted the offensive to General Gutor, who had in his favour the facts that the Austro-Hungarian armies were as war-weary as the Russian, and that the German Head-quarters Staff, who were well acquainted with the extent of Russian disaffection, and had indeed been instrumental in inspiring and organizing it, were sceptical about the possibility of such an attack. Furthermore, in the sectors from Brody to the Dniester and beyond, where Lechitsky had halted in 1916, the Russians had a considerable numerical superiority—54 divisions to 30 composite Austro-Hungarian, Turkish, and German divisions, and were well equipped, well posted, and well supplied.

General Gutor directed three armies. The Eleventh, under General Erdelli, was to act along an 11-mile front from a point north-west of Tarnopol, and to get astride the railway which leads from Tarnopol through Zloczow to Lemberg. The Seventh Army, under General Belkovitch, facing Brzezany, was to cross the Zlota Lipa River, where von Bothmer had made his stand, and was then to wheel north-eastward in the same direction as Erdelli's army, with which it was to get in touch. If this movement succeeded, the combined armies were to advance towards Bobrka and the railway from Halicz to Lemberg. The whole of this task was through difficult country. Far to the south General Korniloff, with the Eighth Army, was entrusted with the turning movement. He was to overrun the Halicz region and to obtain control of the railway thence to Lemberg. If this wide turning movement succeeded, and if Belkovitch also did well, the Austro-Hungarian armies would be outflanked and in danger of being rolled up, while pinned down in the north by Erdelli.

The venture was audacious, and the Russian commanders scarce dare trust their men. The Seventh Army's task in assaulting the Zlota Lipa line was such as would have tried the bravest and most loyal troops. The attack began on 1st July, and Belkovitch's men advanced bravely enough, protected by good artillery. In the first assault they took the river line and

2000 prisoners. But between the Zlota Lipa and its tributary Tseniow was a death-trap, and the Russians were caught in a murderous cross-fire. The day was not lost; but at this critical moment occurred an incident which was symptomatic of Russia's disorders, and was the death sentence of Russia's continuance as a combatant. A division which might have turned the scale refused to advance.



Russia's Last Effort in 1917: map showing approximately the farthest line reached by the attacking armies, and the Russian positions after the retreat

The ground won was with difficulty held, and in the days that followed, the Germans, awakened to an unexpected danger, steadily reinforced the weak point, while Russian battalions were refusing to stay in the front line. The situation reacted on Erdelli's troops farther north, where the Eleventh Army had done well at very little cost, and had captured 6000 prisoners by 3rd July. It became less difficult each day for the German directing staff to hold this attack in check, and it was stopped by 6th July. On the Dniester, Korniloff's army began to advance on this day—on which, according to plan, the enemy should have had all their attention concentrated on Erdelli and Belkovitch. Korniloff did very well. On 6th to 7th July he felt his way forward from Stanislau to Dolina, and on 8th July, joining battle with von Bothmer, broke down resistance at Jezupol with ease, and sent forward his best arm, his cavalry, to the River Lukwa, 8 miles behind von Bothmer's first-line defences. Realizing his danger, von Bothmer counter-attacked, but was again borne down, and Korniloff's van reached the Lukwa. In two days' fighting Korniloff had broken through on a 30-mile front, and his main body, in the wake of General Chermiroff's fighting division, poured into the plains of the Dniester. Theoretically a decisive victory had been won. It was in fact indecisive, because the leader's shock troops had been used up, and the situation was crumbling from within. His troops got intoxicated, mutinied, and he could use them no further.

But the rot now became dangerous to the point of mortality in the Eleventh Army of Erdelli. On 20th July, following a strong German counter-attack between Pienaki and Batkow, which was nearly the most northerly point of the advance, the 607th Mlynoff Regiment left the trenches voluntarily. They ran away, leaving the other regiments to bear the brunt of the attack. The breach widened as the Russians opened the gate. The German-Austrian attack, spreading to Zborow, found a Russian division ready to throw down its arms, and in a day the German-Austrian wedge was thrust in between the Eleventh and Seventh Russian Armies. The disaster was complete and irreparable. The command of the army group was hastily transferred from Gutor to Korniloff, but neither Korniloff's ruthless discipline nor Brussiloff's genius could alter the essentials of the situation, which were that the Russian armies would not fight, and were fleeing in panic.

All attempts to stop the flight were useless. On the night of 20th July the breach was 20 miles wide; on 21st July German guns were shelling

Brussiloff's head-quarters at Tarnopol; on 23rd July the remnants of the Russian armies were retreating to the Sereth amid scenes of drunken brutality as disgraceful as any that the war has recorded, and only to be compared with those that were to become common in Russia and Siberia in the struggle to establish Bolshevism. Farther north the Russian front imitated the cowardice and treachery at Tarnopol. On 25th July whole Russian army corps deserted the Dvinsk front, on which depended the safety of Riga. On the same day Korniloff was compelled to begin the relinquishment of the ground he had won with the Eighth Army. Stanislau was abandoned; Kolomea followed on 27th July; Czernowitz went on 31st July; and the loss of the Bukovina followed that of Galicia.

For a time Korniloff seemed to have a chance of restoring coherence to some part of the Russian armies. He succeeded in wringing permission from Kerensky to enforce discipline. But the military-political understanding between these two, though it appeared to fail because of Kerensky's suspicions of Korniloff, whose arrest he ordered (on the 11th Sept.), was never a possibility. Russia was sick unto death. Her soldiers demanded peace; her peasants and townspeople asked for bread, and turned to Lenin and Trotsky, who promised both. During the summer there were many attacks by the Germans on the Riga front, which they used as a training-ground for their troops; and fighting of a similar character took place on the Russo-Roumanian front. But on 16th Oct. the Germans, capturing Oesel Island in the Dvina, took the first step to the subjugation of the northern armies, and continued to take numbers of willing prisoners on the Riga front during the rest of that month. The bulletins of the fighting are contradictory and obscure, but by 3rd Nov. Russians and Germans were fraternizing on the Riga front, and on 20th Nov. hostilities ceased. Lenin demanded (1st Dec.) the surrender of General Dukhonin, the then Commander-in-Chief, who was murdered two days later—the day after the negotiations between the Bolsheviks and the German peace delegation began at Brest-Litovsk.

British Front in the West, 1917

On the Western front it had been expected that the heavy hammering to which the Germans had been subjected during the battle of the Somme would be carried on in concert under the direction of Sir Douglas Haig and

General Foch. But a change was made in the French High Command, General Joffre retiring, and his place being taken by General Nivelle, who had done so well at Verdun, while Foch was relegated to the task of preparing against a possible thrust of the Germans through Switzerland. Unity of command was not achieved, except in appearance. Nivelle's plan was to strike at the German centre; Haig was to aid him by simultaneous attack, though Haig's own prepossessions were in favour of freeing the Channel ports by a burst from the Ypres salient. In the result, neither plan succeeded. Nivelle failed because neither the French Government nor a section of the French soldiery would bear a repetition of the losses incurred in his thrust at the Chemin-des-Dames; Haig failed because he had neither the time nor the weather in which to drive his last blows home in the autumn of the year.

A contributory cause of the comparative failure of the Franco-British plan of campaign in 1917 was the want of perception of the intention of the Germans to withdraw from the positions in the Somme area which they had defended so stubbornly in 1916, and which the Allies were preparing to render untenable or to batter down. The want of perception was not complete, but both British and French plans were upset by the suddenness and extent of the withdrawal, which the Germans effected with much less loss than they should have been forced to sustain. The first symptom of the general withdrawal was discovered in March, when portions of St. Pierre Vaast Wood, near the junction of the French and British lines, were found to be evacuated. By 17th March the German voluntary retirement was in full swing, and their forces ruined everything as they retreated. By 17th March, also, the British front from Roye to Arras was moving forward, and on 2nd April the Fifth Army was within two miles of St. Quentin, while the Fourth Army on 5th and 6th April was at Ronssoy and Lempire. While the British armies were pushing towards the Cambrai-St. Quentin line the French were pushing on a 30-mile front from the north of the Upper Somme, towards the new German line from St. Quentin, behind Soissons, in front of the St. Gobain plateau, the Forest of Coucy, and the Chemin-des-Dames. Behind the new line Nivelle matured reconstructed plans for the great French attack towards Laon.

Of the new fortress line (the Hindenburg and Drocourt-Quéant line) which the Germans had constructed and continued to improve, the La Fère-Laon

position and the Chemin-des-Dames were the southern bastion, and the Vimy Ridge the north-western pillar. Sir Douglas Haig's preconceived plan had been to attack the Arras front, not in order to assault this line, but as preliminary to the Ypres salient thrust farther north. Nevertheless, the plan could be adapted and it was prosecuted. Preparations on a large scale, equivalent to building a counter-fortress front, had been made for the Arras operations, and the greatest precautions were taken to lend the attack all the support which mines and artillery could give. Two armies, the First (General Horne) and Third (General Allenby), were prepared for this action. Horne's army made its attack on the Vimy Ridge, with the Canadian Corps as shock troops, on 9th April, and the attack was extended on a 12-mile front from Hénin-sur-Cojeul, south-east of Arras, to Givenchy-en-Gohelle, north of Arras. The Canadians took the whole of Vimy Ridge, except its northern end, the conquest of which was completed next day. Five villages fell into British hands and 6000 prisoners. Subsequent days saw the extension of the victory, but though Vimy village, Givenchy-en-Gohelle, and other important points were taken, the fortified villages of Hénin and Wancourt held out by dint of machine-guns, and prevented the possibility of the Third Army's joining hands with the Fifth Army beyond the third line of the German defences until it was too late. The whole of the expected gains were therefore not realized, but the possession of the Vimy Ridge was invaluable, and became a most important factor in stemming Ludendorff's rush in 1918, when he attacked the Third Army after destroying the Fifth.

The British attacks did not end on 11th April, as they should have done, but were continued here, as well as at other portions of the British line to its junction with the French armies, in order to lend assistance to Nivelle while his attack in Champagne was in progress. It was a very costly procedure, and its scope may be inferred from the statement that on 23rd April a "second phase of the battle of Arras" began; another 12-mile front east of Arras was launched on 3rd May; and there were bloody encounters about Bullecourt or Fontaine-les-Croiselles on 7th and 12th, 15th and 16th, and 21st May. Some 20,000 German prisoners were captured in these preliminary spring operations, but the drainage in casualties to the British armies was heavy, and more damaging still was the loss of time by the postponement of Field-Marshal Haig's major plan farther north.

This, however, was at last begun on 7th June by the assault on the Messines-Wytschaete Ridge, which, under the name of the 'Battle of Messines' denotes one of the most completely successful actions fought in that year. It was undertaken by the Second Army (General Plumer), and the preparation for it, including the mining of the ridge, had been as near perfection as possible. The attack was launched; the mines which blew the German front line to pieces were exploded on the morning of 7th June at ten minutes past three. Nine miles of front were stormed and 6400 prisoners taken. In the next week the number of prisoners was considerably augmented; German counter-attacks were beaten off; and the captured position enlarged and firmly consolidated.

From this time forward the operations of the British armies may be envisaged as an attempt to enlarge the great bulge of the Ypres salient by fighting their way up to and along the ridges which enclosed it, so as to force the Germans to relinquish their hold on the coast near Nieuport, Zeebrugge, and Ostend. One very awkward spoke was put in the British wheel by an attack (10th July) on the extreme coastal sector of Lombaertzyde and the mouth of the Yser, by which any combined sea and land attack that might have been projected by the British was discounted. Most of the positions were recovered, but the amphibious plan had perforce to be postponed.

The putting into effect of Sir Douglas Haig's major plan, after the preliminary step of the capture of the Messines Ridge had been taken, did not operate till 31st July, when the 'Third Battle of Ypres' began, with a combined British and French attack on a 15-mile front beyond Boesinghe, on the Yser-Ypres Canal, to Zillebeke. The attack was conducted by the First French Army (General Anthoine); the bulk of the fighting fell on the Fifth British Army (General Gough). Twelve villages were taken and 5000 prisoners. There was, and could be, no break-through. The Germans rallied to a counter-attack, and were able to do so because their defences, their concrete pill-boxes, and their machine-gun effectiveness could, and did, hold up attacks before they progressed too far.

The rest of the British campaign in 1917 till it was arrested by the torrential rains of a wet October, and by the mud of the impossible declivities, may be summed up as a series of desperate forward thrusts which exacted each its

toll of prisoners, ground, and positions, but none of which succeeded in its object of inflicting a lethal injury on the German resistance. In the end these attacks had to cease while the last fragment of coveted ridge, the Passchendaele spur, was still not won, because on that ridge, as elsewhere, though blood had been poured out like water, and losses endured with unflinching fortitude, flesh and blood could do no more. The chief actions were as follows:

15th Aug.—British attack on wide front from north-west of Lens to Bois Hugo, north-east of Loos. Enemy's position penetrated to 1 mile depth.

16th Aug.—Franco-British attack on 9-mile front north of Ypres-Menin road. British carry Langemarck.

15th to 21st Sept.—Second phase of third battle of Ypres. 3000 Germans captured.

4th Oct.—British advance on 8-mile front, anticipating German attack east of Ypres. 3000 German prisoners.

9th Oct.—Third phase of third battle of Ypres. One mile advance on Passchendaele Ridge. 2000 prisoners.

6th Nov.—British attack on Ypres Ridges. Canadians capture Passchendaele.

These attacks were interspersed with costly minor encounters, and by the repulse or endurance of counter-attacks. The battle may be said to have closed by stress of weather in mid-November.

It was followed on 20th Nov. by a British attack of an altogether different kind, in an unexpected quarter—at Cambrai. Here the Third Army, under General Byng, made an attack on a 10-mile front between St. Quentin and the River Scarpe—tanks being employed for the first time in large numbers to lead the advance without a preliminary bombardment. It was a complete surprise, and all but a complete success. The 'Hindenburg Line' was broken, numerous villages and 8000 prisoners were taken. If the cavalry had been up, as they ought to have been, a first-class disaster might have been

inflicted on the Germans, and their railway communications at this point broken. But the cavalry were late; the next few days were spent in consolidation by peculiarly hard fighting; and on 30th Nov. the Germans counter-attacked and subjected the British defences of the newly-created salient at Bourlon Wood and Moeuvres to almost as severe a surprise as they had sustained ten days before. The British lost a number of prisoners, and had eventually to evacuate most of their hard-won positions.

French Front in the West, 1917

The history of the French armies in 1917 is largely the history of Nivelle's frustrated attempt to pierce the German centre between Soissons and Rheims in April; the pause necessitated by the fact that it was felt impossible to press the French armies too hard or too soon after the disappointment of the attenuated success at the Chemin-des-Dames, where the chief sacrifices of the attack took place; and the efforts on a more moderate scale by General Pétain, who succeeded Nivelle, to win tactical victories at a moderate cost on the terrain acquired in the April adventure. The second battle of the Aisne, as Nivelle's offensive was called, was planned over a length of 50 miles from La Fère, on the Oise, round the edges of the Forest of Gobain and Coucy, to Laffaux, thence below the line of the Chemin-des-Dames ridge and behind the Aisne to the crossing of the river at Berry-au-Bac, and Brimont, near Nogent-l'Abbesse, to the Moronvilliers heights on the other side of Rheims.

The attack on the larger part of this line began on 16th April; that on the Moronvilliers sector on 17th April. The attack on the 16th was extremely costly; it succeeded in only part of its objectives, but it captured 11,000 prisoners. The next day, one of pouring rain, improved on the positions won, especially at the western end of the Chemin-des-Dames, and on the 18th and 19th Nivelle so far enlarged his successes as almost to give them the appearance of a great victory. But the French bolt had been shot, and, to put it plainly, Nivelle was not encouraged to go on. By the 28th he had taken 28,000 prisoners, 175 guns, and some of the strongest points on the heights of the Aisne; but everywhere the positions had been only half-won, and the temper of the French army as a whole had suffered too severe a test. Nivelle was succeeded by Pétain and his plan was abandoned. The rest of the year was occupied by General Pétain in very skilled attempts, named

limited offensives, to repel the Germans from disputing some of the positions won and to enlarge the French gains at other points. The chief engagements were as follows:

5th May.—The French, co-operating with the British on a 20-mile front north of the Aisne, take Craonne and 6000 prisoners.

20th Aug.—French carry enemy defences north of Verdun on 11-mile front; 6000 prisoners. By 28th Aug. the French were back at their original Verdun positions.

23rd Oct.—French advance on the Aisne north-east of Soissons on 5-mile front; 8000 prisoners. By 25th Oct. further 3000 prisoners and 160 guns were taken.

The Balkans, 1917

In the Balkans the military situation during 1917 remained much as the end of 1916 had left it, the Germans, as already mentioned, completing their conquest of Roumania, and the Allies remaining comparatively inactive in the field while they cleared up the extremely unsatisfactory situation in Greece. The chief operations consisted of a Franco-Serbian attack north of Monastir, and attacks by General Milne on the British front in the Struma valley; but though the situation remained in the field virtually unchanged, the political situation was vitally affected in 1917 by the deposition of King Constantine in favour of his second son, Alexander—the king being induced to abdicate on 12th June—and the formation of a new Government under M. Venizelos. From 30th June, Greece, having formally severed diplomatic relations with Germany, was at length added to the list of countries fighting on the Allies' side.

Italian Campaign, 1917

Italy, who, meantime, had proclaimed Albania an independent state under Italian protection, and occupied Yanina in June, 1917, had committed herself wholly to the Allies' cause in 1916 by declaring war on Germany on 28th Aug. It was in Aug., 1917 that Ludendorff began his preparations for the terrific blow which was to lead to the Italian disaster at Caporetto in October. During the earlier months of the year the initiative had remained

with the Italians, and Cadorna had used it in a series of offensives which, while carrying him farther across the Carso towards Trieste, and winning seemingly impregnable positions in the Trentino, together with upwards of 30,000 prisoners and 140 guns, were indecisive, and left his exhausted troops—reduced by some 150,000 casualties—ripe for the blow which Ludendorff had prepared for them. The Russian Revolution and the collapse of Roumania freed Austria-Hungary at the same time from any anxieties on her Galicia front, so that she could concentrate all her energies and the bulk of her armies against the Italians. When the enemy's great counter-stroke was delivered on 24th Oct., the control of the campaign was taken over by the German High Command. Ludendorff had been training picked troops in special tactics—to be developed at their full strength on the Western front in the following year—and von Below was transferred from the French front to take command. Following an intense bombardment, the Austro-German troops were launched against the Second Italian Army between Zaga and Auzza, in deep formations so closely packed that the way could not be lost amid the prevailing snow and rain. Taken by surprise, and seriously demoralized in parts by Bolshevik propaganda and enemy intrigue, the Second Italian Army, which had hitherto distinguished itself by its splendid courage, found the whole of its left wing giving way before the impact, thus opening a gap for the enemy 20 miles wide over the Julian and Carnic Passes. Having shattered both the first and second Italian lines at Caporetto and Vodil Vrh, the Germans and Austrians surged forward from the Tolmino bridge-head until the retreating Italians, becoming entangled with their own reserves, broke in disorder. Although the right wing of the Second Army held, and many heroic efforts were made by isolated units, General Cadorna decided that it was impossible to save the situation from irretrievable disaster except by a general retreat to the Piave line. This was ordered on 26th Oct., when the broken fragments of the Second Army, as well as the Italian Third Army, began the great retreat which will be remembered as one of the finest achievements in Italian military history. The territorial gains of over two years' fighting had already been lost, and the conquering invaders reached within 15 miles of Venice, but they never succeeded in making the disaster irremediable. General Cadorna's scathing Order of the Day denouncing the units of the Second Army which had let the enemy through, and the new sense of national unity inspired by the military crisis, did their work. By 8th Nov. the bulk of the Italian armies, now in orderly

retreat, were across the Piave, and two days later were ready to turn on their pursuers. The danger was not yet over; the enemy succeeded in carrying several further heights dominating the Venetian plain between the Piave and the Brenta; but by 21st Nov., when a last German attack was made on the Monfenara Ridge, and defeated, the invaders were for the time being held. They claimed some 250,000 prisoners and 1800 guns of every calibre, besides immense quantities of munitions. They had also succeeded in diverting French and British divisions from the Western front, the Allies sending what assistance they could to stave off the defeat of Italy. The French and British troops arrived in time for the winter struggle, which began on 4th Dec., by which time General von Below had returned to France—being succeeded in command of the Austro-German operations by General Krobatin—and General Cadorna had been transferred to the Allied War Council, his place as Italian Commander-in-Chief being taken by General Diaz. Two British divisions, under Cavan, took up their positions in the Italian line at Montella; the French divisions at Monte Tomba; and they served to relieve the pressure while the enemy was making his last efforts to break through before the long-delayed snows put an end to the struggle for the year.

Naval War in 1917

At sea, where the British Grand Fleet was now commanded by Sir David Beatty in place of Sir John Jellicoe, who had succeeded Sir Henry Jackson as First Sea Lord in the previous November, the year 1917 passed without a single outstanding engagement. Since Jutland, the German High Seas Fleet had run no risk above water of again seriously challenging Britain's sovereignty of the sea, though her submarine campaign was pursued with ever-increasing vindictiveness. Risking rupture with the United States and other neutrals whose shipping and subjects were thus exposed to wanton attack, she inaugurated 'unrestricted' U-boat war on 1st Feb. "Give us two months of this", said the German Foreign Secretary to the American Ambassador in Berlin, "and we shall end the war and make peace." The argument was that the Allies' losses in tonnage, already more than they could bear, would increase to such an extent that they would be starved into submission. The United States, with President Wilson as spokesman, replied by severing diplomatic relations with Germany, but it was not until 5th April that she formally declared war against her. Cuba followed suit on 5th

April; Panama three days later; Brazil on 2nd June. Germany could afford the risk of offending the smaller American republics, but her defiance of the United States was a fatal blunder. She relied on her submarines to prevent the transport of American troops across the Atlantic—at least until they were too late to affect the issue. The first American contingents crossed unharmed and arrived in France on 26th June. It was not until the following year, however, that the new American army was ready to throw its weight into the scales on the Western front. The naval resources of the United States were at once placed at the Allies' disposal, the American destroyer squadron in particular being of great service in helping in the protection of trade off the Irish coast. Proof of the closeness of co-operation between the British and United States navies was afforded in June, 1917, when Vice-Admiral Sims, commanding the United States Naval Forces in European Waters, was given the command of the Irish station during the absence of Vice-Admiral Bayley on sick leave. No foreign naval officer had ever previously held the command of British ships, as well as his own, off the British coast.

Apart from the relentless submarine campaign, which every week exacted heavy toll yet never brought the Allies within measurable distance of the starvation-point to which the Germans had been so sure of reducing them, the enemy's operations at sea were restricted to destroyer and torpedo-boat raids in the Channel from Zeebrugge. Some of these did a certain amount of damage to patrol boats, and bombarded Ramsgate, Broadstairs, and Margate (27th Feb. and 26th April) with little effect save the death of women and children. One memorable incident in these minor operations in 1917 was the raid on Dover on the night of 20th-21st April, when 6 German destroyers, after firing a number of rounds inland, were caught on their way back by the British destroyers *Broke* (Commander R. G. E. Evans) and *Swift* (Commander A. M. Peck)—the advance ships of the British destroyer guard in the Straits of Dover. The *Swift*, which was leading, dashed between two of the retreating destroyers and, turning, sent one of them to the bottom with a torpedo. The *Broke* rammed the third vessel, and while the two ships were locked, an old-fashioned hand-to-hand fight took place on the *Broke's* forecastle, in which the German crew were beaten back. Two minutes later the *Broke* wrenched herself free and the German destroyer sank. Ten

German officers and 108 men were rescued at the close of this dashing affair.

1918 on the European Fronts

First Phase of Ludendorff's Offensive.—The Bolshevik betrayal of the Allies at Brest-Litovsk, the treaty of which took Russia irrevocably out of the conflict, and released Germany's Eastern forces for a concentrated assault in the West, gave the Central Powers their greatest opportunity of winning the war since their first hopes were shattered on the Marne in 1914. The crushing defeat of Italy at Caporetto afforded them further grounds for confidence. America, it was true, had thrown in her lot on the side of the Allies, but Germany counted on striking her decisive blow before the American troops could arrive in sufficient numbers to matter; and she had not yet lost faith in her submarines. All through the winter of 1917-8 Ludendorff, the German Commander-in-Chief, was secretly training his troops in the new tactics which were to bring open warfare into full play again, equipping them with Teutonic completeness, and massing guns and ammunition proportionate to the task in view.

The Allies on the Western front were meantime forced to remain on the defensive until such time as the American reinforcements should arrive in sufficient numbers to enable them to regain the initiative. Since April, 1917, the British army had borne the chief burden of the war in the West, and "the bloody struggles to conquer the Flanders ridges"—the words are those of Sir Douglas Haig himself—as well as the prolonged fight at Cambrai, "had left the army at low ebb in regard both to training and numbers". In view of the expected German offensive, it became imperative to fill up the ranks as rapidly as possible, and place the line in a sound state of defence. Late in Jan., 1918, Sir Douglas Haig took over a new stretch of French line, extending the front of the Fifth Army to cover the village of Barisis, 7 miles south of the Oise. The additional line, taken over somewhat against Haig's judgment, and giving the Fifth Army, which stretched on the left as far as Gouzeaucourt, no less than 42 miles to guard, extended the British front, all told, to 125 miles. The whole of this had to be greatly strengthened and supported by prepared positions to which the troops could retreat when the expected German drive took place, for it was regarded as inevitable that some dent must result in the Allies' line where the colossal blow was dealt.

This constructional work called for every man who could be spared for the task, and seriously interfered with the necessary training of the troops in new tactics of defence.

The months preceding the 'hammer blow' were marked by intense raiding activity on both sides, chiefly undertaken to procure information, but sometimes to secure useful positions for subsequent events. The most important of these included the sanguinary struggles for Bullecourt in the early days of January, in which the Australians greatly distinguished themselves, and German attacks at Dixmude (6th March), and in the region of Houthulst Forest and the Menin road (8th March), for positions destined to play their part in the new attempt to reach the Channel ports. By the middle of February, when 28 additional German divisions had arrived from the Russian front, and 6 from Italy, and great supply dumps were springing up in all directions behind the German lines—but particularly opposite the British Third Army at Cambrai and the British Fifth Army to the south of it—Sir Douglas Haig had no doubt as to what was to come. The only questions were "Where?" and "When?"

All strategical considerations pointed to an attack on the Fifth Army south of Arras, with the object of separating the British and French armies and seizing the centre of communications at Amiens. Neither the British nor the French Head-quarters Staff, therefore, was taken unawares when the great offensive began on 21st March. Both had worked out plans to meet it. More than half Sir Douglas Haig's infantry and the whole of his cavalry were allocated to this sector's defence, and General Pétain had arranged to send a French army corps to their assistance in case of need. The final dispositions of the Germans were carried out with the utmost secrecy: sunken roads, bivouacs, and every device of camouflage being employed to conceal their last stages of concentration. Even so, Sir Douglas Haig learnt from his Intelligence Department on 19th March that the enemy was putting the finishing touches to his impending attack, and that it would be launched by the 21st, if not before.

It was heralded, in fact, at 5 a.m. on the 21st by an intense bombardment in a thick mist which made it impossible for the British batteries to render effective aid to the battered first-line trenches. The onslaught was organized in two parts, the northern advance being directed against Byng's Third

Army from the Sensée River to the Cambrai road, and the southern attack from the Flesquières salient opposite Cambrai to St. Quentin. No fewer than 40 German divisions—nearly half a million men—specially trained for the new offensive, were launched against this southern half, and of these more than half were directed against the 16,500 yards of front held by Gough's Fifth Army nearest St. Quentin. All told, the German drive consisted of 64 divisions on the opening day of the offensive. To meet it the British had but 19 infantry divisions in line and 10 in reserve, with cavalry. From first to last the Germans employed in this attack some 78 divisions—exceeding in numbers the total fighting strength of the whole of the British armies in France. By the 9th of April, when the Germans, foiled in the opening move of their supreme offensive, had shifted the spear-head of their assault to Flanders, the total number of British divisions employed both in cavalry and infantry did not exceed 46.

Some part of the line was bound to give before the terrific impact of the infantry attack which followed the bombardment on 21st March. The plans made for repairing the breach in co-operation with the French broke down for a time because the British Fifth Army, attacked by a far greater force than had been anticipated—23 German divisions having been massed as secretly as possible in order to bring them into position at the critical hour—had been forced back sooner than was expected. There was no time either for the British reserves or the French reinforcements to repair the breach before the assaulting divisions under von Hutier, who had established his reputation for this form of operation on the Eastern front, were through it and extending it in all directions.

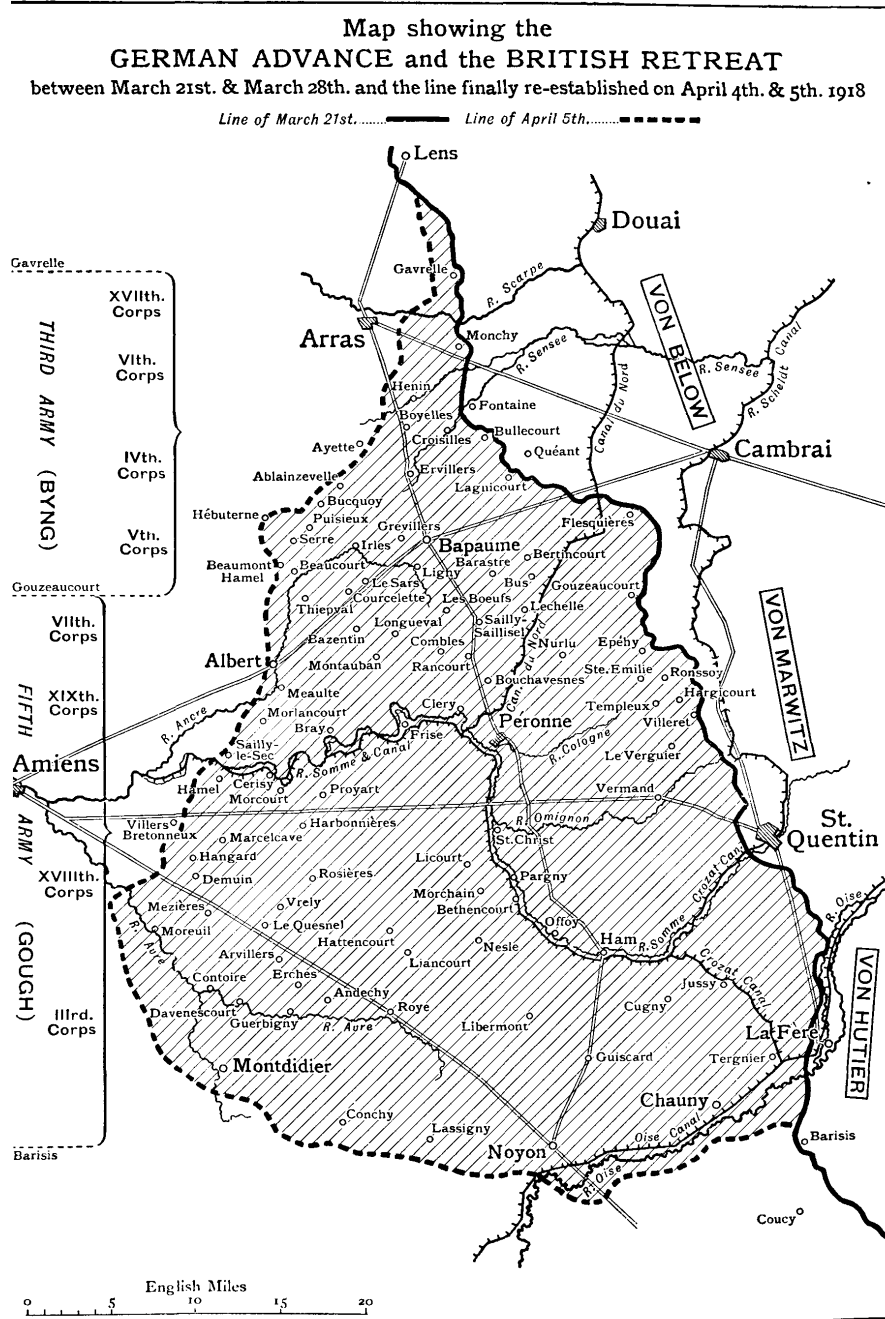
There were, in point of fact, two serious breaches along the 42-mile front held by the Fifth Army. One, the less dangerous of the two, was south of the Oise, where the line was held more lightly than elsewhere, owing to the marshes, which had been relied on to make any considerable attack at this point unlikely. As ill luck would have it, a long spell of dry weather had made the ground easily passable, and the Germans, well aware of the position, swept across it in overwhelming force, like a tidal wave. Heroic stands were made for the forward redoubts and battle positions, but the whole of the ground south of the St. Crozat Canal was so submerged in the flood that by nightfall there was nothing for it but to withdraw the divisions of the 3rd Corps, which had been defending it, to the line of the Somme

Canal. Nevertheless, though the Germans made their greatest progress on the 21st at this point, it was not their most dangerous thrust, being nearest to French reinforcements.

The chief danger-point was farther north, on the Fifth Army's left, below the Flesquières salient. By noon, with the fog still so thick that it was impossible to see 50 yards ahead, the Germans had advanced as far as Ronssoy, inside the second zone of the British defensive positions, together with Hargicourt and Villeret to the south. This opened a gate to the third line 3 miles wide, and before the day was over the enemy had pushed towards this as far as Templeux-le-Gerard. But for the stubborn defence of Epéhy, to the north, and Le Verguier, to the south, the breach would have been perilously widened on the following day, but at both these points the German advance was temporarily checked. It could not be stayed long. Supplied with an overpowering weight of men to crush through anywhere, von Hutier was ready to pay the price exacted for every success by British artillery-fire at short range, and British machine-gun posts held to the last: and when St. Emilie and Hervilly finally fell on the 21st, Epéhy and Le Verguier could only hold out long enough for the general line of defence to be withdrawn from them. The retreat thence, hard-pressed as it was, left the Fifth Army's centre with a sagging flank to the south, of which the on-coming Germans did not fail to take full advantage. Thenceforward the tide swept on for days in ever-increasing volume, all the reinforcements that Sir Douglas Haig or Pétain could send serving only to stop gaps here and there. Ham, Bapaume, and Péronne had fallen by the 24th, and about two-thirds of the territory wrested from the Germans in 1916 regained by them. Germany was announcing to all the world that the 'Kaiser's Battle', as the emperor himself had caused it to be named, had already been won. To drive the news home, the enemy, on 23rd March, began bombarding Paris with long-range guns capable of firing 70 miles. Nesle and Noyon were the next to go, and by 26th March—save at Albert, which held out until the following day—the Allies were back beyond the line from which they started in 1916.

Elsewhere the enemy's progress along the 60-mile battle front had been slow and costly. He had least success in the north against the British Third Army, partly because the positions held were stronger, partly because his heaviest and most persistent blows had been reserved for the Fifth Army. Some isolated gaps were made on this front, but nothing beyond repair, and

the ground lost was not vital. It is impossible in the space at our command to follow all the complications of attack and counter-attack in the fateful days which followed, until, by 26th March, the Germans were within a dozen miles or so of Amiens, with the British Fifth Army still retiring before them in a state of disintegration. At this critical juncture, when the reserves had all been thrown in, General Gough adopted a suggestion made by General Grant, Chief Engineer of the Fifth Army, that a last line of defence—a forlorn hope to save Amiens—should be formed from stragglers, army school personnel, tunnelling companies, Canadian and American engineers, anyone, in short, who could be roped in. The command of this heterogeneous force, after being organized by General Grant and posted according to General Gough's instructions, was handed over to General Sandeman Carey. 'Carey's force', as it came to be called, aided by the 1st Cavalry Division, which was rushed across the Somme from the north at the same time, earned a special tribute from Mr. Lloyd George in the House of Commons for the magnificent fight which it put up in this last line of defence.



The German effort was becoming spent, though by broadening out the salient the enemy continued to press back the French as well as the British. The French Third Army, sent to the assistance of the British Fifth Army, played a lion's part in preventing him from extending his gains too dangerously in the south. On the 28th he concentrated his main energy against the stubborn British Third Army, which, conforming to the

retreating line of the Fifth Army on its right, had fallen back to new battle positions, but in good order. Fresh shock-divisions were brought up to break this northern pivot of the British defence, and, after the usual full-dress bombardment, were launched as before in continual waves of assault. This time, however, there was no fog to handicap the British gunners, who were given the opportunity of a lifetime when they opened fire from hidden positions on serried ranks of German infantrymen, marching shoulder to shoulder at point-blank range. Six times the advance was renewed, and as many times mowed down, and when a final attempt was made, after a second bombardment in the afternoon, it met with similar failure. The Germans had shot their bolt. Their appalling losses on the 28th told on all their subsequent efforts in this first and greatest of Ludendorff's offensives in 1918.

It was on the 28th that General Gough relinquished the command of the British Fifth Army, General Rawlinson (Fourth Army) succeeding to the task of extricating its shattered divisions. Two days earlier the long-needed decision had been made by which the command of the Allied armies passed into the hands of General Foch as Generalissimo, who thenceforward, until the end of the war, held supreme control. Though several anxious days were to pass before the Allies could breathe freely again in the Amiens area, the position hourly improved as reinforcements, French and British, arrived on the scene. Counter-attacks recovered some of the ground on 30th March, and when the Germans resumed their advance towards Amiens on a more limited scale on 4th and 5th April, their losses were out of all proportion to their gains.

Second Phase of Ludendorff's Offensive

It is probable that the operations of 4th and 5th April were designed chiefly to pin the British armies to the southern area, while Ludendorff, finding his road barred to Amiens, prepared to strike a fresh blow in the north in a decisive bid for the Channel ports. Though well aware of a possible thrust in this direction, Sir Douglas Haig had been compelled to draw heavily on his Flanders front for reinforcements during the exhausting battle for Amiens, in which as many as 46 out of his total 58 divisions had been engaged. By the end of the first week in April the bulk of the British troops holding the Flanders line had passed through the furnace of the southern

battlefield and were sadly in need of rest and reinforcement. Had the ground been in its usual condition of slush and mire at this season of the year they could have been relied upon to hold up any advance, but a dry spring had prepared the path for a German advance, and as soon as this was seen to be imminent it was reluctantly decided voluntarily to evacuate the Passchendaele salient, won at such frightful cost in the closing months of the previous year. Steps were also taken to relieve the Portuguese troops^[2] who, though not seriously engaged, had been too long in the trenches south of the salient. Before either of these plans could mature the Germans upset both by launching their great attack at 4 a.m. on 9th April.

As in the opening move against the Fifth Army on 21st March, the assault—launched by the army of General von Quast in the direction of Festubert-Armentières against the northern portion of the front held by the British First Army (General Horne)—was favoured by an impenetrable early-morning fog. Through it came five columns of troops like the prongs of a fork, with an army corps as the central point to thrust into the weak spot where the Portuguese were sandwiched in between the British 40th and 55th Divisions. Bursting through the Portuguese sector, the attack spread swiftly to north and south—especially to the north, where the 40th Division, feeling the thrust which pierced the Portuguese line, was forced back on its right flank to the line of the Lys, 3½ miles in its rear. The rest of the 40th, with reinforcements from the 34th Division, formed a new line between Fort Rompre and Bois Grenier, covering Erquinghem and Armentières from the south, and held it the rest of the day. Had the southern pivot also given way when the Portuguese sector was broken the consequences would have been fatal to any hope of checking the German onrush. But the 55th Division never budged after its left flank, borne back by the first assault, succeeded in forming a defensive flank between Festubert and Le Touret, and the importance of its stout defence through the battle, as Sir Douglas Haig bore witness in his dispatches, could not be overestimated. This line was strengthened later in the day by the 21st Division, which, together with the 50th Division—both just relieved from the Somme fighting—had been hurried up as soon as the attack developed.

Next day the battle spread to the north, blazing up along the right of the British Second Army (General Plumer), the army of General Sixt von Armin attacking in another early-morning mist between Armentières and

Hollebeke. The two German armies now acted in concert, and together pushed their advantage until the Lys was crossed in the south and the Messines Ridge carried in the north, with Laventie, Ploegsteert, and a dozen other historic landmarks in between. Outflanked on both sides, Armentières had perforce to be evacuated. Messines was recaptured by the South Africans, but had to be abandoned when the enemy's advance in the south pushed almost as far as Neuve Eglise. On the 11th, when the enemy continued to extend his gains with seemingly endless reinforcements, and had crossed the Lawe, a tributary of the Lys, Sir Douglas Haig issued his famous Order of the Day to his troops, which, while it reflected the gravity of the situation, inspired them to fight it out:

"Every position must be held to the last man.... With our backs to the wall, and believing in the justice of our cause, each one of us must fight on to the end. The safety of our homes and the freedom of mankind depend alike upon the conduct of each one of us at this critical moment."

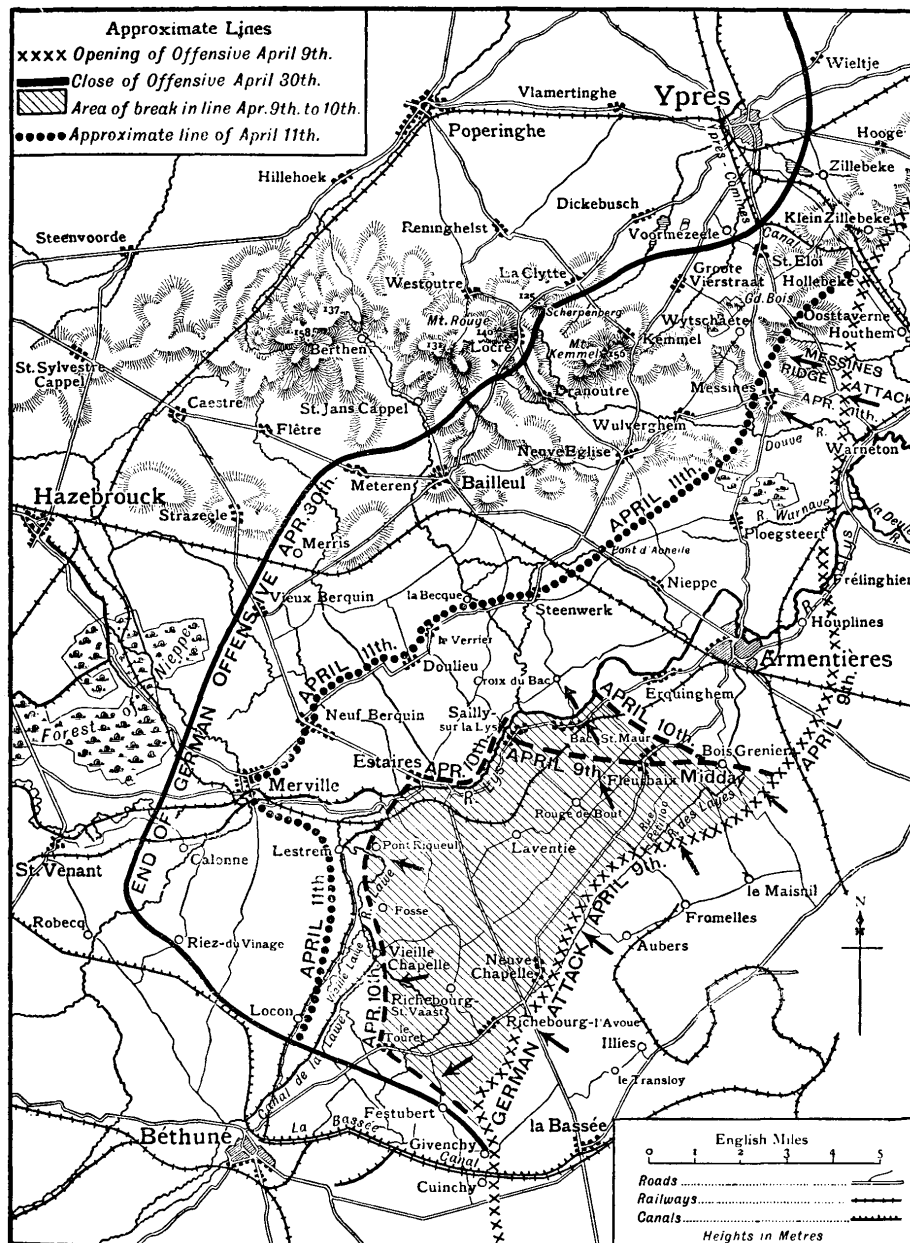
The appeal, with its promise also of speedy reinforcements from the French army—"moving rapidly and in great force to our support"—heartened the battle-worn divisions when they sorely needed its encouragement. The pressure was still too great to prevent the Germans from taking Neuve Eglise on the 14th—after a struggle lasting two and a half days, from house to house—or Merris a day earlier, which brought them within 4 miles of Hazebrouck, where the 1st Australian Division, just detrained, kept them at bay; or Bailleul on the 15th. At midnight on the 15th the British line fell back to the defences between Meteren and Dranoutre, a move involving the simultaneous withdrawal from the Passchendaele Ridge.

These were the darkest hours of the last great battle for Ypres. There were already signs that the German advance, having again failed to break a way through by sheer weight and numbers, was slowing down, and the promised French reinforcements were close at hand. These took over the sector from Meteren to Wytschaete, where, on the 25th, before they had consolidated their positions, they were made to bear the brunt of a fresh German blow, delivered with 9 fresh divisions, from Bailleul to the Ypres-Comines Canal. The French fought desperately to save Kemmel Hill, commanding Ypres some six miles away; but in five hours the Germans had captured both the

village and the crest of the hill. With the Ypres salient now dominated both from Kemmel and Messines it became necessary still further to shorten the line round Ypres. This was accordingly redrawn on the night of 26th-27th April through Pilkem, Wieltje, Zillebeke Lake, and Voormezele.

One more effort was made by the Germans to push right through before the end of the month—on the 29th, when, in a fresh attack in force on the Franco-British front, they succeeded in reaching as far as Locre, behind Kemmel. But the French, who were in no mood to repeat their experience at Kemmel, flung the enemy back with sanguinary losses. The heaviest casualties of the Germans that day, however, were against the British 21st, 49th, and 25th Divisions, whose artillery—like the gunners of the Third Army on 28th March—had the range of them as they advanced in mass formation, and blew them to pieces. Only one of the waves of German infantry succeeded in reaching the British positions, where bomb and bayonet completed its destruction. This marked the last serious attempt on Germany's part to seize the Channel ports.

Though Ludendorff had failed to reach his objectives either on the Amiens or Flanders front he still had a sufficient superiority of force to retain the initiative. With every incentive to compel a decision before the new American army, which was now arriving at the rate of something like 150,000 a month, could enter the field in full strength, Ludendorff had either to throw up the sponge or strike again at the earliest moment in one final effort to beat the Allies to their knees. Meantime the obvious policy on the Franco-British part was to maintain an active defence until their own and American reinforcements made a counter-offensive possible. In the minor operations which marked this period of waiting, the Australians added Villers-Bretonneux to their battle honours. Villers-Bretonneux, which lies on the edge of the ridge facing Amiens—only 8 miles away—had been rushed and captured by the Germans in a surprise attack in thick fog on 23rd April. Before daybreak the next morning the Australians had surrounded the German garrison, and the end of a fierce house-to-house conflict left the place in British hands again, together with nearly 1000 prisoners.



Germany's Last Bid for the Channel Ports: approximate positions of the Allies' line before and after Ludendorff's offensive in April, 1918

Third Phase of Ludendorff's Offensive

Though every day added to the danger of delay, it was not until the end of May that the German army, its plans disorganized by its unexpectedly heavy commitments in the Amiens and Flanders battles, was launched on the third and final phase of Ludendorff's great offensive with a sudden

attack on the Aisne front in the direction of Paris. It was along this front that the 9th British Corps (General Sir A. Hamilton-Gordon), consisting of the 8th, 21st, 25th, and 50th Divisions, and subsequently reinforced by the 19th Division, had been sent to the French armies under General Pétain for much-needed rest after sharing to the full the honours and sacrifices of the earlier battles. By their side were crack French divisions which had also earned the rest which it was felt they could count upon along the main stretch of the Chemin-des-Dames, holding as it did some of the strongest natural defences along the whole battle front. The French had taken months in the previous year, and spent countless lives, to recapture these positions.

The very unlikelihood of choosing such a formidable sector decided Ludendorff to select it for his dramatic attack on 27th May, moving up his specially trained divisions of shock troops at the last moment—with all possible secrecy, accompanied by other surprises in the vast number of guns and aeroplanes brought into action, as well as the largest fleet of German tanks which the enemy had ever employed. Only on the very eve of the new advance did the French learn of the impending blow—too late to avert disaster. Outnumbered by 6 to 1 the Allies, British and French alike, were borne back by the onrush of picked troops as soon as the preliminary bombardment ceased. Helped as usual by a thick early-morning mist the armies of von Böhn, von Hutier, and von Below—all nominally under the German Crown Prince—had carried the whole of the Chemin-des-Dames ridge by nightfall, and were fighting on the Aisne. Within two days they were not only across the Aisne on an 18-mile front, but had swept on to the Vesle, and were even across that river west of Fismes—a depth of 12 miles from their starting-point.

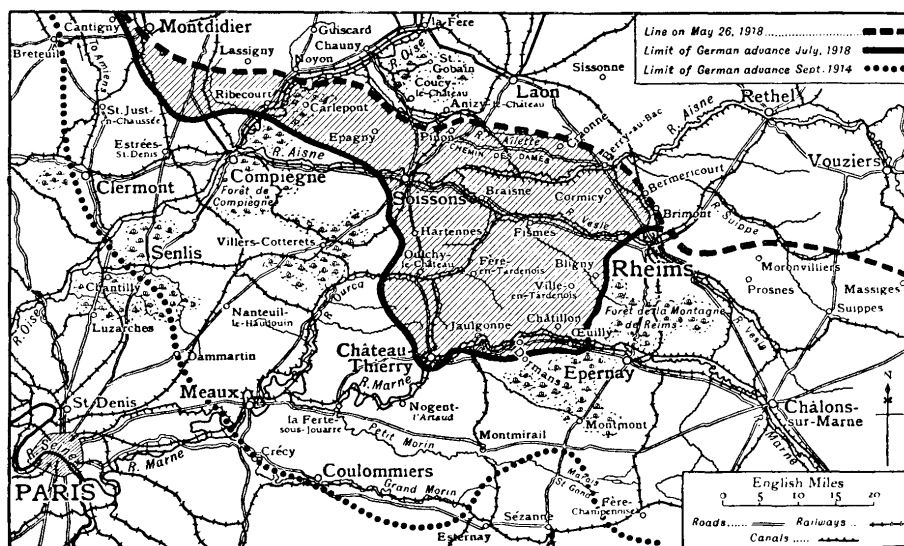
Reserving his stoutest resistance for the flanks, Foch strove hard to save Soissons, but it fell again into the enemy's hands on the 29th, by which date Rheims, on the Allies' right, was also pressed so hard that its outlying positions were carried on the 30th, and only the devoted bravery of Berthelot's Fifth Army—with Gouraud guarding the Champagne front on its right—saved the battered city. The British and French divisions which had meantime been driven back from the Chemin-des-Dames to the Aisne, and from the Aisne to the Vesle—the Allies often fighting side by side in hopeless rear-guard actions against immensely superior numbers—were now holding a line between the Vesle and the Ardre while the Germans

continued to plunge deeper and deeper into the big salient which they had formed towards the Marne. The shattered divisions of the British 9th Corps now formed part of Berthelot's Fifth French Army guarding Rheims, and subsequently played a large part in repelling the enemy's attack on the north-east side of that city. In the words of General Berthelot himself: "They have enabled us to establish a barrier against which the hostile waves have beaten themselves in vain. This none of the French who witnessed it will ever forget."

By 31st May the German advance in the centre had spread as far as the Marne at Château-Thierry, extending thence along a 10-mile front to Dormans. It was at Château-Thierry on this day that the Americans began to play their part in the battle, linking up with a French colonial division on the south bank of the river and preventing the enemy from crossing. New French units were also coming into line, blocking the Germans' path south of Soissons on the road to Villers-Cotterets, as well as at Château-Thierry. Foiled in both these directions, Ludendorff turned to the Ailette front in order to get more elbow-room, and flattened out the French front between Soissons and Noyon. Again unable to make much further headway against the fierce French counter-attacks, he carried the battle still farther to his right, attacking between Noyon and Montdidier on 9th June in the hope of linking up the new Marne salient with the one already formed at Amiens, and so advancing on one immense front. The new 'drive' was again entrusted to von Hutier, whose 25 divisions, employing the same shock tactics as before, swept forward at first to a depth of some 5 miles in the centre; but, held on the wings, were fiercely counter-attacked on the 11th on their exposed right flank and robbed of most of their gains.

On the following day the German War Minister declared that "Foch's so-called Army of Reserve exists no more"; the truth being that Foch, with time and an ever-flowing stream of reinforcements on his side, was gradually becoming master of the situation, and could afford to wait until Ludendorff gave him the opportunity he wanted. Ludendorff, on the other hand, with the pick of his troops 'pocketed' in the great Marne salient, was forced to make another forward move or withdraw them. He made one more attempt on Rheims, three divisions being ordered on 18th June to take it at all costs; but the whole attack was an expensive failure. For the rest of the month, when the weather broke, and during the first half of July,

Ludendorff left most of the fighting to the Allies while he prepared for one last herculean effort to burst through their line. On 28th June Foch felt his way towards his counter-stroke by a preliminary advance between Villers-Cotterets and the Aisne, when he won back over a mile of useful territory and took over 1000 prisoners. On 4th July further minor victories were recorded along both the French and British fronts, the British success being at Hamel, where American units celebrated Independence Day by helping the Australians to recover that fiercely-contested stronghold, with 1500 prisoners.



The First and Last Advances on Paris: map showing approximately (by the shaded area) the limit of the German gains in the final phase of Ludendorff's offensive in 1918, and (by the dotted line) the limit reached in the 1914 advance

On 15th July Ludendorff launched his final effort on a 50-mile front on each side of Rheims. This time the Allies were warned of its direction in time. On the left, where immediate success was vital to the whole plan, the attack was flung into disorder at the very beginning by a deluge of shells from Gouraud's guns before even the German bombardment started; and when, this over, the attacking divisions of von Einem and von Mudra advanced, they found that Gouraud's army, save for volunteer garrisons in concrete forts, had returned undamaged to its main battle positions, to reach which they had to face a concentrated fire that tore their ranks to pieces. Some 50,000 German troops were admitted to have fallen that day before

Gouraud's army. With the failure of the advance in Champagne the German attack on the right, where Italian as well as British and American troops were now fighting side by side with the French, was unavailing, though the line south-west of Rheims was pressed back some 3 or 4 miles, and eight divisions under von Böhn succeeded by 17th July in crossing the Marne at a number of points between Fossoy and Dormans.

Foch's Counter-stroke

They were only allowed to remain south of the Marne long enough for Foch to convert these river crossings into a death-trap. For Foch had now decided that the moment had come when the Germans, exhausted by their advance, were least in a condition to resist a counter-stroke aimed at their flank. The flank which Foch selected for attack was that on the western side of the salient created, from its most northerly point on the Aisne near Soissons, to its southern extremity, Château-Thierry on the Marne, where the symptoms of the exhaustion of the German momentum had been furnished by the ability of American and other contingents to resist further advance. In the earlier half of July a ceaseless stream of men and guns had flowed up from the French side to take cover in the forest of Villers-Cotterets on their flank, in preparation for the blow to come; and by 18th July two French armies were assembled along the 27-mile flank, that of General Mangin aligned between the Aisne and the Ourcq, which bisected the salient, and that of General Degoutte from the Ourcq to the Marne. Mangin's army contained some of the finest French shock divisions as well as two famous British ones, the 34th and the 15th, and a number of keen American troops. Degoutte's army had the more awkward task, judged by the country over which it had to travel, but Foch's plan here, as elsewhere, was to put his best fighting material where it would pierce farthest, and hold the enemy elsewhere. By the same token the army of General Berthelot, with two other supporting British divisions, was entrusted with the task on the other side of the salient, from Rheims to Épernay, not of thrusting at the Germans but of holding them hard.

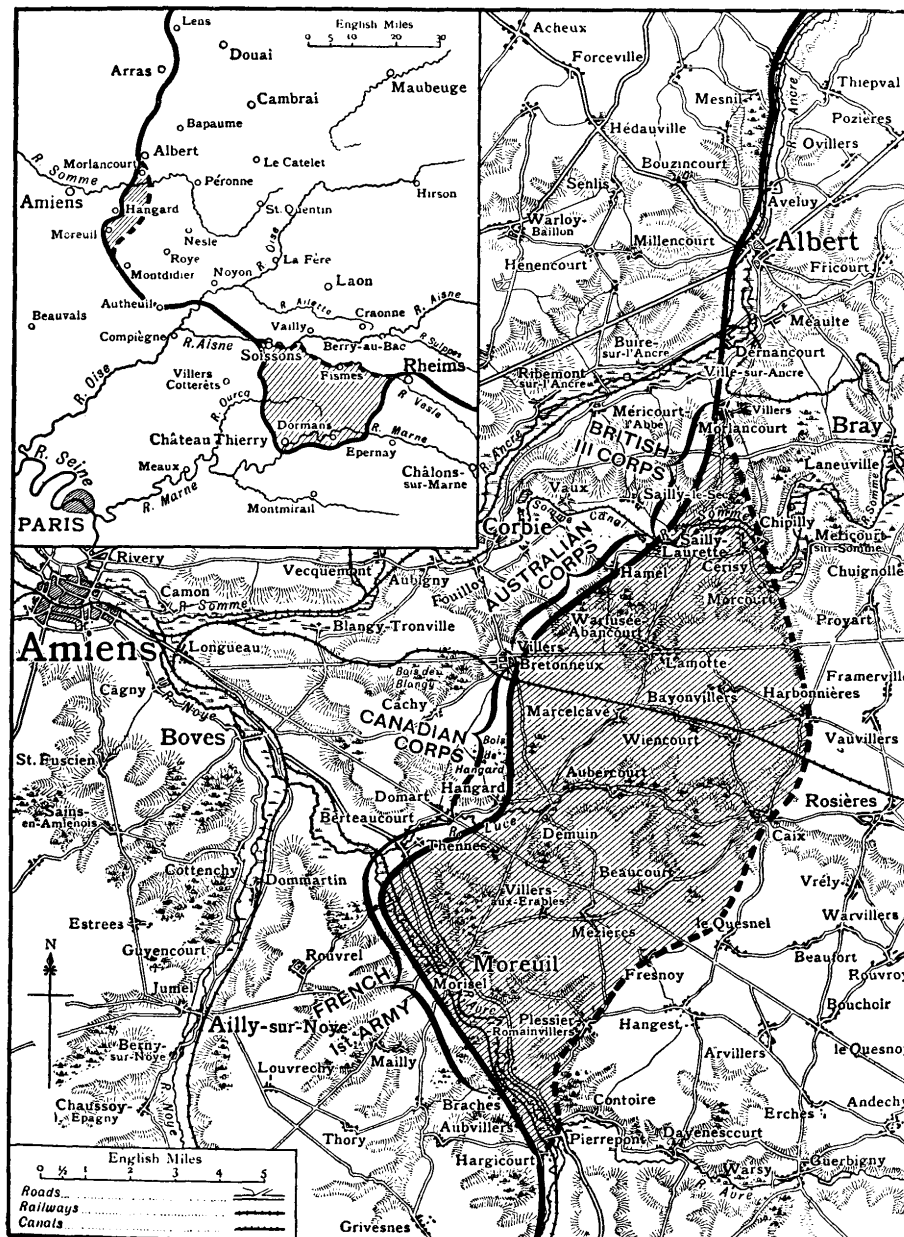
Mangin's army was ordered to strike with all its force. It was equipped with new and speedier 'whippet' tanks, and its immediate onset was masked by the accident of a July thunderstorm on the eve of the fighting. On the morning of 18th July it went forward with nothing but a barrage, but with

the effect of a thunderbolt, and its average advance on that day was 5 miles, with Fontenoy and the plateau of Pernay on the Aisne firmly secured. Degoutte's army went forward for 2 miles over difficult country. A blow of immense significance had been struck along the whole length of the salient's vulnerable side. Next day Mangin's movement continued; he tightened his hold on the Aisne and swung his right wing farther along the Ourcq so as to bring the whole line of the one good north-and-south road in the salient under the fire of his guns, and thus to hamper the German movements terribly. That alone would have forced the enemy to begin a retreat from the Marne, while it might yet be possible. Meanwhile Degoutte was advancing also, and was forcing the Germans away from the neighbourhood of Château-Thierry. The German commander, von Böhn, was not slow to recognize the implications of the situation into which Foch had forced him, and gave orders to recross the Marne. He was in time, but his retreating troops were roughly handled at the crossings, and despite all his attempts to hold up the attack on his western flank by counter-attack, the pressure of Mangin and Degoutte, added to that of de Mitry's army, which was now following him back over the Marne, became every hour more dangerous. By 20th July not only was the Marne itself in process of being abandoned by von Böhn but Château-Thierry had fallen. Degoutte's army was 3 miles north of it. De Mitry's divisions had secured ample crossings for future movement and Berthelot's mixed forces of French, British, and Italians had begun a disconcerting attack on the eastern side of the salient. In three days the Germans lost 20,000 prisoners, and, what was more significant, 450 guns. The first *riposte* of the French Generalissimo had been delivered. It was to be followed in unending succession by others.

The German Commander-in-Chief had to gain time. It was no easy thing to withdraw his 600,000 men crowded between the Aisne and the Marne, but he was obliged to support them lest the salient should collapse too suddenly. He aimed a counter-stroke at Gouraud's army east of Rheims, but the blow spent itself in the air, and Foch replied by setting in motion the army of General Debeney, where it stood opposite to that of General von Hutier, between Montdidier and Noyon. But he had not yet gathered the full fruits of the strength of his positional assault between the Aisne and the Marne, and did not in the least allow pressure here to relax. Mangin seized Oulchy-le-Château on the Ourcq (25th July); and on 26th July Gouraud, on the

other side of Rheims, recovered the ground he had ceded under the German pressure. By Sunday (28th July) the Allied attack in the Aisne-Marne salient had swept convergingly on to the line of the Ourcq, and with that achievement the most important episode in the opening of Foch's campaign was consummated. Soissons fell to Mangin on 1st and 2nd Aug., a signal that the work was done, and Foch was now free to prosecute his larger plan of delivering successive blows at points where they would disperse and use up the German reinforcements most effectively.

The Allies' Victorious Offensive, 1918



The Freeing of Amiens: Map illustrating the recapture of the city's outer defence line on the opening day of the Battle of Amiens, 8th Aug., 1918.

Marshal Foch, after consultation with the British Commander-in-Chief, had desired the commanders of all armies, British and French, to prepare plans of action and to be ready to put them into operation at short notice. He now more particularly addressed inquiry to Sir Douglas Haig as to his willingness to undertake a continuous offensive towards the German centre. Sir Douglas Haig assented to Marshal Foch's representations as to the

superior advantages to be gained from an attack there, and, while continuing a show of preparation in the Ypres area, where Ludendorff, on an estimate of the psychology of the British commander, would expect the counter-attack to come, and was already taking steps to reduce its effectiveness by masked withdrawals, transferred forces steadily to Rawlinson's Fourth Army on the Somme. This army, and the First French Army under Debeney on the right, both directed by Haig, were set in motion on 8th Aug. on a 16-mile front from Morlancourt to Moreuil. The thrust was successful beyond expectation. The British Fourth Army, on the right wing, went through the German divisions of von Marwitz (Second Army) like paper, regaining the old outer-line defences of Amiens; and Debeney's men crushing the resistance of von Hutier's Eighteenth German Army, and reaching Fresnoy and Plessier, where they linked up with General Humbert's Third French Army on the road to Roye. The captures of the day amounted to 17,000 prisoners and 500 guns, an unmistakeable symptom that the German power of resistance was shaking.

On 9th Aug. Rawlinson pushed on still farther; on 10th Aug. General Humbert prolonged Debeney's still attacking line and took Montdidier, and a number of villages. These three armies continued to eat into the enemy's positions and to pin a number of German divisions down till 20th Aug., while Mangin's army at Soissons moved *en échelon* to take up contact with Humbert's right. Meanwhile Ludendorff, fully aware now that the initiative had passed out of his hands, and that the best course that lay open before him was a 'strategic retirement', began to effect one stage of it in the Ypres and Lys district under the direction of General Sixt von Armin, whose withdrawal was followed vigilantly by the British forces; and another stage in the German salient on the Ancre, where General von Below's Seventeenth Army was stationed. Von Below withdrew on the Bapaume line from Serre, Beaumont-Hamel, and Bucquoy to the shelter of the sector of the Hindenburg line behind it (13th, 14th, and 15th Aug.).

But whereas in 1917 Ludendorff had disconcerted both British and French Commanders-in-Chief by a sudden withdrawal on the Bapaume-Péronne line of the Somme, he was not now allowed to withdraw without injury. Haig's battle of Bapaume (21st Aug. and following days) was designed in two stages, the first of which brought up Byng's Third Army to a position in which it was aligned with Rawlinson's Fourth Army, and the second of

which saw the Third and Fourth Army attack von Below in combination. The combined pressure of these two armies was continuously successful, though the Germans fell back stubbornly in many places. By 30th Aug. Bapaume was once again in British hands, and the line of attack was threatening the strongholds of the Hindenburg line, while its extension ran through Heudecourt and Fremiecourt to Cléry. Péronne fell to the Australian Corps by a most gallant feat of arms on 1st Sept. A more strategically significant victory was gained on the same day when the capture of Bullecourt, followed by that of Riencourt and Cagnicourt, opened up the first crevice in the ramifications of the Hindenburg defences known as the Drocourt-Quéant switch line. The battle of Bapaume drove thirty-five German divisions from the old Somme battlefield, and captured 34,000 men and 27 guns.

The crevice in the Drocourt-Quéant defences was still further widened on 1st Sept., when six British divisions of Horne's First Army, including two Canadian, attacked behind tanks a 5-mile front occupied by eleven German divisions and captured Dury Ridge and Quéant, together with 16,000 men and 200 guns. So far, therefore, from Ludendorff's strategic retreat being conducted 'according to plan', it cost the Germans, between 21st Aug. and 9th Sept., some 53,000 men and 470 guns; the French had been able to occupy Ham and Chauny, while the British were going forward; and General Sixt von Armin was forced cautiously to retire from the Ypres salient.

During these operations by the British armies Foch had never relaxed pressure with the three armies of Debeney, Humbert, and Mangin, while still threatening an advance beyond the Vesle in the deflating Aisne salient, west of Rheims, and preparing new blows elsewhere. At the beginning of September the position of the French armies of the centre, won by continuous fighting, was as follows: Debeney had crossed the Somme, taken Ham, and was threatening St. Quentin; Humbert was close to Tergnier and was pointing towards La Fère; Mangin was back in Coucy-le-Château and held the railway thence to Soissons; Degoutte was spreading from Soissons along the Aisne. These threats left General Ludendorff no choice but to shorten his line where he could do so with least risk. He decided on the Vesle front, where General de Mitry, with French and Americans, had been engaging his Seventh and Ninth Armies, and began to retire thence on

4th and 5th Sept. on a 19-mile front. The Americans occupied the Aisne thereupon from Condé to Viel-Arcy, and on 7th Sept. General Mangin crowned his long campaign at the Chemin-des-Dames by taking the ruined Fort de Condé. A week later Allemant and Laffaux Mill fell, and once again the French troops came in sight of Laon. Humbert and Debeney, both pushing forward, embarrassed Ludendorff in his intention of moving divisions to meet a new British movement known thereafter as the battle of Epéhy.

This battle was the preliminary movement in that great attack on the Hindenburg line which, more than any other single action, was the decisive 'blow at the heart' of the German defensive plan. The British advance, viewed as one movement, was made towards Cambrai, which was the northern bastion of the German defensive lines, as La Fère and Laon were the twin southern pillars. On 2nd Sept. the Third Army began a local attack on a 5-mile front which captured Havrincourt and Trescault, while on the extreme right of the Fourth Army the 9th Army Corps and the Australian Corps began a movement which by 17th Sept. placed them in Maissemy, where the Fifth Army in March had been pierced. These preliminary positions having been secured, the Third and Fourth Armies set in motion their important combined attack (18th Sept.) on a 17-mile front from Gouzeaucourt, through Havrincourt to Holnon Wood, where Debeney's First Army lent assistance. The hardest fighting was at Epéhy, which gives its name to the battle, on the left centre, but by nightfall the German defences had been pierced on a 3-mile front, 12,000 prisoners had been taken, and the British forces brought within striking distance of the main Hindenburg lines.

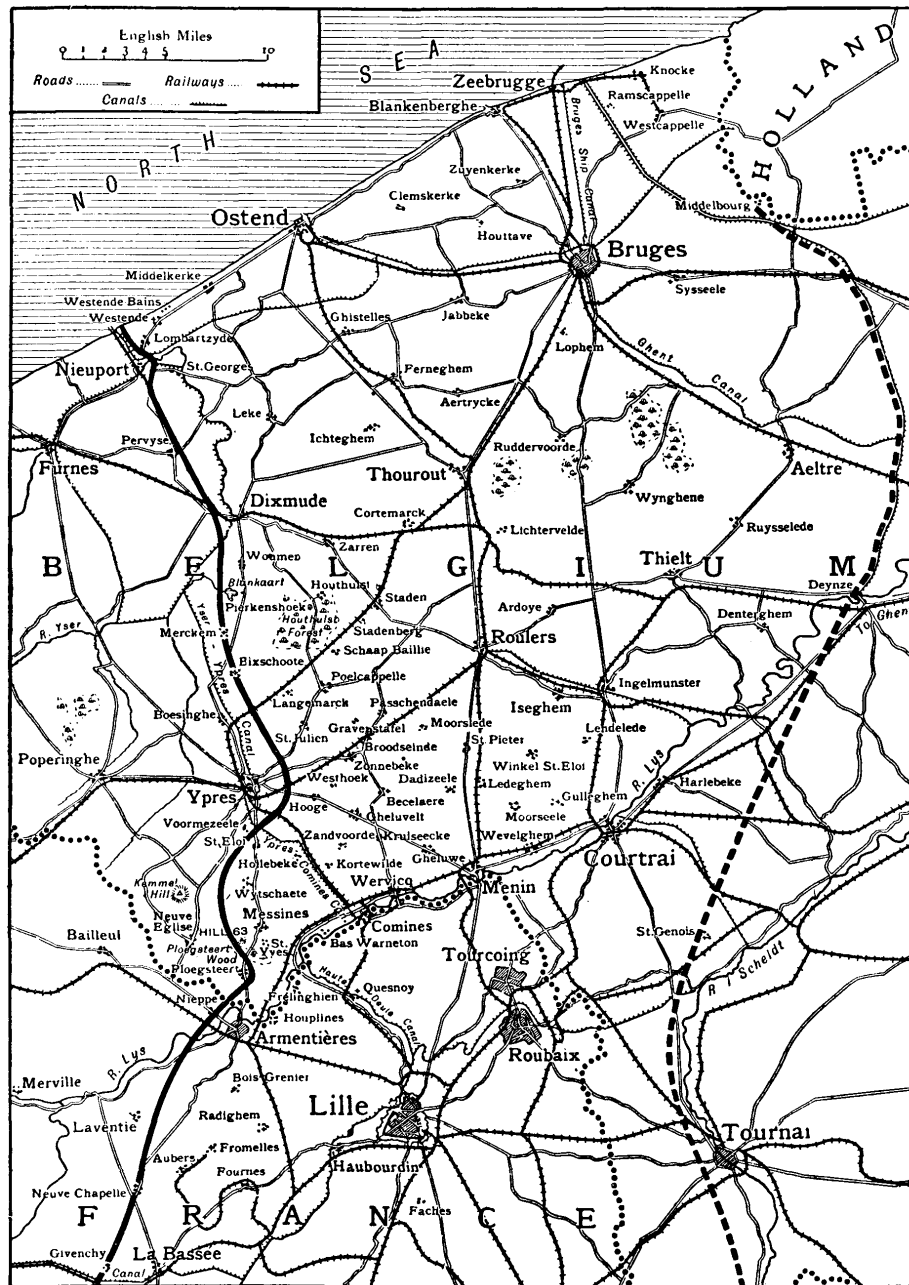
During the weeks in September while the plans for the battle of Epéhy were ripening, Foch had struck hard at another point in the German line, which had appeared invulnerable while the Germans were strong, but was now a menace to Ludendorff's own plans for retirement because it absorbed divisions which he badly needed elsewhere, namely the long-standing salient of St. Mihiel. It was held by seven German and two Austro-Hungarian divisions in September, and Ludendorff had been withdrawing its heavy artillery; but before his plans for withdrawal could be consummated, Foch sent in General Pershing with his young American divisions, aided by two French divisions, at the salient's apex. The

Americans attacked on the two faces of the salient, west and south, the strongest thrust being made by two corps of seven divisions apiece on the southern face. The attack began on 12th Sept., and in thirty hours the salient had disappeared; while in spite of the haste with which the Germans had left it—the firmest resistance was offered by the two Austro-Hungarian divisions on the western face—10,000 prisoners and 450 guns were left behind. This victory, as symptomatic as others of German disorder, freed the Verdun-Commercy railway, and completed the attenuation of Ludendorff's reserves. The 207 German divisions which Ludendorff had commanded at the period of the greatest German strength had fallen to 185, and only 21 reserve divisions remained.

The second stage of Foch's plan now had been reached. The German armies had again been brought back to the line which they had chosen in 1914, after their first rush had recoiled. It was less threatening by the loss of the St. Mihiel salient, but it had been enormously strengthened by four years of engineering. North of the defence line was the railway which, running through Brussels, Mons, Maubeuge, Mézières, Sedan, and Metz, was the chief artery of German communications, and Foch's plan was to cut this artery on either side of the great curve which the German line made when, after coming north to south from the coast, it turned west to east at La Fère. The right half of the thrust was to be made by Gouraud's army at Rheims and the Americans on the Argonne, where they were being steadily accumulated. The more deadly attack was to be made by the First, Third, and Fourth British, and the First French Armies, which should break through Cambrai and St. Quentin towards Maubeuge.

Complementary operations were designed for the armies of Humbert and Mangin at the nose of the curve, and it was expected that, under this comprehensive pressure, Ludendorff would be compelled to withdraw divisions from the coastal sector, where an attack by the British Second Army (Plumer) and the Belgian Army might then be successful against a weakened front. A portion of Degoutte's army was sent northward in readiness for such a blow, and to the Fifth British Army (now commanded by Birdwood) was assigned a task at Lille and Lens similar to that of Humbert and Mangin at St. Gobain.

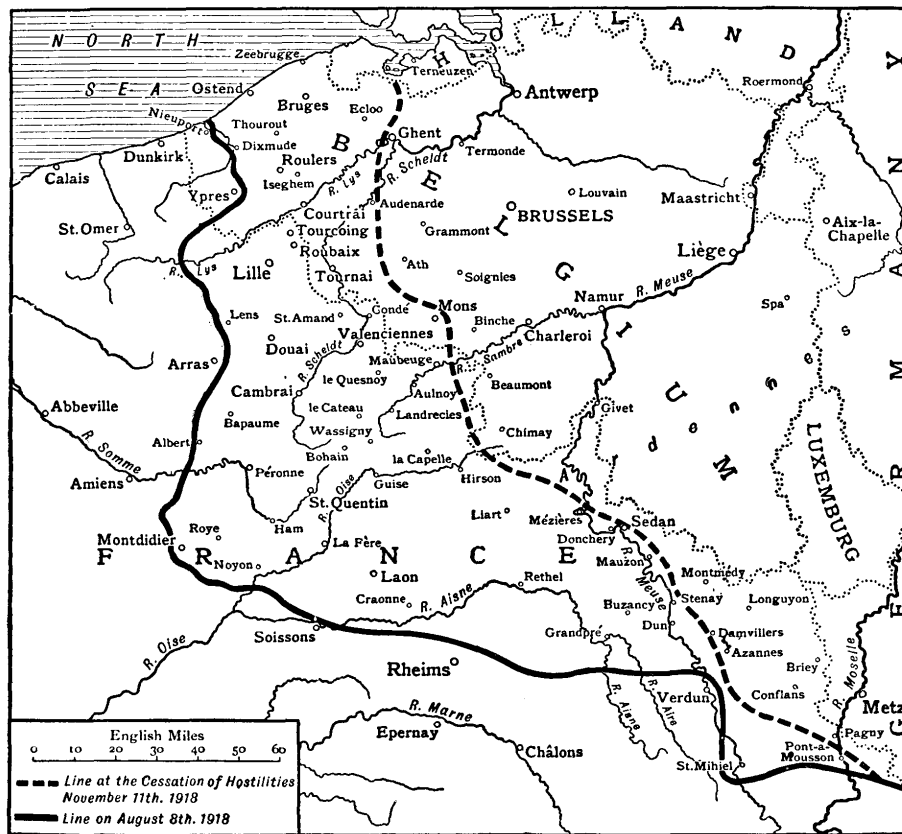
There was a pause of nearly a week, in which the Germans awaited, and the Allies prepared the new move; and then, on 27th Sept., Haig's armies struck what Foch declared to be the blow from which there was no recovery. The battle of Epéhy had given the requisite positions for the attack on that section of the Hindenburg defences which the Germans named the Siegfried line. The plan was to send forward the First (Horne) and Third (Byng) British Armies to clear the way on a line from Sauchy-Lestrée to Gouzeaucourt, seizing the crossings of the Canal du Nord, and so preparing the way for an attack by the Fourth Army. The dangerous movement was accomplished (27th Sept.): the crossings seized, the canal held, and Cambrai threatened. On 29th Sept. the Fourth Army took up the combat, and in a tremendous action along a front of 20 miles, supported by attacks from the other British armies and the First French Army, got across the vital defences of the St. Quentin Canal in the Siegfried zone. The next day the fighting spread furiously along the front of all four armies: the breach was widened; a portion of the Scheldt Canal taken; and by 3rd Oct. the Fourth Army had pierced the Siegfried line vitally. By 9th Oct. the German defences were no longer defences, and in this decisive encounter they had lost 36,000 prisoners and 380 guns.



The Liberation of the Belgian Coast, 1918: map showing approximately the Allies' line on 28th Sept.—represented by the solid line—and on 25th Oct.—represented by the broken line

Elsewhere victories had been won which appear minor only by comparison. American divisions had been transferred to the left bank of the Meuse at Verdun, and, supported by French divisions, had carried Montfaucon and Varennes (10,000 prisoners); on the other side of the Argonne Gouraud was

advancing, and by 1st Oct. was 9 miles from his starting-point below Moronvilliers, and had accumulated 13,000 prisoners and 300 guns.



Map showing approximately the Allied line on 8th Aug., 1918, and on Armistice Day, 11th Nov., 1918

The results of these attacks were at once apparent, as Foch had predicted, in the north. His plan of an attack by Plumer's Second British Army, and by the Belgian Army stiffened by French divisions, was as successful as he had hoped. Under the weight of the Franco-Belgian-British attack on 28th Sept. the thinly held German front melted away. By nightfall the British held all the ridge between Wytschaete and the canal north of Hollebeke; the Belgians were in Houthulst Forest. By the end of September the blood-drenched ridge of Passchendaele, and all its tragic surroundings, had passed into the Allies' hands, and the German grip on the coast had been finally unloosed. This marked the beginning of the end. Since Foch attacked between the Aisne and the Marne on 15th July the Allies had taken a quarter of a million prisoners, 3669 guns, and 23,000 machine-guns. The rest of the

campaign must either be a disaster of the first magnitude to the Germans, or, at best, a painful and ineffective retreat to the line of the Meuse.

What remained of the campaign was the work of clearing up; but this was not an easy task, because, though Ludendorff was aware that victory remained permanently with the Allies, he assured the German Government that delaying actions could be fought till the following spring. Against such an undesirable protraction of the war, Foch was preparing a final stroke in the neighbourhood of Metz with the aid of a Second American Army, which was being organized by General Pershing, who had relinquished the command of the First American Army to General Leggett. But the progress of the First American Army in the Argonne, where the fighting was conducted under circumstances of great difficulty, and where the transport was admittedly defective, was slow; and, lacking the place and the resources for another outflanking blow against the retreating Germans, the Allied armies could do no more than press their retreat.

In that retreat position after position was forced from their hands. Cambrai fell on 9th Oct.; on 15th Oct. von Einem's army was far from the coast; on 17th Oct. Ostend had fallen; Horne's First Army had taken Douai, and Birdwood's Second Army had liberated Lille. King Albert re-entered Bruges on 25th Oct. Farther south the British Third and Fourth Armies were close to Le Cateau on 17th Oct.; and Mangin and the French had re-entered Laon—so long the German Great Head-quarters—on 13th Oct. The story of the rest of the campaign, though it involved much severe fighting in breaking the resistance of German rear-guards, is the record of the steady drive of all the British, French, and Belgian Armies which had produced the German collapse, while the right wing of Foch's greater pincers, comprising the only great new reinforcements he could bring to bear, namely, Gouraud with the Americans on the extreme right, worked its way up for a last decisive blow.

It was a slow operation; but by 4th Nov. Gouraud and Leggett had joined hands north of the Bourgogne Forest; and by 6th Nov. Gouraud entered Rethel and an American division reached Sedan. Thus, though behind schedule time, Foch's right wing approached its decisive position in the first week in November; and in the second week the left wing (British) had occupied Maubeuge. What would have been the consequence had Foch advanced his right wing farther and with effect is a matter for the military

expert. That Ludendorff was in no doubt of its disastrous results to the German armies is shown by his request for an Armistice on 9th Nov.

The Balkans, 1918

Bulgaria, thoroughly war-weary, and dissatisfied with Germany's refusal to give her the whole of the Dobrudja when terms were made with Roumania, had surrendered to the Allies on 30th Sept. The last Balkan campaign had been swift and decisive. No major operations had taken place during the earlier months of 1918, but on 15th Sept. General Franchet d'Esperey, who had succeeded General Sarrail in supreme command of the Allied forces, launched an offensive which rapidly transformed the whole military situation. French and Serbian troops, on the left or Monastir front, with the eager Serbians as the spear-head of the attack, penetrated the Bulgarian positions with an impetuosity which in two days carried them 12 miles behind the enemy's lines on a 22-mile front, and drove a wedge between the First and Second Bulgarian Armies. Meanwhile the British and Greek divisions were engaged in a far more difficult task on the Doiran front, where they suffered heavy losses in storming impregnable positions between the Dopropolje Ridge and Vetrenik, but succeeded in preventing the Bulgarians opposing them from sending reinforcements to their hard-pressed troops along the Monastir front. This 'wing of sacrifice' pinned the Second Bulgarian Army to the Doiran front until it was too late to join its retreating First Army. Realizing the plight in which it stood, it hurriedly evacuated its positions on the night of 21st Sept. and fled in confusion towards Sofia. The pursuit of the Bulgarians was now taken up by all the Allied armies from Doiran to Monastir, the vengeful Serbians in particular harrying the retreating Second Army with a remorseless energy which drove it head-long through Northern Serbia in increasing disaster towards Belgrade, while the British and Greek forces under General Milne entered Bulgaria hard on the heels of the demoralized First Army. By 26th Sept. the Bulgarian politicians realized that the whole position was hopeless, and sent a *parlementaire* under a white flag to the Allied head-quarters. Four days later they signed an armistice at Salonika, handing over complete control of the Bulgarian railways and communications, demobilizing the Bulgarian armies, and surrendering their arms and ammunition. On 4th Oct. King Ferdinand abdicated, and his eldest son reigned in his stead as King Boris III.

Italian Campaign, 1918

In less than a month—on 3rd Nov. to be exact—Austria-Hungary, after experiences in the field similar to those which had fallen to Bulgaria's lot, surrendered to the Italians. This turn of the tide in the Italian campaign in 1918, which amply atoned for the disaster of Caporetto in the preceding year, and crowned Italian arms with triumph, followed a final attempt of the Austro-Hungarian army, now under the direction of General von Arz, to crush the Italian front in conjunction with Ludendorff's great offensive in the West. General von Arz's main attack was delivered on 15th June on a 46-mile front along the Piave, and extended across the mountain positions between the Piave and the Brenta. The two British divisions west of Asiago played a great part in hurling the enemy back in this sector, the French divisions similarly distinguishing themselves on their right. Elsewhere some progress was made at certain points, and the Piave was crossed in two places; but by the third day it was already obvious that the attack had failed. Then the weather broke; rainstorms swept down the hills and turned the Piave, which had been low when the enemy crossed, into full flood, sweeping away a number of his bridges. Hurried efforts were made to get the marooned troops back, and though General Diaz was unable to bring up enough divisions in time to complete their discomfiture, they lost heavily enough in the retreat, their casualties before the battle died down on the other side the river amounting to some 200,000. Biding his time for his own great counter-offensive until Foch could be assured about the situation in the West, where Italian troops distinguished themselves in the operations round Rheims, General Diaz gradually pushed the Austrians back, until by 7th July he had cleared the whole Piave delta. It was not until 24th Oct. that his final blow was delivered. Its success was immediate and overwhelming. Launched on the night of 23rd-24th Oct., the main attack consisted of an advance across the Piave with the Tenth Italian Army—placed under the command of Lord Cavan—now including three British divisions, together with the Eighth and Twelfth Italian Armies. Cavan's force formed the spear-head of the thrust, and ensured the success of the battle by seizing the Island of Grave di Papadopoli in the Piave mid-stream, held by the enemy as an advance post. This was captured in a daring surprise attack by night, without any artillery preparation, and paved the way for the passage of the troops across the swollen river. At the same time the Fourth Italian Army,

with a French division, advanced across the old battle-ground of Asiago and Monte Grappa, where, however, the Austrians counter-attacked, holding up the advance until the whole front collapsed with the triumphant progress of the main attack across the Piave. By 27th Oct. the breach had widened until it spread across the entire front of the three Italian armies, which thereupon swept the plains and mountain heights until all the enemy's positions between the Brenta and the Piave had been regained. The Austrian retreat became a rout. By the end of the month the Italians claimed 50,000 prisoners, had cut the railway between the plains and the mountains at Conegliano, and occupied Feltre. With Germany in similar plight; Turkey and Bulgaria already finished; and her own internal affairs rapidly going from bad to worse, Austria appealed to General Diaz for an armistice. When the end came on 3rd Nov. with the signing of the agreement which involved the demobilization of the Austrian army; the surrender of the Austrian fleet; the occupation by the Allies of the Trentino, the Istrian peninsula, and a portion of the Dalmatian coast and islands, the Italians had just captured both Trent and Trieste, a landing-force having arrived at Trieste for the occupation that very day. The wholesale nature of the Austro-Hungarian surrenders during the closing phase of this decisive campaign may be gauged from the fact that by 3rd Nov. they had amounted to no fewer than 300,000 prisoners and 5000 guns.

Germany accepts Defeat

The complete collapse of the Great War, and with it all the Pan-Germanic dreams of world-power, came with dramatic swiftness. Ludendorff resigned, and though the Kaiser had entreated Hindenburg to make one last stand on the line of the Meuse, his appeal had been in vain. Hindenburg knew the hopelessness of the position, not only of the German army but also of the German home front. Ominous disturbances were breaking out in all parts of the Fatherland, including a mutiny at Kiel. Turkey (30th Oct.) as well as Bulgaria and Austria-Hungary had surrendered, and Germany was in no position to face the Allies alone. The Armistice terms, with all their crushing humiliations, had perforce to be accepted, and were signed on 11th Nov. They included, besides evacuation of territory, the surrender of the bulk of the German navy, 5000 additional guns, 30,000 machine-guns, 3000 trench-mortars, and 2000 aeroplanes. A zone of territory on the Rhine was to be occupied by the Allies, and the Treaties of Brest-Litovsk (signed by

the Bolshevik Government with the Central Powers on 3rd March, 1918, in defiance of Russia's solemn engagement not to make separate peace with them) and Bucharest (forced on Roumania in March, 1917, at the close of von Mackensen's drive) were declared null and void.

Unable to face his subjects, William II abdicated on 9th, slipped across the frontier into Holland on 10th Nov., and on the 28th of that month signed the formal document of his abdication, the Crown Prince, who also sought refuge in Holland, following suit three days later. With the Kaiser fell all the rulers of the German states. On 10th Nov. the Imperial Chancellor, Prince Max of Baden, resigned in favour of Herr Friedrich Ebert, ex-shoemaker, who became 'First Imperial President of the German Republic', with a Government formed from both wings of the Socialist party.

Last year of Naval War

The war had been won at sea as well as on land, though the British navy had far fewer opportunities than the army of getting to grips with the enemy. The hazards of naval warfare never ceased by day or night; and by the end of the war the range of the submarine had extended from the eastern end of the Mediterranean to the American coast. All the reckless efforts of the Germans to win the war by these means failed. With the adoption of new methods to cope with the danger—including the employment of 'Q' boats, or mystery ships—and the invaluable help of the American reinforcements, the monthly shipping losses in the closing year of the war grew progressively less, falling, indeed, from the total of 1,494,473 tons in the September quarter of 1917, to 915,513 tons in the corresponding quarter of 1918. Meantime the new tonnage under construction to make good these losses was as rapidly increasing. The U-boat war, in short, had failed; and the Germans knew it had failed.

The outstanding operation at sea in the closing year of the war was the raid on Zeebrugge and Ostend on St. George's Day under Vice-Admiral Sir Roger Keyes, the object being to block the outlet of the German submarines and destroyers from their depot at Bruges. Since all the neighbouring coast was strongly fortified, the attack was fraught with the gravest danger, and had to take the Germans as far as possible by surprise. The chief ship of the expeditionary force was the old cruiser *Vindictive* (Captain Carpenter),

which, with the ferry-boats *Iris II* and *Daffodil*, were told off to act as shield to the three blocking-ships intended for Zeebrugge—*Thetis*, *Intrepid*, and *Iphigenia*. Two old submarines were also taken, charged with explosives, and ordered to ram themselves below the viaduct connecting Zeebrugge Harbour with the Mole, and then blow themselves up. One of them carried out her orders to the letter; the other's rope parted while she was being towed into position, and she was too late to help. The simultaneous attempt to land, shortly after midnight, followed a bombardment from a squadron of monitors, and was supported by a flotilla of destroyers and a fleet of motor-boats. The *Vindictive* ran alongside the Mole within five minutes of being discovered by the German garrison, and was kept in position by the *Daffodil* and *Iris* while landing-parties jumped ashore to do what damage they could and the blocking-ships were being rammed at the entrance. This successfully accomplished—both the *Intrepid* and *Iphigenia* being blown up in the fairway—the battered *Vindictive*, taking her landing-parties aboard, backed out and returned with her supports to Dover. The first attack on Ostend, which took place the same day, was a comparative failure owing to the undetected removal of a buoy, the two blocking-ships sent for the purpose being sunk outside the harbour. Three weeks later the commander of the *Brilliant* (Commander Godsall), who was in charge of that operation, tried again, this time successfully—the old *Vindictive*, patched up as a blocking-ship, being sunk 200 yards up the channel of Ostend—but Commander Godsall was killed by a shell just after completing his task.

The German navy made one more appearance on the high seas—when, under the terms of the Armistice, 6 battle-cruisers, 10 battleships, 8 light cruisers, 50 destroyers, and all her submarines were surrendered, the bulk of them to the Grand Fleet at Rosyth under Sir David Beatty on 21st Nov. On the following day the captive ships were sent to Scapa Flow, where, exactly seven months later, their German crews, while the British battle-fleet was absent on gunnery practice, scuttled practically every vessel. A week later—on 28th June, 1919—Germany signed the Peace Treaty at Versailles, involving unconditional acceptance of all the Allies' terms.

Turkey and the War

In Aug., 1914, the German war-ships *Goeben* and *Breslau* escaped the British squadron stationed to intercept them near Messina, the escape

arising partly through delay in correcting orders, and partly through the fact that the German commander had no scruple about violating neutrality in the course he sailed through the Straits of Messina. The ships reached the Dardanelles on 11th Aug., and were sold to Turkey. This was a symptom of Turkish relations with Germany rather than a determining cause of Turkey's entrance into the war as an ally of the Central Powers; but many efforts were made to change Turkey's attitude before the British, Russian, and French Ambassadors left Constantinople (1st and 2nd Nov.) and Great Britain declared war on her (5th Nov.). The first act of war undertaken by Turkey was the dispatch to the Caucasus front of three army corps, with a plan of campaign designed by General Liman von Sanders, and commanded by Enver Pasha. They were decisively beaten by the Russian forces at Sarikamish in December. Had the Russians possessed abundance of transport, or had the roads been less difficult, this victory might have been pressed. But Russia wanted all her resources elsewhere, and it was not till 1916, two years later, that she made any serious attempt to carry the war into Turkish territory. Then, under the direction of the Grand Duke Nicholas, and General Yudenitch, who was some years later to come into prominence as the leader of an unsuccessful attempt to reach Petrograd and overthrow the Bolsheviks (1920), a Russian expedition drove back the Turkish forces on Erzerum and captured it (16th Feb., 1915). On 16th April, after another pause to gather transport, the Russians found their way to Trebizond on the Black Sea, and during the next few months spread over the Asiatic Peninsula to Bitlis, Musk, Van, Mosul, Erzingan, and Diarbekr, fighting generally with local success, but with no concerted plan of campaign. The effort expired in the autumn, and was not revived.

Gallipoli Campaign

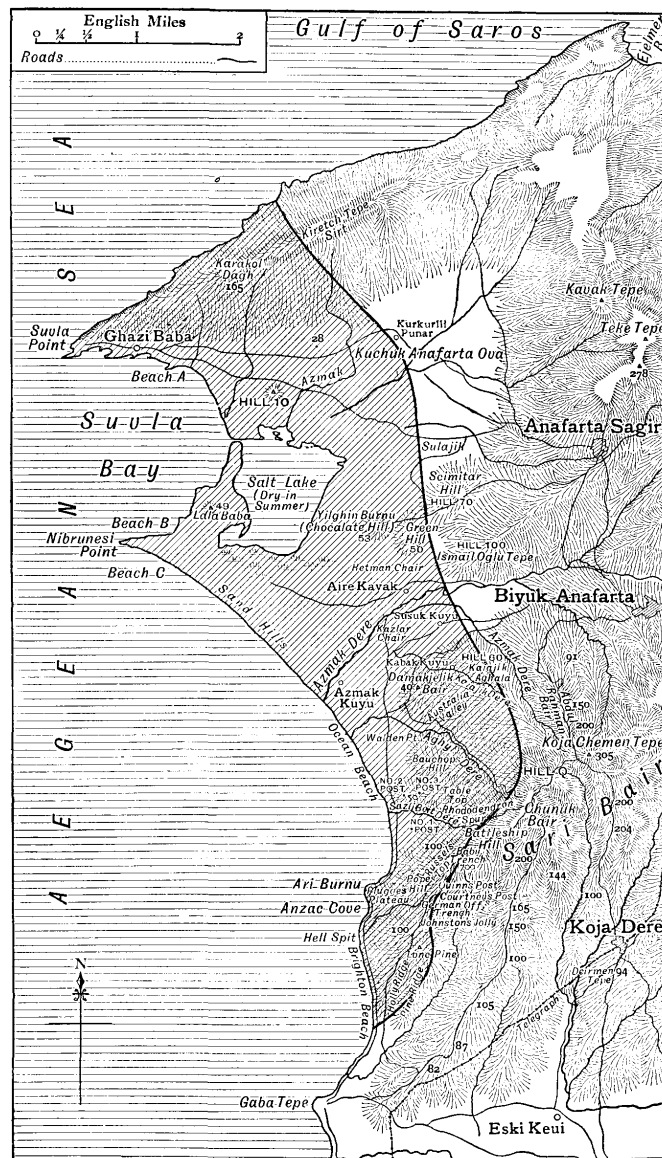
Before, however, the Russians undertook their own expedition, they were urgent in pressing on France and Great Britain the desirability of opening up the Dardanelles so as to bring the Allies into touch with each other through the Black Sea ports. In response to this invitation, the naval possibilities of which were insufficiently considered by the advisers of the British Admiralty, and the prospects of which had been roundly condemned by all who had previously considered the problem, Admiral Carden began a bombardment of the Dardanelles forts with old battleships, British and French, on 19th Feb., 1915. The bombardment was renewed on 25th Feb.

and 6th March. Admiral Carden resigned on 16th March owing to ill-health, and was succeeded by Admiral de Robeck, who favoured the idea of breaking into the Sea of Marmora by rushing the straits. Four big ships, one the *Queen Elizabeth*, engaged the guardian forts, Chanak and Kalid Bahr, at long range (18th March); other ships closed in, and a French squadron penetrated as far as Kephez Point. But the hidden batteries, the mines, and various other devices frustrated the attempt, which was a practical failure by the afternoon. Two good ships, *Irresistible* and *Inflexible*, were put out of action, and two others, *Bouvet* and *Ocean*, sunk. After this Admiral de Robeck accepted the professional view that a fleet operation should be combined with one on land. Lord Kitchener, who had been as little disposed towards the Dardanelles adventure as Lord Fisher, reluctantly consented, and General Sir Ian Hamilton was appointed to command the British expedition, while General d'Amade was in charge of the less numerous French Colonial Corps.

Sir Ian Hamilton's force eventually comprised the 29th Division, the Australian and New Zealand (Anzac) Corps, the East Lancashire Territorial Division, and part of the Royal Naval Division. There was an unfortunate delay in getting the expedition off from its Egyptian base, and the Turks had ample time to prepare for its landing (25th April), which they did not believe to be a possible feat. The landing was effected, nevertheless, with incomparable gallantry, at five beaches on the nose of the Gallipoli Peninsula, and at Gaba Tepe (by the Anzac Corps) farther north. Some 8000 men were got ashore in twelve hours, and an advance was made on the following day (26th April). On 28th April the French Corps, which had made a feint attack on the other side of the Narrows, was brought across to assist in a general advance on the dominating height of Achi Baba. The resulting engagement, the battle of Krithia, revealed the fact that the Turkish field fortifications, the Turkish artillery, and the Turkish numbers had all been too well organized under von Sanders to admit of being rushed. Evidence accumulated that Achi Baba, which was the key to the Dardanelles forts, could only be taken at ruinous cost by a frontal attack, though an attack in force on 6th May went nearer to success than any other, and appeared to fail only through a culminating misfortune of misdirected artillery. Fighting continued in May and June, both here and at Gaba Tepe, where General Birdwood commanded the Anzac Corps. Engagements in

which local advances were made took place on 29th May (Anzac Corps), 4th and 21st June (French Corps), 28th June (29th Division), and though the climate and the insufficient protection inflicted appalling losses on the hard-trying British and French troops, it was on the Gallipoli Peninsula that the flower of the Turkish army also was lost. Of the naval forces the *Queen Elizabeth* was summarily ordered home just before a new Turco-German submarine campaign set in; three more old battleships were torpedoed; but the English submarines retaliated by torpedoing a Turkish battleship, gunboats, and transports in the Straits.

General Gouraud relieved General d'Amade in the middle of May, but was badly wounded by a shell (30th June), and returned to France. General Sir Ian Hamilton, who had unceasingly asked for more guns and more men, received some part of the aid he asked in order to put a new plan into operation. The plan was to land a force at Sari Bair, on the neck of the Peninsula, where it could co-operate with the Anzac Corps in forcing a way to the commanding heights there, and ultimately might take the Turks in the flank. One force was to land at Anzac Cove; two other of the three new divisions sent out to Sir Ian Hamilton were to be landed 4 miles north, at Suvla Bay. The forces at Anzac were landed on 4th-6th Aug., bringing up the numbers of Sir W. Birdwood's command to 36,000. Sir F. Stopford was in command of the force to land at Suvla Bay (6th and 7th Aug.), with Generals Hammersley and Mahon as divisional generals.



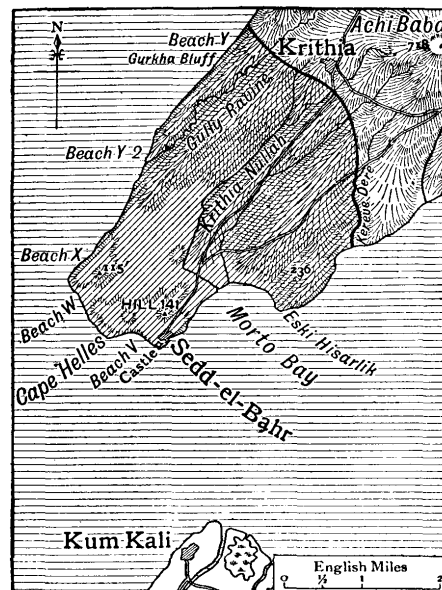
The Suvla and Anzac Line: map showing approximately by the shaded portion the area occupied after the linking up of the two armies in Aug., 1915

The fresh attack on the night of 6th Aug. was not completely successful, and the Anzac columns under General Godley were still short of the ridges at Koja and Chunuk on the night of 7th Aug. The Turks were then alive to the threat, and made continuous counter-attacks, so that by 10th Aug. all the efforts of the combined force of Anzacs, Gurkhas, and English regiments had failed to maintain more than a slippery hold on the ridges. The chief cause of the failure was that the contributory aid of the attack at Suvla Bay

farther north had not been forthcoming, owing to various reasons. Lack of water had exhausted the unseasoned divisions landed in Suvla Bay; misunderstandings among the generals were other reasons; and when the Commander-in-Chief, on hearing of their failure, came up from Zimbros, he met a situation which it was too late to redeem. General Stopford was succeeded by General de Lisle (15th Aug.). Lord Kitchener declined to send further troops from home, and though Sir Ian Hamilton, bringing the veteran 29th Division up to Suvla, with reinforcements of yeomanry, made another attempt on 21st Aug., it was unavailing. After 24th Aug. the British campaign assumed a defensive character, and in spite of Sir Ian Hamilton's representations the War Office declined to waste further men and material on it. On 15th Oct. Sir Ian Hamilton was recalled, and Sir Charles Munro, who reported that the expedition should be abandoned, was supported by Lord Kitchener after personal inspection of the surroundings. The evacuation was carried out with remarkable success on 18th Dec. and 8th Jan. (1916). The cost of the expedition in men was 31,389 killed, 78,749 wounded, and 9708 missing.

Mesopotamian Campaign

The Mesopotamian campaign ran contemporaneously in part with the Dardanelles operations, and in part with the campaign between Erzerum and the Caucasus in Armenia, in which Russia co-operated. A small British force, sent out to the Persian Gulf in 1914, occupied Basra (22nd Nov.) and pushed outposts forward to the junction of the Tigris and Euphrates. General Sir John Nixon took command (Jan., 1915), and in April (11th to 14th), following a Turkish attack, the British force, now a division strong, advanced to Shaiba, and on 31st May took Kurna. Amara, on the Tigris, capitulated to General Townshend (3rd June), who was empowered to push forward an advance column, while another column, under General Gorringe, beat the Turks at Nasiriyeh (24th July). At this time the Russians were on the headwaters of the Euphrates.

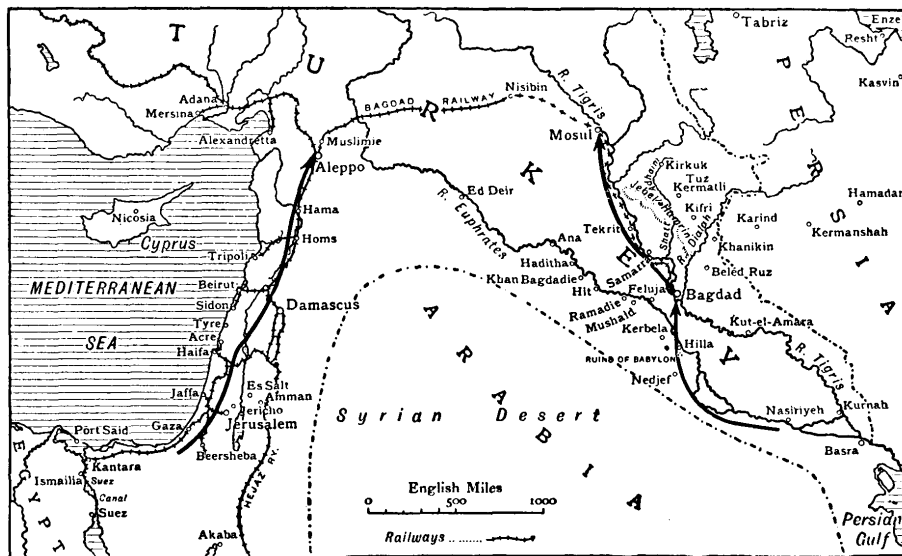


Map showing approximately the area in the Southern Zone evacuated in Jan., 1916

Fired by these successes, General Nixon ordered Townshend to seize Kut if possible. It was so dangerous an enterprise that probably the Turks did not believe that it would be attempted. But Townshend traversed the difficult country in six weeks, arriving within striking distance of the Turkish forces under Nur-ed-Din Bey on 15th Sept., and, in a very capably devised and executed engagement on 27th and 28th Sept., drove them before him. Townshend, against his own better judgment, was then sent forward to attempt the impossible feat of pushing on to the Turkish base of Bagdad. He arrived in front of their advanced position, Ctesiphon, on 22nd Nov., and attacked it with his four brigades, one of which was a reinforcement, and his rather scanty cavalry and artillery. The first line of the defences was carried, but Nur-ed-Din, who had retired on his supports, counter-attacked, and in spite of Townshend's resistance compelled him to retreat (25th Nov.) by way of Azizieh, fighting heavy rear-guard actions all the way to Kut (3rd Dec.). Townshend's force suffered severely and lost part of its river

transport. By 5th Dec. it was invested in Kut, and after two unsuccessful attempts to carry the town by force (8th and 23rd Dec.), the Turks sat down to starve it out. Townshend's mixed force of British and Indians, ill-fed and assailed by beri-beri and scurvy, endured the siege for 145 days, till 29th April, 1916, when it surrendered to Khalil Pasha. The prisoners were at first well treated, but most of them were compelled to march through Mesopotamia, Armenia, and Anatolia to prisoners' camps, and many died from their hardships.

An attempt to relieve Kut failed. It was made by Generals Aylmer and Younghusband, whose relief expedition set out on 3rd Jan., 1916. An engagement was fought on 8th Jan., and a pitched battle at the Shatt-el-Hai on 8th March. This failed, when on the brink of success, through want of luck or enterprise at the critical moment. Several changes were made in the Mesopotamia leadership after the impossibility of relieving Kut had become apparent, and in Aug., 1916, General Stanley Maude took over the command. The attention drawn to the blunders, and the scandals of medical assistance and transport, enabled General Maude to begin his task by rectifying them, and he built light railways from Basra. He made his first move forward on 12th Dec., 1916, and the history of the Mesopotamian campaign then entered on a brighter epoch, a result due to Maude's deliberate, bold, and prudent generalship. Contact began on 5th Jan.; the right bank below Kut was cleared by 15th-16th Jan.; and by 23rd Feb. Kut itself was so far encircled that the Turks had no option but to abandon it with guns and prisoners. Maude resumed his advance on 5th March, and by 11th March had manœuvred the Turks out of Bagdad as he had forced them out of Kut. The integrity of the Mesopotamian Turkish force had been destroyed, and its portions distributed at points believed to be out of Maude's reach. One of these was at Ramadie, but a dashing night-march captured the Turkish commander and 3400 prisoners (28th Sept., 1917). Other successes were gained at Tekrit and the Jebel Hamrin hills before Sir Stanley Maude died of cholera (18th Nov.). He was succeeded by General Marshall, who, in 1918, occupied Hit on the Euphrates, and ended the Mesopotamian campaign with a crushing victory at Mosul just before the Armistice.



Map showing the converging lines of the main British advances in Mesopotamia and Palestine at the points reached on 31st Oct., 1918

Palestine Campaign

Turkey assumed the offensive against Egypt, of which the Sultans were nominally suzerains, by an attempt to force the Suez Canal in Feb., 1915. A force under Djemal Pasha made a dash across the 150 miles of the Sinai Peninsula from Palestine, and was in a position to attack Ismailia and the Bitter Lakes on 2nd Feb. The ample Imperial troops on the spot enabled General Marshall to beat off the attack with ease, but the raiders were not pursued. An attempt on a more concerted scale was made in 1916, though at this time the Turkish garrisons in the Hejaz were having difficulties with the Arabs. A base was established at El Arish, and on 3rd Aug. a well-equipped force of 18,000 advanced to attack Major-General Lawrence's outposts at Romani, in the north of the Sinai Peninsula. The attack was beaten off, and General Sir A. Murray's main body of Australians, New Zealanders, Yeomanry, Territorials, and the 52nd Division coming up, counter-attacked and drove the Turks back 18 miles, inflicting a loss of 4000 prisoners and 4 guns. Between the 6th and 12th Aug. General Murray improved on his victory, and advanced steadily to Mazar (16th Sept.) and El Arish, evacuated by the Turks (20th Dec.). On 22nd Dec. an attack was made by one of the desert columns on Magdhaba, and the whole Turkish force of 3000 destroyed or captured.

On 9th Jan., 1917, the columns under Generals Dobell and Chetwode reached Rafa, the Mediterranean port on the southern borders of Palestine. A pipe line conveying water and a railway followed close on General Murray's advance, and by the middle of March, 1917, these had reached Rafa, enabling an advance to be begun in Palestine, with Gaza as the immediate objective. The first battle of Gaza began on 26th March. It was a failure. Chetwode's mounted forces, including the Australians, moved in a wide detour round Gaza, which was entered while Dobell's infantry attacked the main Turkish positions. But the infantry thrust was not strong enough to penetrate the opposition offered to it, and after two days' fighting the attack was called off. Water was short, and General Murray suspended further attack till 19th April, when, as frequently happened, the Turks, having been reinforced and established in good positions, fought well and beat off the assailants.

Sir A. Murray and General Dobell left the Palestine operations, and General Sir E. Allenby was sent out to organize a subsequent advance when the opportunity was propitious. He was reinforced; the communications were improved through the summer; and on 31st Oct. he surprised the Turks by making a diversion towards Beersheba while demonstrating towards Gaza. Beersheba was rushed by the Australians, and Allenby's right secured. He then sent the 52nd Division to a point between Gaza and the sea while occupying Ali Muntar, and from these two vantage points shelled the Turks out (7th Nov.). It was the decision of the Allied War Council at Versailles in the autumn of 1917 that Allenby, strongly reinforced, should put the Turks out of the war by one decisive blow, since they were already in straits by reason of losses in their numerous campaigns, including those against the Arabs of the Hejaz, which had been organized by the Emir Feisul and Colonel T. E. Lawrence. Accordingly, the Turks were pursued through Ascalon and Ashdod to Jaffa (16th Nov.). Allenby's main columns pressed on more slowly between Beersheba and Hebron (7th Dec.), but entered Jerusalem on 9th Dec., after disposing of an ineffective Turkish stand on the day before. The operations were practically suspended after the Jordan had been crossed at the end of March, 1918, when it became necessary to recall some of the best British divisions to France to meet the last German onslaught in France and Flanders.

When this had been disposed of, Allenby, who had been supplied with Indian divisions, and who, in Sept., 1918, was much superior in forces and artillery to the Turks, resumed his attack (19th Sept.). Facing him were the Turkish Seventh and Eighth Armies in front of Shechem, and the Turkish Fourth Army east of Jordan. A heavy artillery preparation ushered in a frontal infantry attack which tore a gap in the Turkish right and right centre, and through it the cavalry swept north across the Plain of Sharon to reach and seize the passes at Megiddo. Another infantry attack pinned the Eighth Army on the Jordan. The Turks began to retreat too late, for the Imperial cavalry, pouring through the Megiddo passes, crossed the plain of Esdraelon and barred every avenue of retreat on the western side of Jordan. They only missed the capture of General Liman von Sanders at Nazareth by a few hours.

The Turkish armies were in effect destroyed, for on the eastern side of Jordan the Arab cavalry of the Emir Feisul had turned the line of retreat there, and had cut the Hejaz railway running to Damascus. In less than a week the Turkish Seventh and Eighth Armies were wiped out, and 10,000 of the Fourth Army surrendered at Rabboth Amman (29th Sept.). Haifa and Acre on the coast fell as fast as the Imperial cavalry could reach them. Damascus was entered by the Australians on 1st Oct. Beirut fell a few days later, and by the end of October General Allenby had reached Aleppo. The Turkish Cabinet immediately entered into negotiations, Enver Pasha and Talaat Pasha having been replaced. An armistice was signed on 30th Oct. at Mudros, and Turkey was out of the war. It may be added that Talaat Pasha, who was responsible for the massacres of the Armenians, was killed in Berlin by an Armenian student in 1921.

The War in Germany's Colonies

Within a few weeks of the outbreak of hostilities the war had been carried to all the scattered possessions of Germany in Africa, the Pacific, and the Far East. Among the first to fall were German Samoa (occupied by the New Zealand troops on 29th Aug.); the islands of the Bismarck Archipelago and German New Guinea (occupied by the Australians on 11th-13th Sept.); and the more northerly islands of the Pacific, surrendered at the beginning of October to a Japanese squadron. These last were at once handed over by Japan to Australia. Japan, faithful to her alliance with Great Britain, had

declared war against Germany on 23rd Aug., and attacked the formidable German post of Kiao-Chau, 'the key to Northern China', which had been seized by Germany on the flimsiest pretext in 1898. Japan invested Kiao-Chau—garrisoned by 6000 well-armed men, with powerful defences—with a force of 23,000 troops under General Kamio, supported by a small British contingent of South Wales Borderers and Sikhs under General Bernardiston. After a six weeks' siege, and a preliminary bombardment preparatory to launching a final assault, the garrison capitulated (7th Nov.).^[3]

The remainder of Germany's colonies were in Africa, where the campaigns resulting in their conquest varied from a few weeks to the whole length of the war. The shortest was in Togoland, which surrendered unconditionally to a Franco-British force under Colonel Bryant on 27th Aug., 1914, after a brisk little campaign lasting just three weeks. The Cameroons, larger than the Fatherland, took much longer to subdue, and the operations were of a more arduous nature. A false start was made from Nigeria with insufficient forces, which met with disaster towards the end of August. A more adequate expedition was then organized under General C. M. Dobell, in co-operation with French forces under General Aymerich. Duala, the capital, was occupied on 27th Sept., but it was only after desultory fighting for a year and a half that the Germans, realizing the hopelessness of further resistance, left the colony to its fate by retreating into the neighbouring Spanish territory of Rio Muni.

The campaign in German South-West Africa was held up at first by the revolt in South Africa, which had to be crushed before General Botha was free to give his undivided attention to the task which he had offered to undertake in proof of the Union's loyalty. A preliminary move had been made in September by the occupation of the coastal harbours, Luderitz Bay and Swakopmund, from which the Germans retired to concentrate in Windhoek, their capital inland. A 'regrettable incident' occurred on 26th Sept., 1914, when a British patrol was attacked at Sandfontein and a small relief force compelled to surrender. This was followed by news of the revolt of Maritz, who had been partly responsible for the reverse at Sandfontein; and the small civil war which followed in South Africa, though nipped in the bud by the sterling loyalty of most South Africans, served to postpone the conquest of German South-West Africa until the following year.

When the campaign began in earnest in the spring of 1915, the whole south-eastern part of the German colony had already been cleared by Colonel van Deventer, whose desert march with three separate columns in these preliminary operations was one of the outstanding achievements of the campaign. In the great converging march which followed, led in the south by General Smuts, and in the north by General Botha, with columns consisting largely of mounted burghers from the Transvaal and Orange Free State, the Germans were outfought and outmanœuvred throughout. The advance began on 27th April, and by 5th May Botha, building a light railway with supplies behind him, entered Karibib without opposition, and Windhoek a week later, the Germans, 5000 strong but greatly outnumbered, withdrawing the bulk of their forces to the north. Twice they proposed an armistice, offering terms, but the only terms that Botha would agree to implied unconditional surrender. Having rested and refitted his men after this trying trek across country from Swakopmund, he proceeded to enforce these terms by a series of remarkable marching feats, with his own columns in the centre and those of Brits and Myburgh to left and right respectively, starting on 18th June. In less than a week Botha's force had covered 100 miles and captured Otyiwarango; on 1st July, after a brief rest, his infantry were in touch with the enemy's main force, entrenched from Otavi to Tsumeb, with Brits and Myburgh sweeping round on either side. On 8th July all resistance collapsed with the total surrender of the Germans under Colonel Franks, the military commandant. The whole campaign in South-West Africa did not cost us more than 140 lives, and total casualties amounting to 1200.

The conquest of German East Africa was a very different affair. It was opposed by a force which, at its maximum, could muster 25,000 well-drilled troops—2000 of them Europeans—with 60 guns and machine-guns, and sufficient supplies of ammunition; the whole commanded by von Lettow-Vorbeck, a leader of resource and inflexible determination. When, at the outbreak of war, the British cruisers *Astræa* and *Pegasus* bombarded Dar-es-Salaam and destroyed the wireless station, the Germans retaliated by crippling the *Pegasus*, then lying at Mombasa, with their fast cruiser *Königsberg*, which had fled to East Africa after escaping the fate of the rest of von Spee's squadron off the Falkland Islands. They also raided the Uganda railway, occupied Taveta on the British East African border, and

threatened an advance on Mombasa along the coast. This was checked by the arrival of British reinforcements, naval and military, the naval units of which forced the *Königsberg* to seek the shelter of the Rufigi River, where, as already stated, she was afterwards destroyed.

A second expeditionary force under General Aitken was brought from India in the closing months of the year, and ordered to land at the northern German port of Tanga, with the object of cutting off the enemy troops operating on the British border. The magnitude of the task had been gravely under-estimated. Landing on 4th Nov. in difficult bush country—familiar enough to the defenders, but an impenetrable maze to the landing force—the attempt ended in disaster, costing some 800 casualties before the expedition re-embarked. Another blow was dealt by von Lettow-Vorbeck early in 1915 (19th Jan.) when he recaptured Jassin, gallantly held to the last by Indian troops under Colonel Ragbir Singh; but in the same month he lost one of his chief ports on Victoria Nyanza; in the following May other British forces captured Sphinxhaven after an action by armed steamers; and a blockade was declared of the German East African coast, where the Island of Mafia, off the mouth of the Rufigi River, was also seized. Several blockade-runners succeeded in getting through to von Lettow-Vorbeck with much-needed supplies of ammunition; and with Britain's hands full to overflowing with other campaigns, the main German forces in East Africa had perforce to be left until Botha had completed the conquest of South-West Africa, and the Union was free to lend a hand in expelling the Germans from the last and most valuable of their colonies.



Map illustrating the blockade of the German East African Coast—the shaded area—and the military operations early in 1915

When at length the new expeditionary force was nearly ready, the command was given to General Smith-Dorrien, but he was forced to relinquish it through ill-health shortly after landing in Africa. His place was filled by General Smuts, who arrived on the scene in Feb., 1916, and started his campaign in the following month by clearing the enemy from the British borderland and the Kilimanjaro district. Much valuable spade-work had been previously done in this direction by General Tighe as a preliminary to sweeping the enemy's main forces southwards, while the Belgians from the Belgian Congo cleared his north-western province of Ruanad, and British forces drove his garrisons from the shores of Victoria Nyanza. The first stage in the new campaign was closed when Smuts established his headquarters at Moshi, where he reorganized his force in three divisions, one (British and Indian) under Hoskins, the other two (South African) under van Deventer and Brits. Van Deventer's force had the hardest task in clearing the Germans from Kondona Irangi on his march to the central railway, von

Lettow-Vorbeck making his one great defensive stand in this direction. Van Deventer, however, was too quick for him; beat off his counter-attack after the capture of Kondoa Irangi (19th April, 1916); and struck hard when he attempted to bar the progress of Smuts's other two columns, which, after clearing the Gare and Usambara mountains in May and June, had pushed into the Nguru hills from Kangata. By this time the Germans realized that they were outmatched both in strategy and numbers, and the bulk of them would probably have shortened the war but for von Lettow-Vorbeck, who was determined to hold out, if possible, until the fate of Germany's 'place in the sun' had been decided on the battlefields of Europe. More than once it seemed as though he could not escape the wide net which Smuts flung out to trap him, but he proved as elusive as De Wet in the South African War, usually escaping with his diminishing forces through tracks unknown to his pursuers.

All the railways and ports were lost during the remainder of 1916. Dar-es-Salaam, the capital, surrendering on 3rd Sept. Towards the end of that year the Belgians from the Congo under General Tombeur drove the enemy from Tabora, on the central railway; British troops under General Northey helped in the converging movement from Northern Rhodesia—which had been invaded by the Germans at the beginning of the war—by advancing as far as Iringa. In the extreme south some Portuguese had also joined in the movement by an advance across the Rovuma River, but were forced back by a German column, and punished by raiding parties in their own territory.

The campaign seemed practically over at the beginning of 1917, when General Smuts was summoned to Great Britain to share in an imperial conference, and General Hoskins was left to account for the remainder of the enemy, now hemmed in on nearly all sides in the south-eastern corner of the colony, with von Lettow-Vorbeck's head-quarters at Mahenge. Torrential rains came to the Germans' rescue, and little progress had been made before Hoskins, called away to another theatre of war, was succeeded by General van Deventer in May (1917). Guerrilla warfare continued throughout the year, in which both sides suffered heavily, but it was not until 1st Dec. that van Deventer could report that the former Germany colony was at length clear of the enemy. Forced out of Mahenge (occupied by the Belgians on 9th Oct.), the Germans retired in two main bodies towards the Portuguese frontier, one under Tafel, which was cut off and compelled to surrender on

26th Nov., and the other under von Lettow-Vorbeck, which contrived to escape across the border and continue the war in Mozambique for nearly another year. Allied columns pursued the remnants unceasingly, but could never get to real grips with them in the difficult bush country between the Rovuma and the Zambesi. Towards the end of September, 1918, von Lettow-Vorbeck dashed back, recrossed the Rovuma, and coolly marched across the south-western corner of German East Africa into Northern Rhodesia, where he was being finally rounded up when news arrived of the Armistice. It was not, however, until 12th Nov. that the German leader was able to comply with the Armistice terms of unconditional surrender on his part, tendering his submission to the British magistrate at Kasama. In recognition of their "gallant and prolonged resistance", von Lettow-Vorbeck and his officers were permitted by General van Deventer to retain their swords, while the European rank and file were allowed to carry their arms as far as Dar-es-Salaam.

BIBLIOGRAPHY: Earl French, *Despatches*; Earl Haig, *Despatches* (edited by Lieutenant-Colonel J. H. Boraston); Sir Julian Corbett, *The Official History of the War: Naval Operations*; Viscount Jellicoe, *The Grand Fleet, 1914-1916*, and *The Crisis of the Naval War*; Filson Young, *With the Battle Cruisers*; F. A. Mumby and others, *The Great World War* (9 vols.); *The Times History and Encyclopedia of the War* (21 vols.); A. F. Pollard, *A Short History of the Great War*; Lord Edward Gleichen, *Chronology of the War*; Thomas G. Frothingham, *A Guide to the Military History of the World War, 1914-1918*; Sir James Willcocks, *With the Indians in France*; W. T. Massey, *How Jerusalem was Won*, and *Allenby's Final Triumph*; John Masefield, *Gallipoli*; Sir Ian Hamilton, *Gallipoli Diary*; Sir Charles Townshend, *My Campaign in Mesopotamia*; V. J. Seligman, *The Salonica Side-Show*; Angus Buchanan, *Three Years of War in East Africa*; Boyd Cable, *Airmen o' War*; J. F. C. Fuller, *Tanks in the Great War*; *Germany's Violation of the Laws of War, 1914-1915* (translated from the French Official Records); *Ypres, 1914* (published by order of German General Staff, translated by G. C. W.); Theobald Bethmann-Hollweg, *Reflections on the World War* (translated by G. Young); General Ludendorff, *My War Memories, 1914-1918*; Marshal von Hindenburg, *Out of my Life* (translated by F. A. Holt); Count Czernin, *In the World War*.

Eurotium. See *Plectascineæ*.

Eurydice (ū-rid'ī-sē), in Greek mythology, the wife of Orpheus (q.v.).

Euse'bius, of Cæsarea, the father of ecclesiastical history, a Greek writer, born in Palestine about A.D. 265, died about 340. About 315 he was appointed Bishop of Cæsarea. When the Arian controversy broke out, Eusebius showed considerable sympathy with Arius. At the Council of Nicæa (A.D. 325), when Arian doctrines were condemned, he took a leading part. His ecclesiastical history (*Historia Ecclesiastica*) extends from the birth of Christ to 324. Amongst his other extant works is a life of Constantine the Great, which may be said to continue his ecclesiastical history to within a few years of the writer's own death.

Eusporangiate Ferns, those in which the sporangium is a massive organ arising from several cells, whereas in leptosporangiate ferns it is a more delicate structure derived from a single superficial cell. The bulk of living ferns are leptosporangiate, the eusporangiate class comprising only two living families, viz. the Marattiaceæ and Ophioglossaceæ, as well as many extinct types. The eusporangiate types include the more primitive ferns; all the other main groups of Pteridophytes, and the seed-plants are also eusporangiate.

Eustachian Tube, in anatomy, a canal leading from the pharynx to the tympanum of the ear.

Eustachio, Bartolomeo, Italian physician and anatomist, born soon after 1500, died about 1574. He devoted himself to medical science and in particular to anatomy, which he much enriched by his researches. Amongst his discoveries were the *eustachian tube* and the eustachian valve of the heart.

Eustatius, St., or **St. Eustache Island**, a Dutch island in the W. Indies, one of the Leeward Islands, 11 miles north-west of St. Christopher's, pyramidal in form; area, 8 sq. miles. Sugar, cotton, and maize are raised; but the principal production is tobacco. The climate is healthy, but earthquakes are frequent. Pop. 1431.

Euter'pe, (1) one of the Muses, considered as presiding over lyric poetry, the invention of the flute being ascribed to her. She is usually represented as a virgin crowned with flowers, having a flute in her hand. (2) In botany, a

genus of palms, natives of South America, sometimes nearly 100 feet in height.

Euthanasia, literally, a painless or easy death, a term often used in connection with the theory or proposal that it should be lawful to administer drugs to bring about a painless death, in the case of persons suffering from painful and hopeless diseases.

Eutro'pius, Flavius, a Latin historian, who flourished about A.D. 360. His abridgment of the history of Rome (*Breviarium Historiæ Romanæ*) is written in a perspicuous style, and is often read, or rather spelled out word by word, by beginners in Latin. It is of little or no authority as a history.

Eutyches (ū'ti-kēz), a Greek heresiarch who lived in the fifth century after Christ. He was superior of a monastery near Constantinople, and his heresy consisted in maintaining that after the incarnation there was only a divine nature in Christ under the appearance of a human body. The doctrines of Eutyches were condemned by the Council of Chalcedon in 451, and he was expelled from his monastery. He died not long afterwards. His followers were often called Monophysites (Gr. *monos*, single, *physis*, nature) as well as Eutychians.

Euxanthine, a substance supposed to be derived from the bile or urine of the buffalo, camel, or elephant. It comes from India under the name of purree or Indian yellow, and is used as a pigment.

Eux'ine (*Pontus Euxīnus*), the ancient name for the Black Sea.

Evangel'ical, a term often used to qualify certain theological views, especially strict views on the question of the atonement, justification by faith, the inspiration and authority of the Scriptures, and allied doctrines. In England the so-called Low Church party is evangelical in its views. In a more general sense the word implies a peculiar fervency and earnestness in insisting on such doctrines as regeneration and redemption. The 'Evangelical Church' is the official title of the Established Church of Prussia, formed in 1817 by the union of Lutherans and Calvinists.

Evangelical Alliance, an association of members of different sections of the Christian Church, organized in London in 1845, to lend its influence in favour of evangelical doctrines, religious union and liberty, and against

superstition and unbelief. The Alliance, incorporated in 1912, has branches throughout the world, the American branch being especially strong, and has held meetings at Paris, Berlin, Amsterdam, Geneva, New York, Copenhagen, Florence, and London. A week of united prayer is held in London in the early part of January each year. The organ of the Alliance, *Evangelical Christendom*, is published in London.

Evangelical Association, a body of American Christians, chiefly of German descent, established about the beginning of the last century. In form of government and mode of worship it generally agrees with the Methodist Episcopal Church.

Evangelical Union, the name of a religious sect, also familiarly known as the Morisonians, from the Rev. James Morison, its originator. It took rise in Scotland in 1840, and three years afterwards organized itself as a separate Christian denomination. The Morisonians maintain the universality of the atonement, combining with this the doctrine of eternal personal and unconditional election, and denying that anyone will be condemned for Adam's fall. In point of government the individual Churches are independent. The body has now only a few adherents, the majority having united in 1896 with the Congregationalists.

Evan'gelists, the writers of the history or doctrines, precepts, actions, life, and death of Christ; in particular, the *four evangelists*, Matthew, Mark, Luke, and John. The ancient symbols of the four evangelists are: for Matthew, *a man's face*; for Mark, *a lion*; for Luke, *an ox*; and for John, *a flying eagle*.

Ev'ans, Sir Arthur John, born in 1851, son of Sir John Evans. He was educated at Harrow, Oxford, and Göttingen, is a distinguished archæologist, as well as an investigator into the history and affairs of Eastern Europe, and in particular has made important investigations and discoveries in Crete. He has travelled in Finland, Lapland, and the Balkan countries, and has written works connected with these travels and with his archæological researches. From 1884 to 1908 he was keeper of the Ashmolean Museum, Oxford. He was knighted in 1911, and was president of the British Association 1916-7. Among his works are: *Through Bosnia*, and *Scripta Minoa*.

Evans, Sir George de Lacy, a British general, born at Moig, in Ireland, in 1787, died 9th Jan., 1870. After some years of service in India, he joined the army of Wellington in the Peninsula in 1812, where he served with distinction. In 1814 he was sent to America, and was present at the battles of Bladensburg and New Orleans, returning to Europe in time to take part in Waterloo. In 1830 and 1831 he was elected member for Rye, and in 1833 for Westminster. In 1835 he was appointed to the command of 10,000 troops raised in Britain on behalf of the Queen of Spain. Under the training of Evans this force became an excellent army, and several times defeated the Carlists. During the Crimean War he distinguished himself as commander of the second division of the English army, and received the thanks of the House and other honours.

Evans, Sir John, English archæologist, born in 1823, died in 1908, was an active member of a firm of paper-makers, but eventually retired from business and devoted himself chiefly to scientific pursuits, being distinguished as a geologist, numismatist, and antiquarian. From 1878 till 1896 he was treasurer of the Royal Society; in 1897 he was president at the Toronto meeting of the British Association. He was created a K.C.B. in 1892. His chief works are: *The Ancient Stone Implements, Weapons, and Ornaments of Great Britain and Ireland*; and *The Ancient Bronze Implements, Weapons, and Ornaments of Great Britain and Ireland*.

Ev'anston, a city of the United States, Illinois, 13 miles from Chicago. It is pleasantly situated on Lake Michigan. It is the seat of the North-Western University and other institutions. Pop. 37,215.

Ev'ansville, a town in the United States, in Indiana, pleasantly situated on a height above the Ohio. It contains some handsome buildings, including custom-house and post office. Coal and iron abound in the vicinity, and there are numerous factories, flour-mills, and iron-foundries, and a large shipping trade. Pop. 85,264.

Evaporation, the process of changing a liquid into vapour, which goes on at the surface of the liquid. When evaporation takes place into a closed space above the liquid, a state is reached in which the space is saturated with vapour, and no further change is apparent. Certain conditions influence the rate at which evaporation takes place in the open, such as the extent of

the exposed surface, the dryness, pressure, and temperature of the air, and the presence or absence of wind. The farmer looks for a drying wind to dry his corn-sheaves before they are carted from the field. Evaporation is always accompanied by an absorption of heat by the vapour, namely the latent heat necessary for the change of state from liquid to vapour. This heat is abstracted from the liquid, which, as a consequence, falls in temperature, and neighbouring bodies may also be cooled. A bather emerging from the water experiences this cooling as the water evaporates from his skin. In hot countries, water is kept cool by being placed in porous earthenware jars: the evaporation which takes place at the outer surface prevents the water in the jar from becoming too warm. The continual evaporation from land and water surfaces is the cause of the presence of water in the atmosphere, whether in the form of invisible vapour or of cloud, mist, fog, rain, hail, or snow. The cooling caused by evaporation is applied in industry in the case of the ammonia compression machine, in working refrigerating plant for the preservation of food-stuffs, and in the manufacture of artificial ice. When evaporation takes place at the surface of a dry solid, the process is called sublimation, and it may be observed in the cases of iodine, camphor, and ice.

Ev'elyn, John, an English writer of the seventeenth century, born at Wotton, in Surrey, 31st Oct., 1620, died there, 27th Feb., 1706. After completing his course at Oxford, he studied law at the Middle Temple, visited various parts of the Continent, and in 1659 prepared the way for the Restoration. He published numerous works, amongst which are: *Sculptura, or the History and Art of Chalcography*; *Sylva, or a Discourse of Forest Trees*; treatises on gardening and architecture. But by far his most important work is his memoirs, comprehending a diary and correspondence, which are interesting contributions to the history of the time. Evelyn's *Diary* was first published in 1818, and an edition by Austin Dobson appeared in 1906.—Cf. H. Maynard Smith, *The Early Life of John Evelyn*.

Evening-primrose, *Cenothëra*, a genus of plants, nat. ord. Onagraceæ. *C. biennis*, an American species common in cottage gardens, is not infrequent as an escaped plant in England.

Ev'erest, Mount, the highest summit of the Himalayas and the loftiest mountain in the world, 29,002 feet. It was named after Sir George Everest

(1790-1866), chief of the great Indian trigonometrical survey and Surveyor-General of India. In 1921 an expedition set out to climb the mountain.

Ev'ere'tt, Alexander Hill, an American diplomatist, born at Boston in 1792, died at Canton in 1847. After studying at Harvard, in 1809 he accompanied John Quincy Adams to St. Petersburg (Petrograd) as Secretary of Legation. He afterwards filled successive diplomatic posts in the Netherlands, Spain, and elsewhere. He was the author, amongst other works, of *Europe, or a General Survey of the Present Situation of the Principal Powers* (1822); and a similar work on America.

Everett, Edward, an American statesman and author, brother of the preceding, born at Dorchester, Massachusetts, 11th April, 1794, died at Boston, 15th Jan., 1865. After travelling for some years in Germany and England, he returned to America in 1819 to occupy the chair of Greek literature at Harvard. He became editor of the *North American Review*, and entering the political world became successively member of Congress, Governor of Massachusetts, and Minister Plenipotentiary in England (1841-5). In 1845 he was appointed president of Harvard College; in 1852 Secretary of State; in 1853 a Senator. He wrote poems; a *Defence of Christianity*, and his speeches and orations have been published.

Evergreen, a plant that retains its verdure through all the seasons, as the fir, the holly, the laurel, the cedar, the cypress, the juniper, the holm-oak, and many others. Evergreens shed their old leaves in the spring or summer, after the new foliage has been formed, and consequently are verdant through all the winter season. They form a considerable part of the shrubs commonly cultivated in gardens, and are beautiful at all seasons of the year.

Everlasting-flowers, a name applied to certain plants which, when dried, suffer little change in their appearance. The plants to which this name is peculiarly applied belong to the genus *Helichrysum*, but it is also given to members of allied genera, such as *Antennaria* or *Gnaphalium*.

Everlasting-pea, a popular name for *Lathyrus latifolius*, cultivated in flower-gardens, and belonging to the same genus as the sweet-pea.

Eversion of the Eyelids (Ectropion), is a turning outward of the eyelids, and it may be congenital or acquired. The latter follows severe infection of

the hair-follicles of the eyelashes.

Evesham (ēvz'am), a town in England, in the county and 15 miles S.E. of Worcester, beautifully situated on the Avon, and giving name to a parliamentary division of the county. It was the seat of a monastery as early as the eighth century. Simon de Montfort was defeated by the Royal troops at Evesham on 4th Aug., 1265. Pop. 8685.

Eviction, the dispossession of a person from the occupancy of lands or tenements. The term occurs most commonly in connection with the proceedings by which a landlord ejects his tenant for non-payment of rent or on determination of the tenancy. In the case of eviction of tenants in Ireland, generally for non-payment of rent, the tenants are frequently readmitted as care-takers, or under some other title. The Rent Restrictions Act, 1920, operative until 1923, severely curtails a landlord's common law rights to recover possession of certain premises at the time when, but for the Act, the tenancy would expire.

Evidence is that which makes evident, which enables the mind to see truth. It may be (a) *intuitive*, i.e. resting on the direct testimony of consciousness, of perception or memory, or on fundamental principles of the human intellect; or it may be (b) *demonstrative*, i.e. in a strict sense, proofs which establish with certainty as in mathematical science certain conclusions; or it may be (c) *probable*, under which class are ranked *moral evidence*, *legal evidence*, and generally every kind of evidence which, though it may be sufficient to satisfy the mind, is not an absolutely certain and incontrovertible demonstration.

In jurisprudence evidence is classified into that which is *direct* and *positive* and that which is *presumptive* and *circumstantial*. The former is that which is proved by some writing containing a positive statement of the facts and binding the party whom it affects; or that which is proved by some witness, who has, and avers himself to have, positive knowledge thereof by means of his senses. Whenever the fact is not so directly and positively



Everlasting Flower
(*Helichrysum sentii*)

established, but is deduced from other facts in evidence, it is *presumptive* and *circumstantial* only. The following are the leading rules regarding evidence in a court of law:

(1) The point in issue is to be proved by the party who asserts the affirmative. But where one person charges another with a culpable omission this rule will not apply, the person who makes the charge being bound to prove it. (2) The best evidence must be given of which the nature of the thing is capable. (3) Hearsay evidence of a fact is not admissible. The principal exceptions to this rule are—death-bed declarations, evidence in questions of pedigree, public right, custom boundaries, declarations against interest, declarations which accompany the facts or are part of the *res gestæ*, &c. (4) Insane persons and idiots are incompetent to be witnesses. But persons temporarily insane are in their lucid intervals received as witnesses. Children are admissible as witnesses as soon as they have a competent share of understanding and know and feel the nature of an oath and of the obligation to speak the truth.—BIBLIOGRAPHY: Sir J. F. Stephen, *Digest of the Law of Evidence*; W. M. Best, *Law of Evidence*.

Evidences of Christianity. These may be divided broadly into two great classes, viz. *external evidences*, or the body of historical testimonies to the Christian revelation; and *internal evidences*, or arguments drawn from the nature of Christianity itself as exhibited in its teachings and effects, in favour of its divine origin. The first Christian apologies—those of Justin Martyr, Minucius Felix, and Tertullian, written in the second century—were mainly intended as justifications of the Christian religion against the charges of atheism and immorality commonly made at that time. Of a more philosophical kind, and dealing more comprehensively with the principles of religion and belief in general, are the works of Origen, Arnobius, and Augustine in the centuries immediately succeeding. During the Middle Ages, the scientific representation of Christianity is mostly the work of the schoolmen occupied in welding Aristotelian or Platonic philosophy with the fabric of Christian dogmatics, or writing attacks on the Jewish and Mohammedan faiths.

In the sixteenth and seventeenth centuries the influences of the Renaissance and the Reformation gave rise to a spirit of inquiry and criticism which developed English deism as represented by Herbert and Hobbes in the

seventeenth century, and Collins and Bolingbroke in the eighteenth. The general position of English deism was the acceptance of the belief in the existence of God, and the profession of natural religion along with opposition to the mysteries and special claims of Christianity. It was in confutation of this position that the great English works on the evidences of Christianity of Butler, Berkeley, and Cudworth were written. In France the new spirit of inquiry was represented by Diderot, D'Holbach, and the encyclopedists in general, who assailed Christianity mainly on the ground that it was founded on imposture and superstition, and maintained by sacerdotal trickery and hypocrisy. No reply of any great value was produced in the French Church, although in the previous age Pascal in his *Pensées* had brought together some of the profoundest considerations yet offered in favour of revealed religion. The nineteenth century has been distinguished by the strongly rationalistic spirit of its criticism. The works of such writers as Strauss, Bauer, and Feuerbach, attempting to eliminate the supernatural and the mysterious in the origin of Christianity, have been answered by the works of Neander, Ebrard, and Ullmann on the other side. The historical method of investigation, represented alike by the Hegelian school and the Positivists and Agnostics in philosophy, and by the Evolutionists in science, is the basis of the chief attacks of the present time against the supernatural character of Christianity. The tendency of all the critics is to hold that while Christianity is the highest and most perfect development to which the religious spirit has yet attained, it differs simply in degree of development from any other religion. Notable amongst later apologists of Christianity have been Paley (*Natural Theology*), Chalmers (*Natural Theology*), Mansel, Liddon, and others.—BIBLIOGRAPHY: J. R. Illingworth, *Reason and Revelation*; A. Garvie, *Handbook of Christian Apologetics*.

Evil, Origin of. The difficulty of the question lies mainly in this, that the existence of evil in the world seems inconsistent with the view that it was created and is maintained by an omnipotent and beneficent creator. The various theories on the subject have all sought to elude this difficulty either by the supposition of some principle of evil equally eternal with that of good, or by regarding evil as having only a relative existence, being a kind of good in an imperfect and immature stage. Perhaps the oldest theory upon this subject is that of Parseeism, or the religion of Zoroaster, according to which there were two original principles, one good (Ormuzd) and the other

evil (Ahriman). This is the doctrine that is now very often spoken of as Manichæism, from the fact that it was adopted by Manes, who attempted to engraft it on the doctrines of Christianity. In contradistinction to this dualistic theory with reference to the origin of evil stand the Monistic theories of Brahmanism and Platonism. According to the Brahmanic doctrine of the emanation of all things from one original being (Brahma), this original being was regarded as the sole true existence, and the phenomenal world, with all the evils appearing in it, was held to be mere illusion. Similarly, Plato held that the good was the essence of all things, and that the evil and imperfect contained in them had no real existence. The theory enunciated by Leibnitz in his *Theodicée* resembles that of Plato. In that work he assigns to the evil existing in the world created by God, which he holds to be the best of all possible worlds, a merely relative existence. According to Plato, all that we call evil is only evil to us because we do not see it in relation to the rest of the universe, for in relation to the universe it is not evil but good, and accordingly cannot be evil in its own nature. Another view on the subject is that which neither assigns to the evil principle (as it does to God or the good principle) an original existence, nor denies the real existence of evil, but ascribes it to the exercise of man's free-will. Besides the theoretical problem of the origin of evil, there is the practical one of the elimination of evil which forms the subject of *Ethics*.—BIBLIOGRAPHY: H. Rashdall, *The Theory of Good and Evil*; B. Bosanquet, *The Value and Destiny of the Individual*.

Evil Eye, a power which, according to an old and widespread superstition, resides in some people of doing injury to others by a mere look, or a look accompanied by certain words or charms. This belief, common amongst the ancients, is still prevalent among the more ignorant classes in Italy, Russia, Andalusia, the Highlands of Scotland, and other places. The Finns, Lapps, and Scandinavians, the Arabs and the Turks are all firm believers in the evil eye.

Evolute. The evolute of a curve is the curve which is the envelope of all its normals or the locus of all its centres of curvature. The first curve is called the involute of the second. These names are given to the curves because the end of a stretched thread unwound from the evolute will describe the involute.

Evolution, a term introduced into biological writings in the early part of the eighteenth century to denote the mode of generation of living things. At first it was used in the same sense as we now apply the word development, more especially with reference to the process whereby the germ of an animal or plant becomes transformed into the adult organism; but it is now used in biology for the process by which more complex plants or animals have been derived from a series of less specialized ancestors by transformation. In accordance with the teaching of modern biology, all living creatures are the progeny of one original group of microscopic unicellular organisms, different branches of which during many millions of years have become diversely modified in structure and function to form the vast multitudes of diverse species of plants and animals with which we are acquainted.

The idea of a transformation of one type of being into another is extremely ancient, and its origin was in all probability genetically related to the primitive conceptions which have survived to the present day in totemism (q.v.) and such myths as the story of the were wolf. For once it was believed that a totem-animal, like a cow or a pig, could give birth to human beings, or that the Great Mother could at will assume a great variety of living forms, ranging from a mollusc or a grain of barley to a higher mammal, it was a comparatively simple step to arrange these potential 'ancestors' in a series, and provide mankind with a mythical genealogy. It is possible that such beliefs may have suggested to the Greek philosophers, such as Aristotle, speculations as to the process by which custom and change of habit might modify the structure of animals. But it was not until the end of the seventeenth century that the trend of philosophical speculation, associated with the growing understanding of natural processes, started lines of investigation which, after many failures, eventually brought forth Charles Darwin's *Origin of Species* in 1859, and established once for all the fact that different species of animals and plants have been produced by the differentiation of the progeny of the same ultimate ancestors. The history of these events was admirably summarized by Huxley in 1878 in an article *Evolution in Biology*, republished in *Darwiniana* (1893, p. 187); in this account due credit is given to the pioneers of the eighteenth and early nineteenth centuries, such as Lamarck, and to Darwin's co-discoverer of the hypothesis of natural selection, Alfred Russell Wallace.

At the present time there is a general consensus of opinion among serious biologists, and, in fact, most educated men, as to the reality of evolution; but there is wide divergence of opinion as to the exact pedigrees of the various groups of plants and animals, and especially as to the mechanisms whereby the processes of transformation have been effected.

The evidence that establishes the proof of evolution is of manifold kinds. "The gradations of structure, from extreme simplicity to very great complexity, presented by living things, and of the relation of these graduated forms to one another. The existence of an analogy between the series of gradations presented by the species which compose any great group of animals or plants, and the species of embryonic conditions of the highest members of that group. Large groups of species of widely different habits present the same fundamental plan of structure; and parts of the same animal or plant, the functions of which are very different, likewise exhibit modifications of a common plan. Structures are found in a rudimentary or apparently useless condition in one species of a group which are fully developed and have definite functions in other species of the same group. These considerations, when studied in conjunction with the facts of the geological succession of the forms of life, of geographical distribution, and the effects of varying conditions upon living organisms, establish the truth of evolution" (Huxley).

The full meaning of these statements will be better understood if a concrete example is studied, and perhaps the case of man and his ancestry is most instructive for this purpose. The fact that man has a vertebral column, a brain and nervous system, a heart and blood-vessels, digestive and other systems of organs, built up in accordance with the arbitrary plan which is shared also by all mammals, birds, reptiles, amphibians, and fishes, proclaims that man belongs to the vertebrate group of animals, and that all such vertebrata must originally have sprung from the same common ancestors. The possession of four limbs with five fingers or toes on each, and a host of identical arrangements of bones, muscles, nerves, &c., in these limbs, reveals the fact that all the four-limbed creatures or Tetrapoda represent one group which developed from some fish-like ancestor to become an amphibian. The discovery of fossilized remains of extinct animals reveals that the fishes are much older than the amphibians, and that a number of intermediate stages demonstrate the process of gradual

transformation which converted one group of fishes into four-limbed, semi-terrestrial amphibians. Geological evidence also proves that the reptiles came definitely later than the amphibians, and that only one small group of very primitive amphibians shared in the progressive modifications of brain, limbs, and organs of circulation, &c., to become reptiles—creatures able to live wholly on the dry land, and capable of a wider range of activities than the Amphibia. From the primitive reptiles were derived not only the highly specialized forms that have survived to the present day as lizards, tortoises, snakes, &c., creatures that differ profoundly from their earliest reptilian ancestors, but also the ancestors of birds and the ancestors of mammals. One particular group of primitive reptiles is known, from fossilized remains found in South Africa, that reveals many of the distinctive peculiarities of mammals not shared by other reptiles; and it is now certain that these cynodonts—so called from their dog-like teeth—include the parents of the mammals. The fact that all the Mammalia are provided with glands which in the female supply milk for the nutrition of their young, that they have a hairy coat, that they have a highly developed brain more fully adapted for learning by experience than is the case in other vertebrates, that they have limbs capable of a much more varied and active range of skilled movements, and a host of identical transformations of viscera, muscles, nerves, and vessels prove the common ancestry of mammals from some very primitive cynodont reptile. Different mammals have been specialized in structure for amazingly varied modes of life, on land, under the ground, in trees, in the air, or in rivers or the sea. Of the terrestrial animals some have been modified for fleetness, like the horse and the antelope; others for strength, like the elephant; others again, like the lion and the tiger, to prey upon their weaker relatives. At the dawn of the age of mammals one particular group was able to survive without any of the profound alterations of the structure of the limbs which such creatures as the horse and the ox, the elephant and the whale, the tiger and the bat had to adopt to avoid extinction, and retained the primitive type of limbs with their fingers and toes which became the most useful and plastic instruments for performing skilled movements and acquiring experience and knowledge as soon as the brain was sufficiently advanced in structure and capability to put these instruments to their full use. The group of mammals which delayed the time of specialization until it was able to profit by its greater adaptability was the Prosimiæ, the ancestors of the apes and man. These small creatures for a

long time lived a life of obscurity in trees without submitting to those extreme adaptations of structure which are found in most arboreal and flying mammals. But the cultivation of their powers of vision, and the acquisition of skill in the use of their primitive but plastic hands, guided by vision, eventually conferred upon some of these Prosimiæ vastly enhanced powers of skilled action and of learning by experience and of acquiring knowledge, which culminated in the attainment of the supreme power of discrimination distinctive of human intelligence.

The fact that man belongs to the same order (Primates) as the apes is proved not merely by the possession of a body which in most respects is identical in structure with such of them as the gorilla and the chimpanzee, of a similar process of development characterized by identical stages up to a certain stage, but also by the fact that the blood of man and the apes react towards one another as those of relatives, and in a way not shared by the blood-reactions of other mammals. The apes, also, are subject to certain human diseases from which other mammals are immune. Man shares with the anthropoid apes (gorillas, chimpanzees, orangs, and gibbons) so many peculiarities which differentiate all of them from the tailed monkeys that there can be no doubt that the human family was derived from a primitive anthropoid ape, possibly a species that lived in the foothills of the Himalayas in Miocene times, as is suggested by fossils recently discovered by Dr. Pilgrim, director of the Geological Survey of India.

The vestigial remains of the muscles, blood-vessels, &c., of the tail reveal the fact that man's Primate ancestry began with a tailed form. In fact, the human embryo actually possesses a tail for some weeks of its existence. If India reveals the fossilized remains of a variety of Miocene anthropoid apes closely akin to the ancestors of man, the gorilla, the chimpanzee, and the orang, the Egyptian Fayum has provided the evidence of the origin of the anthropoid apes in Oligocene times as very diminutive creatures distantly akin to the gibbons, but bearing very obvious indications in the form of their teeth of an affinity with the Prosimian sub-order Tarsioidea, a very interesting group of Eocene Primates found in a fossilized condition in North America and France, one of the members of which has survived in the peculiar Spectral Tarsier still found living in the forests of Borneo, Java, and the Philippines. The detailed study of the structure and development of Tarsius, and comparisons with other Primates and mammals of other orders,

provide the information necessary to fill in the gaps left in the geological record, and enable us to sketch out the general scheme of man's ancestry, and to appreciate the nature of the factors which determined the evolution of such an intelligent mammal as man.

Within recent years the increasing knowledge of embryology and comparative anatomy, and the recovery of fossilized remains of vast numbers of hitherto unknown animals, has established the truth of evolution and the exact line of ancestry of many animals. Professor Osborn's work on the evolution of the horse, and Dr. C. W. Andrews's revelation of the ancestry of the elephant, are striking recent illustrations of the exactness of the demonstration palæontology can give of the past history of mammals. Dr. Robert Brown and Professor D. M. S. Watson have given conclusive proofs of the origin of mammals and birds from primitive reptiles, and the latter zoologist has pushed back the ancestry of these higher vertebrates still further, and shown how the reptiles were derived from primitive Amphibia, and the changes that occurred in vertebrate anatomy when certain fishes crawled out of the water and developed into four-footed Amphibia. All of these conclusions are matters of fact and not of theory, even if we are still in the dark as to the exact mechanism whereby the variations which the forces of evolution use in effecting transformations were themselves brought about. Within recent years there has been a revival of interest in the problem whether characters acquired by parents as the result of their individual experience can be transmitted to their offspring. For the last thirty years biologists have been influenced by the teaching of Weismann that nothing happening to the parents can affect the morphological capabilities of the germ plasm from which the next generation is derived; but recent research suggests that this negative doctrine is too rigid, and makes it probable that certain influences brought to bear upon the parents may be transmitted also to the offspring. If this is so, it opens one avenue of explanation of how structural modifications are effected and the possibility of evolution is created. But at the present moment the whole question is being actively investigated and discussed.

Just as every complex animal can be shown to be derived during development from a simple microscopic cell or egg, so the study of evolution reveals the fact that all animals were originally derived from microscopic unicellular animals known as protozoa, which are with

difficulty distinguishable from unicellular plants, from which all the varied forms of complex vegetable life were derived. It is equally certain that these unicellular plants and animals are themselves only the specialized descendants of common ancestors—unspecialized unicellular organisms which are neither strictly vegetable nor animal. But we are quite in the dark as to the processes whereby these most primitive living organisms were evolved from inorganic matter, and how they acquired these peculiar properties of growth and differentiation and their powers of reproduction, commonly called vital, which are their distinctive characteristics. Bibliographical references to most of the matters mentioned in this article will be found in the Presidential Address to Section H of the British Association for the Advancement of Science, Dundee meeting (see Report of British Association, 1912).

Ev'ora, a town in Portugal, capital of the province of Alemtejo, 75 miles east of Lisbon. It is an ancient place, poorly built, and its walls, citadel, and forts are all in a state of ruin. It has a Roman aqueduct still serviceable, a Gothic cathedral, and an ecclesiastical seminary. Pop. 17,900. The district of Evora has a pop. of 144,300.

Evremond, or **Evremont**. See *St. Evremond*.

Evreux (ev-reu), a town of N.W. France, capital of the department of Eure, in a fertile valley on the Iton. It is an ancient town with narrow streets and has many fine buildings, including an ancient Gothic cathedral. The town was frequently occupied by the English in the fifteenth century. Pop. 18,950.

Ewald (ā'va'lt), Georg Heinrich August von, a German Orientalist and Biblical critic, born at Göttingen 16th Nov., 1803, died there 5th May, 1875. After studying at the university of his native town, in 1827 he became extraordinary, in 1831 ordinary professor of theology, and in 1835 professor of Oriental languages. In 1837 he lost his chair at Göttingen on account of his protest against the king's abrogation of the liberal constitution, became professor of theology at Tübingen, but in 1848 returned to his old chair at Göttingen. When Hanover was annexed by Prussia in 1866 he became a zealous defender of the rights of the ex-king. Among his chief works are the following: *Complete Course of the Hebrew Language*, *The Poetical Books*

of the Old Testament, History of the People of Israel, Antiquities of the People of Israel. The *History* is considered his greatest work.

Ewald (ā'va'lt), Johannes, Danish poet, born at Copenhagen in 1743, died in 1781. After studying theology at Copenhagen University he ran away and enlisted in the Prussian service, which he soon deserted for the Austrian. On his return to Copenhagen an elegy which he wrote on the death of Frederick V of Denmark was received with general admiration, and awoke in himself the consciousness of poetic talent. His reputation rapidly increased with the publication of his tragedies, *The Death of Balder* (English translation by George Borrow), *Adam and Eve*, and *Rolf Krage*; and his odes and songs, notable amongst which are: *King Christian* and *Liden Gunver*. Ewald, who had dissipated habits, died in utter poverty. His collected works were published in 1914.

Ewart, James Cossar, zoologist, was born at Penicuik, Midlothian, in 1851, studied medicine at Edinburgh University, graduated in 1874, and was soon after appointed demonstrator of anatomy in the university. From 1875 to 1878 he was Conservator of the museums of University College, London, in the latter year took his M.D. degree, and from 1878 to 1882 was professor of natural history in Aberdeen University, being then appointed to the natural history chair at Edinburgh. Since then (having been also connected with the Scottish Fishery Board for about ten years) he has devoted much attention to the question of fish-culture and preservation, and has visited North America, Denmark, and Norway for purposes of investigation. He has also carried out experiments in the hybridization of zebras and horses. His publications include: *The Locomotor System of the Echinoderms* (with G. J. Romanes, 1881); *The Natural and Artificial Fertilization of Herring Ova* (1884); *On Whitebait* (1886); *On the Preservation of Fish* (1887); *The Electric Organ of the Skate* (1888-9); *The Cranial Nerves and Lateral Sense-organs of the Elasmobranchs* (1889-91); *The Development of the Limbs of the Horse* (1894); *The Penicuik Experiments* (1899); *Guide to Zebras, Hybrids, &c.* (1900); *Multiple Origin of Horses and Ponies* (1904).

Exalbuminous Seeds, those which, when ripe, contain no endosperm, this having been entirely absorbed into itself by the developing embryo. Opposed to albuminous seeds. See *Cotyledons*.

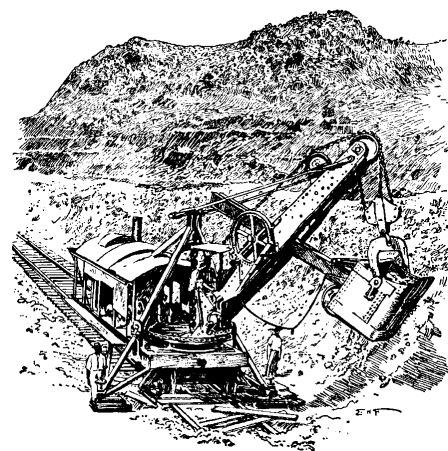
Examiner of Plays, a British official and censor of plays, who acts for the Lord Chamberlain, under whose jurisdiction the theatres are placed. No play can be produced without the sanction of the examiner, to whom a copy of every new play intended for production must be sent seven clear days before the first performance. The examiner either grants or refuses his licence, and frequently insists upon an alteration of the text. The abolition of this censorship of plays is a subject which in recent years has given rise to much discussion in the theatre-loving world.

Exanthemata (eruption of the skin), a term applied to infectious diseases with skin eruptions, accompanied by general disturbances. The term includes scarlet fever, measles, German measles, smallpox, chicken-pox, and others.

Exarchate (egz-är'kāt), a name of a province or territory under an *exarch*, or viceroy. In the sixth century after Christ Justinian formed the middle part of Italy into a province of the Eastern Empire, and gave the government of it to an *exarch*. (See *Ravenna*.) Exarch was also the title of an ecclesiastical grade in the Greek Church, inferior to the patriarchs but superior to the metropolitans. Among the modern Greeks an *exarch* is a deputy of the patriarch, who travels about in the provinces and visits the bishops and churches.

Exambion, in Scots law, the name given to the contract by which one piece of land is exchanged for another.

Excavation, the process of removing soil or rock in engineering or exploration works, such as for docks, retaining-walls, railway cuttings, canals, foundations, &c. On a small scale, or in situations unsuitable for machinery, it is performed by hand, the soil being first loosened by the pick, and then shovelled into barrows. In rocky soils, drilling and explosives may be employed. In large works power shovels or 'steam-navvies' are employed, which are essentially cranes carrying a large shovel, or



Excavating the Culebra Cut, Panama Canal

a system of steel buckets of the dredger type. These carry their own means of propulsion, and run on temporary rails laid down as the work proceeds. They work against the face of the excavation, and load directly into bogies or wagons.

Excavations. The forgotten history of the remote past has been reconstructed by those scientists who have explored the sites of ancient seats of civilization. Egyptian and Babylonio-Assyrian investigations date from the middle of the eighteenth century, but it was not until the middle of the nineteenth century, when modern scholars first penetrated the secrets of the lost languages (see *Cuneiform Writing* and *Hieroglyphics*) that the ancient civilizations were rendered more or less articulate. As the texts were being deciphered with increasing ease and accuracy, a basis was provided for archæology, and it became possible to frame chronological systems. During the latter years of the nineteenth and the early years of the present century, excavators, philologists, and ethnologists have provided a fairly continuous and detailed history of man from 3000 B.C. till classical times, thus bridging a gulf which used to be misty with doubtful legends and traditions. The most dramatic excavations were those begun by Heinrich Schliemann in 1871 at Hissarlik, the site of ancient Troy. On the hillock he dug through nine successive settlements. He afterwards excavated at Mycenæ and Tiryns in Greece and found evidence of a high pre-Hellenic culture. Following up the clues thus afforded, Sir Arthur Evans and others, excavating in Crete during the opening years of the present century, discovered abundant relics, including palaces and towns, of the earliest Ægean civilization now referred to as Minoan (see *Crete*). Of late years Central and Western Europe have yielded evidence of the 'drift' of Minoan culture to outlying parts. Excavations in Russian Turkestan and Chinese Turkestan have revealed traces of ancient culture centres dating back beyond 2000 B.C. In the Americas the excavators have thrown considerable light on the pre-Columbian civilizations of Peru, Central America (Maya), and Mexico, but the picture-writings have not been deciphered. See *Babylonia*; *Crete*; *Egypt*; *Troy*.

Ex'cellency, a title given to ambassadors and plenipotentiaries, governors of colonies, the President of the United States, of France, &c.

Excess Profits Duty, devised in 1915 to meet the extraordinary expenditure occasioned by the war, is a tax upon the profits of certain trades and businesses carried on in the United Kingdom, or owned or carried on abroad by persons resident in the United Kingdom, in so far as these profits, after deduction of a specified allowance, exceed a pre-war standard. The main exceptions are agriculture, offices and employments, and professions where personal qualifications predominate and only small capital is necessary. In some cases, e.g. estate agencies, where a portion of the profits arises from professional skill, only the portion otherwise arising falls into charge. The pre-war standard is an alternative one at the option of the taxpayer. Firstly, it may be a profits standard—the average of the profits of any two of the last three pre-war years, or if there have been only two pre-war years, then the average profits of those years, or the actual profits of the last year, or if there has been only one pre-war year, then the actual profits of that year. Where the average profits of the last three pre-war years are 25 per cent less than the average of the three years immediately preceding them, the taxpayer may take the average of any four of those six years. Secondly, it may be, and as a general rule where the business has not had one full pre-war year must be, a percentage standard, calculated at the appropriate rate on the capital in the business at the end of the last pre-war trade year, or where there has not been one pre-war year, then on the average amount of capital employed (*a*) during the year or accounting period in question, or (*b*) in respect of periods ending after 31st Dec., 1919, during the first accounting period. The percentage standard of sole traders, partnerships, and private companies may in respect of accounting periods ending after that date be increased by £500 per annum for each working proprietor, but not so as to exceed £750 per annum each. The general free allowance is £200, increased in 1920 to £500 for new or re-opened businesses of ex-service men. A further abatement is now granted where profits do not exceed £4000, with the result that liability cannot arise unless profits exceed £832.

The rate of duty, at first 50 per cent, was raised to 60 per cent in 1916, and to 80 per cent in 1917. Reduced to 40 per cent in 1919, it was again increased to 60 per cent by the Finance Act, 1920. It was abolished in 1921. Undoubtedly it was injurious to trade and an incentive to wasteful expenditure, but an effective substitute for supplying the unparalleled

financial needs of the State resulting from the war apparently could not be devised.

'Munitions Levy' is the counterpart of Excess Profits Duty in its application to Government-controlled establishments for the production of munitions of war.

In the fiscal year 1919-20 the tax produced £290,045,000, while the estimate for the year 1920-1 was £220,000,000.

Exchange, a place in large commercial towns where merchants, agents, bankers, brokers, and others concerned in commercial affairs meet at certain times for the transaction of business. The institution of exchanges dates from the sixteenth century. They originated in the important trading cities of Italy, Germany, and the Netherlands, from which last-named country they were introduced into England. The Royal Exchange of London was established by Sir Thomas Gresham in 1556. In some exchanges only a special class of business is transacted. Thus there are stock exchanges, corn exchanges, coal exchanges, cotton exchanges.

Exchange, in commerce, that species of transactions by which the debts of individuals residing at a distance are cancelled by order, draft, or bill of exchange, without the transmission of specie. Thus, a merchant in London who owes £100 worth of cotton goods in Glasgow gives a bill or order for that amount which can be negotiated through banking agencies or otherwise against similar debts owing by other parties in Glasgow who have payments to make in London. The creditor in Glasgow is thus paid by the debtor in Glasgow, and this contrivance obviates the expense and risk of transmitting money. The process of liquidating obligations between different nations is carried on in the same way by an exchange of foreign bills. When all the accounts of one country correspond in value with those of another, so that there is an even balance, the exchange between the countries will be *at par*, that is, the sum for which the bill is drawn in the one country will be the exact value of it in the other. Exchange is said to be *at par* when, for instance, a bill drawn in New York for the payment of £100 sterling in London can be purchased there for £100. If it can be purchased for less, exchange is *under par* and is against London. If the purchaser is obliged to give more, exchange is *above par* and in favour of London. Although the

numerous circumstances which incessantly affect the state of debt and credit prevent the ordinary course of exchange from being almost ever precisely at par, its fluctuations are confined within narrow limits, and if direct exchange is unfavourable between two countries this can often be obviated by the interposition of bills drawn on other countries where an opposite state of matters prevails. See *Bill of Exchange*.—BIBLIOGRAPHY: G. J. Goschen, *Theory of Foreign Exchange*; H. Withers, *Money Changing*.

Exchange, Deed of, in English law, an original common law conveyance for the mutual transfer of real estate. It takes place between two contracting parties only, although several individuals may be included in each party; and the parties must take an equal estate, as fee-simple for fee-simple, legal estate for legal estate, copyhold for copyhold of the same manor, and the like.

Excheq'uer (Fr. *échiquier*, chess-board), in Britain, the department which deals with the moneys received and paid on behalf of the public services of the country. The public revenues are paid into the Bank of England (or of Ireland) to account of the Exchequer, and these receipts as well as the necessary payments for the public service are under the supervision of an important official called the Comptroller and Auditor General, the payments being granted by him on receipt of the proper orders proceeding through the Treasury. The public accounts are also audited in his department. The Chancellor of the Exchequer, who must be a member of the House of Commons, is the head of the Treasury Department. When the Prime Minister is a member of the House of Commons, he sometimes holds the office of Chancellor of the Exchequer.

Exchequer, Court of, an ancient English court of record, established by William the Conqueror, and intended principally for the care and collection of the royal revenues. It was one of the supreme courts of common law, and is said to derive its name from the chequered cloth, resembling a chess-board, on which the sums were marked and scored with counters. The judges of this court were the chief baron and five junior or *puisné* barons. This court was abolished by the Judicature Act of 1873, and its jurisdiction transferred to the High Court of Justice. In Canada there is a Court of Exchequer for the Dominion.

Exchequer and Audit Department, a department of the English Civil Service charged with the functions of auditing the accounts of all other departments.

Exchequer Bills, bills of credit issued by authority of Parliament as a means of raising money for temporary purposes. They are of various sums—£100 or any multiple—and bear interest at a rate fixed for every half-year according to the rate ruling in the money-market at the time. These bills pass from hand to hand as money, and form part of the public unfunded debt of Great Britain. *Exchequer bonds* are similar, but they run for a definite number of years (six at most) at a fixed rate of interest.

Excise, an inland duty or impost laid on commodities produced and consumed within a country, and also on licences to manufacture and deal in certain commodities. Excise duties were introduced into England by the Long Parliament in 1643, being then laid on the makers and vendors of ale, beer, cider, and perry. Being found to be a convenient and productive source of revenue, they continued to gain ground, and in 1919 yielded £59,663,000. In Britain the excise includes duties on spirits and beer, licences on dogs, guns, carriages, servants, plate, railways, game, &c. In 1917 an excise duty was laid on entertainments, matches, and table waters. Spirits and beer yield over £44,000,000.

Excitomotor action, the action of nerves distributed to muscular organs, the stimulation of which leads to movement. Thus, irritation of a nerve supplying a muscle will lead to contraction of the muscle by excitomotor action, and irritation of certain nerves distributed to blood-vessels will lead to contraction of the vessel by acting on its muscular coat.

Exclusion, Bill of, a bill introduced into the British Parliament during the reign of Charles II for the purpose of excluding the Duke of York (afterwards James II), he being a Roman Catholic, from the throne.

Excommunication, the exclusion of a Christian from the communion and spiritual privileges of the Church. Excommunication was a recognized penalty among the Jews (*John*, ix, 22), and was practised early by the Christian Church. A distinction gradually arose between a lesser and a greater excommunication, the former being a suspension from Church privileges, the latter a formal expulsion excluding from all communion with

the faithful. In the Middle Ages the Popes often excommunicated whole cities and kingdoms. In such a case all religious services ceased, and the grave inconveniences thus caused made excommunication a formidable weapon in the hands of the Pope, till with frequent abuse it lost its force. Besides excommunication an extreme degree of denunciation called *anathema*, and cutting the offender off from all the hopes and consolations of the Christian faith, is used in the Roman Catholic Church. Both Luther and Calvin were in favour of the right of excommunication by the ministers of the Church. In the Church of England both the less and the greater excommunications are recognized.—Cf. E. Taunton, *The Law of the Church*.

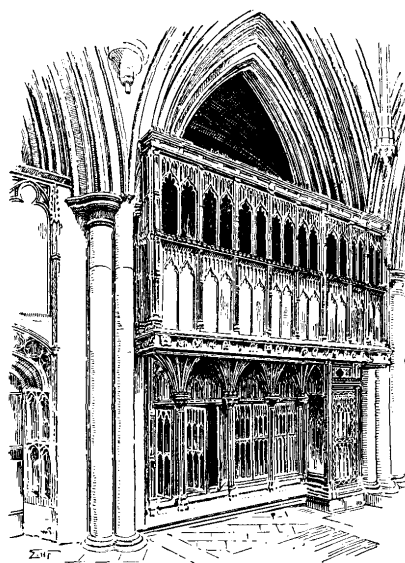
Excre'tion, in physiology, the separation and carrying off of waste matter from some organ of an animal body, a function performed by the lungs, kidneys, and the skin, besides the action of the intestinal canal.

Excubito'rium, in mediæval churches, a gallery where public watch was kept at night on the eve of some festival, and from which the great shrines could be seen.

Exe, a river of England, which rises in Exmoor, in the county of Somerset, and after a southerly course of about 50 miles falls into the English Channel at Exmouth.

Execution, in law, is a judicial writ grounded on a judgment of the court by which the writ is issued, and is granted for the purpose of carrying the judgment into effect, by having it executed. Execution is granted by a court only upon the judgments given by the same court, not upon those pronounced by another. See *Elegit*; *Fieri facias*.

Executioner, the official who carries into effect a sentence of death, or inflicts capital punishment in pursuance of a legal warrant. In England the duty of executing the extreme sentence of the law devolves upon the sheriff, and in Scotland on the civic magistracy, but in practice the duty is performed by another in presence of these functionaries. In the reign of



Excubitorium, St. Alban's Abbey

James I Gregory Brandon was the executioner of London, and hence the name Gregory has often been employed to designate executioners. In France the executioner is styled Monsieur de Paris.

Exec'utive, that branch of the government of a country by which the laws are carried into effect or the enforcement of them superintended. The term is used in distinction from the *legislative* and the *judicial* departments, and includes the supreme magistrate, whether emperor, king, president, or governor, his cabinet or ministers, and a host of minor officials.

Exec'utor, in law, is one appointed by a man's last will to carry its provisions into execution after the testator's death. The testator may, by the English law, appoint any person of sound mind and discretion, though otherwise under some legal disabilities as to contracting and transacting business in general, such as a married woman, or a minor. The duties of executors and of administrators are, in general, the same, the difference of the two depending mostly on the mode of appointment, the executor being nominated by the testator, the administrator being appointed by the Judge of Probate. In Scotland an executor appointed by will is styled executor *nominate*, and by authority of the court executor *dative*.

Exequa'tur (Lat., 'Let him accomplish'), a written recognition of a consul or commercial agent issued by the Government to which he is accredited, and authorizing him to exercise his powers.

Ex'eter, a city, river-port, parliamentary county, and municipal borough of England, county town of Devon, on the left bank of the Exe, 10 miles north-west of its outlet in the English Channel. It is pleasantly situated on the summit and slopes of an acclivity rising from the river, and has handsome squares, terraces, and streets. The chief architectural feature is the cathedral (founded in 1112), a long, low building with fine west front, unique in having two towers forming its transepts, and only these two. The city has remains of the old castle and old walls, Guildhall, Albert Memorial College, training college, and St. Michael's Church. There are iron-foundries, works for agricultural implements and paper, and 'Honiton' lace is made. By a canal vessels of 300 tons can reach the city. The largest vessels remain at Exmouth. Exeter is a place of remote antiquity, having been a British settlement long prior to the invasion of the Romans, by whom it was called

Isca Damnoniorum. It was taken by William the Conqueror in 1068. It long returned two members to Parliament, but lost one of them in 1885. Pop. 59,608.

Exeter College, Oxford, a college, originally called Stapledon Hall, founded in 1314 by Walter de Stapledon, Bishop of Exeter, who made a foundation for a rector and twelve fellows. In 1404 Edmund Stafford, Bishop of Exeter, added two fellowships and obtained leave to give the college its present name.

Exeter Hall, a large building on the north side of the Strand, London, opened in 1831, and capable of containing over 3000 persons. In it the 'May Meetings' of the several religious societies were held. It was sold in 1907 by the Young Men's Christian Association to a firm of caterers, who demolished it in 1908. The Strand Palace Hotel now stands on its site.

Exfoliation is the scaling off of dead tissues; thus there may be exfoliation of the skin, of bone, or of other parts from their living surroundings.

Exhibition, a fixed sum given for a term of years from the funds of a school, college, or university, generally upon the result of a competitive examination. An exhibition, no matter what its pecuniary value, is less of a distinction than a scholarship. At Cambridge an exhibitioner has no standing in his college different from that of the ordinary 'pensioner', while a scholar is on the Foundation of his college.

Exhibitions. The earliest recorded 'exhibition', in the modern acceptance of the word, is that in which, for one hundred and eighty days, Ahasuerus "shewed the riches of his glorious kingdom" at Shushan (*Esther*, i, 2-4). The first European exhibition was held at Venice in 1268; while the great fairs of Leipzig and Nijni-Novgorod partook of the same nature. But the real forerunner of the modern exhibition was that held in London by the Society of Arts in 1756, when carpets, china, and similar artistic objects were displayed. This was followed in 1761 by an exhibition of agricultural machinery. In 1797 France held a display of *objets d'art* at St. Cloud; there was another two years later at the Louvre; while a third in 1802 is memorable for the first issue of an official catalogue, and for the presence of Montgolfier the aeronaut, and Jacquard the inventor of the loom that bears his name, among exhibitors. Besides the exhibition of Irish industries,

held at Dublin (1829), and the Birmingham exhibition of metal-work (1849), numerous displays were held in both Europe and America before the first International or 'Great' Exhibition of 1851. This memorable display was under the active patronage of Prince Albert, and for its accommodation the Crystal Palace was erected in Hyde Park. Its success gave the exhibition movement an impetus which produced examples at New York and Dublin (1853), both of which proved financial failures; at Munich and Melbourne (1854); Paris (1855); and at South Kensington, where was held the second International Exhibition (1862). Five years later, after Constantinople, Oporto, and Agra had, among other places, held displays, came the great Paris Exhibition, planned on a colossal scale; though the main building in the Champ de Mars was stigmatized as 'a gasometer' by Napoleon III. Vienna had a magnificent but financially disastrous exhibition in 1873, and in 1876 Philadelphia celebrated the centenary of the Independence of the United States by a display. In 1878 Paris held another exhibition, for which the Trocadero was built. A series held at South Kensington (1871-4) had only a moderate success, but great popularity was attained by the 'Fisheries', 'Health', 'Inventions', and Colonial Exhibitions held in the years 1883-6. Edinburgh was the scene of a forestry display in 1884, and at the same time New Orleans opened an exhibition which continued till the following year. The Paris Exhibition of 1889 was notable for the 'side-shows', which included the Eiffel Tower; while the financial result of that held in 1900, which covered 550 acres of ground and admitted thirty-nine million people, was far from satisfactory. Meanwhile Chicago had in 1893 celebrated the four hundredth anniversary of Columbus's discovery of America by a huge 'World's Fair'. The present century opened inauspiciously with the assassination of President McKinley at the Pan-American Exhibition at Buffalo (1901); Glasgow had a thoroughly successful exhibition the same year; St. Louis was the scene of another in 1904; while London organized the Franco-British Exhibition in 1908. This last owed much to the late Imre Kiralfy, who had directed many previous displays, and was the designer of the 'White City' at Shepherd's Bush, where in 1914 was held the Anglo-American Exposition. After the conclusion of the European War various plans for new exhibitions were put forward, such as the British Empire Exhibition, which is to be held in London in 1923, and which is intended to foster imperial interests.

Ex'ile, a punishment by which a person is compelled to leave the city, province, or the country where he has previously resided. It is a punishment for State criminals.

Exmoor', a wild and hilly district of England, in the extreme south-west of Somersetshire, extending also into Devonshire, formerly a forest.

Ex'mouth, a town of England, in Devonshire, 10 miles S.S.E. of Exeter, at the mouth of the Exe. It is picturesquely situated, and is one of the best-known sea-bathing places on the Devonshire coast. The chief industries are lace-making and the fisheries. Pop. 13,614.

Exmouth, Edward Pellew, Viscount, a British naval officer, born at Dover in 1757, died 23rd Jan., 1833. He went to sea at the age of thirteen, served as midshipman in the *Blonde* frigate during the American War, and greatly distinguished himself at Lake Champlain. In 1782 he was made a post-captain for a brilliant action in the *Pelican*, and on the outbreak of the war in 1793 was appointed to the command of the frigate *La Nymphe*. From this time till the peace in 1802 he was employed on active service. In 1804, on the resumption of hostilities, he was sent to take the chief command on the East India station, in the *Culloden*, of seventy-four guns; and here he remained till 1809, when he had attained the rank of vice-admiral. His next appointment was the command of the fleet blockading the Scheldt. In 1814 he was made Baron Exmouth with a pension of £2000 per annum. In 1816 he was sent with a fleet to punish the Dey of Algiers for outrages committed, and to force him to give up his Christian captives and abolish Christian slavery. Along with some Dutch war vessels he bombarded the city for eight hours, and inflicted such damage that the Dey agreed to every demand. Three thousand Christian slaves were thus restored to liberty. Lord Exmouth was made a viscount and received honours from several of the European sovereigns, and the freedom of the City of London. In 1821 he retired into private life.

Exoascineæ, a family of parasitic ascomycetous Fungi, distinguished by the absence of any definite fruit-body, the asci being produced in a layer on the surface of the host. The best known are *Exoascus Pruni*, the cause of the mal-formed fruits called 'bladder-plums' or 'pocket-plums', and *E. turgidus*,

which produces the abnormal tufts of branches on silver birch known as 'witches' brooms'.

Exobasidiineæ, a family of parasitic basidiomycetous Fungi, resembling the Exoascineæ in most respects, but producing basidia in place of asci, a remarkable instance of parallel evolution. The commonest British species is *Exobasidium Vaccinii*, which is frequent on cowberry (*Vaccinium Vitis-Idæa*) in Scotland. *E. vexans* is the cause of a serious disease of the tea-plant called 'blister-blight', which is very destructive to the Assam plantations.

Ex'odus (Gr. *exodos*, a going out), the name given in the *Septuagint* to the second book of the *Pentateuch*, because it describes the departure of the Israelites from Egypt. The contents of the book are partly historical, describing the departure of the Israelites from Egypt, and partly legislative, describing the promulgation of the Sinaitic law. One of the difficulties connected with this book is that, according to Scriptural chronology, the residence of the Israelites in Egypt was only 215 years, and it seems incredible that in this time "the threescore and ten souls" who accompanied Jacob to Egypt could have become the two and a half millions who left with Moses.

Exogamy (Gr. *exo*, outside, and *gamos*, marriage), a term applied to the custom of allowing marriages only between members who do not belong to the same group. The study of exogamy is practically a branch of ethnology. The opposite of exogamy is endogamy, or prohibition of marriage outside the tribe. Exogamy, which among other causes may be ascribed to a desire of forming useful alliances with hostile tribes, is practised among Australian aborigines, Mongols, and American Indians, and the custom is widely distributed in various forms in all stages of civilization.—BIBLIOGRAPHY: Sir J. G. Frazer, *Totemism and Exogamy*; E. A. Westermarck, *The History of Human Marriage*.

Exogenous Plants (*eks-oj'e-nus*), or **Exogens**, old names for Dicotyledons (q.v.).

Exogenous Structures, in botany, are those which arise from superficial tissues of the parent organ, as stem-branches and leaves. Opposed to endogenous structures.

Ex'orcism (Gr. *exorkizein*, to expel with an oath), the casting out of evil spirits by certain forms of words or ceremonies. An opinion prevailed in the ancient Church that certain persons, those particularly who were afflicted with certain diseases, especially madness and epilepsy, were possessed by evil spirits; this was called *demoniac possession*. Over such persons forms of conjuration were pronounced, and this act was called *exorcism*. There were even certain men who made this a regular profession, and were called *exorcists*. Exorcism still forms a part of the beliefs of some Churches. In the Roman Catholic Church exorcist is one of the inferior orders of the clergy. —Cf. Sir J. G. Frazer, *The Golden Bough*.

Exosto'sis, in medicine, an excrescence or growth from one of the bony structures of the body. It is generally found at the end of long bones near the joints, and in connection with the skull.

Exothermic Compounds are compounds which evolve heat during their formation; these are usually stable compounds, as they must be supplied with a corresponding amount of heat for their decomposition.

Exot'ic, belonging to foreign countries; a term used especially of plants. Exotic plants are such as belong to a soil and climate entirely different from the place where they are raised. They are nearly always greenhouse or hothouse plants.

Expansion, in physics, the increase of the dimensions of a body caused by a change of temperature. In general, bodies expand with rise of temperature, solids expanding least and gases most. This change in the dimensions of solids with temperature is expansion in length, area, and volume. These changes are referred to a unit called the coefficient of expansion, or expansion of unit length, area, or volume for 1° rise. For example, the coefficient of linear expansion of iron is .000011; that is, a yard, say, of iron rod becomes longer by this fraction of a yard for each degree centigrade that its temperature is raised. The coefficient of volume expansion of a solid is three times its linear coefficient. Crystals have different rates of expansion along the three crystalline axes. Fused silica or quartz expands so slightly that it may be plunged when red-hot into water without being cracked. Invar, a nickel-steel alloy, which also has an extremely small coefficient of expansion, is used in making clock pendulums, which are

unaffected by change of temperature. The observed expansion of liquids is affected by the expansion of the vessel containing them, and the apparent expansion of a liquid is thus always less than its real expansion. The expansion of water with rise of temperature is irregular; water contracts from 0° to 4°C., and thereafter expands at an increasing rate until the boiling-point is reached. The expansion of mercury and other liquids is employed in thermometry. Gases when kept at constant pressure expand by about $\frac{1}{273}$ rd part of their volume at 0°C. for each degree rise.

Expectation, in the doctrine of chances, the value of any prospect of prize or property depending upon the happening of some uncertain event. A sum of money in *expectation* upon a certain event has a determinate value before that event happens. If the chances of receiving or not receiving a hundred pounds, when an event arrives, are equal, then, before the arrival of the event the expectation is worth half the money.—*Expectation of life*, the number of years which a man or woman of any age may be expected to live. To calculate this with any reasonable degree of probability, it is necessary to have particulars of a large number of individuals. The earliest observations were based upon the records of the registers of certain towns. Out of a number of people alive at one date the numbers still living at the end of successive years were found, and an estimate of the probable duration of life was made. With increased facilities for collecting statistics of a larger number of lives, this estimate has since been modified. It is necessarily influenced by the progress of civilization, improved methods of sanitation, increased knowledge of the best methods of treating diseases, and other causes, but the collected statistics are now so numerous that the actuaries of life assurance offices have been able to prepare tables of mortality from which the cost of life premiums and the price of annuities are calculated.

Expectorants are drugs used to increase and liquefy the secretion in the lungs and air-passages. The most effective are ipecacuanha, squills, apomorphine, ammonium carbonate, and potassium iodide. Some of these are usually present in the many cough mixtures so widely used.

Experiment, an operation designed to discover some unknown truth, principle, or effect, or to establish it when discovered. It differs from observation in the fact that the phenomena observed are, to a greater or less

extent, controlled by human agency. Experiment distinguishes the modern method of investigating nature, and to it we owe the rapid strides made in chemistry and physics.

Expert (Lat. *experiri*, to test), a person eminently skilled in any particular branch or profession; specifically, a scientific or professional witness who gives evidence on matters connected with his profession, as an analytical chemist or a person skilled in handwriting.

Exploits', River of, a river which traverses nearly the whole of Newfoundland from S.W. to N.E., and falls into the Bay of Exploits. It is about 150 miles long, and is navigable for steamers for 12 miles.

Explosion, a sudden violent outburst accompanied by a loud noise, and giving rise to an impulsive wave which spreads outwards from the place of explosion. Generally applied to the very rapid combustion of explosive substances which, under the influence of heat or shock, are resolved with extreme rapidity into gaseous form. Substances are termed low explosives or high explosives according as they are set off by combustion or detonation.

Explosive Mechanism, in botany, (1) in flowers, an arrangement for the transference of pollen to an insect-visitor by a sudden movement of floral organs. (2) In fruits, an arrangement for the forcible expulsion of seeds from a fruit brought about in various ways (see *Sling-fruits*). The spores of Ferns, ascospores, and some conidia are also liberated explosively.

Explosives. An explosive is a substance or mixture of substances which, by the action of a blow or of heat, can be converted very easily and suddenly into a more stable substance or substances, usually gaseous, with the simultaneous liberation of a large amount of heat. Explosives are divided into classes according to the uses to which they are put, but the line of demarcation is not always very clear. The industrial and blasting powders may be either low or high explosives. A 'low' explosive explodes by the application of heat, and burns more or less uniformly and slowly, and projects neighbouring objects to a distance. A 'high' explosive explodes under a blow, and the whole of the substance is instantly transformed, and instantly exerts its maximum pressure, creating a violent disturbance in a limited area without necessarily projecting substances to any great distance.

Service explosives, for naval and military purposes, are divided into propellants and high explosives. Sporting powders are specially modified propellant powders. Examples of these classes of explosives, with notes on their composition, are given below.

Low Explosives.—The best examples of the low explosive are gunpowder and similar mixtures. The constituents of the 'gunpowder' explosives are generally not explosive alone, but only when mixed. A 'gunpowder' mixture contains carbon or carbonaceous matter like wood-meal, hydrocarbons, starches, and sugars, &c., which burn owing to the presence of highly oxygenated substances like peroxides, chlorates and perchlorates, nitrates, permanganates, chromates and dichromates, all of which convey the necessary oxygen. In addition, there usually is present some very easily ignited substance like sulphur or sulphides, or phosphorus or phosphides, &c. As compared with other explosives, gunpowder or black-powder has certain advantages. It is cheap, easily ignited, insensitive to shock, and stable at moderately high temperature; it burns regularly, and its residue is non-corrosive. But it is weak in power, and produces much smoke. It is excellent for armour-piercing shell and for rings of time-fuses. Gunpowder made in different countries varies in composition, but for rifle, cannon, and sporting powders it, usually contains 74 to 75 parts of saltpetre, 9 to 14 parts of sulphur, 12 to 16 parts of charcoal. For blasting powders less saltpetre and more charcoal is used. Charcoal is made by the carbonization of wood. In England dogwood, alder, and willow woods are used; in Germany alder and willow are used; in France black alder and also white alder, poplar, aspen, birch, and hazel; in Switzerland hazel wood; in Spain oleander, yew, willow, hemp stems, and vine; in Italy hemp stems. The wood is generally carbonized in iron retorts. The product is allowed to cool out of contact with air, else it may inflame. Wood burnt for ordnance powders gives a yield of 20 to 30 per cent charcoal; that for small-arms gives a yield of 40 per cent. The charcoal contains from 68 to 85 per cent carbon, from 2.8 to 3.7 per cent hydrogen, from 12 to 27 per cent oxygen, and may have up to 5 per cent ash. The saltpetre is found naturally in Chile, India, and in other countries, and is refined by crystallization from water. It is a colourless, crystalline solid. Sulphur, a pale-yellow solid, melting-point 113° C., boiling-point 444.5° C., is found in nature, and is refined to a purity of 99.5 per cent and over. It has a low ignition temperature of 261°

C., and makes the powder burn more readily. Under the pressure of the press and the incorporating mill it flows and cements the minute particles of charcoal and saltpetre together. The three ingredients are ground, mixed, sieved, incorporated or mixed in drums or mills, broken down, and then pressed, corned or granulated, and glazed. Cannon powders receive an addition of graphite to reduce the rate of burning. The powder is then dried in a stove, finished in a reel to get rid of the last traces of dust, and blended. In the United States powder for blasting contains sodium nitrate instead of potassium nitrate. The powder is cheaper and stronger, but is hygroscopic. Sprengsaltpetre is largely used in Stassfurt salt-mines, where a mild explosive is required, and consists of 75 parts by weight of sodium nitrate, 10 parts by weight of sulphur, and 15 parts by weight of brown coal. It is cheap, and does not produce poisonous fumes. Bobbinite is largely used in coal-mines. It is black powder with ammonium and copper sulphates, possibly also starch and paraffin wax. When gunpowder explodes, the product consists of 43 per cent gases, 56 per cent solids, and the rest water. The composition of other industrial and blasting powders is given in the sequel.

Propellant Explosives.—The chief propellants are nitrocellulose, also called nitrocotton or guncotton, and nitroglycerine.

Nitrocellulose.—The chief sources of cellulose are wood and cotton. When cotton is plentiful, nitrocellulose is made as follows. Cotton-waste is hand-picked to get rid of string, wood, &c.; it is opened out by a teasing-machine, which tears off small portions at a time, and the cotton is then dried to about 0.5 per cent moisture content. The cotton is then nitrated with 'mixed acid'—a mixture of about 16 per cent nitric and 75 per cent sulphuric acid and about 8 per cent water—at 15° to 25° C. After the nitration, the acid is removed and the nitrocotton boiled up in water to stabilize it. Generally nitrocotton contains about 12 to 13 per cent of nitrogen. Wet nitrocotton is quite safe although it can be detonated, but dry nitrocotton is very dangerous. To-day, paper is usually made from wood-pulp, and when the cotton supplies of Germany were stopped during the European War, nitrocellulose had to be made from wood-pulp via a form of paper crêpe prepared by the Germans from the pulp. For propellant purposes the nitrocotton is 'gelatinized', either alone or mixed with nitroglycerine, and is

then worked up into different forms, such as wire, rods, grains, or tape, when it becomes controllable at will, so that the firing is not dangerous.

Nitroglycerine.—Mixed acid, containing 41 per cent nitric acid and 57.5 per cent sulphuric acid, is brought to 22° C. by cooling coils of brine, and pure glycerine is injected into the acid at such a rate that no glycerine accumulates unchanged, and that the temperature is kept between 15° and 22° C. When all the glycerine has been added, the liquid is allowed to stand, and the nitroglycerine rises to the surface. It is run off to the wash-house, where it is washed free from acid and settled. The process is a dangerous one, and great care must be taken at every stage of the manufacture. The floors of the plant must be free from grit and dirt, no accumulation of liquid should be allowed anywhere, special clothing and rubber boots must be worn, no metallic implements may be used, and the plant should not be handed over for repairs except under the supervision of a responsible person. Nitroglycerine, when absorbed in a porous earth called 'Kieselguhr', is called dynamite. Kieselguhr, or simply guhr, absorbs twice its weight of nitroglycerine; cork charcoal absorbs nine times its weight. Dynamite cartridges are generally exploded by detonators.

Preparation of Cordite; Nitrocellulose Tape (N.C.T.); Ballistite; &c.—For cordite, the nitrocotton, freed from moisture, is mixed with nitroglycerine, and the paste or the cotton itself, if N.C.T. is to be made, is incorporated into a uniform dough with ether and alcohol. Some mineral jelly is added to render the explosive more stable. The dough is pressed through different sizes of dies according to the product desired. For rifle powder fine cords are used; for artillery, thicker cords or flat ribbons of varying thicknesses are required. The cords or tapes from the dies are cut into suitable lengths, the solvents driven off, and the products blended to obtain uniform ballistic quality. For ballistite the nitrocellulose is beaten up with nitroglycerine in water. The paste is freed from water, dried, and worked into horn-like sheets by means of rollers.

High Explosives; Picric Acid.—At the outbreak of the European War the chief high explosive of the Entente Powers was lyddite (in France, mélenite), also called trinitrophenol or picric acid. It is a bright-yellow solid, melting-point 122° C., sparingly soluble in water, and forms easily exploded metallic salts. It is now displaced by trinitrotoluene. Picric acid is

made from phenol or carbolic acid. Phenol is obtained from coal-tar, or made synthetically from benzene. The phenol is sulphonated with strong sulphuric acid, and the phenol-sulphonic acid resulting is nitrated with strong nitric acid at about 100° C. Picric acid separates, and is washed free from mineral acid and dried. It may also be made from benzene without converting it into phenol thus: The benzene is chlorinated and gives chlorbenzene. This is nitrated into dinitrochlorbenzene, and is then treated with caustic soda to give dinitrophenol. This is then further nitrated into trinitrophenol or picric acid. Picric acid has a high melting-point, it must be used pure, is dissolved by water, it attacks metals forming dangerous compounds, and requires troublesome plant for its manufacture. Hence it has been displaced by more suitable substances, notably by trinitrotoluene.

Trinitrotoluene (T.N.T.).—This compound may now be made in a continuous plant. Mononitrotoluene is put in at one end of the plant and comes out as trinitrotoluene. Mixed nitric and sulphuric acid is put in at the end where the T.N.T. is obtained, and emerges, where mononitrotoluene is put in, as waste acid. T.N.T. in the past has also been made discontinuously thus: The toluene is nitrated by mixed acid into either mono- or dinitrotoluene, which is then trinitrated. The conversion into mononitrotoluene was used in the United Kingdom, France, Germany, Canada, and America; the conversion into dinitrotoluene was used in Italy. The T.N.T. emerges liquid, and is passed over a rotating drum internally cold water cooled. A knife strips the thin congealed skin of T.N.T. off. This T.N.T. is only grade 3, and for conversion into grade 1 it must be purified. Formerly this was done by means of an organic solvent, but this dangerous and expensive method has been displaced by treatment either with phenol or sodium sulphite, which gives a grade 1 product. T.N.T. is a very pale yellow solid, melting-point 80.2° C., and therefore, melted by hot water, almost insoluble in water, burns quickly in the air, is inert, and comparatively safe to handle. It has displaced picric acid owing to its superiority, physically and chemically, over that substance.

Tetryl or C.E. (Composition Exploding).—Tetryl, also called tetranitrodimethylaniline, or more correctly trinitrophenylmethylnitramine, is a powerful high explosive, and is thus obtained. One part of dimethylaniline is dissolved in 10 parts of strong sulphuric acid, and the solution allowed to flow into strong nitric acid. The temperature should be

kept below 40° C., else decomposition may occur. After nitration the yellow tetryl separates out, and is filtered off and water-washed till free from acid. It is then dried in hot-air stoves. Tetryl is much more dangerous than T.N.T., and is also more poisonous to handle. No other nitro-bodies were made in England on the large scale during the European War, but on the Continent, owing to the scarcity of raw materials, dinitrobenzene, dinitrotoluene, and nitronaphthalenes, and even less important nitro-bodies, were made. Their manufacture is similar to that of those already described. Probably none of these nitro-bodies so pressed into use is as good as T.N.T.

Detonating Substances.—Though modern explosives are not easily exploded by a blow, they are sensitive to shock of given intensity, and lesser or different shocks will not suffice. The 'detonator' to produce the shock is set into the explosive. A complete shell carries two detonators. One, in the percussion cap, sets off the propellant charge which expels the projectile; the other, in the fuse in the nose of the shell, is ignited by the discharge of the gun, and detonates the high-explosive filling at a set interval after the discharge of the shell. Therefore, the shell can be exploded either in its flight when it is used as shrapnel shell, or on its arrival at its objective when it can be used for small-calibre artillery shell for field-guns, &c., or after its arrival when it is used for heavy howitzer and armour-piercing shell for destroying entrenched works, armoured forts, or ships. The manufacture of detonators is a very dangerous and delicate operation. Some substances (the copper acetylides) explode by a scratch, some (nitrogen iodide) by the touch of a feather or the tread of a fly, some explode even in solution when poured from one vessel into another (diazobenzeneperchlorates). Mercury fulminate is more often employed in the detonator, and is prepared from mercury, alcohol, and nitric acid. It is expensive, and most modern detonators consist of lead azide or salts of styphnic acid, with a layer of T.N.T. in a narrow aluminium cylinder.

The following is a list of the more important explosives, the different groups not being mutually exclusive:—

COAL-MINE EXPLOSIVES

American.

Aetna Coal-mine Powder A, B, C: nitroglycerine explosive.

Black Diamond: 2A, 3A, 6L.F are nitroglycerine explosives; 5, 7, 8, ammonium nitrate explosives.

Carbonite: nitroglycerine 26, barium nitrate 4, potassium nitrate 29, wood-meal or starch flour 40, calcium carbonate 0.25.

Du Pont Permissible: nitroglycerine, ammonium nitrate, common salt.

Eureka: nitroglycerine and hydrated salt.

Monobel: ammonium nitrate, nitroglycerine, wood-meal, alkali chloride.

Red H1-7: ammonium nitrate explosives.

Trogan Coal Powder: contains nitrostarch.

Austrian and Hungarian.

Chloratit: during war was used in coal-mines.

Dynammon: ammonium nitrate, potassium nitrate, red charcoal.

Pannomite: nitroglycerine, collodion cotton, ammonium nitrate, dextrin, glycerine, nitrotoluene, alkali chloride.

Titanite: Ammonium nitrate, trinitrotoluene, curcuma charcoal.

Belgian.

Alsilite: ammonium nitrate, trinitrotoluene, ferrosilicon-aluminium, salt.

Baellnite: ammonium-nitrate, trinitrotoluene.

Densite: alkaline nitrates, trinitrotoluene, dinitrotoluene, and ammonium chloride.

Favier Explosives: mixtures in varying proportions containing ammonium nitrate, nitronaphthalene, paraffin, and resin; higher

nitrated naphthalenes, potassium nitrate, and tetryl may be present. Manufactured by the French Government as Explosifs N or Favier or Grisounites. Grisounites-couche for coal-mines have theoretical explosion temperatures of 1500° C., Grisounites-roche of 1900° C. Ammonite, Westfalite, Bellite, Roburite are explosives of this type; other ammonites, Bellite Nos. 2 and 4, Faversham powder, and negro powder have ammonium or sodium chloride added.

British.

Ammonite: Favier type; ammonium nitrate 75, dinitronaphthalene or other nitro-body, salt 20.

Bellite: ammonium nitrate and metadinitrobenzene; salt and starch may be added.

Bobbinite: the only gunpowder explosive allowed in England, not allowed in foreign mines; alkali nitrate, carbohydrates, wax may be added. Shatters coal less than high explosives.

Cambrite: a Nobel carbonite plus 8 per cent of a cooling agent.

Denaby Powder: ammonium nitrate, alkali nitrate, T.N.T., ammonium chloride.

Dynobel: nitroglycerine 15, collodion cotton 0.5, nitro-body 3, ammonium nitrate 46, wood-meal 5.5, salt 29.5, magnesium carbonate 0.5. Limit charge, 18 to 30 ounces. Swing of ballistic pendulum, 2.35 inches.

Monarkite: ammonium and sodium nitrate, nitroglycerine, nitrocotton, starch, mineral jelly, salt.

Monobel: ammonium nitrate, nitroglycerine, wood-meal, salt, alkali chloride, magnesium carbonate; nitro-body may be present.

Negro Powder: Grisounite type; ammonium nitrate 88, T.N.T. 10, graphite 2.

Rex Powder: nitroglycerine 12, salt 20, wood-meal 8, ammonium nitrate 60. Charge, 20 ounces. Swing of pendulum, 2.61 inches.

Rippite (Super): nitroglycerine, nitrocotton, potassium nitrate, borax, alkali chloride.

Roburite: ammonium nitrate 61, T.N.T. 16, salt 23.

Stomonal: nitroglycerine, ammonium nitrate, sodium nitrate, wood-meal, wheat-flour, salt, ammonium oxalate.

Thames Powder. Similar to above.

Danish.

Aerolite: ammonium nitrate 78.1 per cent, potassium nitrate 7.5 per cent, sulphur 8.75 per cent, tar 2.5 per cent, sago-meal 1.25 per cent, manganese dioxide 1.25 per cent, resin 0.6 per cent.

Poudre Blanche Cornil: ammonium nitrate, alkali nitrate, nitronaphthalene, lead chromate.

French.

Favier Explosives. See under *Belgian* explosives.

Grisounite. As above.

Grisoutine or Grisou Dynamite. It is a mixture of ammonium nitrate and blasting gelatine. It is the only explosive except Grisounite allowed in the more dangerous French mines.

Naphthalite (Grisou): potassium chlorate 80 per cent, aryl hydrocarbons, 12 per cent nitrocompounds not trinitro-bodies, paraffin, fatty oils, flour and other organic substances. May contain alkali chlorides and up to 4 per cent blasting gelatine.

German.

Albit (Wetter): a chlorate explosive replacing nitrates; scarce during the war.

Astralit (Wetter): ammonium nitrate explosive containing some blasting gelatine. Many varieties, which may also contain rape-oil and potato-meal. Has been used in trench howitzers.

Carbonit. Numerous compositions come under this name and are made in different countries. Consists chiefly of nitroglycerine and metallic nitrates. Arctic carbonite, a low-freezing mixture, contains 15.5 per cent nitroglycerine, 10.5 per cent nitrohydrocarbon, 42 per cent potassium nitrate, 31.7 per cent wood-meal, and 0.3 per cent calcium carbonate.

Chloratzit: potassium chlorate or perchlorate, aryl nitro-bodies, resins, and carbohydrates. For coal-mines add cooling agents.

Detonit: ammonium nitrate, charcoal, meal, 4 per cent blasting gelatine, neutral salts.

Donarit: ammonium nitrate 80 per cent, trinitrotoluene 12 per cent, rye-flour 4 per cent, nitroglycerine 4 per cent. The standard in Germany for sensitiveness of ammonium nitrate explosives.

Dorfit: ammonium nitrate, trinitrotoluene, flour, salt, alkali nitrate.

Dynamit: nitroglycerine 75 per cent, Kieselguhr 25 per cent.

Gehlingerit: ammonium nitrate, trinitrotoluene, flour.

Permonit: a perchlorate explosive.

Tremonit: contains gelatinized dinitroglycerine, pea-flour, and salt.

SOME BLASTING EXPLOSIVES

Ammonal: ammonium nitrate 80 to 90 per cent, aluminium 4 to 18 per cent, charcoal 2 to 6 per cent. The more violent mixtures contain some trinitrotoluene in addition. Has been used in grenades and by Austrians in trench-howitzer bombs. Not suitable in underground workings owing to poisonous gases evolved.

Astralit: a mixture of ammonium nitrate and blasting gelatine. May contain wood-meal, trinitrotoluene, paraffin-oil. Has been used for

projectiles.

Carbodynamite: nitroglycerine absorbed in cork charcoal instead of guhr.

Gelignite: nitroglycerine 56 to 63 per cent, nitrocotton 4 per cent, wood-meal 7 per cent, potassium nitrate 27 per cent, calcium carbonate 0.2 per cent.

Oxyliquit: liquid oxygen absorbed in a porous combustible material. Used in construction of Simplon Tunnel. Very cheap; safe after misfire because oxygen evaporates off.

Perdit: German mining, demolition, and rifle grenade explosive. Ammonium nitrate 76 per cent, potassium perchlorate 6 per cent, wood-meal 2 per cent, dinitrotoluene 16 per cent.

Rendarock: a brand of American dynamite.

Sprengel Explosives: one or both of the substances to be liquid, and mixing to occur shortly before firing. Nitric acid, alkali chloride, nitrogen peroxide on nitrobenzene, nitronaphthalene, carbon bisulphide, petrol, picric acid. Panclastite, Prométhée, Rack-a-Rock, are examples of this class.

Tonite: gun-cotton and barium nitrate.

SOME HIGH EXPLOSIVES

Alumtol: ammonium nitrate, trinitrotoluene, aluminium powder. Used for trench mortars, bombs, &c.

Amatol: a mixture of ammonium nitrate and trinitrotoluene, used for shell-filling. Called by Germans Füllpulver. The Germans did not develop this explosive as much as the Entente Powers, who effected great economy of trinitrotoluene.

Blastine: ammonium perchlorate, sodium nitrate, dinitrotoluene, paraffin-wax. It evolves hydrochloric acid gas.

Blasting gelatine contains 93 per cent nitroglycerine and 7 per cent nitrocotton. It is the most powerful explosive in common use.

Crèsylite is a French explosive, and contains picric acid and nitrated cresol.

Fumyl is a smoke-producing explosive, and contains trinitrotoluene and ammonium chloride. It was used to open poison-gas shells.

Granatfülling (Shell Filling): a term used by the Germans.

Granatfülling C/84 is picric acid.

Granatfülling C/02 is trinitrotoluene.

Other substances were used by them for shell filling, such as trinitroanisole, dinitrobenzene, hexanitrodiphenylamine, and hexanitrophenylsulphide.

Lyddite: a term for picric acid formerly used in shells.

T.N.T. is trinitrotoluene or trotyl.

Toxol is a mixture of trinitrotoluene and trinitroxylene.

Triplastite is a plastic high explosive containing 70 per cent nitrotoluenes, 8 per cent nitrocotton, 29 per cent lead nitrate. Used for shell filling.

SOME MISCELLANEOUS EXPLOSIVES

Anilite: Sprengel type. French liquid explosive used in aerial bombs.

Centralite: not an explosive but a stabilizer and regulator. It is dimethyldiphenylurea.

Collodion Cotton: low nitrated nitrocotton, soluble in a mixture of ether and alcohol. It dissolves in nitroglycerine and liquid nitro-body, preventing their exudation.

Gelatine Dynamite: a mixture of blasting gelatine with potassium nitrate and wood-meal. It may also contain calcium and magnesium

carbonate and mineral jelly.

Gelignite is similar to Gelatine Dynamite, but contains less Blasting Gelatine.

Halakite: contains potassium chlorate, ammonium nitrate, trinitrotoluene or other nitro-body; may contain nitrocotton, sodium nitrate, and wood-meal.

Pyrocollodion: highly nitrated, soluble gun-cotton. Adopted by the United States.

White Gunpowder: a mixture of potassium chlorate, potassium ferrocyanide, and sugar. Very sensitive, and only used in the laboratory.

PROPELLANTS FOR SHOT-GUNS

Amberite: insoluble nitrocotton 18.6 per cent, nitrates 28 per cent, soluble nitrocotton 46 per cent, vaseline 6 per cent.

Du Pont Smokeless Powder: nitroglycerine 10 per cent, ammonium nitrate 67.5 per cent, wood-pulp 8 per cent, salt 15 per cent (for coal-mines). Soluble nitrocotton 46 per cent, metallic nitrates 2.2 per cent (for shot-guns).

E.C. Powder: insoluble nitrocotton 44 per cent to 48 per cent (Empire Powder), soluble nitrocotton 30 per cent to 34 per cent, metallic nitrates 14 per cent to 9 per cent, vaseline 6 per cent to 7 per cent, camphor 4.6 per cent.

Ideal Powder: made by Nobels.

Neonite. Similar to the above compositions, but containing 73 per cent of insoluble nitrocotton, 9 per cent soluble nitrocotton. It is also made for rifled small-arms especially for rim-fire rifles.

New Explosives Company Smokeless Powder. Similar to above.

Rifleite: insoluble nitrocotton 1.7 per cent, soluble nitrocotton 82.5 per cent, nitro-body 4.8 per cent. The nitrocellulose is made from curcuma.

Ruby Powder: a cheap non-solvent powder, 46 per cent insoluble nitrocotton, 4 per cent soluble nitrocotton.

Smokeless Diamond and Stoumarkel Smokeless are similar to above.

PROPELLANTS FOR RIFLED FIRE-ARMS

Amide Powder: ammonium nitrate, potassium nitrate, charcoal. Has also been used in German artillery.

Ammonpulver: ammonium nitrate and charcoal. Has been used by Austrian artillery, and lately reintroduced by the Germans.

Ballistite: equal parts of nitroglycerine and soluble nitrocotton with some mineral jelly.

Cordite: the principal smokeless powder of the British Empire.

Indurite: gun-cotton and nitrobenzene. Abandoned by U.S. navy.

Neonite: a gelatinized powder. Contains nitrocellulose insoluble and soluble, metallic nitrates, and vaseline.

Noddite: a strip sporting-rifle powder containing nitroglycerine, nitrocellulose, mineral jelly.

Rottweil Smokeless Powder: a gelatined powder containing camphor and diphenylamine.

See also *Grenade; Shell; Torpedo; Fireworks; Rockets*.

Expo'nent. In algebra a^3 denotes three a 's multiplied together; a^n means that n of the letter a are to be multiplied. These numbers or letters placed immediately above and to the right of another number or letter are called exponents, and indicate the power to which the number or letter is raised. Exponents can be fractional or negative, in which case new interpretations can be found. On the assumption that $a^m \times a^n = a^{m+n}$ for all values of m and n , $a^{1/n}$ is interpreted as the n th root of a , a^{-n} as the reciprocal of a^n .

Exponential Theorem. If $a^x = N$, x is said to be the logarithm of N to the base a . There are two bases of logarithms in common use, the base 10 and the Napierian base e . The exponential theorem states that the value of e^x is given by the infinite series $1 + x + x^2/(1.2) + x^3/(1.2.3) + \dots + x^n/(1.2.3..n) +$, &c. Putting x equal to 1, $e = 1 + 1 + 1/(1.2) + 1/(1.2.3) + 1/(1.2.3.4) +$, &c. e can be expressed to any number of decimal places by working out the value of the terms on the right-hand side. It is an incommensurable number which to five decimal places is equal to 2.71828. See *Logarithm*.

Exports. See *Foreign Trade*.

Ex Post Facto, in law, a term designating something as done after and bearing upon something previously done; thus a law is said to be *ex post facto*, or retrospective, when it is enacted to punish an offence committed before the passing of the law.

Exposure, the situation of a building, &c., with respect to sun and wind; 'aspect' with regard to the quarter of the heavens. A house facing south-west or south-east will be found much healthier than one facing due north; and as it is also warmer, less fuel will be required for heating purposes.

Expressed Oils, in chemistry, are those which are obtainable from bodies only by pressing, to distinguish them from mineral and essential oils, which last are, for the most part, obtained by distillation.

Extension, (1) in physics and metaphysics, that property of a body in virtue of which it occupies a portion of space. (2) In logic, *extension* is the extent of the application of a general term, that is, the objects collectively which are included under it; thus, the word figure is more extensive than triangle, circle, or parallelogram; European more extensive than French, Frenchman, German, &c. Matter and mind are the most extensive terms of which any definite conception can be formed. Extension is contrasted with *comprehension* or *intension*.—Cf. Titchner, *Text-Book of Psychology*.

Extincteur (eks-tan-teur), an apparatus for the extinction of fire, consisting of a metallic case containing water and materials for generating carbonic acid. When required, the materials are brought into contact by pushing a rod which breaks a bottle containing acid, the gas mixes with the water, and the

pressure generated is sufficient to project the water charged with the gas to a distance of 40 or 50 feet.

Ex'tract, a term to denote all that can be dissolved out of a substance by a specified menstruum, such as water, alcohol, ether, &c. In modern pharmacy the term is applied to two kinds of preparation from vegetables. One is got by digesting the plant in water or other solvent, and evaporating or distilling away the excess of solvent until the extracted matter is sufficiently inspissated. The other is got by bruising the plant in a mortar, separating the juice, warming it until the green colouring-matter separates, and filtering it off. The juice is next heated until the albumen coagulates, and again filtered. The juice is now evaporated to a syrup, the green colouring-matter added and well mixed, and the evaporation is thereafter continued until the required concentration is attained. Extracts must be capable of being redissolved, so as to form a solution like that from which they were derived. Extracts are used in cookery, medicine, and the manufacture of perfumery.—*Extract of Meat* (*extractum carnis*) is a soft, yellowish-brown solid, or very thick syrup, which is employed as a portable soup. It is now manufactured on the large scale by processes proposed by Liebig.

Extradi'tion (Lat. *ex*, out, and *tradere*, to hand over), the act by which a person accused of a crime is given up by the Government in whose territories he has taken refuge to the Government of which he is a subject. Conventions have been entered into by Britain with almost all civilized countries for the apprehension and extradition of persons charged with particular offences, especially those of the most heinous stamp, such as murder, robbery, embezzlement, arson, rape, and piracy. The Extradition Act of 1870 makes special provision that no criminal shall be surrendered for a political offence, and that the criminal shall not be tried for any but the crime for which he was demanded. Other British Extradition Acts are those of 1873, 1895, and 1906. See *International Law*.—Cf. Sir E. Clarke, *The Law of Extradition*.

Extravagan'za, in music or the drama, a species of composition designed to produce effect by its wild irregularity and incoherence; differing from a burlesque in being an original composition and not a mere travesty.

Extravasa'tion (Lat. *extra*, beyond, and *vas*, vessel), an escape of some fluid, as blood or urine, from the vessel containing it. *Blood extravasation*, in contusions and other accidents, is when blood-vessels are ruptured by the injury, and the blood finds its way into the neighbouring tissues. In some accidents to the urethra and bladder extravasation of urine is a very serious occurrence.

Extreme Unction has been, since an early period, one of the seven sacraments of the Roman Catholic Church. It is performed in cases of mortal disease by anointing in the form of a cross, the eyes, ears, nose, mouth, hands, feet, and reins (in the case of males). It is administered after confession and the eucharist, and is believed to remove the last stains of sin. It can only be administered by a bishop or priest, and is not applied in the case of young children or excommunicated persons.

Exu'ma, Great and Little, two of the Bahama Islands. The former is 30 miles long and 3 miles wide, and has a good harbour. Pop. 3465.

Eyalet (ī'a-let), a former administrative division of the Turkish Empire, subdivided into sanjaks or provinces, and kazas or districts. It was ruled by a pasha, and gave place to the vilayet on the reorganization of the empire in 1871.

Eyck (īk), Hubert and Jan van, brothers, famous painters of the old Flemish school, born at Maaseyck, Hubert in 1366, Jan probably about 1385. They lived first at Bruges, whence the younger brother is called John of Bruges, and afterwards at Ghent, to which they removed about 1420. Here they executed the celebrated *Adoration of the Lamb* for the Cathedral of Ghent; a painting which, in its different parts, contains above three hundred figures, and is a masterpiece. It was in two horizontal divisions, comprising ten panels, of which only the two central ones remain at Ghent, the others being at Berlin. Hubert did not live to see it completed. He died at Ghent (1426), as did also his sister Margaret, who was likewise a painter (1431). Jan finished the work in 1432, and returned to Bruges, where he remained till his death, which took place in 1441, and executed several excellent pieces. His reputation became very great even during his lifetime, by his share in the introduction of oil-painting; the original invention of which has been incorrectly ascribed to him by many. Jan van Eyck also introduced

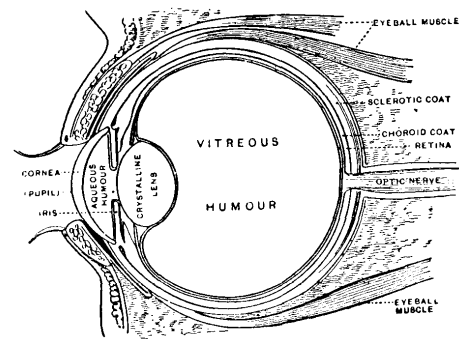
improvements in linear and aerial perspective, and in painting upon glass. Three portraits by Jan van Eyck, *The Scholar*, *The Man with a Turban*, and *Giovanni Arnolfini and his Wife*, are in the National Gallery, London; the *Madonna of the Chancellor Rollin* is in the Louvre.—Cf. W. H. J. Weale, *John and Hubert van Eyck*.

Eye, the organ of vision, is an instrument presenting some analogy to a photographic camera. It is provided with a lens for focussing images upon a sensitive membrane called the *retina*. The walls of the globe of the eye are formed principally of two fibrous membranes: one white and opaque—the *sclerotic* (Gr. *sklēros*, hard)—which envelops two-thirds of the globe posteriorly; and the other transparent, and resembling a horny plate, whence its name, *cornea* (Lat. *corneus*, horny). The sclerotic is a tough fibrous coat, and is the part to which the phrase 'white of the eye' is applied. In the front of the globe the sclerotic is abruptly transformed into the transparent portion (the cornea), which is circular, and which forms a window through which one can see into the interior. A membrane, the *conjunctiva*, so named because it unites the eye to the lid, spreads over the anterior portion of the globe, and then folds back on itself and lines the internal surface of the eyelids. On the internal surface of the sclerotic is a vascular membrane called the *choroid*. This is essentially the blood-vessel coat of the eyeball. About the place where the sclerotic passes into the cornea the choroid becomes continuous with the *iris*, a round curtain, the structure seen through the cornea, differently coloured in different individuals. In its centre is a round hole, the *pupil*, which appears as if it were a black spot. The *iris* forms a diaphragm suspended in the space behind the cornea which is filled with clear fluid, the *aqueous humour*. The *iris* consists of a framework of connective tissue, and its posterior surface is lined by cells containing pigment which gives the colour to the eye. In its substance are bundles of involuntary muscular fibres, arranged in a ring round the margin of the pupil. In a bright light the circular fibres contract and the pupil is made smaller; but in the dark these fibres relax and allow the pupil to dilate more or less widely, thus allowing more luminous rays to enter the eye. Just behind the pupil is the *crystalline lens*, resembling a small, very strongly magnifying glass, convex on each side, though more so behind. Minute bands of muscle in the choroid can alter the convexity of the lens, and thus adapt its focus to the position of objects at different distances. The large

space in the globe of the eye that intervenes between the retina lining its inner surface and the lens in front is occupied by a transparent jelly-like mass called the *vitreous body*. The internal surface of the choroid, or rather the pigmentary layer which covers it, is lined by the retina or nervous tunic upon which the images of the objects that we see are focussed. The retina is a very complex mechanism consisting of certain elements which are stimulated by light, and others that transmit the effects of such stimulation as a nervous current to the optic nerve which carries it to the brain. It is composed of cones and cylinders or *rods*, joined together like the stakes of a palisade, perpendicular to the plane of the membrane, and forming by their free extremities a mosaic, each microscopic division of which is about 0.0008 of a line in diameter according to Helmholtz, and represents a section of a rod. These rods and cones are believed to be the agents by whose aid the waves of light become transformed into the stimulus of a sensation. The ocular globe is put in motion in the orbit by six muscles, grouped two by two, which raise or lower the eye, turn it inward or outward, or on its antero-posterior axis. In these movements the centre of the globe is immovable, and the eye moves round its transverse and vertical diameters. These three orders of movements are independent of each other, and may be made singly or in combination, in such a manner as to direct the pupil towards all points of the circumference of the orbit. Each eye is furnished with two eyelids, moved by muscles, which shield it from too much light and keep it from being injured. They are fringed with short, fine hairs called eyelashes; and along the edge of the lids is a row of glands similar to the sebaceous glands of the skin. The eyebrows, ridges of thickened integument and muscle, situated on the upper circumference of the orbit and covered with short hairs, also regulate to some extent the admission of light by muscular contraction. In reptiles, some fishes (sharks, &c.), in birds, and in some mammals a third eyelid or nictitating membrane is present, and can be drawn over the surface of the eye so as to clear it of foreign matters, and also to modify the light. The lachrymal apparatus is composed of, firstly, the *lachrymal gland*, which lies in a depression of the orbital arch; secondly, the *lachrymal canals*, by which the tears are poured out upon the conjunctiva a little above the border of the upper lid; thirdly, the *lachrymal ducts*, which are destined to receive the tears after they have bathed the eye, and of which the orifices or *lachrymal points* are seen near the internal ends of the lids; fourthly, the *lachrymal sac*, in which the

lachrymal ducts terminate, that empties the tears into the *nasal canal*. The tears, by running over the surface of the conjunctiva, render it supple and facilitate the movements of the globe and eyelids by lessening the friction. The influence of moral or physical causes increases their secretion, and when the lachrymal ducts do not suffice to carry them off they run over the lids.

Vision.—The retina renders the eye sensible of light, and we may therefore consider it as the essential organ of vision. The function of the other portions is to focus the luminous rays on the surface of the retina, a condition necessary for distinct vision and the clear perception of objects. The visual impressions are transmitted from the retina to the brain by means of the optic nerve. The two optic



Section of the Human Eye

nerves converge from the eyes toward the centre of the base of the brain, where there is a partial interlacement of their fibres in such a manner that a portion of the right nerve goes to the left side of the brain, and a part of the left nerve to the right side; this is called the *chiasma* or *commissure* of the optic nerves. The principal advantage of having two eyes is in the estimation of distance and the perception of relief. In order to see a point as single by two eyes we must make its two images fall on corresponding points of the retinas; and this implies a greater or less convergence of the optic axes according as the object is nearer or more remote. To accommodate the eye to different distances the lens is capable of altering itself with great precision and rapidity. When we look at a near object, the anterior surface of the lens bulges forward, becoming more convex the nearer the object; the more distant the object the more the lens is flattened. When the transparency of the cornea, the crystalline lens, or any of the contents of the globe of the eye is destroyed, either partially or entirely, then will partial or total blindness follow, since no image can be formed upon the retina; but although all the media and the cornea be perfectly transparent, and retain their proper forms, which likewise is necessary to distinct vision, yet injury or inactivity of the optic nerve, or injury of the parts of the brain with which it is connected, may produce disturbance of vision or total blindness. Defective vision may also arise from the crystalline lens being so convex as to form an image before the rays reach the retina (a defect known as short sight or myopia), in which case distinct vision will be procured by interposing a concave lens between the eye and the object of such a curvature as shall cause the rays that pass through the crystalline lens to meet on the retina; or the lens may be too flat, a defect which is corrected

by convex lenses. In old age, and in fact in most people after about forty-five years of age, the elasticity of the lens becomes reduced, and convex lenses become necessary to make it possible to focus near objects. This condition is known as *presbyopia*. In the lower organisms the organs of sight appear as mere pigment spots. In higher forms, simple lenses or refracting bodies occur. Insects, crustaceans, &c., have large masses of simple eyes or ocelli aggregated together to form compound eyes—the separate facets or lenses being optically distinct, and sometimes numbering many thousands. In the cephalopods well-developed eyes presenting a distant analogy in structure to those of the highest animals are found; and in all vertebrate animals the organ of vision corresponds generally to what has been described, though they vary much in structure and adaptation to the surroundings of the animal.—Cf. J. Herbert Parsons, *Diseases of the Eye*.

Eye (ā), a municipal borough, England, county Suffolk, 19 miles north of Ipswich. Up till 1885 it sent a member to Parliament, and it still gives its name to a parliamentary division of East Suffolk. Pop. 1781.

Eyebright (*Euphrasia officinālis*), a small plant belonging to the nat. ord. Scrophulariaceæ, which is common in Britain and most parts of Europe, in North Asia, &c. It is an annual, half-parasitic on grass-roots, from 3 to 8 inches high, often much branched. The whole plant has a bitter taste. Under the name of *euphrasy* it formerly enjoyed a great reputation in diseases of the eyes.

Eyemouth, a fishing-town of Berwickshire, Scotland, at the mouth of the Eye, an important place in the thirteenth century. Pop. 2477.

Eye-piece, in a telescope, microscope, or other optical instrument, the lens, or combination of lenses to which the eye is applied. As the use of a single lens limits the field of view, eye-pieces are generally formed from two lenses or lens systems, called the eye-lens and the field-lens, placed at a distance apart. The Ramsden or positive eye-piece consists of two lenses of equal focal length placed at a distance apart equal to two-thirds of the focal length of either lens. The object looked at requires to be a short distance in front of the field-lens. In the Huyghens or negative eye-piece the focal lengths of the field- and eye-lenses are in the ratio 3 to 1, and the distance between the lenses is twice the focal length of the eye-lens. As a result the

image formed by, say, the objective of a telescope is in focus when between the two lenses.

Eylau (ī'lou), a small town, about 28 miles distant from Königsberg, in Prussia, famous for a bloody battle fought between Napoleon and the allied Russians and Prussians, on the 7th and 8th of Feb., 1807. Both sides claimed the victory. The loss of the Allies was about 20,000 men, while that of the French must have been considerably greater.

Eyre (ār), Edward John, Australian explorer and colonial governor, born in Yorkshire 1815, died in 1901. He went to Australia in 1833, in 1839 discovered Lake Torrens, and in 1840 explored its eastern shores and the adjacent Flinders Range. He then commenced his perilous journey along the shores of the Great Australian Bight, and reached King George's Sound, in Western Australia, a distance of 1200 miles, with a single native boy, having left Adelaide more than a year before. In 1845 he published *Discoveries in Central Australia*. After filling several governorships, he was appointed Governor of Jamaica in 1862. In 1865 he was confronted with a negro rebellion, which he crushed with some severity, and was recalled. On his return to England John Stuart Mill and other so-called humanitarians took measures to try him for murder, but failed. Tennyson and Carlyle were among his most strenuous defenders.

Eyre, Lake, a large salt-water lake of South Australia. Area about 4000 sq. miles, but it is subject to great fluctuations in size.

Eze'kiel (Heb. *Yehezgēl*, 'God shall strengthen'), the third of the great prophets, a priest, and the son of Buzi. He was carried away when young (about 599 B.C.) into the Babylonian captivity. His prophetic career extended over a period of twenty-two years, from the fifth to the twenty-seventh year of the captivity. The *Book of Ezekiel* contains predictions made before the fall of Jerusalem, 586 B.C. (chaps. i-xxiv); prophecies against some of the neighbouring tribes (chaps. xxv-xxxii); prophecies concerning the future of Israel (xxxiii-xxxix); and a series of visions relating to the circumstances of the people after the restoration.

Ezra, a celebrated Jewish scribe and priest. Under his guidance the second expedition of the Jews set out from Babylon to Palestine in the reign of Artaxerxes I, about 458 B.C. The important services rendered by Ezra to his

countrymen on that occasion, and also in arranging, and in some measure, it is believed, settling the canon of Scripture, are specially acknowledged by the Jews, and he has even been regarded as the second founder of the nation. Josephus states that he died in Jerusalem; others assert that he returned to Babylon, and died there at the age of 120 years. *The Book of Ezra* contains an account of the favours bestowed upon the Jews by the Persian monarchs, the rebuilding of the temple, Ezra's mission to Jerusalem, and the various regulations and forms introduced by him. It is written partly in Hebrew and partly in Chaldee; this has led some to conclude that it is the work of different hands.

F

F, the sixth letter of the English alphabet, is a labio-dental articulation, formed by the passage of breath between the lower lip and the upper front teeth. It is classed as a surd spirant, its corresponding sonant spirant being *v*, which is distinguished from *f* by being pronounced with voice instead of breath, as may be perceived by pronouncing *ef*, *ev*. The figure of the letter **F** is the same as that of the ancient Greek digamma, which it also closely resembles in power. As a mediæval Roman numeral **F** stands for 40, and with a bar above it, it is 40,000. **F**, in music, is the fourth note of the diatonic scale.

Faam-tea, or **Faham-tea**, a name given to the dried leaves of the *Angræcum fragrans*, an orchid growing in the Mauritius and in India, and much prized for the fragrance of its leaves, an infusion of which is used as a stomachic and as an expectorant in pulmonary complaints.

Faber, Frederick William, D.D., a theologian and hymn-writer, the nephew of George Stanley Faber, born at Durham in 1814, died 26th Sept., 1863. In 1845 he became a convert to Roman Catholicism, and founded the oratory of St. Philip Neri, afterwards transferred to Brompton.

Fabian Society, a Socialist organization, founded in 1888, whose object, as defined by the basis which members are required to sign, is the nationalization of all land and industrial capital for the benefit of the whole community; this result to be attained, not by any violent upheaval, but by educating the minds of the masses and gradually extending the control of the State over the factors of production. Its policy, as expounded in *Fabian Essays* (1889) by a number of its early members, is frankly opportunist, and contemplates the use of existing political machinery and the acceptance of any measure of reform which will further the ultimate aims of the society. The name is derived from that of the Roman general, Quintus Fabius Maximus, known as Cunctator from the cautious tactics by which he ultimately defeated Hannibal. Prominent members of the society at different times have been Bernard Shaw, Sidney Webb, H. G. Wells, and J. A.

Hobson. The society has branches in Great Britain, the Colonies, and America, and has issued a number of publications, notably *Fabian Tracts*. A research department recently established has done useful work. The growth of the labour movement has rather diminished the importance of the society, and has led to some secessions from its ranks.

Fabii (fā'bi-ī), an ancient and renowned family of Rome, who, having undertaken the duty of defending Roman territory against the incursions of the Veientes, established themselves at a post on the River Cremera. Being drawn into an ambush, they were killed to a man (477 B.C.). A boy who happened to be left in Rome became the second founder of the family. Among its celebrated members in aftertimes was Fabius Maximus, whose policy of defensive warfare was so successful against Hannibal in the second Punic War (218-201 B.C.); and Quintus Fabius Pictor, who lived about the same time and wrote a history of Rome in Greek, thus being the earliest Roman historian.

Fable (Lat. *fabula*, narrative), in literature, a term applied originally to every imaginative tale, but confined in modern use to short stories, either in prose or verse, in which animals and sometimes inanimate things are feigned to act and speak with human interests and passions for the purpose of inculcating a moral lesson in a pleasant and pointed manner. The fable consists properly of two parts—the symbolical representation and the application, or the instruction intended to be deduced from it, which latter is called the *moral* of the tale, and must be apparent in the fable itself. The oldest fables are supposed to be the Oriental; among these the Indian fables of Pilpay or Bidpai, and the fables of the Arabian Lokman, are celebrated. Amongst the Greeks, Æsop is the master of a simple but very effective style of fable. The fables of Phædrus are a second-rate Latin version of those of Æsop. In modern times Gellert and Lessing among the Germans, Gay among the English, the Spanish Yriarte, the Italian Pignotti, and the Russian Ivan Krylov, are celebrated. The first place, however, amongst modern fabulists belongs to the French writer La Fontaine. R. L. Stevenson wrote a collection of fables.—Cf. Walter Jerrold, *The Big Book of Fables*.

Fabliaux (fab'li-ō; O.F. *fabliaus*, Lat. *fabella*, dim. of *fabula*, story), in French literature, the short metrical tales of the Trouvères, or early poets of the Langue d'Oil, composed for the most part in the twelfth and thirteenth

centuries. These productions were intended merely for recitation, not for singing, and had as their principal subjects the current gossip and news of the day, which were treated in a witty and sarcastic way. The fabliaux lashed not only the clergy and nobility in their degeneracy, but even mocked the religious chivalrous spirit, and the religious and knightly doctrines and ceremonies.

Fabre, Jean Henri, French entomologist, born at Sainte-Leone, Aveyron, in 1823, died 11th Oct., 1915. The son of very poor parents, he received a free education at Rodez, and then went to the normal school at Vaucluse, and at the age of eighteen he began his career as teacher. He was in charge of a primary school, and in his spare time studied mathematics and physics. He subsequently became professor of physics at the College of Ajaccio, and his interest in insects having in the meantime been aroused, he turned his entire attention to entomological pursuits. His reputation as a naturalist increased, and his work was praised by Darwin. He was particularly noted for the remarkable patience with which he investigated the life-history of insects, and for his minute and painstaking observations. His works first appeared in the *Annales des sciences naturelles* from 1855-8, and were afterwards amplified in his *Souvenirs Entomologiques* in 10 volumes, published between 1878 and 1907. He was made a chevalier of the Legion of Honour, and in 1912 the French Government granted him a pension. His works were translated into English by A. Texeira de Mattos.

Fabria'no, an episcopal city of Italy, province of Ancona. Pop. 23,750.

Fabricius, Gaius (with the cognomen Luscinius), an ancient Roman, celebrated on account of his fearlessness, integrity, moderation, and contempt of riches. After having conquered the Samnites and Lucanians, and enriched his country with the spoils, of which he alone took nothing, he was sent on an embassy to Pyrrhus, King of Epirus, who tried in vain to corrupt him by large presents. When consul in 279 B.C., Fabricius delivered up to Pyrrhus his treacherous physician, who had offered to poison his royal master for a sum of money. In gratitude for the service the king released the Roman prisoners without ransom. In 275 B.C. Fabricius was chosen censor. He died about 250 B.C.

Fabricius, Johann Albrecht, a German scholar, born at Leipzig in 1668, died in 1736. He became professor of rhetoric and moral philosophy at Hamburg, and published many learned works, amongst which are his *Bibliotheca Latina*, *Bibliotheca Ecclesiastica*, and *Bibliotheca Antiquaria*.

Fabricius, Johann Christian, German entomologist, born 1743, died 3rd March, 1808. After studying at Copenhagen, Leyden, Edinburgh, and under Linnæus at Upsala, he obtained the post of professor of natural history in the University of Kiel. In 1755 appeared his *Systema Entomologiæ*, which gave to this science an entirely new form. In 1778 he published his *Philosophia Entomologica*, written upon the plan of the well-known *Philosophia Botanica* of Linnæus.

Facciolati (fa't-cho-lă'tē), Jacopo, Italian classical scholar, born 1682, died 1769; professor in the University of Padua. The most important work with which he was connected was the *Totius Latinitātis Lexicon*, compiled by Forcellini under his direction and with his co-operation.

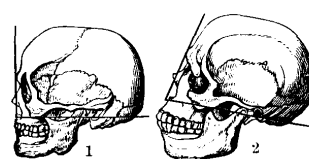
Face, the front part of the head, the seat of most of the sense-organs. The bony basis of the face, exclusive of the thirty-two teeth (these not being in the strict sense bones), is composed of fourteen bones, called, in anatomy, the *bones of the face*. The anterior part of the brain-case (frontal bone) also forms an important feature of the face. Of all these bones the lower jaw only is movable, being articulated with the base of the skull. The other bones are firmly joined together and incapable of motion. In most mammals the jaws project much more than in men, and form the prominent feature of the face, while the forehead recedes. See *Facial Angle*.

Face'tiæ, humorous sayings, witticisms, jests. There have been many collections of such. Amongst the most notable are the *Asteia* (Jests) of Hierocles, an old Greek collection, the *Liber Facetiarum* of Poggio Bracciolini, and Joe Miller's *Jest-Book*.

Facial Angle, an angle of importance in the method of skull measurement introduced by Camper, the Dutch anatomist, who sought to establish a connection between the magnitude of this angle and the intelligence of different animals and men, maintaining that it is always greater as the intellectual powers are greater. Suppose a straight line drawn at the base of the skull, posteriorly across the external orifice of the ear to the bottom of

the nose, and another straight line from the bottom of the nose, or from the roots of the upper incisors, to the most prominent part of the forehead, then both lines will form an angle which will be more or less acute. In apes this angle is only from 45° to 60°; in the skull of a negro, about 70°; in a European, from 75° to 85°. In another mode of drawing the lines the angle included between them varies in man from 90° to 120°, and is more capable of comparison among vertebrate animals than the angle of Camper. Though of some importance in the comparison of races, this angle is fallacious as a test of individual capacity.

Facial Nerve, a motor nerve which supplies the muscles of expression on either side of the face. Injury to this nerve produces *facial paralysis*, the result of which is that the affected side is smooth, unwrinkled, and motionless, the eyelids are wide open and cannot be closed, and the muscles of the sound side having it all their own way drag the mouth to that side.



Facial Angle.
1, European. 2, Negro.

Factor, in arithmetic, is any number which divides a given number without a remainder, thus 3, 5, 7 are all factors of 105. In algebra, any expressions multiplied together to form a product are said to be factors of the product; for example, $x + 1$, $x + 2$, $x + 3$ are factors of $x^3 + 6x^2 + 11x + 6$.

Factor, in commerce, an agent employed to do business for another in buying or selling, or in the charge of property. A factor differs from a broker in holding a wider and more discretionary commission from his employer, in being able to buy and sell in his own name, and in having a lien on goods for his outlay. The difference, however, depends so much upon the usage of the particular trade, or upon the special instructions constituting the agency, that no exact line of demarcation can really be drawn between them. The term factor has in common usage generally given place to the terms agent and broker, the former applied in the more general, the latter in the more restricted sense. It is still retained in some special cases, as in that of house-factors and factors on landed property in Scotland, who have charge of the letting and general management of house property, farms, &c.; called in England estate agents.

Fac'tory (from *factor*), a name which appears originally to have been given to establishments of merchants and factors resident in foreign countries; it now more commonly signifies a place in which the various processes of a particular manufacture are carried on simultaneously. The rapid growth of factories in this sense is a comparatively recent development of industry, resulting from the free use of machinery and the consequent subdivision of labour. Amongst the advantages of the factory system are generally counted: first, increased productiveness arising from the minute division of labour; second, the mechanical accuracy and the cheapness of the product turned out by machinery; third, the facilities for union and co-operation for common improvement afforded by bringing large masses of workmen together. But this last consideration is probably more than counter-balanced by the smaller amount of independent intelligence called forth in the individual worker, through the monotony of the minutely subdivided operations. Decided disadvantages of the factory system are the unhealthiness of the crowded rooms, where the air is full of deleterious elements; and the increasing demand on the labour of women and children, interfering as it does with the economy of domestic life.—BIBLIOGRAPHY: R. W. Cooke-Taylor, *Factory System and Factory Acts*; B. L. Hutchins, *A History of Factory Legislation*.

Factory Acts, Acts passed for the regulation of factories and similar establishments. Considering that women and children were not qualified fully to protect themselves against the strain of competition, the British legislature has passed a series of Acts to regulate the conditions of their employment in factories. The immediate occasion of the first Act passed to regulate factory employment in England was the outbreak of an epidemic disease which committed great havoc among the younger persons employed in factories in the district round Manchester at the beginning of the nineteenth century. An Act was passed (1802) in which provision was made for the regular cleansing and ventilation of mills and factories, and also for limiting the hours of labour to twelve daily. In 1819 an Act followed which prescribed an hour and a half for meals in the course of a working day, and prohibited children under nine years of age being employed in factory work at all. Various Acts were passed up to 1878, when a general Factory and Workshop Act was passed, consolidating the previous series of statutes. Its scope was extended by a further series of enactments, until in the year 1901

the last general Act was passed, which consolidates and amends all previous legislation. The Act contains general provisions regarding drainage, sanitary conveniences, overcrowding, ventilation, fencing of dangerous machinery, &c. Factories are distinguished from workshops as making use of steam or other mechanical power. In textile factories the hours of labour for women and young persons (the latter between 14 and 18 years of age) are restricted to 10, but only 6½ on Saturday and 56 in the week. In non-textile factories and workshops the hours may be 10½ per day and 60 per week at most. Children (of 12 to 14 years) were not allowed to be employed more than 6½ hours on any one day. (The Education Acts now prohibit almost entirely the employment of 'school children' in factories and workshops.) Provision is made for a certain number of annual holidays. Special provisions for particular kinds of factories are made, and under these the employment of females and young persons is regulated in bleaching- and dyeing-works, lace-factories, manufactories of earthenware, lucifer matches, percussion caps, cartridges, blast-furnaces, copper-mills, forges, foundries, manufactories of machinery, metal, india-rubber, gutta-percha, paper, glass, tobacco, letterpress printing, bookbinding, &c. The Act of 1901 included laundries carried on by way of trade or for the purposes of gain. An Act of 1907 extended the Act of 1901 to laundries carried on as ancillary to another business or incidentally to the purposes of any public institution. A short Act passed in 1911 gave power to make regulations applicable to cotton-cloth factories. Certain exceptions in regard to working over-time are provided for; thus women may sometimes work 14 hours a day. So far there has been no direct interference in any of the Factory Acts with the labour of adult male persons; but it is obvious that indirectly the position of the male labourer also is affected by legislation of this sort.

Fac'ulæ (Lat. *facula*, a torch), bright markings on the sun's disc, i.e. portions more brilliant than the general surface. They are supposed to be parts of the luminous surface, or photosphere, which are elevated to a greater height, and therefore suffer less absorption of their light in its passage through the overlying gases and vapours. Like the spots, they are in a state of constant change, and exhibit a similar periodicity in numbers and extent.

Faculties, Court of, in English law, a jurisdiction or tribunal belonging to the archbishop. It does not hold pleas in any suits, but has power to grant

licences or dispensations, such as, to marry without banns, or to remove bodies previously buried.

Faculty, the members, taken collectively, of the medical or legal professions; thus we speak of the medical faculty, and (in Scotland) of the faculty of advocates. The term is also used for the professors and teachers collectively of the several departments in a university; as, the faculty of arts, of theology, of medicine, or of law.

Faculty, in law, is a power to do something, the right to do which the law admits, or a special privilege granted by law to do something which would otherwise be forbidden.

Fæces, the excrementitious part evacuated by animals. It varies, of course, with different species of animals, according to their diet. The main constituents are unassimilable parts of the food, on which the digestive process has no effect, and other portions, quite nutritious, but which have escaped digestion, also certain waste matters, &c. In disease the composition varies extremely.

Faed (fād), John, R.S.A., artist, born in Kirkcudbrightshire in 1820, died in 1902. He showed artistic talent at an early age, and in 1841 went to Edinburgh to study. Some years later he acquired a considerable reputation, and was elected to the Royal Scottish Academy in 1851. Among his principal works are: *Shakespeare and his Contemporaries*; *An Incident of Scottish Justice*; *The Morning after Flodden*; *A Wappenshaw*; two series of drawings illustrating *The Cotter's Saturday Night* and *The Soldier's Return*; *John Anderson, My Jo*; *Auld Mare Maggie*; *The Gamekeeper's Daughter*; *The Hiring Fair*.

Faed, Thomas, R.A., younger brother of the preceding, born at the same place in 1826, died in 1900. He studied in Edinburgh, where at an early age he became known as a clever painter of rustic subjects. In 1852 he settled in London, where he won a high reputation. The subjects he painted are for the most part domestic or pathetic, and in these he invented and told his own story, and that with a success that emulates Wilkie. Among his principal works are: *Sir Walter Scott and his Friends* (1849), *The Mitherless Bairn* (1855), *The First Break in the Family* (1857), *Sunday in the Backwoods* (1859), *His Only Pair* (1860), *From Dawn to Sunset* (1861), *The Last o' the*

Clan (1865). A number of Faed's works have been engraved in large size, and have been very popular.

Faenza (fa'-en'za), an episcopal city of N. Italy, in the province of and 19 miles south-west of Ravenna. It is supposed to have been the first Italian city in which the earthenware called *faience* (q.v.) was introduced. The manufacture still flourishes here, and there is also a considerable trade in spinning and weaving silk. Pop. (commune), 40,164.

Fagaceæ, a nat. ord. of apetalous Dicotyledons, all trees and shrubs, mostly natives of temperate regions. It includes the beeches and oaks, and the sweet chestnut (*Castanea*).

Fagging, a custom which formerly prevailed generally at most of the English schools, and is still practised at Eton, Winchester, Harrow, Rugby, and one or two other places. It consists in making the junior boys act as servants or 'fags' in the performance of multifarious menial offices for elder boys, such as carrying messages or preparing breakfast for their master, in return for which the elder boy accepts a certain responsibility for keeping order, and becomes the recognized adviser and protector of his 'fags'.

Faggot-vote, a term formerly applied in Britain to a vote procured by the purchase of property so as to constitute a nominal qualification without a substantial basis. Faggot-votes were chiefly used in county elections for members of Parliament. The way in which they were usually manufactured was by the purchase of a property which was divided into as many lots as would constitute separate votes, and given to different persons, who need not be resident members of the constituency. The practice disappeared after the Reform Act of 1884.

Faguet, Émile, French literary historian, critic, and journalist, born at La Roche-sur-Yon in 1847, died in 1916. He became professor of poetry at the Sorbonne in 1897. Endowed with a keen power of analysis and a wealth of original ideas, he was one of the most brilliant French critics of the nineteenth century. Whilst praising the literature of the seventeenth century, Faguet somewhat depreciated the writers of the eighteenth century. Among his numerous works are: *Le théâtre contemporain* (1880-91), *Dix-huitième siècle* (1890), *Seizième siècle* (1893), *Drame ancien et drame moderne*

(1898), *Histoire de la littérature française* (1900), *Propos littéraires* (1902), *Initiation into Literature*, and *Initiation into Philosophy*.

Fagus. See *Beech*.

Fahlerz (fāl'erts), or grey copper ore, is of a steel-grey or iron-black colour. It occurs crystallized in the form of the tetrahedron, also massive and disseminated. Its fracture is uneven or imperfectly conchoidal. Specific gravity, 4.5-5.1. *Tetrahedrite*, the typical species, is composed of copper, sulphur, and antimony. Part of the copper is often replaced by iron, zinc, silver, or mercury, and part of the antimony by arsenic.

Fahlunite, a mineral of a greenish colour occurring in six-sided prisms. It is a pseudomorph after iolite, and consists mainly of hydrous aluminium silicate. It takes its name from Fahlun in Sweden.

Fahrenheit (fä'rèn-hīt), Gabriel Daniel, German physicist, known for his arrangement of the thermometer, was born at Danzig in 1686, died in 1736. Abandoning the commercial profession for which he had been designed, he settled in Holland to study natural philosophy. In 1720 he effected a great improvement by the use of quicksilver instead of spirits of wine in thermometers. He invented the Fahrenheit scale (see *Thermometer*), and made several valuable discoveries in physics. He was elected a fellow of the Royal Society of London in 1724.

Faidherbe (fā-derb), Louis Léon César, a French general, born in 1818, died in 1889. He entered the army in 1840, served in Africa and the West Indies, was appointed Governor of Senegal in 1854, and afterwards of a district in Algiers from 1867 to 1870. After the fall of Napoleon III, he was summoned by the Government of the National Defence to France and appointed commander of the army of the north. He fought some bloody but indecisive battles with the Germans under Manteuffel and Goeben. After the war he was elected to the Assembly by Lille, his native place, but on the triumph of Thiers retired from politics to private life. He wrote *Épigraphie Phénicienne*, and valuable monographs on Senegal, the Sudan, and other parts of Africa.

Faience (fa'-yens'), imitation porcelain, a kind of fine pottery, superior to the common pottery in its glazing, beauty of form, and richness of painting.

Several kinds of faience are distinguished by critics. It derived its name from the town of Faenza, in Italy, where a fine sort of pottery called *majolica* was manufactured as early as the fourteenth century. The majolica reached its greatest perfection between 1530 and 1560. In the Louvre, the Musée de Cluny, the British and Victoria and Albert Museums, at Berlin, and at Dresden are rich collections of it. The modern faience appears to have been invented about the middle of the sixteenth century, at Faenza, as an imitation of majolica, and obtained its name in France, where a man from Faenza, having discovered a similar kind of clay at Nevers, had introduced the manufacture of it. True faience is made of a yellowish or ruddy earth, covered with an enamel which is usually white, but may be coloured. This enamel is a glass rendered opaque by oxide of tin or other suitable material, and is intended not only to glaze the body, but to conceal it entirely. See *Pottery*.—Cf. M. L. Solon, *The Old French Faience*.

Failly (fa`-yē), Pierre Louis Charles Achille de, French general, born in 1810, died in 1892. He distinguished himself in the Crimean War, and commanded a division against the Austrians in 1859. He was the means of introducing the Chassepôt rifle into the French army, and commanded the troops which dispersed Garibaldi's irregulars at Mentana. At the outbreak of the Franco-German War Failly received the command of the 5th Corps, but was very unfortunate or unskilful in his organization of operations. His masterly inactivity in the early weeks of the war caused great popular indignation in France. Sedan ended his career as a soldier.

Failsworth, a town of England, in Lancashire, 4 miles north-east of Manchester, with cotton-mills. Silk-weaving and hat-making are also carried on. Pop. 16,972.

Fainéants (fā-nā-a`n` ; Fr., 'do-nothings'), a sarcastic epithet applied to the later Merovingian kings of France, who were puppets in the hands of the mayors of the palace. Louis V, the last of the Carolingian dynasty, received the same designation.

Fainting, or **Syncope**, a sudden suspension of the heart's action, of sensation, and the power of motion. It may be produced by loss of blood, pain, emotional disturbance, or organic or other diseases of the heart. It is to be treated by placing the patient on his back in a recumbent position or even

with head slightly depressed, sprinkling cold water on his face, applying stimulant scents to the nostrils, or anything which tends to bring back the blood to the brain. The admission of fresh cool air and the loosening of any tight articles of dress are important.

Fairbairn, Patrick, Scottish theologian, born 1805, died 1874. He became a minister of the Established Church, but joined the Free Church at the disruption in 1843. In 1853 he was appointed professor of divinity in the Free Church College, Aberdeen, and in 1856 principal of the Free Church College, Glasgow. Among his works are: *Typology of Scripture*; *Jonah: his Life, Character, and Mission*; *Ezekiel; Prophecy*; *Hermeneutical Manual*; *Pastoral Epistles of St. Paul*. He edited and wrote extensively for the *Imperial Bible Dictionary*.

Fairbairn, Sir William, British civil engineer, born at Kelso, Roxburghshire, in 1789, died 18th Aug., 1874. He was apprenticed as an engine-wright at a colliery in North Shields, and commenced business on his own account in Manchester with James Lillie in 1817, where he made many improvements in machinery, such as the use of iron instead of wood in the shafting of cotton-mills. About 1831, his attention having been attracted to the use of iron as a material for shipbuilding, he built the first iron ship. His firm became extensively employed in iron shipbuilding at Manchester and at Millwall, London, and had a great share in the development of the trade. He shares with Stephenson the merit of constructing the great tubular bridge across the Menai Strait. Fairbairn was one of the earliest members of the British Association for the Advancement of Science, of which he was president in 1861 and 1862. He was created a baronet in 1869. Sir William wrote many valuable professional books and papers, amongst which are: *On Canal Steam Navigation* (1831); *Iron: its History, Properties, and Manufacture* (1841); *Application of Iron to Building Purposes* (1854); *Iron Ship-building* (1865). His brother Sir Peter, born 1799, died 1861, had also great mechanical ability, and founded large machine-works at Leeds.

Fairfax, Edward, the translator into English verse of Tasso's *Jerusalem Delivered*, was the natural son of Sir Thomas Fairfax of Denton, and born in the last quarter of the sixteenth century. He settled at Newhall, in the parish of Fewston, Yorkshire, to a life of studious leisure. The first edition of his

translation bears the date of 1600. One or two eclogues by him also remain. He died in 1635.

Fairfax, Thomas, Lord, parliamentary general during the English Civil War, born in 1611 at Denton, in Yorkshire, died at Nun Appleton, Yorkshire, 12th Nov., 1671. He was the son and heir of Ferdinando, Lord Fairfax, to whose title and estates he succeeded in 1648. After serving in the Netherlands with some reputation, he returned to England, and on the rupture between Charles I and the Parliament joined the forces of the latter. In 1642 he was appointed General of the Horse, and two years later held a chief command in the army sent to co-operate with the Scots. In 1645, on the resignation of the Earl of Essex, Fairfax became general-in-chief of the Parliamentary army. After the victory at Naseby he marched into the western counties, quelling all opposition, put down the insurgents in Kent and Essex in 1647, and captured Colchester. In April, 1649, he was occupied along with Cromwell in suppressing revolt in the army; but positively declined to march against the Scottish Presbyterians. He was a member of Cromwell's first Parliament. He co-operated in the restoration of Charles II, being one of the committee charged to secure his return.

Fairford, a town in Gloucestershire, England, 8 miles east by south of Cirencester, with a church the twenty-eight windows of which are filled with beautiful stained glass, formerly ascribed to Albert Dürer, but now known to have been designed and executed in England. Fairford was the birth-place of John Keble. Pop. 1410.

Fair Head, a basaltic promontory on the north coast of Ireland, County Antrim, rising to the height of 636 feet.

Fairies and Elves. The fairies of folk-belief must be distinguished from the fairies of imaginative literature. Shakespeare, for instance, drew upon the fairy-lore of living tradition to create a new fairy mythology (as in *A Midsummer Night's Dream*) which became a literary convention. In the fairy stories of Hans Andersen the folk-material was similarly used in a free and individual manner. A distinction must likewise be drawn between the Celtic fairies and the Teutonic elves. The former are mainly females, like the nymphs of Homer, ruled over by a fairy queen, while the latter are mainly males, ruled over by an elf-king. Mab, the fairy queen, had no

consort in British fairy literature until Oberon, King of the Fairies, was imported from mediæval romance. His name 'Auberon', anciently 'Alberon', is identical with that of the German elfin king 'Alberich'. Indeed, the very names 'fairy' and 'fay' were introduced into these islands from abroad. More than one class of supernatural beings referred to in Gaelic as the *side* (Irish) or *sith* (Scottish), and pronounced *shee*, are now called 'fairies'. These include the Danann deities of Irish mythology and 'the mothers' (Y Mamau) of Welsh folk-lore. The word *side* or *sith* has the secondary meaning of 'peace', and refers to the silence of death and the silence of fairy movements. It may also be translated as 'supernatural', 'Otherworld', or 'unearthly'. Mysterious diseases that come in epidemics and afterwards disappear are referred to as diseases of the side or sith. Cat demons are *cait shith*, the cuckoo is *eun sith*, the mythical 'water horse' is *each sith*, a monstrous dog that passes over land and sea by night is *cu sith*, while the 'will-o'-the-wisp' is *teine sith* ('supernatural fire'). In Iceland *side* refers to the dwellings (earth mounds, &c.) of the Dananns, &c., as well as to the supernatural inhabitants. The fairies of folk-belief always come from the west on eddies of wind, and cannot be seen except by those who have 'second sight', or those whose eyes have been anointed with a green balsam possessed by fairies. Sometimes the fairies render themselves visible to all, but one who grasps the garment of a fairy finds his hand closing on nothing. The usual height of fairies is about 3 feet, but they have power to shrink and pass through a crack in a door. They may also assume great stature. The Danann *side* of Ireland are of human or above human height. In Scotland 'green ladies' are of ordinary human size. The chief fairy colours are blue (the eyes), golden (the hair), and green, red, and grey (for clothing). Occasionally fairy beings are white and black. A black fairy with a red spot above the heart is referred to in Scottish stories, but is rare. He can be slain by piercing the red spot. The *side* or *sith* may be attired entirely in green with red caps, or have red cloaks and green skirts. A beautiful fairy queen may suddenly transform herself into an ugly old hag with black and white face and garments, as did the fairy who carried off Thomas the Rhymer to the Underworld. The dead were supposed to go to Fairyland, the Pagan Paradise. Those who died before their time were doomed to visit their former haunts as 'green ladies', i.e. green ghosts, until their measure of life was completed. Stories that tell of visions of the dead in the Underworld refer to them feasting and dancing, or reaping corn and plucking fruit in

well-watered valleys. The resemblance of the Celtic Agricultural Paradise to the Otherworld of the Egyptian Osiris, which was originally situated under the ground, is of special interest. Both in the Underworld Paradise and on the 'Isles of the Blest' (the Celtic *Avalon* or 'apple land' and 'Land of the Ever-Young') is a tree of life, which may be an apple tree, a hazel tree, or a rowan tree. The apples, nuts, or berries confer longevity on the gods and the souls of human beings that partake of them. On those human beings who have won their favour, the fairies bestow weapons, implements, musical instruments, songs, tunes, and medicines, and the power to work charms and foretell future events. In the Underworld, fairies engage in metal working and other industries. Sometimes they visit houses, and spin and weave with supernatural skill and speed in a single night big bales of clothing material, or make beautiful garments. Fairies possess gems, gold, silver, and copper in their underground dwellings. Cornish miners hear them working in their mines. The 'banshee' (Ir. *ben-side*) is a Fate who is seen washing the blood-stained clothing, or the 'death clothes', of those who are doomed to die a sudden death. She either howls, or sings a weird song, or can be heard 'knocking' as she strikes the clothing with a beetle during the washing, when a tragedy is at hand. Fairy women of great beauty have human lovers, but vanish for ever after a few meetings, with the result that their lovers become demented. Fairy men (*fer-side*) likewise upset the minds of girls. The fairies abduct human children, leaving 'changelings' in cradles, or carry off wives to act as 'wet nurses' or midwives. Men who die suddenly are supposed to be transported to Fairyland. King Arthur, the Rev. Mr. Kirk of Aberfoyle, Thomas the Rhymer, and others were removed to the Fairy Paradise. Among the Celtic *side* are *pixies*, *geniti-glinni* (valley genii), *Bocanachs* (male goblins), *Bananachs* (female goblins), *Demna aeir* (spirits of the air), &c. Fairies may appear in animal forms, chiefly as beautiful birds. Elves are workers in metals, like Wayland Smith, and guardians of treasure who assume the forms of fish, otters, serpents, &c. Black elves dwell under the ground, and white elves haunt the air and the sea. Sea-elves are 'nikkers'. The Greek 'Fates', like the Celtic fairies, spin, weave, and embroider wedding and other garments in a single night for those they favour, and sometimes appear in groups of three, as old hags, to foretell tragic events or work spells. Celtic 'women of the *side*' sometimes appear in groups of three. The nereids are, like the fairy ladies, beautiful and capricious, and are likewise invariably blue eyed and golden haired. 'Nereid

born' refers to the changeling idea. Nereids travel on whirlwinds, and, like the Celtic fairies, cause spinning spirals of dust on highways. They confer gifts on mortals, and accept offerings of food. Lacon the poet sings, "I will set a great bowl of white milk for the nymphs". In the Scottish Highlands the milk offering was poured on the ground for those 'under the earth', or into a hollowed stone (*clach-na-gruagaich*). The Indian 'nagas' have power to change from serpent to human form, or to appear as half-human, half-reptile beings. Like the Celtic fairies, they have been referred to by some writers as aborigines who hid from invaders in earth-houses, in forests, and among the hills. In this connection P. C. Roy, the translator of the *Mahābhārata*, writes: "Nagas are semi-divine and can move through air and water and ascend to high heaven itself when they like, and have their home at Patala (the Underworld). To take them for some non-Aryan race, as has become the fashion with some ... is the very height of absurdity.... None of these writers, however, is acquainted with Sanskrit, and that is their best excuse." The fairies and elves of China and Japan resemble those of Europe. In Polynesia there are fairy-like beings. They are called *Patupaiarehe*, and dwell in lonely places, appearing only at night. Human beings receive gifts from them, or knowledge of how to make nets, weapons, &c. The changeling idea is as prevalent as in Europe. It is of special interest to find that the Polynesian fairies have, like the Celtic, fair hair and white skins. Other peoples believe in the existence of fairy-like beings. See *Folklore*.—BIBLIOGRAPHY: T. Keightley, *Fairy Mythology*; E. S. Hartland, *Science of Fairy Tales*; Delattre, *English Fairy Poetry from the Origins to the Seventeenth Century*; H. A. Giles, *China and the Chinese*; J. G. Lawson, *Modern Greek Folklore and Ancient Greek Religion*; Sir George Grey, *Polynesian Mythology*; P. W. Joyce, *Social History of Ancient Ireland*.

Fair Isle, an island lying nearly midway between the Orkney and Shetland Islands, 3 miles long by 2 broad. It is inaccessible except at one point, and rises to the height of 711 feet. Some grain is grown, but the surface is mostly in pasture. The men engage in fishing; the women knit a well-known variety of hosiery, introduced, it is said, by Spaniards who escaped from the wreck of one of the vessels belonging to the Spanish Armada. There are two lighthouses. Pop. 147.

Fair Oaks, Battle of, fought at Fair Oaks, in Virginia, 7 miles east of Richmond, between the Confederates under General Johnston and the Union troops under General M'Clellan, 31st May, 1862. The loss on each side was nearly 6000 men; the result was indecisive.

Fairs (Lat. *feria*, holiday, connected with *festus*, feast), periodical meetings of persons having goods or wares for sale in an open market held at a particular place, and generally for the transaction of a particular class of business. The origin of fairs is obviously to be traced to the convenience of bringing together at stated times the buyers and sellers of the stock-produce of a district. In Europe the numerous festivals of the Church afforded the most favourable opportunity for the establishment of these markets. This association is indicated in the German name of a fair, which is identical with that used for the ceremony of the mass. In the Middle Ages fairs were of great importance, and were specially privileged and chartered by princes and magistrates, public proclamation being made of their commencement and duration. But modern facilities of communication have much diminished the necessity for periodical markets, and it is now chiefly amongst agriculturists that they are of much importance, large agricultural meetings being held in various districts for the sale of cattle and horses, and for the exhibition of agricultural implements. There are also, especially in Scotland, a considerable number of hiring fairs for farm servants. In the less developed commerce of the East, however, they still retain much of their ancient importance and magnitude. In Europe the most important fairs of the present day are those at Leipzig and Frankfort-on-the-Main in Germany, at Lyons in France, and Nijni-Novgorod in Russia. The last is, indeed, the largest fair in the world. The fairs of Great Britain now mostly consist of the weekly market-days of country towns and the agricultural meetings already mentioned. In many places the old fair-days are still kept, but are now merely an assemblage of penny-theatres, peep-shows, and such amusements. Amongst the fairs which were once celebrated saturnalia, or rather bacchanalia, may be mentioned Donnybrook Fair in the county of Dublin; Bartholomew and Greenwich Fairs, London; and Glasgow Fair. The first three are now extinct. Fairs in the sense of markets are unknown in the United States, but the term is usually given to ladies' fancy bazaars, collections of fine art or the higher industries for public exhibition.—Cf. C. Walford, *Fairs, Past and Present*.

Fair Trade, an economical policy advocated by many in Britain, which, while not opposed to free trade in principle, would meet the prohibitory tariffs that foreign countries may put on British goods by placing equally heavy duties on goods sent from these countries to Britain. See *Free Trade*.

Fairweather, Mount, on the west coast of North America, in Alaska territory. It rises to the height of 14,900 feet, and is covered with perpetual snow.

Fairy Ring, a circle, or part of a circle of grass, of a darker colour and more luxuriant growth than the surrounding herbage, superstitiously associated with fairy revels. Actually it is due to the growth of a subterranean fungus-mycelium, which gradually spreads outwards from a central point of origin, the older parts dying and serving as manure for the grass, which appears even more vigorous than it really is by contrast with that on the outermost edge of the ring, where the living mycelium has a bad effect upon the grass-roots. The commonest fairy-ring fungus is *Marasmius oreades*, the fairy-ring champignon.

Faith, the assent of the mind to the truth of what is declared by another, resting on his authority and veracity, either without other evidence or on probable evidence of any kind. In a special sense the term faith is used for the assent of the mind to what is given forth as a revelation of man's relation to God and the infinite, i.e. a religious faith. In Christian theology we have: first, *historical* or *speculative faith*, or belief in the historic truthfulness of the Scripture narrative and the claims of Scripture to an inspired and supernatural origin; second, *evangelical* or *saving faith*, that emotion of the mind (as Dwight defines it) which is called trust, or confidence exercised towards the moral character of God, and particularly of the Saviour.—Cf. W. R. Inge, *Faith and its Psychology*.

Faith-healing. The tenets of the Peculiar People and of other believers in healing by faith differ from the views of Christian Scientists in this respect: that, while the latter hold pain and disease to be illusions of the imagination, the faith-healer admits their existence, but affirms the possibility of their removal by non-scientific means. Some make use of anointing with oil, while others hold prayer and the laying-on of hands to be the only requisites. Faith-healing traces its source to the raising of the apparently

dead, the curing of the sick, the restoration of sight to the blind, and other recorded miracles of Christ; thence through the miracles of the disciples and their successors, down to the performances of Dorothy Trudel in Switzerland, and the displays of Dowie in London (1904). Faith-healers flourish extensively in Sweden and America, while even in England 'Bethshans' (houses of cure) have been established. Faith, or some may say credulity, attributes the alleged cures to supernatural agency; science sees in them the action of 'suggestion', with an exalted and emotional state of mind in the patient, more especially when surrounded and encouraged by a crowd of expectant and credulous lookers-on. Excessive reliance on his or her own powers has not seldom brought a faith-healer within the menace of the law, owing to neglect to employ a qualified medical man, and the consequent death of the sufferer. In a broad sense of the term the belief in healing by faith has been at the root of 'touching' for the King's Evil, a practice followed by several English and French sovereigns (see *Macbeth*, iv, 3, 141); of the value placed on relics; of alleged cures effected at such 'holy places' as St. Winifred's Well in Wales, and the miraculous grotto at Lourdes; and of the firm belief, not yet entirely extinct, in many rustic remedies, such as the removal of children's ailments by immuring a live shrew in a cleft ash tree.—BIBLIOGRAPHY: F. Podmore, *Mesmerism and Christian Science*; G. B. Cutten, *Three Thousand Years of Mental Healing*.

Fakirs (fa' -kērz'; literally 'poor men'), a kind of fanatic met with chiefly in India and the neighbouring countries, who retire from the world and give themselves up to contemplation. They are properly of the Mohammedan religion, but the term is often used for a mendicant of any faith. They are found both living in communities and solitary. The wandering fakirs gain the veneration of the lower classes by absurd penances and self-mutilations.

Falaise (fa' -lāz), a town, France, department of Calvados, picturesquely situated on a rocky precipice (Fr. *falaise*) 23 miles S.S.E. of Caen. It contains several objects of interest, among others the ruined castle of the Dukes of Normandy, where William the Conqueror was born. Pop. 6850.

Falashas, inhabitants of Amhara, in Abyssinia, who claim descent from Jewish emigrants during the reign of Jereboam. See *Abyssinia*.

Falckenstein, Edward Vogel von, a Prussian general, born in 1797, died in 1885. In 1813 he entered the Prussian army, distinguishing himself at the battles of Katzbach and Montmirail. In 1848 he served in the Holstein campaign, and he acted as colonel and chief of staff in the war with Denmark in 1864. In the war of 1866 he commanded the Seventh Army Corps. On the outbreak of the Franco-German War in 1870 he was appointed military governor of the maritime provinces.

Falcon (fa''kn), a name of various birds of prey, members of the family Falconidæ. The falcons proper (genus *Falco*), for strength, symmetry, and powers of flight are the most perfectly developed of the feathered race. They are distinguished by having the beak curved from the base, hooked at the point, the upper mandible with a notch or tooth on its cutting edge on either side, wings long and powerful, the second feather rather the longest, legs short and strong. The largest European falcons are the jerfalcon or gyrfalcon proper (*Falco gyrfalco*), a native of the Scandinavian Peninsula, and the Iceland falcon (*F. islandus*); to which may be also added the Greenland



Greenland Falcon (*Falco candicans*)

falcon (*F. candicans*). Between these three species much confusion at one time prevailed, but they are now distinctly defined and described. These three Arctic falcons are often referred to the special genus *Hierofalco*. In the Greenland falcon the prevailing colour at all ages is white, in the Iceland falcon dark. The latter more nearly resembles the true gyrfalcon of Norway, which, however, is generally darker, rather smaller, but with a longer tail. The average length of any of these falcons is about 2 feet. The Greenland species used to be the most highly prized by falconers. Its food consists chiefly of ptarmigans, hares, and water-fowl. It is found over a wide range of northern territory. The peregrine falcon (*F. peregrinus*) is not so large as the jerfalcon, but more graceful in shape. It chiefly inhabits wild districts, and nestles among rocks. It preys on grouse, partridges, ptarmigans, pigeons, rabbits, &c. Its flight is exceedingly swift, said to be as much as 150 miles an hour. The peregrine falcon was one of those most frequently used in falconry. Other British falcons are the hobby (*Hypotriorchis subbuteo*), formerly a great favourite for the chase of small game when

falconry was in fashion; the merlin or stone-falcon (*Æsalon regulus*), small but swift and spirited; the kestrel or wind-hover (*Tinnunculus alaudarius*), one of the most common British falcons. The term falcon is by sportsmen restricted to the female, the male, which is smaller and less courageous, being called *tiercel*, *tersel*, *tercelet*, or *falconet*. See *Falconry*.

Falco'ne, Aniello, Italian painter, born in 1600, studied along with Salvator Rosa under Spagnoletto. His paintings, consisting chiefly of battle-pieces, are masterpieces, but very rare. He died in 1665.

Falconer (fa'k'nér), Hugh, Scottish naturalist, born in 1808, died, 1865. After having graduated in arts at Aberdeen and medicine at Edinburgh, he went to India as a surgeon in 1830. Here he made valuable geological researches, and turned his attention to the introduction of tea cultivation. In 1837 he accompanied Barnes's second mission to Cabul. He visited England in 1843 and published an illustrated descriptive work entitled *Fauna Antiqua Sivalensis* (Ancient Fauna of the Siválik Hills). He returned to India in 1848, where he had been appointed superintendent of the botanic gardens at Calcutta. In 1855 he returned to England, where he died.

Falconer, William, poet and writer on naval affairs, born at Edinburgh in 1732. He went to sea in the merchant service, was wrecked, and wrote a poem (*The Shipwreck*) descriptive of the incidents, published in 1762. He then entered the navy, and was rated as midshipman on board the *Royal George*. In 1769 he published a *Universal Marine Dictionary*. The same year he sailed for Bengal as purser of the *Aurora* frigate, which is believed to have foundered at sea.

Falcon'idæ, a family of diurnal birds of prey, in which the destructive powers are most perfectly developed. The family includes the different species of eagles as well as the hawks and falcons properly so called, and comprises the sub-families Gypaëtinae (lammergeiers), Polyborinae (carrion hawks), Accipitrinae (hawks and harriers), Aquilinae (eagles), Buteoninae (kites and buzzards), and Falconinae (falcons).

Falconry (fa'kn-ri), also called *hawking*, the pursuit of game by means of trained hawks or falcons. Falconry is a sport of great antiquity in Asia, having been followed in China as early as 2000 B.C. In Europe it was, during the Middle Ages, the favourite amusement of princes and nobles,

and, as ladies could take part in it, became very general. Charlemagne passed laws in regard to falconry, while in Germany Henry the Fowler and the Emperor Frederick the Second were greatly addicted to it, and the latter wrote a work on the subject. In France it reached its greatest popularity under Francis I, whose grand falconer controlled an establishment of fifteen nobles and fifty falconers, at an annual cost of about 40,000 livres. In Britain the sport was practised before the Norman Conquest, but became still more popular after it, and till about 1650 enjoyed the prominence now held by fox-hunting. One of the most interesting of English works on the subject is that which forms the first part of the *Boke of St. Albans*, first printed in 1481. George Turberville's *Booke of Faulconrie or Hawking* (1575), and Simon Latham's *The Faulcon's Lure and Cure* (1633), may also be mentioned. Though the invention of fire-arms gradually superseded this amusement, it is not yet entirely extinct. The Duke of St. Albans is still hereditary grand falconer, and presents the king with a *cast* (or pair) of falcons on the day of his coronation. In Persia and other Eastern countries hawking is still in favour. The game hunted includes hares and rabbits, and, in the East, gazelles; with herons, wild geese, and many smaller birds. The training of a hawk is a work requiring great patience and skill, the natural wildness and intractable nature of the birds being very difficult to overcome. When a hawk suffers itself to be hooded and unhooded quietly, and will come to the trainer's hand to receive food, its education is considered far advanced, and the work of accustoming it to the *lure* may be proceeded with. The *lure* may be a piece of leather or wood, covered with the wings and feathers of a bird, and with a cord attached. The falcon is fed from it, and is recalled from flight by the falconer swinging the lure round his head with a peculiar cry. When the bird has been taught to obey the lure, it is next practised in the art of seizing its game, being initiated with prey fastened to a peg, and flown later at free game. When fully trained and being used for sport, the falcon is kept hooded until actually required to fly. Among the many technical terms connected with falconry may be mentioned that of *mew* (= moult), from which is derived the familiar name *mews*, originally places where hawks were kept while moulting.—BIBLIOGRAPHY: J. E. Harting, *Bibliotheca accipitraria*; Salvin and Broderick, *Falconry in the British Isles*; E. B.



Falcon with
Hood

Michell, *The Art and Practice of Hawking*; H. Cox, C. Richardson, and G. Lascelles, *Coursing and Falconry* (The Badminton Library).

Fald'stool (O.H.Ger. *falden*, to fold, and *stol*, chair), a folding stool provided with a cushion for a person to kneel on during the performance of certain acts of devotion, especially a kind of stool placed at the south side of the altar, on which the Kings of England kneel at their coronation. The term is also given to a small desk at which the litany is enjoined to be sung or said.

Faler'nian Wine, an ancient wine of great repute amongst the Romans. It was made from the grapes grown on Mount Falernus in Campania. It was strong and generous, probably much resembling modern sherry.

Falie'ro, Marino, Doge of Venice, born in 1274, commanded the troops of the republic at the siege of Zara in Dalmatia, where he gained a brilliant victory over the King of Hungary. He succeeded Andrea Dandolo, 11th Oct., 1354, was accused of the design of overthrowing the republic and making himself sovereign of the state, and beheaded 17th April, 1355. The last scenes of his life are depicted in Byron's tragedy of *Marino Faliero*.

Fal'kirk, a parliamentary burgh of Scotland, in Stirlingshire, 21½ miles west by north of Edinburgh. The older portion of it is old-fashioned and irregularly built. There are several modern suburbs. In the town or its vicinity are the Carron Ironworks, the Falkirk Foundry, and others works, collieries, chemical-works, and distilleries. Falkirk is connected with the port of Grangemouth by a railway 3 miles long. The Trysts of Falkirk, held on Stenhousemuir, 3 miles to the N.N.W., are the largest cattle-fairs in Scotland. Falkirk is an old town, with many historical associations. In the neighbourhood was fought the battle of Falkirk in 1298 between Sir William Wallace and Edward I, the Scots, who were much inferior in numbers, being defeated. About 1 mile south-west from the town the Highlanders under Prince Charles defeated the Royal forces under General Hawley, 17th Jan., 1746. *Stirling and Falkirk Burghs* return one member to the House of Commons. Pop. of Falkirk, 33,312.

Falkland (fa'k'land), Lucius Gary, Viscount, an English man of letters, born about 1610. His father being then Lord-Deputy of Ireland, he was educated at Trinity College, Dublin. After passing a short time abroad, he

devoted himself to a life of retirement and literary studies, chiefly residing at his seat at Burford, near Oxford, which he made a kind of academy for the learned men of the neighbouring universities. In 1639 he joined the expedition against Scotland; and in 1640, his peerage being Scottish, he was chosen member of the House of Commons for Newport, in the Isle of Wight. In the first instance he warmly supported the Parliament, but doubts of the ultimate objects of the parliamentary leaders caused him to modify his attitude; and in 1642 he accepted from Charles I the office of Secretary of State. When hostilities began, he embraced decidedly the cause of the king, though he desired peace rather than victory. He was slain at the battle of Newbury, 20th Sept., 1643. He left behind him a work entitled *A Discourse on the Infallibility of the Church of Rome*, several pamphlets and published speeches, and a few poems, but nothing that explains the universal praises bestowed on him by contemporaries.

Falkland (fa'k'land), an ancient royal burgh of Scotland, county of Fife, 21 miles north of Edinburgh. It was once the residence of the Scottish kings, and possesses remains of an ancient palace and some curious old houses. There was formerly a castle here, in which David, eldest son of Robert III, was starved to death by order of his uncle the Duke of Albany, but no trace of it now remains. Falkland Palace was garrisoned by Rob Roy in 1715. Pop. 781.

Falkland Islands, an island group belonging to Great Britain, in the South Atlantic Ocean, about 300 miles east of the Straits of Magellan. They consist of two larger islands, East Falkland and West Falkland, containing respectively about 3000 and 2300 sq. miles, with a great number of smaller ones surrounding them; total area, 6500 sq. miles. They are hilly and boggy, entirely destitute of trees, but covered with a variety of grasses very nutritive for the sheep and cattle the rearing of which is the principal industry. Fish and sea-fowl abound. Wool, frozen meat, hides, and tallow are the chief exports; value in some years £600,000. The climate is equable and very healthy. The Falkland Islands were discovered by Davis on the 14th Aug., 1592. In 1710 a French vessel from St. Malo touched at them, and named them Îles Malouines. Settlements were afterwards formed on them by the French, Spaniards, and British alternately, but the British have ultimately retained possession of them. They now form a Crown colony which has a Governor and other officers appointed by the Government. Port

Stanley, in East Falkland, is a thriving settlement, and has now a wireless station. During the European War the Germans suffered a naval defeat off the Falkland Islands in Dec., 1914. Pop. of the group, 3275.

Fal'lacy (Lat. *fallax*, apt to mislead), in logic, is when an argument is used as decisive of a particular issue, which in reality it does not decide. Properly a fallacy is a fault in the form of reasoning (see *Logic*), but the term is applied also to faults in the substance of the argument such as the *petitio principii*, or proving one proposition by assuming another which is identical with it; *ignoratio elenchi*, or mistaking the point at issue; *post hoc ergo propter hoc*, or arguing as if sequence were the same thing as cause and effect.

Fallières, Clément Armand, eighth President of the French Republic, born at Mézin, department of Lot-et-Garonne. The son of peasants, he studied law, was mayor of Nérac for some years, and in 1876 was elected to the Chamber of Deputies. Under-Secretary of State in the Ministry of the Interior in 1880, he became Minister of the Interior in 1882, Minister of Public Instruction from 1883 to 1885, and subsequently Minister of Justice and Public Instruction. He entered the Senate in 1890. In 1899 he became President of the Senate, and on 16th Jan., 1906, was elected President of the Republic. During his tenure of office Fallières displayed decided democratic principles and a tendency towards the Left. It was due to his initiative that a Ministry of Labour was formed in 1909. His term of office ended in Jan., 1913. Fallières was one of France's most democratic Presidents.

Fall of Bodies. All bodies on the earth, by virtue of the attraction of gravitation, tend to the centre of the earth. A ball held in the hand presses downward; if dropped, it descends vertically; if placed on an inclined plane, it rolls down, in doing which it presses the plane with a part of its weight. In the air bodies fall with unequal velocities, a piece of paper, for instance, more slowly than a ball of lead; and it was formerly thought that the velocity of the fall of bodies was in proportion to their weight. This error was attacked by Galileo, who, experimenting with balls of different substances which he dropped from the tower of Pisa, was led to the conclusion that the resistance of the air acting on different extents of surface was the cause of the unequal velocities, and that in a vacuum all bodies

would fall with the same velocity. The truth of this last proposition was first demonstrated by Newton in his celebrated 'guinea-and-feather' experiment, where a guinea and feather are shown to fall side by side in the vacuum of the air-pump. This experiment proves that the force of gravitation in bodies is proportional to their inertia, that is, to their mass. The laws of falling bodies, that is of bodies falling freely in a straight line and through a distance short in comparison with that of the earth's centre, are the following:

1. When a body falls from rest it acquires velocity at the rate of about 32.2 feet per second every second. This number, which represents the acceleration due to the force of gravity, varies slightly with the locality, increasing from the equator to the poles, and diminishing as we recede from the surface of the earth. (See *Gravity*.) At the end of five seconds, therefore, the body would be found to be moving at the rate of 5×32.2 , that is, 161 feet per second.

2. The space fallen through in the first second is half of 32.2, that is, 16.1 feet; and the space fallen through in any given time is found by multiplying the square of the number of seconds by 16.1. Thus in three seconds a body falls 9×16.1 feet, or 144.9 feet.

3. The square of the velocity acquired by falling through any number of feet is found by multiplying twice that number by 32.2. Thus if a body falls 9 feet, the square of the velocity acquired is $2 \times 32 \times 9$, or 576 if we take 32 instead of 32.2; and taking the square root of 576, we find that a velocity of 24 feet is acquired in a fall of 9 feet.

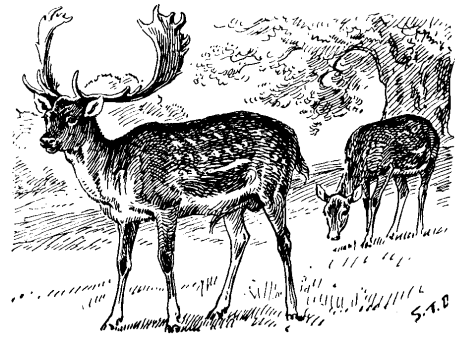
4. When a body is projected vertically upward with a given velocity, it continues to rise during a number of seconds found by dividing the number that expresses the velocity of projection by 32.2; and it rises to a height found by dividing the square of that number by 2×32.2 , or 64.4. For a machine used in verifying the laws of falling bodies, see *Attwood*.

Fall of Man, a commonly received doctrine of Christianity, founded upon the historical narrative contained in the third chapter of the book of *Genesis*, together with the allusions to the same matter in other parts of Scripture. Adam, having eaten of the forbidden fruit, is said to have fallen; and the relation of mankind in general to this fall is stated by St. Paul in the words:

"By one man's disobedience many were made sinners" (*Rom. v. 19*). Thus, in the fall of Adam, all men are held to have fallen and to have contracted 'original sin', alienating them from God and rendering them morally inadequate. The doctrine of the fall does not stand alone in Scripture. It is argued by some interpreters that in the original sentence pronounced on the transgressors there is contained the promise of a redemption, and that the whole scope of Scripture is directed to the development of this promise, and of the divine scheme of providence associated with it.

Fallopian Tubes, in anatomy, are two ducts each of which opens by one extremity into the womb, at either angle of the fundus, and terminates at the other end in an open trumpet-shaped mouth, which receives the ovum as it escapes from the ovary and transmits it to the womb. They are named after Fallopius or Fallopio, an Italian anatomist of the sixteenth century, who first recognized their functions.

Fallow Deer, a European and Western Asiatic deer, the *Cervus dama*. It is smaller than the stag, of a brownish-bay colour, whitish beneath, on the insides of the limbs, and beneath the tail. The horns, which are peculiar to the male, are very different from those of the stag; they are not properly branched, but are broader towards the upper part, and divided into processes down the outside. A simple snag rises from the base of each, and a similar one at some distance from the first. It was introduced at an early period into Britain, possibly by the Romans, and is kept in many English parks.



Fallow Deer (*Cervus dama*)

Fallow Land, ground that has been left uncultivated for a time, in order that it may recover itself from an exhausted state. Strictly speaking, fallow ground is left altogether without crops; but in agricultural usage strict fallow is not always adopted, and the term fallow is applied to various modes of treatment, of which at least three distinct varieties are recognized: *bare fallow*, *bastard fallow*, and *green-crop fallow*. Bare fallow is that in which the land remains completely bare for a whole year; in bastard fallow it is ploughed up and worked after the removal of a spring or summer crop,

preparatory to the sowing of a root or forage crop, to occupy the ground during autumn or winter; in green-crop fallow the land is sown with a root-crop, such as turnips or potatoes, placed in rows far enough apart to admit of the intermediate spaces being stirred, pulverized, and cleaned, during its growth, by horse or hand implements.

Fall River, a city and port, Bristol County, Massachusetts, United States, on an arm of Narraganset Bay, on Taunton River, 53 miles s.s.w. of Boston. It is at the head of deep-water navigation, and the terminus of a line of steamers from New York. It contains several handsome streets, and has extensive cotton, woollen, and calico-printing factories, as well as ironworks. Pop. 129,828.

Falmouth, a seaport and municipal borough of England, in Cornwall, 250 miles w.s.w. of London. There is a good harbour there, with a fine roadstead affording excellent refuge for shipping. Falmouth was at one time an important packet station, but is now chiefly a port of call, its principal trade being in supplies and stores for shipping. Falmouth and Penryn together give name to a parliamentary division of the county, returning one member to Parliament. Pop. 13,318.

False Imprisonment, the unlawful imprisonment or detention of any person. Every confinement of the person is imprisonment, whether in a common prison or a private house, or even by forcibly detaining one in the streets or highways. The law punishes false imprisonment as a crime, besides giving reparation to the party injured, through an action of trespass.

False Personation (English law). All forms of false personation, for the purpose of obtaining the property of others, are made penal by express statute. To personate the owner of any share, stock, or annuity, &c., is felony, and renders the offender liable to penal servitude for life, or to a modified term of penal servitude or imprisonment. The false personation of voters at an election is a misdemeanour punishable with imprisonment and hard labour, for a term not exceeding two years.

Falset'to (It.) applies, in singing, to the notes above the natural compass of the voice. It is also called the *head* or *throat* voice, in contradistinction to the *chest* voice, which is the natural one. The falsetto voice is produced by tightening the ligaments of the glottis.

False Weights and Measures. The using of false weights and measures is an offence at law punishable by fine. By various British statutes standards are provided for weights and for measures of capacity or dimension, and all contracts of sale, &c., are referred to such standards unless there is a special agreement to the contrary. See *Weights and Measures*.

Fal'ster, an island belonging to Denmark, situated at the entrance of the Baltic, east of Laaland, from which it is separated only by a narrow strait; flat, well watered, and wooded; productive in grain, pulse, potatoes, and, above all, fruit; area, 183 sq. miles. The principal town is Nykjöbing. Pop. 37,460.

Falun, or **Fahlun** (fä'lu'n), a town of Sweden, on Lake Runn, 130 miles north-west of Stockholm. It has an excellent mining-school, museums, and mineralogical collections. Within the town boundary is the famous Falun copper-mine, formerly the richest in Sweden, and worked for 500 years. Pop. 11,966.

Fama Clamo'sa ('a clamant report'), in the ecclesiastical law of Scotland, is a public report imputing immoral conduct to a clergyman, licentiate, or office-bearer of the Church. When the fama has become so notorious that it cannot be overlooked, the presbytery, after due inquiry, and if no particular party comes forward to institute a process, usually appear as accusers themselves.

Famagos'ta, or **Famagusta**, a seaport on the east coast of Cyprus. It is of remote antiquity, was an important place during the Middle Ages under the Lusignan kings of Cyprus and the Venetians, but, after being captured by the Turks in 1571, it declined. It has improved under the British, and has got a new harbour. Pop. 5327.

Famati'na, a district and mountain range in the Argentine Republic, province of La Rioja, rich in copper; highest summit, the Nevada de Famatina, 19,758 feet high.

Familiar Spirits, demons or evil spirits supposed to be continually within call and at the service of their masters, sometimes under an assumed shape, sometimes attached to a magical ring, or the like, sometimes compelled by

magic skill, and sometimes doing voluntary service. We find traces of this belief in all ages and countries, under various forms.

Family, in zoological classification, a group of species more comprehensive than a genus and less so than an order, a family usually containing a number of genera, while an order contains so many families. Family names usually terminate in -īdæ (after Latin patronymics, such as Æacīdæ, sons or descendants of Æacus). In botany it is sometimes used as a synonym of natural order.

Family Compact, the name given to an alliance organized by the Duc de Choiseul, first minister of Louis XV, between the various members of the Bourbon family, then sovereigns of France, Spain, the Two Sicilies, Parma, and Piacenza, mutually to guarantee each other's possessions. It was signed 15th Aug., 1761, and entailed on Spain a war with England.

Famine, an extreme scarcity of food affecting considerable numbers of people at the same time. Its causes are either natural, such as crop failures due to disease or to excessive or deficient rainfall, the effect of these being aggravated when the crop concerned is one on which the population mainly depends; or political and economic, such as war, or defects in the organization of production and distribution. In the Early and Middle Ages famines were frequent; but the rapidity of modern communication and transport made famines rare in Europe, until the conditions caused by the Great War produced great scarcity in Central Europe. In Ireland the years 1814, 1816, 1822, 1831, 1846 were marked by failure of the potato crop, and in the last-mentioned year the dearth was so great that £10,000,000 were voted by Parliament for relief of the sufferers. India has been the seat of many great famines, which recur at more or less regular intervals; but of late the British officials have been successful in organizing preventive and relief measures, such as improvement in railways and irrigation, the multiplication of industries, and the institution of a famine insurance grant. Amongst the more recent famines are that in North-West India (1837-8), in which above 800,000 perished; that in Bengal and Orissa (1865-6), when about a million perished; that in Bombay, Madras, Mysore (1877); that of 1896-7; and that of 1900 in Bombay, Punjab, &c., perhaps the most serious on record, when the Government spent £10,000,000 in relief. In China a great famine took place in 1877-8, in which over 9 millions are said to have

perished; another took place in 1888-9, owing to the overflow of the Yellow River.

Fan, the name of various instruments for exciting a current of air by the agitation of a broad surface. (1) An instrument made of wood or ivory, feathers, thin skin, paper, variously constructed and mounted, and used by ladies to agitate the air and cool the face. As an article of luxury the fan was well known to the Greeks and Romans. Fans are said to have been introduced into England from Italy in the reign of Henry VIII, and Queen Elizabeth was very fond of them. There is a collection of fans at the Victoria and Albert Museum, London. (2) Any contrivance of vanes or flat discs revolving by the aid of machinery, as for winnowing grain, for cooling fluids, urging combustion, or assisting ventilation, is also so called.

Fanar'ïots, or **Phanariots**, the inhabitants of the Greek quarter, or Phanar, in Constantinople, particularly the noble Greek families resident there since the times of the Byzantine emperors. The dragoman or interpreter of the Porte and other high officials used to be taken from their number. They have now mostly lost their influence at Constantinople, and have in many cases transferred themselves to Athens.

Fanat'icism, a term applied more particularly to the extravagance manifested in religious matters by those who allow themselves to be hurried away by their fancy and feelings, to the adoption not only of wild enthusiastic views, but also of inordinate and not infrequently persecuting measures. By an extension of the term it is also sometimes applied to other forms of extravagance.

Fancy, a term approaching imagination in meaning. In its general acceptation it refers both to the forms of the imagination and to the mental faculty which produces them; but it is used frequently for the lighter or more fantastic forms of the imagination, and for the active play of that faculty which produces them. See *Imagination*.

Faneuil Hall (fan'ū-il), a public building in Boston, famous as the place where stirring speeches were made at the outbreak of the war for American independence. It was built between 1740 and 1742 by a Huguenot merchant named Peter Faneuil.

Fanfare (Sp. *fanfárria*, brag), a short, lively, loud, and warlike piece of music, composed for trumpets and kettle-drums. Also small, lively pieces performed on hunting-horns, in the chase.

Fan-foot, a name given to a North African lizard of the genus *Ptyodactylus* (*P. lobatus*), one of the geckoes, much dreaded in Egypt for its supposed venomous properties.

Fanning Island, a coral island in the centre of Polynesia, lat. 3° 51' N., long. 159° 22' W. Since 1888 it belongs to Britain, is a landing-place of the Canada-Australia cable, and the stretch from this to Vancouver, 3458 miles, is the longest in the world. The island was discovered by Edmund Fanning in 1798. Area, 15 sq. miles; pop. about 200. It forms one of a small group sometimes called Fanning Islands.

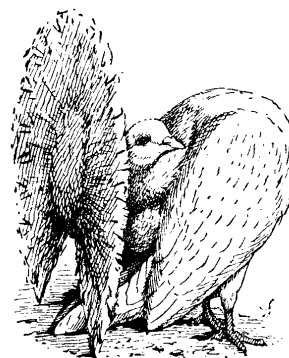
Fano, a seaport of Italy, on the Adriatic, province of Pesaro e Urbino, 29 miles north-west of Ancona. It is a handsome town, and has a triumphal arch erected to Augustus, and other antiquities. Pop. (commune), 25,000.

Fan-palm, a name sometimes given to the talipot palm or *Corýpha umbraculifera*, a native of Ceylon and Malabar. See *Talipot Palm*.

Fans, an African race of people inhabiting the region of the west coast about the Gaboon River and the Ogoway. They are an energetic race, skilled in various arts, and are rapidly increasing in numbers (about 300,000). They are cannibals, but contact with Europeans is leading them to give up the practice.

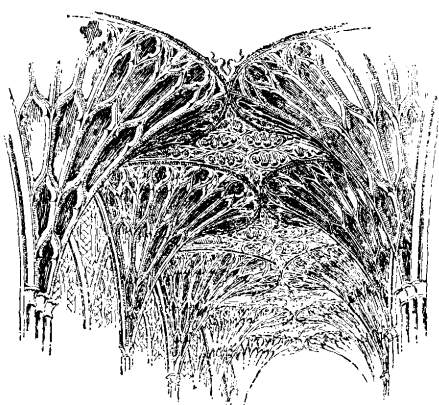
Fanshawe, Sir Richard, an English diplomatist, poet, and translator, born in 1608, died at Madrid in 1666. He studied at Cambridge, was secretary of the English Embassy at Madrid, and took the Royal side on the outbreak of the Civil War in 1641. He was made a baronet in 1650, was taken prisoner at Worcester, but permitted to go at large on bail. After the Restoration he was employed on several diplomatic missions, and in 1664, as Ambassador at Madrid, negotiated a peace between England, Spain, and Portugal. His poetical abilities were above mediocrity, as is evinced by his translations of the *Lusiads* of Camoens, the *Pastor Fido* of Guarini, the Odes of Horace, and the fourth book of the *Æneid*.

Fan-tail, a variety of the domestic pigeon, so called from the fan-like shape of their tails. Also a name applied to certain birds (species of *Rhipidura*) of the fly-catcher family, native to India, Australia, and New Zealand.



Fan-tail Pigeon

Fantees', a people of West Africa inhabiting the coast district of the Gold Coast Colony, between the Ashantis and the sea. They were at one time the most numerous and powerful people situated immediately on the Gold Coast seaboard; but their power was almost entirely broken after 1811 by repeated invasions of the Ashantis, and they have since lived under British protection. The soil is fertile, producing fruits, maize, and palm-wine.



Fan-tracery Vaulting, Gloucester Cathedral

Fan-tracery, in architecture, elaborate geometrical curved work, which spreads over the surface of a vaulting, rising from a corbel and diverging like the folds of a fan. Fan-tracery vaulting is much used in the Perpendicular style, in which the vault is covered by ribs and veins of tracery, of which all the principal lines diverge from a point, as in Henry VII's chapel, Westminster Abbey.

Farad, the practical unit of capacity for electricity, in the electromagnetic system of units. The capacity of a conductor or condenser whose potential is raised by one volt when given a charge of one coulomb. This unit is too large for most purposes, and capacities are usually expressed in microfarads (q.v.).

Far'aday, Michael, one of the greatest of English chemists and physicists, was born in humble circumstances at Newington Butts, near London, 22nd Sept., 1791, died 25th Aug., 1867. Early in life he was apprenticed to a bookbinder in London, but occupied himself in his leisure hours with electrical and other scientific experiments. Having been taken by a friend to Sir Humphry Davy's lectures, he attended the course, and became so

interested that he decided to abandon his trade. With this end he sent his notes of the lectures to Sir Humphry Davy, who was so struck with the great ability they showed that he appointed him his assistant at the Royal Institution. In 1829 he became lecturer at the Royal Military Academy at Woolwich, and in 1833 he was appointed to the newly established chair of chemistry at the Royal Institution. It was while in this office that he made most of his great electrical discoveries. His communications to the *Philosophical Transactions* were published separately in three volumes (1839, 1844, 1855). In 1832 he received the honorary degree of D.C.L. from Oxford, and was made an honorary member of the Academy at Berlin. In 1835 he received a pension of £300 a year from Lord Melbourne. As an experimentalist Faraday was considered the very first of his time. As a popular lecturer he was equally distinguished, and used to draw crowds to the Friday evening lecture at the Royal Institution. Amongst his published works we may mention the following: *Researches in Electricity* (1831-55), *Lectures on Non-metallic Elements* (1853), *Lectures on the Forces of Matter* (1860), *Lectures on the Chemical History of a Candle* (1861).—BIBLIOGRAPHY: J. Tyndall, *Faraday as a Discoverer*; S. P. Thompson, *Michael Faraday: his Life and Work*.

Faradization, or **Faradism**, the medical application of the induced currents which Faraday discovered in 1831.

Farallo'nes, a group of small islands in the Pacific, about 30 miles from the entrance to the Bay of San Francisco.

Faran'dola, an exciting dance popular amongst the peasants of the south of France and the neighbouring part of Italy. The men and women, placed alternately and facing different ways, form a long line winding out and in with a waving motion.

Farce, a subdivision of comedy, characterized chiefly by exaggeration and lack of rational character drawing. Farce stands in the same relationship to comedy as melodrama does to tragedy. Many farces commence with an impossible postulate, such as *The Comedy of Errors*. Jonson's *Silent Woman* is one of the best English farces. Gilbert's *Engaged* and Foggerty's *Fairy* are notable modern examples.

Farcy, a disease to which horses are liable, intimately connected with glanders, the two diseases generally running into each other. It is supposed to be a disease of the absorbents of the skin, and its first indication is generally the appearance of little tumours called farcy buds on the face, neck, or inside of the thigh. By an order in Council animals affected with farcy must be destroyed.

Fardel-bound, a term applied to cattle and sheep affected with a disease caused by the retention of food in the maniplies or third stomach, between the numerous plaits of which it is firmly impacted. Over-ripe clover, vetches, or rye-grass are liable to produce the disease.

Fareham, a town of England, in Hampshire, at the north-west extremity of Portsmouth harbour, giving name to a parliamentary division of the county. It has building-yards, potteries, and brickworks, and a considerable trade. Pop. 10,066.

Farel, Guillaume, one of the earliest and most active of the Swiss reformers, was born in 1489 in Dauphiny, died in 1565. At an early period he was led by his intercourse with the Waldenses to adopt similar views. After preaching in various parts of Switzerland, he came to Geneva, where he was so successful at the religious conferences of 1534 and 1535 that the Council formally embraced the Reformation. He was instrumental, also, in persuading Calvin to take up his residence in Geneva. At attempt on the part of the two reformers to enforce too severe ecclesiastical discipline was the cause of their having to leave the city in 1538. Farel took up his residence at Neufchâtel, where he died.

Fargo, a town of N. Dakota, United States, on the Red River of the North and the N. Pacific Railroad. Pop. 14,330.

Faria y Sousa, Manuel de, Portuguese historian and poet, born 1590, of an ancient and illustrious family, died about 1649. Among his writings are: *Discursos Morales y Politicos*, *Epitome de las Historias Portuguesas*; *Comentarios sobre la Lusiada*; and a collection of poems.

Faribault, a town of Minnesota, United States, 53 miles south of St. Paul's. Here are the State asylum for the deaf, dumb, and blind, and an episcopal divinity college. Pop. 9000.

Faridpur (fa-rēd-pör'), a district of India, in Eastern Bengal; area, 2267 sq. miles; pop. 1,937,650. Chief town, Faridpur, on the Mará Padmá. Pop. 11,000.

Fari'na, a term given to a soft, tasteless, and commonly white powder, obtained by trituration of the seeds of cereal and leguminous plants, and of some roots, as the potato. It consists of gluten, starch, and mucilage.

Farinel'li, Carlo, an Italian singer, born at Naples in 1705, died in 1782. His true name was Carlo Broschi, and to develop his vocal powers he was made a eunuch. He sang in Vienna, Paris, and London with the greatest success. On visiting Spain, where he intended only a brief sojourn, he found King Philip V plunged in a profound melancholy. He succeeded in rousing him from it by the powers of his voice, and became his prime favourite and political adviser. But the penalty of his advancement was that for ten years he had to sing every night to his royal master the same six airs. On his return to Italy, in 1762, he found himself almost forgotten, but continued to exercise a splendid hospitality in his country house, near Bologna.

Fari'ni, Luigi Carlo, an Italian statesman and author, born in 1812, died 1st Aug., 1866. He studied medicine at Bologna, and practised as a physician. He became known as a nationalist and patriot in the political movements of 1841, had to leave the country for a time, but returned and was made a member of the Reform Ministry at Rome during the disturbances of 1848. Disapproving equally the views of the old Conservative and the extreme Republican party, he went to Piedmont, where he was elected a Deputy, and fought with great energy both in pamphlets and in Parliament on behalf of Cavour and the Piedmontese Constitutionalists. After the peace of Villafranca, he was chosen dictator of the duchies of Parma and Modena, and was mainly instrumental in inducing them to unite with the Piedmontese monarchy. His *History of the Papal States from 1814 to 1850* is well known. In 1862 he became President of the Ministry, but lost his reason in 1863.

Farmers-general (Fr. *Fermiers généraux*), private contractors, to whom under the old French monarchy was let out the collection of various branches of the revenue, poll-tax, duties on salt and tobacco, and customs. These contractors made enormous profits on the farming of the public

revenues. A revenue collected in this way not only imposed a much heavier burden on the people, but the merciless rigour of irresponsible and uncontrolled exactors subjected them to hardships and indignities to which they could not submit without degradation. In 1790 the system was suppressed by the Constituent Assembly, and many of the farmers-general were sent to the guillotine by the Revolutionary Tribunal.

Farne, or Ferne, Islands, a group of seventeen islets, England, separated from the coast of Northumberland by a channel about $1\frac{3}{4}$ miles wide. They have been the scene of some disastrous shipwrecks, including that of the *Forfarshire* in 1838. (See *Darling, Grace*.) There are two lighthouses. Pop. 15.

Farnese (fa`r-nā`ze), an illustrious family of Italy, whose descent may be traced from about the middle of the thirteenth century, and which gave to the Church and the Republic of Florence many eminent names, amongst which the following may be mentioned: Pietro Farnese (died 1363), a general of the Florentines in the war against Pisa; Alessandro, who became Pope as Paul III (1534-49), and whose gifts to his natural son Pier Luigi of the duchies of Parma and Piacenza laid the foundation of the wealth and greatness of the family; Ottavio (1520-85), son and successor of Pier Luigi, spent a long and peaceful reign in promoting the happiness of his subjects. Alessandro (1546-92), elder son of Ottavio, became famous as a most successful general of the Spaniards in the wars with the Netherlands and France. Ranuzio (1569-1622), son of Ottavio, was a gloomy and suspicious tyrant. The line became extinct with Antonio in 1731. The name of the Farnese is associated with several famous buildings and works of art. The *Farnese Palace*, at Rome, was built for Pope Paul III, while he was cardinal, by Sangallo and Michel Angelo. It now belongs to France, and is occupied by the French Embassy. Its sculpture gallery was formerly very celebrated, but the best pieces have been removed to Naples, including the following: the *Farnese Bull*, a celebrated ancient sculpture representing the punishment of Dirce, discovered in 1546 in the Baths of Caracalla at Rome; *Farnese Hercules*, a celebrated ancient statue of Hercules by Glycon, found in the Baths of Caracalla in 1540; *Farnese Flora*, a colossal statue of great merit, found in the Baths of Caracalla; *Farnese Cup*, an antique onyx cup, highly ornamented with figures in relief.

Farnham, a town of England, county of Surrey, 3 miles s.w. of Aldershot; a well-built place. North of the town is Farnham Castle, the residence of the Bishops of Winchester. The staple trade is in hops. Farnham was the home of Swift's 'Stella' (Hester Johnson). Pop. 12,133.

Farnworth, a manufacturing and mining town of Lancashire, England, 3 miles from Bolton. Pop. (urban district), 27,901.

Faro, a seaport of Portugal, province of Algarve, 62 miles s.e. of Cape St. Vincent. It is surrounded by Moorish walls, and has a convenient harbour. Its trade is considerable. Pop. 12,680.

Faro, a promontory forming the north-east point of Sicily at the entrance to the Strait of Messina. The point is strongly fortified, and on it there is a lighthouse over 200 years old.

Faroe Islands (fā'rō; Dan. *Färöer*, 'Sheep Islands'), a group of islands in the North Atlantic, lying between Iceland and Shetland. They belong to Denmark, and are twenty-one in number, of which seventeen are inhabited. The islands generally present steep and lofty precipices to the sea. Barley is the only cereal that comes to maturity; turnips and potatoes thrive well. There is no wood, but plenty of excellent turf, and also coal. The inhabitants are chiefly engaged in fishing and the rearing of sheep. Thorshavn, in Strömö, the largest island, is the seat of government. Pop. 18,000.

Farquhar (fär'ka'r), George, Irish playwright, was born in Londonderry in 1677, and died in 1707. He was for a short time at Trinity College, Dublin, but was, according to one account, sent down for making a profane though clever joke on the miracle of walking on the sea. He became an actor, but left the stage after inadvertently injuring a fellow-actor, owing to his forgetting to substitute a stage-sword for the genuine article. He produced his first comedy, *Love and a Bottle*, in 1698. It is a lively and amusing comedy, and was well received. *The Constant Couple* (1699) was also successful, as was its sequel *Sir Harry Wildair*. His other best-known plays are *The Recruiting Officer* (1706), and his masterpiece *The Beaux' Stratagem* (1707), written when he knew that death was fast approaching him. Farquhar was in dire poverty most of his life; he had a commission in the army for a while, but gave it up owing to some false hopes of promotion held out by the Duke of Ormond. He increased his embarrassments by

marrying in 1703 a penniless woman who had fallen in love with his appearance, and pretended to be an heiress. Although he lived and died in great distress, his gaiety never flagged; and *The Beaux' Stratagem* is one of the most mirthful comedies of the time.

Farquhar was a great playwright, and not much of a literary man. His comedies are all good acting comedies. He had been an actor himself, and so was much more closely in touch with the stage than the aristocratic Congreve. Farquhar stands above his contemporaries by reason of his realism. He did not go to other dramatists for his characters, but went straight to life. Indeed, in several cases his plays seem to have been in part autobiographical; the bard was the hero of the story. His plots are well constructed, especially his later ones. His characters are most of them genial rogues, and while he is no Puritan his morality compares very favourably with the cynical indecency of his contemporaries. His influence upon Fielding, and therefore upon the rise and development of the English novel, was great, as he introduced a return to real models, and eschewed artificiality. Personally Farquhar was a most loveable man, and he appears to have lived and died a very gallant gentleman.

Farragut, David Glascoe, admiral of the United States, born in 1801, died 13th Aug., 1870. He entered the navy as midshipman at the age of eleven, was promoted to a lieutenancy in 1821, and was actively engaged in his profession up till 1851, when he was appointed assistant inspector of ordnance. In 1855 he received a commission as captain. In 1861 he was appointed to command the expedition against New Orleans, undertaken on the formation of the Confederacy, and sailed in January of the following year. New Orleans surrendered to the combined attack of the land and naval forces on 28th April, and Farragut proceeded to Vicksburg, which he attacked unsuccessfully. In consequence of his success at New Orleans he was promoted to the rank of vice-admiral, and placed in command of the blockading squadron of the Gulf of Mexico. In Aug., 1864, he attacked the Confederate fleet in the Bay of Mobile, and forced it to surrender, thus making the fall of Mobile merely a question of time. After this exploit he was made admiral, a grade which had not hitherto existed in the United States navy.

Far'rant, Richard, one of the earliest English composers of music. Very little is known of his history. He was a gentleman of the Chapel Royal in 1564, and subsequently organist and choir-master. He is supposed to have died about 1580. His music, which is ecclesiastical, is distinguished by purity, simplicity, tenderness, and elevation. The anthems *Call to Remembrance*, and *Hide not Thou Thy Face*, composed by him, are well known.

Farrar, Frederic William, English divine, son of a clergyman, born in Bombay, 7th Aug., 1831, died 22nd March, 1903. He graduated at Cambridge, 1854, was assistant master at Harrow in 1855, headmaster of Marlborough College in 1871, Archdeacon of Westminster, 1883, and Dean of Canterbury, 1895. He wrote various popular theological works and works of fiction, and was Bampton Lecturer in 1885. Among his principal works are: *The Life of Christ* (1874), *Life of St. Paul* (1879), *The Early Days of Christianity* (1882), *Lives of the Fathers* (1889), *Darkness and Dawn*.

Fars, or **Farsistan**, a maritime province in the south-west of Persia, abutting on the Persian Gulf. It is mountainous, but has many rich and well-cultivated districts. The most important products are grain, fruit, wine, oil, cotton, tobacco, silk, cochineal, and attar of roses. The manufactures include woollen, silk, and cotton goods; and in these and other articles an active trade is carried on, chiefly with Hindustan. Pop. estimated at 1,700,000.

Farsan, two islands on the east side of the Red Sea on the coast of Yemen, called respectively Farsan Kebîr and Farsan Segîr.

Farthingale, or **Fardingale**, an article of ladies' attire worn in the days of Queen Elizabeth, and closely resembling the more recent crinoline. It was formed of circles of whalebone hoops, and protruded more at the waist than the Victorian crinoline.

Farukhabad, or **Farrakhabad** (*far-ak-ä-bäd'*), a city in Agra division, United Provinces of British India, 2 or 3 miles from the Ganges, a handsome, well-built town, with avenues of trees in many of its streets. Pop. 59,647.

Fasces (fas'sēz), in Roman antiquities, a bundle of polished rods, in the middle of which was an axe, carried by lictors before the superior magistrates. The number of fasces and lictors varied with the dignity of the magistrate. In the city the axe was laid aside.

Fas'cia (Lat., a bandage), in anatomy, signifies any thin sheet of fibrous tissue, such, for example, as the covering which surrounds the muscles of the limbs and binds them in their places.

Fascination (Lat. *fascinare*, to charm), the exercise of an overpowering and paralysing influence upon some animals attributed to certain snakes. Squirrels, mice, and the smaller birds are said to be the most subject to this power; but the fact is far from clearly explained, and is not perhaps even sufficiently demonstrated. Most of the accounts agree in describing the animal fascinated as having a painful consciousness of its danger, and the power exercised over it, but to be unable to resist the desire to approach the fascinator. It is probable, however, that the real explanation of the phenomenon is to be found in the influence of the intense emotion of fear upon the muscles.



Farthingale

Fascinēs (fa-sēnz'), in field engineering, bundles of boughs or rods from 6 to 18 feet in length and usually 1 foot in diameter, used in raising batteries, strengthening parapets, or revetting slopes. The twigs are drawn tightly together by a cord, and bands are passed round them at the distance of 2 feet from each other. Very long thin ones are called *saucissons* or *battery-sausages*.

Fasho'da, a station in the Anglo-Egyptian Sudan, on the Bahr-el-Abiad or White Nile, 400 miles south of Khartoum, and about 70 miles north-east of the confluence of the Sobat with the Nile. In July, 1898, it was occupied by a French force under Colonel Marchand, but some months later was claimed by the British for Egypt. The affair threatened to involve the two countries in war, but ultimately the French evacuated the place, which was then formally occupied by Sudanese troops. It has been renamed Kodok.

Fast-and-loose, a cheating game sometimes played at fairs by gipsies, and also called 'prick the garter'. A belt or strap is doubled and rolled up with the double in the middle of the coils; it is then laid on a board, and the dupe is asked to catch the double with a skewer, when the gambler takes the two ends and looses it or draws it away, so as always to keep the skewer outside the doubled end. The game is mentioned four times in Shakespeare, e.g. *Antony and Cleopatra*, iv, 12, 28.

Fasti (Lat.), among the Romans, registers of various kinds; as, *fasti sacri*, calendars of the year, giving the days for festivals or courts, being a sort of almanac.

Fasting, the partial or total abstinence of mankind and animals from the ordinary requisite supply of aliment, by which it is to be understood that quantity which is adapted to preserve them in a healthy and vigorous condition. It would appear that various warm-blooded animals are capable of sustaining total abstinence much longer than human beings. Cats and dogs have survived for several weeks without nourishment of any kind, but it is probable that few human beings could survive such deprivation for more than a week. The use of water without solid food enables life to be sustained much longer than it could otherwise be.

Fasts, temporary abstentions from food, especially on religious grounds. Abstinence from food, accompanied by signs of humiliation and repentance or grief, is to be found more or less in almost all religions. Among the Jews fasts were numerous, and we find many instances of occasional fasting in the Old Testament. Herodotus says that the Egyptians prepared themselves by fasting for the celebration of the great festival of Isis. So in the Thesmophoria at Athens, and in the rites of Ceres at Rome, it was practised. The Church of Rome distinguishes between days of fasting and of abstinence. The former are: (1) the forty days of Lent; (2) the Ember days, being the Wednesday, Friday, and Saturday of the first week in Lent, of Whitsun week, of the third week in September, and of the third week in Advent; (3) the Wednesdays and Thursdays of the four weeks in Advent; (4) the vigils or eves of Whitsuntide, of the feasts of St. Peter and St. Paul, of the Assumption of the Virgin, of All Saints, and of Christmas Day. When any fasting day falls upon Sunday, it is observed on the Saturday before. The Greek Church observes four principal fasts: that of Lent, one beginning

in the week after Whitsuntide, one for a fortnight before the Assumption, one forty days before Christmas. In the East, however, the strict idea of a fast is more preserved than in the West. The Church of England appoints the following fixed days for fasting and abstinence, between which no difference is made: (1) the forty days of Lent; (2) the Ember days at the four seasons; (3) the three Rogation days before Holy Thursday; (4) every Friday except Christmas Day. The Church, however, gives no directions concerning fasting.—BIBLIOGRAPHY: L. Duchesne, *Christian Worship*; J. Dowden, *The Church Year and Kalendar*; article *Fasting* in Hastings' *Encyclopædia of Ethics and Religion*.

Fat, an oily concrete substance, a compound of carbon, hydrogen, and oxygen, deposited in the cells of the adipose or cellular membrane of animal bodies. In most parts of the body the fat lies immediately under the skin. Fat is of various degrees of consistence, as in tallow, lard, and oil. It is generally white or yellowish, with little smell or taste. It consists of esters of glycerine with fatty and other acids, and these are generally termed glycerides. The commonest of these are stearin, a waxy solid, palmitin, a softer solid, and olein, an oil. Fats are insoluble in water. When boiled with caustic alkalies, e.g. caustic soda, they are decomposed (saponified), yielding an alkali salt of the fatty acid (soap) and glycerine. The consistency of any natural fat depends on the proportions in which these three substances are present, e.g. mutton suet consists mainly of stearin, and olive oil of olein. In the body fat serves as a packing, and helps to give roundness of contour. Being a bad conductor of heat, it is useful in retaining warmth, but its chief function is that of nutrition.

Fa'talism, the belief in fate, or an unchangeable destiny, to which everything is subject, uninfluenced by reason, and pre-established either by chance or the Creator. Fatalism existed among the Hebrews, Greeks, and Romans, and is still prevalent among Mohammedans. The fact that many events in man's life seemed to be inevitable gave rise to the belief in fatalism. Amongst notable historical examples of the belief in fate may be mentioned the old Greek conception of a fate which stood behind the gods themselves as a controlling power; the Mohammedan fatalism, which regards all things great and small as inexorably predetermined, so that no accident is possible. Fatalism is to be distinguished both from *determinism* and *predestination*.

Fategarh (fat-e-gar'), a town, United Provinces of India, on the Ganges, now municipally united with Farukhabad; the scene of a massacre of upwards of 200 Europeans during the Mutiny of 1857. Pop. 12,500.

Fatehpur (fat-e-pör'), a town of India, in district of the same name, Allahabad division, United Provinces, 50 miles S.E. of Cawnpore. Pop. 16,939.—The district has an area of 1639 sq. miles, and a pop. of 686,400.

Fatehpur Sikri, a town of India, district of Agra, United Provinces. It was the favourite residence of the Emperor Akbar, who enclosed and fortified it. It now chiefly consists of a vast expanse of magnificent ruins enclosed by a high stone wall some 5 miles in circuit. Pop. 6132.

Fates (in Lat. *Parcæ*, in Gr. *Moirai*), in Greek and Latin mythology, the inexorable sisters who spin the thread of human life. The appellation *Clotho* (the spinner) was probably at first common to them all among the Greeks. As they were three in number, and poetry endeavoured to designate them more precisely, *Clotho* became a proper name, as did also *Atrōpos* and *Lachēsis*. *Clotho* means she who spins (the thread of life); *Atropos* signifies unalterable fate; *Lachesis*, lot or chance; so that all three refer to the same subject from different points of view. They know and predict what is yet to happen. *Lachesis* is represented with a spindle, *Clotho* with the thread, and *Atropos* with shears, with which she cuts it off. We find also in the northern mythology three beautiful virgins, the *Nornen*, who determine the fate of men. Their names are *Urd* (the past), *Varande* (the present), and *Skuld* (the future).

Fatherlasher, or **Bull-head**, a fish of the genus *Cottus* (*Cottus bubālis*), from 8 to 10 inches in length. The head is large, and is furnished with several formidable spines. The fish is found on the rocky coasts of Britain, and near Newfoundland and Greenland. In the latter regions it attains a much larger size, and is a considerable article of food.

Fathers of the Church, The.

1. *The term 'Fathers'.*—This term, as used in the sense of spiritual parents of the Christian faith and life, appears to have become current in the fourth century. It was so used by Christian teachers, who cited as authoritative the great teachers and guides who were their predecessors. By the 'Fathers' they

meant, specifically, the earlier writers who carried on the work of instruction which was begun by Peter and John and the rest of the Apostles. As employed nowadays, the term has a great fluidity of meaning. In the widest sense it signifies all ecclesiastical writers (i.e. all writers within the Christian Church who treat of matters of Christian belief and practice) belonging to the older post-Apostolic period. In the narrower and more frequent sense it signifies only those ecclesiastical writers of the older post-Apostolic period who conform, more or less, to the Catholic tradition. As St. Vincent of Lérins lays it down, "Those alone should be named 'Fathers' who have been staunch in the communion and faith of the One Catholic Church, and have received ecclesiastical approbation as teachers".

2. *Fathers and Doctors*.—To such among the Fathers as were regarded as the most eminent the distinguishing title of 'Teachers' (*doctores*) was given. Thus in the Western Church Ambrose, Jerome, Augustine, and Gregory I were the four great Teachers or Doctors; while in the Eastern Church a similar position was assigned to Athanasius, Basil, Gregory of Nazianzus, and Chrysostom. But others also have been acknowledged as Doctors, as Hilary of Poitiers, Cyril of Alexandria, John of Damascus, and our one English Father (born out of due season) the Venerable Bede—not to speak of the application of the term to some of the mediæval Schoolmen.

3. *The Patristic Period*.—While it is universally agreed that the Apostolic Age is succeeded by the Age of the Fathers, there is a difference of view as to when the Age of the Fathers terminates. Gregory I (the Great) is usually regarded as the last of the Latin or Western Fathers and the first of the Schoolmen, and John of Damascus as the last of the Greek or Eastern Fathers. But where the term 'Fathers' is broadly used to designate the older Church writers in general, the tendency is—and it is logically defensible—to extend the Patristic period far beyond the Age of the Great Fathers (325-451), and to include among the later Fathers many mediæval writers. Thus the Abbé Migne, who in the middle of last century issued a monumental edition of the original Greek and Latin texts of the Fathers, carried the Latin Fathers as far as Innocent III in the beginning of the thirteenth century, and the Greek Fathers down to the Council of Florence and the fall of Constantinople in the middle of the fifteenth century.

4. *Division and Classification.*—The broadest division of the Fathers is according to locality, and is into Eastern and Western. To this the division according to language, into Greek and Latin, largely corresponds. But it is to be remembered that in the early Patristic Age, or before Tertullian, Latin was not used by ecclesiastical writers. It is also to be remembered that among the Eastern Fathers there were writers in Syriac, Armenian, and Coptic, as well as Greek. Another broad and general division is into ante-Nicene, Nicene, and post-Nicene; which is according to the principle that the Council of Nicæa (A.D. 325) marks the transition from a simple and unsystematized to a unified doctrinal testimony. But it is usual in Church history, while observing the aforesaid general divisions, to arrange the Fathers in certain historical groups, representing for the most part distinct schools of thought. There are, however, great names that cannot be conveniently treated under any historical group, such names as Irenæus, Athanasius, Jerome, Leo the Great, and Gregory the Great. Keeping this in view, we might classify the Fathers in accordance with the following scheme: (1) the *Apostolic Fathers* (the best known of whom are Clement of Rome, Ignatius, and Polycarp), who received their title not only as being younger contemporaries and perhaps personal disciples of Apostles, but also for their nearness and faithfulness to the Apostolic tradition; (2) the *Greek Apologists* (the most notable of whom is Justin Martyr), who sought to defend Christian truth on rational and philosophical grounds against both Jew and pagan; (3) the *Alexandrians* (outstanding among whom are Clement and Origen), who greatly furthered the development of Christian theology in general, but whose names are specially associated with the allegorical and mystical type of Scriptural interpretation; (4) the *North African School* (to which Tertullian and Cyprian belong), who shaped Christian Latinity, as well as the theology and ecclesiastical polity of the West; (5) the *Cappadocians* (in which group the most prominent members are Basil, Gregory of Nazianzus, and Gregory of Nyssa), who caught up the theology of Athanasius, providing it with well-defined terms, and so laying broad the foundations of the Greek orthodoxy; (6) the *Antiochians* (among whom Chrysostom, Theodore of Mopsuestia, and Theodoret are the greatest), who were opposed to the Alexandrian mysticism and held by the literal and historical mode of Scriptural interpretation; (7) the *Western Nicene Group* (counting in their number eminent teachers like Hilary of Poitiers and Ambrose), who followed the Alexandrians in their exegetical

method, and in their dogmatic theology Athanasius and the Cappadocians; (8) the *School of Augustine*, in which the Western theological tradition set by Tertullian and Cyprian culminated; (9) the *School of Lérins* (leading members of which are Hilary of Arles and Vincentius), which attempted to mitigate the extreme Augustinian doctrines of sin and grace.

5. *Value of Patristic Study*.—Among the Fathers are many great thinkers and writers (not to say orators, organizers, and statesmen) who should be studied for their own sake, and for the influence they have wielded. We would only indicate here some of the various uses of patristic study. (1) The *student of the Bible* turns to the Fathers, and especially the earlier of them, for light upon the problem of the true or original text of the Bible—although very few of the Fathers knew the Hebrew tongue, and only Origen and Jerome can throw direct light upon the Old Testament text. To the Fathers also, especially great exegetes like Origen, Chrysostom, Jerome, and Augustine, the Biblical student turns for light upon the meaning of the sacred text, and for knowledge of the history of its interpretation. (2) The *student of Church history* finds first-hand material in the Fathers for the older post-Apostolic period. This material is supplied in the tractates and letters of the Fathers generally. But patristic writers from Eusebius downwards furnish us also with formal histories of the Church of both a general and special character. Patristic histories, as indeed all histories, are to be used with critical caution. And not only do the Fathers inform us as to the course of events; we are dependent upon them for our knowledge of the development of creed and liturgy, ritual and order, and other Christian institutions. (3) The *student of ecclesiastical dogma and Christian theology in general* cannot dispense with the study of the Fathers. The patristic was the formative and, in a sense, conclusive period of Christian theology. In the ancient Greek theology the idea of God was developed, and in the so-called Nicene Creed and the Chalcedonian Definition the doctrines of the Trinity and the Person of Christ respectively received their final Greek dogmatic expression. Landmarks in the history of this dogmatic development are the names of Origen, Athanasius, Basil and the Gregories, Theodore of Mopsuestia, and Leo the Great. In the ancient Latin theology, in accordance with the more practical genius of the Westerns, the doctrine of man was developed, and of sin and grace. With this anthropological or soteriological, as distinguished from the other more strictly theological movement, the

names of Tertullian, Cyprian, and Augustine are principally associated. It was left to the mediaeval theologians to work out the doctrine of the Work of Christ.—BIBLIOGRAPHY: F. W. Farrar, *Lives of the Fathers*; E. Leigh-Bennett, *Handbook of the Early Christian Fathers*; S.P.C.K., *The Fathers for English Readers* (a series of biographies); also *Early Church Classics* (a series of translations); H. B. Swete, *Patristic Study* (1902: an excellent introduction to the field of patristic learning, with useful bibliographies); W. Bright, *The Age of the Fathers*.

Fat'imate Dynasty, a line of caliphs claiming descent from Fatima, the favourite daughter of Mohammed, and of Ali her cousin, to whom she was married. In the year 909 Abu-Mohammed Obeidalla, giving himself out as the grandson of Fatima, endeavoured to pass himself off as the Mahdi or Messiah predicted by the *Koran*. Denounced as an impostor by the reigning Caliph of Bagdad, he fled into Egypt, became Caliph of Tunis, and soon conquered all Northern Africa from the Straits of Gibraltar to the borders of Egypt. His son wrested Egypt from the Abbasides in 970 and founded Cairo. The Fatimite dynasty was extinguished in 1171, on the death of Al Adid, the fourteenth caliph, and a new line began with Saladin.

Fatty Acids, the homologues of formic and acetic acid; so called because the members first studied were obtained from fats and oils, e.g. butyric acid from butter, stearic acid from stearin, palmitic acid from palm-oil. These acids are present united with glycerol in the fats as glycerides, and are obtained from them by saponification with superheated steam or mineral acids, when the fatty acid is liberated, floats to the surface, and glycerol remains in solution. They are all monobasic acids; the lower members are colourless liquids, and the higher members from $C_7H_{15}COOH$ upwards are colourless solids. The general formula for the series is $C_nH_{2n+1}COOH$ (where n = the number of carbon atoms in the alkyl group).

Fatty Degeneration, an abnormal condition found in the tissues of the animal body, in which the healthy protoplasm is replaced by fatty granules. It is a sign of defective nutrition, and is common in old age, affecting the muscles, the heart, arteries, kidneys, &c. It is accompanied by great muscular flabbiness and want of energy, the sufferer looking at the same time fat and comparatively well.

Fatty Tissue, in anatomy, the adipose tissue, a tissue composed of minute cells or vesicles, having no communication with each other, but lying side by side in the meshes of the cellular tissue, which serves to hold them together, and through which also the blood-vessels find their way to them. In the cells of this tissue the animal matter called fat is deposited.

Faubourg (fō-bör; Lat. *foris*, outside, beyond, and *burgus*, borough), a suburb of French cities; the name is also given to districts now within the city, but which were formerly suburbs without it. Thus the *Faubourg St. Germain* is a fashionable quarter of Paris in which the ancient nobility still resides.

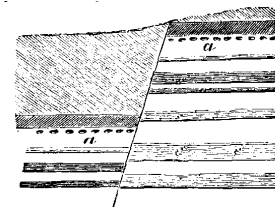
Fau'ces (Lat., 'jaws'), in anatomy, the throat, the slightly constricted communication between the posterior part of the mouth cavity and the pharynx. The tonsils are lodged in the fauces at the sides of the root of the tongue.

Faucigny (fō-sē-nyē), a district of France, department of Haute Savoie, one of the loftiest districts of Europe, being partly traversed by the Pennine Alps.

Fau'cit, Helena, Lady Martin, was born in 1816, died in 1898. She was the daughter of Mrs. Faucit the actress, and made her debut at the Theatre Royal, Richmond, in 1833, as Juliet in *Romeo and Juliet*. She first appeared in London at Covent Garden as Julia in *The Hunchback*, in which she gained a decided success. One of the most important members of Macready's company during the Shakespearean revivals of 1837, she created the heroine's part in Lord Lytton's *Lady of Lyons*, *Money*, and *Richelieu*, and in Browning's *Strafford*, *Blot on the Scutcheon*, and *Colombe's Birthday*. She was married to Sir Theodore (then Mr.) Martin in 1851, after which she but rarely appeared on the stage except for charitable purposes. In 1879 she appeared as Beatrice at the opening of the Shakespeare Memorial Theatre at Stratford-on-Avon. Lady Martin wrote a volume *On Some of Shakespeare's Female Characters*.

Fault, in geology, a fracture of strata, accompanied by a sliding down or an upheaval of the deposits on the one side of the fracture to a greater distance than the other. Faults are frequently recognizable in coal-beds, the miner coming unexpectedly upon an abrupt wall cutting off the seam. The angle

this makes with the plane of the bed he is working usually indicates whether he must look up or down for its continuation on the other side of the fracture; but *reversed faults* occur, in which the strata on one side have been pushed up the slope of the plane of fracture. In mines these faults often serve for natural drains. The cut shows at *a a* the change of position in strata caused by a fault. This is called the *throw*, and is measured vertically.



Fault in Geology

Faun, one of a kind of rural deities or demi-gods believed in among the Romans, inhabiting the forests and groves, and differing little from satyrs. Their form was principally human, but with a short goat's tail, pointed ears, and projecting horns; sometimes also with cloven feet. There are some famous antique statues of fauns, *The Dancing Faun* at the Uffizi in Florence (restored by Michel Angelo); *The Dancing Faun* at Naples; *The Faun* (of *Praxiteles*?) at the Capitoline Museum, Rome; and *The Sleeping Faun*.

Fauna (from *faun*, q.v.), a collective word signifying all the animals of a certain region, and also the description of them, corresponding to the word *flora* in respect to plants.

Faust, or **Faustus**, Doctor John, a celebrated dealer in the black art, who lived in Germany, early in the sixteenth century. There is really a substratum of fact beneath the Faust legend; there actually was a charlatan of this name who lived in the sixteenth century. He seems to have been a pretentious and vicious egomaniac. A vast amount of legend, however, has gathered round his name in Germany. According to some accounts he was born in Suabia, others make him a native of Anhalt, others of Brandenburg. In his sixteenth year he went to Ingolstadt and studied theology, became in three years a *magister*, but abandoned theology, and began the study of medicine, astrology, and magic, in which he likewise instructed his familiar Johann Wagner, the son of a clergyman at Wasserburg. After Dr. Faust had spent a rich inheritance, he, according to tradition, made use of his power to conjure up spirits, and entered into a contract with the devil for twenty-four years. A spirit called *Mephistopheles* was given him as a servant, with whom he travelled about, enjoying life in all its forms, but the evil spirit finally carried him off. Even yet Dr. Faustus and his familiar Wagner play a

conspicuous part in the puppet-shows of Germany, and the legend forms the basis of Goethe's well-known drama *Faust*, and furnishes the libretto for Gounod's famous opera of the same name. As early as 1590 Christopher Marlowe made the legend the subject of his masterpiece *Doctor Faustus*, the last scene of which is one of the most dramatic in all literature.—Cf. H. B. Cotterill, *The Faust-legend and Goethe's Faust*.

Fausti'na, the name of two Roman ladies: (1) Annia Galeria Faustina (died A.D. 141), the wife of the Emperor Antoninus Pius; and (2) her daughter, who was married to the Emperor Marcus Aurelius (died A.D. 175). Both were accused of dissolute conduct.

Favart (fa'-vär), Charles Simon, creator of the serio-comic opera in France, born 1710, died in 1792, the son of a pastry-cook. His poetical reputation rests principally on his numerous productions for the *opéra aux Italiens*, and the comic opera. He also wrote *Mémoires et correspondance littéraires* (1808). He was the director of a company of itinerant actors which followed Marshal Saxe into Flanders. His wife, Madame Favart, was a famous singer, comic actress, and dancer, and helped in the composition of her husband's plays.

Fa'versham, a seaport of England, county Kent, on a branch of the Swale, giving name to a parliamentary division of the county. It is a very ancient place, and has manufactures of brick, cement, and gunpowder. Faversham Creek is navigable up to the town for vessels of 200 tons. Pop. 10,870.

Favre (fävr), Jules, a French politician, born 21st March, 1809, at Lyons, died in 1880. He studied law, and after distinguishing himself at the Lyons Bar came to Paris in 1835, where he became famous as a defender of political prisoners. On the outbreak of the revolution of 1848 he became secretary to Ledru-Rollin. He was a leader of the party of opposition to the President Louis Napoleon; and after the *coup d'état* (1851) he retired from political life for six years, till in 1858 his defence of Orsini for the attempt on the life of the emperor again brought him forward. From this time he again became an active leader of the Republican opposition to the emperor. On the fall of the empire he became Vice-President of the Government of National Defence and Minister of Foreign Affairs. As such he conducted the negotiations for peace with Prince Bismarck, and signed the Treaty of Paris

at Frankfurt on 10th May, 1871. But though he showed great energy, and was very eloquent, his operations both in the matter of the armistice and the peace showed a lack of skill and judgment. He resigned his office in July, 1871.

Favus is a disease due to a fungus, and affects the hair, hair-follicles, and skin, usually of the scalp. It produces rounded cup-shaped crusts, and may lead to very extensive destruction of the hair. Cats and mice are affected by the disease, and are frequently responsible for spreading it. The X-rays are the most effective treatment.

Fawcett, Henry, an English politician and economist, born at Salisbury in 1833, died 6th Nov., 1884. He was educated at Cambridge, studied law for a while at the Middle Temple, but soon renounced it. In 1858, when out partridge shooting, he met with an accident which inflicted on him total blindness. Undiscouraged, however, by his deprivation, he gave his attention to economic studies. In 1863 he was elected to the chair of political economy at Cambridge. In 1865 he was elected member of Parliament for Brighton, which he represented till the general election of 1874, when he was elected for Hackney. He became Postmaster-General in the second Gladstone administration, and effected many reforms in his department. In 1883 he was made Lord Rector of Glasgow University. Amongst his principal writings are: *A Manual of Political Economy*, *Lectures on the Economic Position of the British Labourer*, and articles on Indian finances.

Fawcett, Millicent Garrett, wife of the preceding, born 1847, shared her husband's studies, and published: *Political Economy for Beginners*, *Some Eminent Women of Our Time*, *Life of Queen Victoria*, and *Five Famous French Women*. She is also known as a prominent advocate of all measures for the educational and political advancement of women, and wrote *Women's Suffrage* (1912).

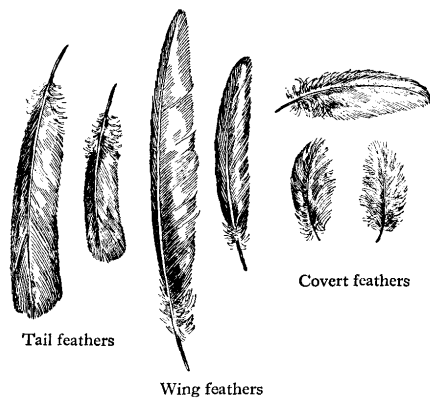
Fayal (fī-a'l'), an island belonging to Portugal, one of the Azores. It is of a circular form, about 10 miles in diameter. The climate is good, and the air always mild and pure. The soil is very fertile, producing in abundance wheat, maize, flax, and almost all the fruits of Europe. It exports a great

quantity of oranges and lemons. The chief place is Villa Horta or Orta. Pop. 22,385.

Fayoum (fa´-yöm´), a province of Middle Egypt, a little to the west of the Nile, surrounded by the Libyan Desert; area about 670 sq. miles. The soil is alluvial, and, in the north, particularly fertile. Fayoum is irrigated by canals coming from the Canal of Joseph, and that from the Nile, and is one of the most fertile provinces of Egypt. Here lay the ancient Labyrinth and the artificial Lake Moeris. On the west lies Lake Birket-el-Kurun. The chief town, Medinet-el-Fayoum, is connected with Cairo by a railway. Pop. of province, 441,583.

Feast of the Dedication of the Temple, a Jewish feast instituted by Judas Maccabæus in 164 B.C. It lasted eight days, and was a time of general rejoicing, when the people—old and young—carrying palm branches, met together in their synagogues to hold services of thanksgiving and commemoration. Every house was illuminated, and even the temple at Jerusalem was lighted up. In certain of its observances it resembles the Feast of the Tabernacles. Some authorities think that Christmas was celebrated in December by the ancient Church because that was the date of the Feast of Dedication. It is mentioned in *John*, x, 22.

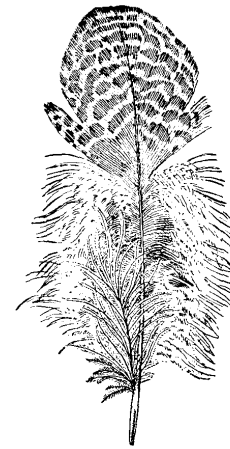
Feather-grass, the popular name of *Stipa pennata*, a native of dry places in the south of Europe. The rigid leaves roll up in dry air like those of marram-grass; the awns are exceedingly long, feathered to the point, and hygroscopic, curling up spirally when dry, and uncurling when moistened; these movements of the awn serve to bury the fruit. *S. tenacissima* is the esparto-grass used in paper-making.



Types of Feathers from a Gull

Feathers, the form which the dermal appendages assume in birds, agreeing in mode of development, but differing in form from hairs and scales. The feather consists of a stem, horny, round, strong, and hollow in the lower part, called the *quill*, and in the upper

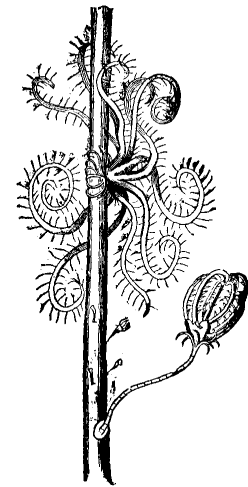
part, called the *shaft*, filled with pith. On each side of the shaft is a web composed of a series of regularly arranged fibres called *barbs*. The barbs and shaft constitute the *vane*. On the edges of the barbs are set the *barbules*, which interlock with those of adjacent barbs, and thus give strength to the vane. Feathers are of four chief kinds. (1) *Quill feathers* of the wing (*remiges*) and tail (*rectrices*); the former are attached to the hand and forearm. (2) *Contour feathers*, which determine the external form and are attached to certain areas of the skin; those overlapping the quills are known as *wing-coverts* and *tail-coverts*. (3) Small soft *down feathers*. (4) *Hair-like feathers* (*filoplumes*). The plumage of birds is of characteristic colours, due either to pigments or physical structure (metallic feathers), and commonly having a protective function by harmonizing with the surroundings (especially in females), or, when of bright kind, playing a part in courtship (especially in males). The feathers of birds are periodically changed, generally once, but in some species twice a year. This is called *moulting*. When feathers have reached their full growth they become dry, and only the tube, or the vascular substance which it contains, continues to absorb moisture or fat. When, therefore, part of a feather is cut off, it does not grow out again; and a bird whose wings have been clipped remains in that condition till the next moulting season, when the old stumps are shed and new feathers grow out. If, however, the stumps are pulled out sooner (by which operation the bird suffers nothing), the feathers will be renewed in a few weeks or even days. The feather is a very strong formation, not readily damaged, the arch of the shaft resisting pressure, while the web and



Feather from the Back of an Argus Pheasant

fine fibres yield without suffering. Being a bad conductor of heat, it preserves the high temperature of the bird, while it is so light as to be easily carried in flight. It is rendered almost impervious to wet by the oily fluid which most birds secrete at the base of the tail. Feathers form a considerable article of commerce, particularly those of the ostrich, heron, swan, peacock, and goose, for plumes, ornaments, filling of beds, pens, and other purposes.

Feather-star, one of the stalkless echinoderms belonging to the Crinoidea. A well-known type is the rosy feather-star (*Antedon rosacea*), not uncommon in British seas, and consisting of a central body or disc, from which proceed five radiating arms, each dividing into two secondary branches, so that ultimately there are ten slender rays. Each arm is furnished on both sides with lateral processes so as to assume a feather-like appearance. It is fixed when young by a short stalk, but exists in a free condition in its adult state.



Feather-star holding on by its grasping threads and its larvæ attached by stalks

Featherstone, an urban district or town in the W. Riding of Yorks, England, 2 miles west by south of Pontefract; inhabitants work chiefly in the collieries. Pop. 14,839.

Feb'rifuge is an agent used to lessen fever. Antipyrine, quinine, and salicylic acid, are well-known examples of drugs used as febrifuges, while cold baths and cold sponging are the most effective of other methods.

Febro'nianism, in Roman Catholic theology, a system of doctrines antagonistic to the admitted claims of the Pope, and asserting the independence of national Churches, and the rights of bishops to unrestricted action in matters of discipline and Church government within their own dioceses. The term is derived from Justinus *Febronius*, a *nom de plume* assumed by John Nicholas von Hontheim, Archbishop of Trèves, in a work entitled *De Statu Ecclesiæ et legitima Potestate Romani Pontificis* (On the State of the Church and the Legitimate Power of the Roman Pontiff), published in 1763.

Feb´ruary (from the Roman *Februa*, a festival of expiation or purification), the second month in the year, having twenty-eight days, except in leap-year, when it has twenty-nine. This latter number of days it had originally among the Romans, until the Senate decreed that the seventh month should bear the name of Augustus, when a day was taken from February and added to August to make it equal to July in number of days.

Fécamp (fā-kän; Lat. *Fiscanum*, derived from *Ficus Campus*, Fig Plain), a seaport of France, department of Seine-Inférieure, 23 miles north-east of Havre. It is one of the best ports in the Channel, and has many vessels employed in the cod, herring, and mackerel fisheries. Pop. 17,383.

Fechter (fesh-tār), Charles Albert, French actor and dramatist, born in 1824, died in America in 1879. His first appearance on the stage was at the Salle Molière, after which he made a short tour of Italy with a travelling French company. Returning to Paris, he appeared between 1844 and 1856 at different Parisian theatres, and in 1857 he was joint-director of the Odéon. In 1860 he came to London, and at once achieved great success as Ruy Blas and Hamlet at the Princess's Theatre, characters in which he departed widely from stage traditions. He subsequently leased the Lyceum, and afterwards the Adelphi, acting youthful and melodramatic parts with striking power. From 1870 to 1878 he lived in the United States, but his experiences as a manager in New York were not successful.

Fed´eral, or **Federalist**, an appellation in America given to those politicians who wanted to strengthen the central government, in opposition to those who wished to extend the separate authority of each individual state. Hence in the Civil Wars of 1861-5 the term *Federals* was applied to the Northern party.

Federal Government, government by the confederation of several united states, self-governing in local matters, but subject in matters of general polity to a central authority, as, for instance, the Swiss Republic, the United States of North America, Mexico, Argentina, Brazil, The Union of South Africa, and Russia since the revolution of 1917. The degree to which such states give up their individual rights as sovereign bodies may be very different.—BIBLIOGRAPHY: Viscount Bryce, *The American Commonwealth*;

Burgess, *Political Science and Comparative Constitutional Law*; Freeman, *History of Federal Government*.

Fee, or **Fief** (A.S. *feoh*, cattle, property), in law, primarily meant a loan of land, an estate held in trust on condition of the grantee giving personal or other service to the prince or lord who granted it. Feudal estates, however, soon came to be regarded as inalienable heritages held on various tenures; hence the term fee came to be equivalent to an estate of inheritance, that is, an interest in land which passes to heirs if the owner die intestate. The amplest estate or interest in land is that of a *fee-simple*, which is also called an absolute fee, in contradistinction to a fee limited or clogged with certain conditions. A fee-simple means the entire and absolute possession of land, with full power to alienate it by deed, gift, or will. It is the estate out of which other lesser estates are said to be carved; such as a *fee-tail* (see *Entail*), which is limited to particular heirs, and subject to certain restrictions of use; and a *base fee*, which ceases with the existence of certain conditions.

Fee-farm, in law, a kind of tenure of land without homage, fealty, or other service, except that mentioned in the feoffment, which is usually the full rent.

Feeling is properly a synonym for sensation, or that state of consciousness which results from the application of a stimulus to some sensory nerve. It is the most universal of the senses, existing wherever there are nerves; and they are distributed over all parts of the body, though most numerous in such parts as the finger-tips and the lines where skin and mucous membrane pass into each other. This universal distribution of feeling is necessary, otherwise parts of the body might be destroyed without our knowledge. The structures which thus apprehend the impressions of contact are papillæ or conical elevations of the skin in which the nerves end, and which are richly supplied with blood-vessels. The term feeling is also used for a general sense of comfort or discomfort which cannot be localized, and it designates states of consciousness which are either agreeable or disagreeable. In a figurative sense the term is also applied to a mental emotion, or even to a moral conception; thus we may speak of a friendly feeling, a feeling of freedom. See *Emotion*.—BIBLIOGRAPHY: A. Bain, *The Emotions and the Will*; T. Ribot, *Psychology of the Emotions*.

Fegatella, a genus of Liverworts, family Marchantiales. *F. conica* is common on moist banks.

Feijoa, a genus of Myrtaceæ, natives of Brazil. The flowers are pollinated by birds, which feed on the juicy petals, a very unusual method.

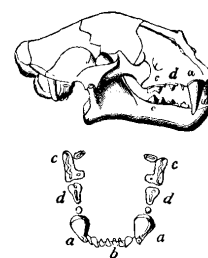
Feisul, or **Feisal**, Emir, King of Irak (Mesopotamia), born in 1887, the third surviving son of Hussein, King of Hejaz. Educated at Constantinople, Feisul held several posts under the Turkish Government, but took an active part in the revolutionary movement which resulted in the deposition of Sultan Abdul Hamid. He then returned to Arabia, where he commanded the Arabs against Ibn Saud, the head of a new religious sect, who threatened his father's emirate. During the European War Hussein sided with the Allies, and Feisul organized and commanded a regular Arab army, which formed Lord Allenby's right wing, and took part in the latter's operations in Palestine. As a reward for his services an independent, or semi-independent, state was established at Damascus under Feisul, and the prince was proclaimed King of Syria in March, 1920. Serious friction, however, arose between the French authorities and the Arabs, and hostilities broke out in July. The French, under General Gouraud, occupied Damascus, compelled the Arabs to recognize the French mandate for Syria, and deposed the new King of Syria. In August, 1921, Feisul became the first Arab king of the new state of Irak (Mesopotamia), set up by the British Government. He was crowned with great splendour at Bagdad on the 23rd of Aug., in the presence of a great gathering of his people and the representatives of the British Government. A personal message from King George V was handed to Feisul, and the British High Commissioner, Sir Percy Cox, formally announced the recognition by Great Britain of the new ruler of Irak.

Felaniche (fel-a' -nē'chā), a town in the Island of Majorca, a very ancient place with Moorish remains. Pop. (commune), 11,400.

Felegyhaza (fā'led-yä-za'), a town of Hungary, 66 miles S.E. of Budapest, with large cattle-markets and an extensive trade in corn, wine, and fruit. Pop. 34,924.

Felicu'di, one of the Lipari Isles, off the north coast of Sicily, 10 miles west of Salina. It is about 9 miles in circuit. The soil is both fertile and well cultivated. Pop. 800.

Felidæ, animals of the cat kind, a family of Carnivora in which the predaceous instincts reach their highest development. They are among the quadrupeds what the Falconidæ are among the birds. The teeth and claws are the principal instruments of the destructive energy in these animals. The incisor teeth are equal; the third tooth behind the large canine in either jaw is narrow and sharp, and these, the carnassial or sectorial teeth, work against each other like scissors in cutting flesh; the claws are sheathed and retractile. They all approach their prey stealthily, seize it with a spring, and devour it fresh. The species are numerous in Europe, Asia, Africa, and America, but none are found in Australia. The family comprehends the lion, tiger, leopard, lynx, jaguar, panther, cheetah, ounce, serval, ocelot, and cat.



Teeth of Felidæ
Skull and
Teeth of the
Tiger. *a*, Canines
or tearing teeth.
b, Incisors or
cutting teeth. *c*,
True molars or
grinding teeth. *d*,
Carnassial or
sectorial teeth.

Felix, Antonius or Claudius, procurator of Judæa and freedman of the Emperor Claudius, is described by Tacitus as unscrupulous and profligate both in his public and private conduct. It was before this Felix that Paul's discourse (*Acts*, xxiv, 25) was spoken. He was recalled A.D. 62, and narrowly escaped condemnation at Rome, on charges which the Jews had lodged against him.

Felix, Marcus Minucius, a distinguished Roman lawyer, who embraced Christianity, and wrote a defence of it in a dialogue entitled *Octavius*. The period when he flourished is uncertain; but Jerome is probably right in placing him about A.D. 230.

Felixstowe, a watering-place in England, on the Suffolk coast, 11 miles south-east of Ipswich, between the mouths of the Orwell and Deben. The steamers which ply between Ipswich and Harwich on the Orwell call at Felixstowe Pier, which is opposite Harwich. Pop. (urban district), 11,655.

Fellah (pl. *fellahin*), an Arabian word meaning 'peasant', and used for the labouring class in Egypt. The fellahs or *fellahin* constitute about three-fourths of the population of Egypt, and are mostly the direct descendants of the old Egyptians, although both their language and religion are now that of their Arabian conquerors. They live in rude huts by the banks of the Nile,

and in past times have suffered much from over-taxation and oppressive rule at the hands of a succession of tyrants, and especially of the Turks before the British occupation of Egypt.

Fella'tah, Fulbe, or Fulahs, a remarkable African race of the negro type, the original locality of which is unknown, but which is now widely diffused throughout the Sudan, where they are the predominant people in the states of Futa-Toro, Futa-Jalon, Bondu, and Sokoto. Though of the negro family, they have neither the deep jet colour, the crisped hair, flat nose, nor thick lips of the negro. In person they are decidedly handsome, and mostly of a light copper colour. They are shrewd, intelligent, and brave, and are mostly Mohammedans. Their influence is continually spreading.

Fel'enberg, Philip Emanuel von, Swiss educationalist, born in 1771, died in 1844. Having devoted himself to the social and intellectual improvement of the peasantry, he purchased the estate of Hofwyl, and established successively an institution for instructing the children of the poorer classes, a seminary for children in the higher grades of life, and a normal school. The pupils were all trained to work in the fields or at the bench, and the product of their labour was sufficient to cover the expenses of their education. Fellenberg's scheme was ultimately so successful as to attract the attention even of foreign Governments. The institutions established by him still exist in a modified form.

Felling, a populous locality in Durham, a little to the south-east of Newcastle, and adjoining Gateshead, consisting of the combined villages of High and Low Felling, and forming an urban district. It contains chemical and other industrial works. Pop. 26,152.

Fel'laws, Sir Charles, traveller and antiquarian, was born in 1799 at Nottingham, died in 1860. He first explored the valley of the Xanthus in Lycia, in 1838, and discovered the remains of the cities Xanthus and Teos. Under the auspices of the trustees of the British Museum he made further explorations in 1839 and 1841, and succeeded in obtaining the marbles now in the Lycian saloon of the museum. He was knighted by the queen in 1845. His principal works are: *The Xanthian Marbles: their Acquisition and Transmission to England*; *Travels and Researches in Asia Minor*; and *Coins of Ancient Lycia before the Reign of Alexander*.

Fellowship, a distinction conferred by some universities, especially those of Oxford and Cambridge, which entitles the holder, called a fellow, to an annual stipend for a certain period. Fellowships in the English colleges commonly range in value from about £150 to £250 or £300 a year, and they all confer upon their holders the right to apartments in the college, and certain privileges as to commons or meals. Formerly they were usually tenable for life or till the attainment of a certain position in the Church or at the Bar, or till marriage; but six or seven years is now a common period during which they may be held, though this may be prolonged in certain circumstances. At Dublin University senior fellows hold their office for life.

Felo de se (Lat., 'a felon in regard to himself'), in law, a person who, being of sound mind and of the age of discretion, deliberately causes his own death. Formerly, in England, the goods of such a person were forfeited to the Crown, and his body interred in an ignominious manner; that is, unless the coroner's jury gave a verdict of unsound mind; but these penalties have been abolished.

Fel'ony, in law, includes generally all crimes below treason and of greater gravity than misdemeanours. Formerly it was applied to those crimes which entailed forfeiture of lands or goods as part of the punishment.

Felsite, or **Felstone**, a hard, compact igneous rock of somewhat flinty appearance, composed usually of quartz and orthoclase felspar intimately mixed, but sometimes of less highly siliceous minerals.

Fel'spar, or **Feldspar**, a very important group of mineral silicates of aluminium, with potassium, sodium, or calcium, ranging from orthoclase, the potassium species, with 64.7 per cent of silica, to anorthite, the calcium species, with only 43.3. Albite, the sodium felspar, has 68.8 per cent of silica, and the species between this and anorthite are regarded as mixtures of albite and anorthite molecules. These molecules probably do not exist as such within the crystals; but the various characters of the species graduate into one another in agreement with the chemical constitution, so that the felspars form an admirable example of the relation of chemical composition, specific gravity, and crystalline and optical features. At the same time orthoclase and microcline are both potassium felspars; yet the former crystallizes in the monoclinic, and the latter in the triclinic system.

All the sodium, sodium-calcium, and calcium species are triclinic, except the rare monoclinic sodium feldspar barroisite. The forms throughout the feldspar series are closely similar, and the hardness is uniform, being just below that of quartz, and about that of a steel file. Feldspar is one of the principal constituents of almost all igneous rocks, such as granite, diorite, and basalt. The alkali species yield kaolin by alteration, and are thus the source of china-clay.

Printed and bound in Great Britain

NOTES

[1] The use of gas, as already pointed out, had been forced on the British by its adoption by the Germans. Ultimately the methods invented by British chemists and physicists outgassed the Germans.

[2] Portugal was drawn into the war on the side of the Allies on 19th March, 1916, when Germany declared war on her, ostensibly because she had requisitioned German merchant ships lying in her harbours, but in reality because an invasion of Mozambique was then becoming necessary to Germany's hard-pressed troops in East Africa.

[3] Japan afterwards assisted the Allies with war supplies—particularly with heavy guns to Russia before the Bolshevik betrayal—subsequently helping to stem the tide of Bolshevism in the Far East, besides contributing with her ships to the defeat of the German submarine campaign in the Mediterranean.

*** END OF THE PROJECT GUTENBERG EBOOK THE NEW
GRESHAM ENCYCLOPEDIA. ESTREMOZ TO FELSPAR ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

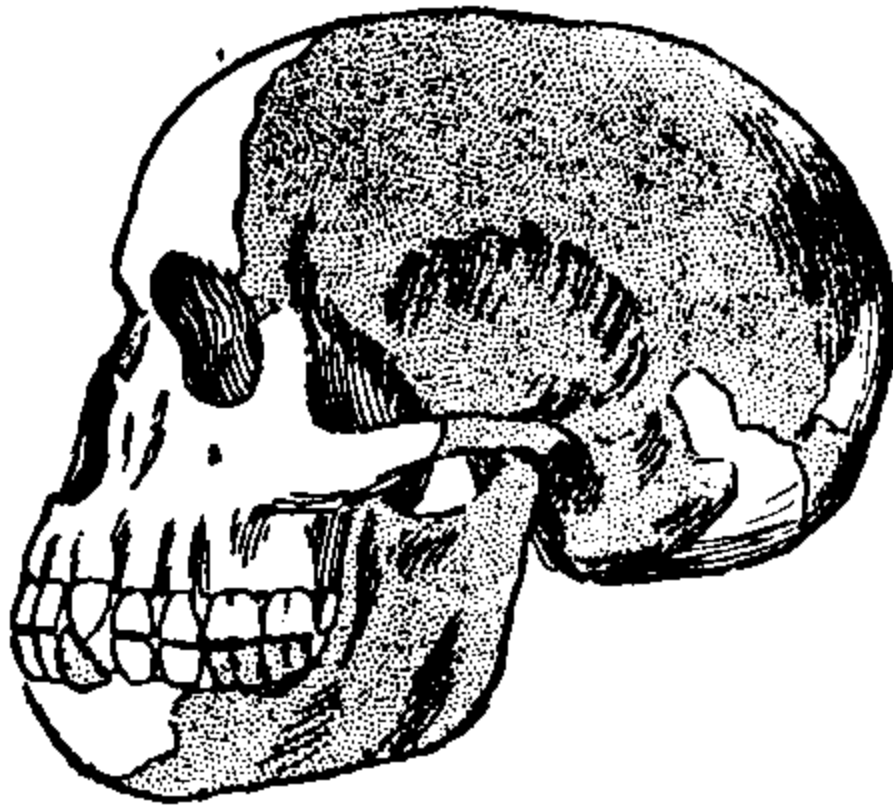
Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.



[back](#)

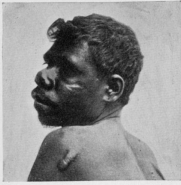


[back](#)

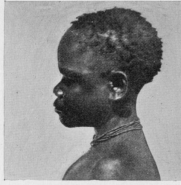


[back](#)

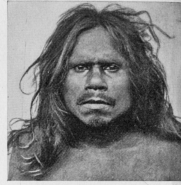
ETHNOLOGY



Australian



Akka Pygmy (Negrito)



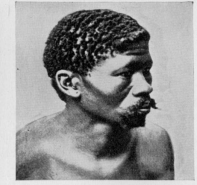
Vedda, Full-face (Pre-Dravidian)



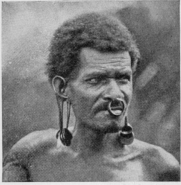
Vedda, Profile (Pre-Dravidian)



Papuan



Bushman



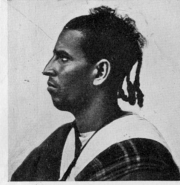
Solomon Islander (Melanesian)



Andaman Islander (Negrito)



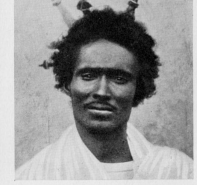
Nubian Girl



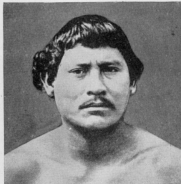
Touareg (Western Hamite)



Zulu (Bantu)



Somali (Ethiopian Hamite)



Carib (Southern American)



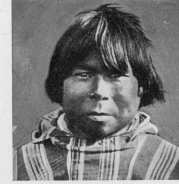
Javanese (Oceanic Mongol)



Maori (Polynesian)



Sikh (Indo-Aryan)



Eskimo



Lapp (Mongoloid)



Chinese Lady (Southern Mongol)



Chinaman (Southern Mongol)



Kalmuk (Northern Mongol)



North American Indian

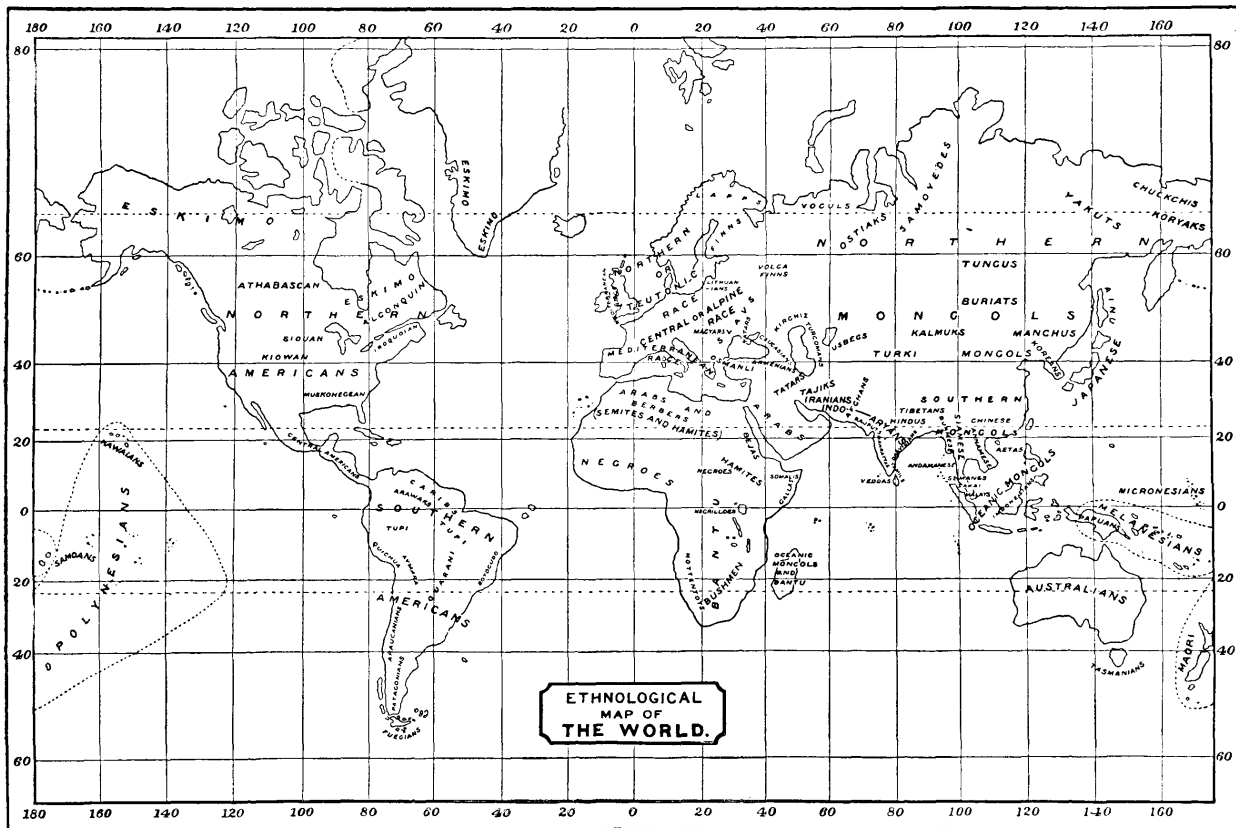


Turk

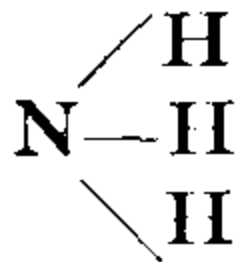


Arab (Semite)

[back](#)



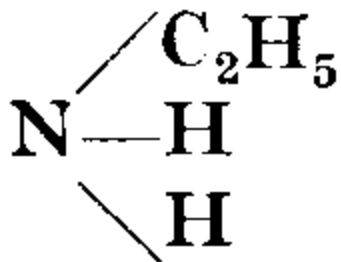
[back](#)



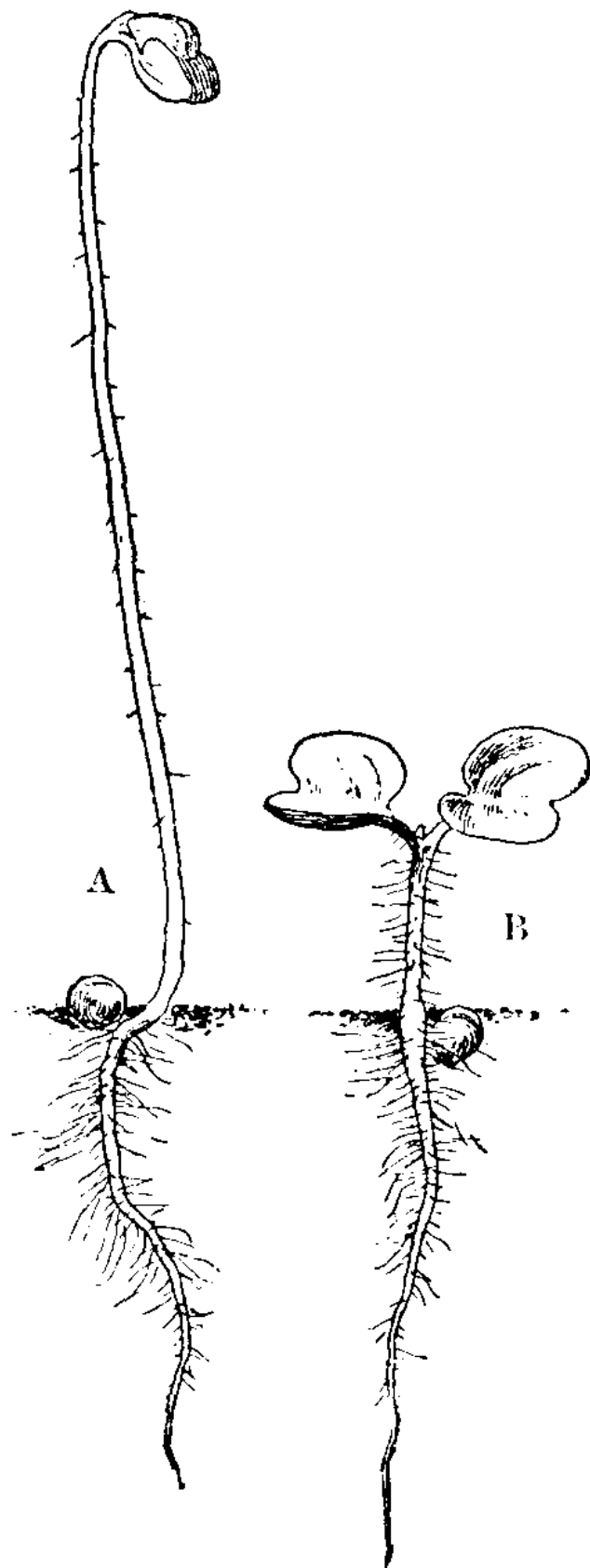
[back](#)



[back](#)



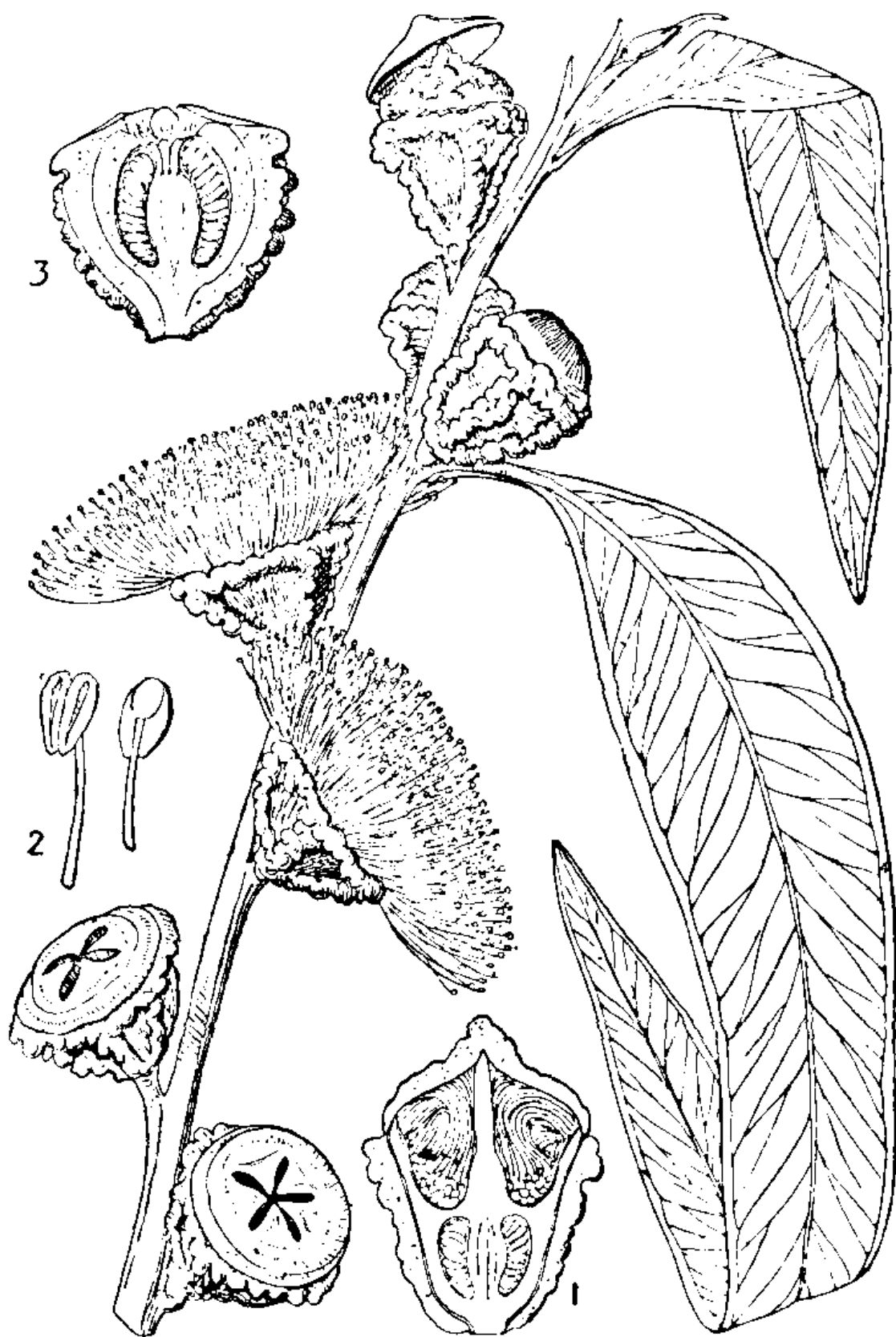
[back](#)



[back](#)



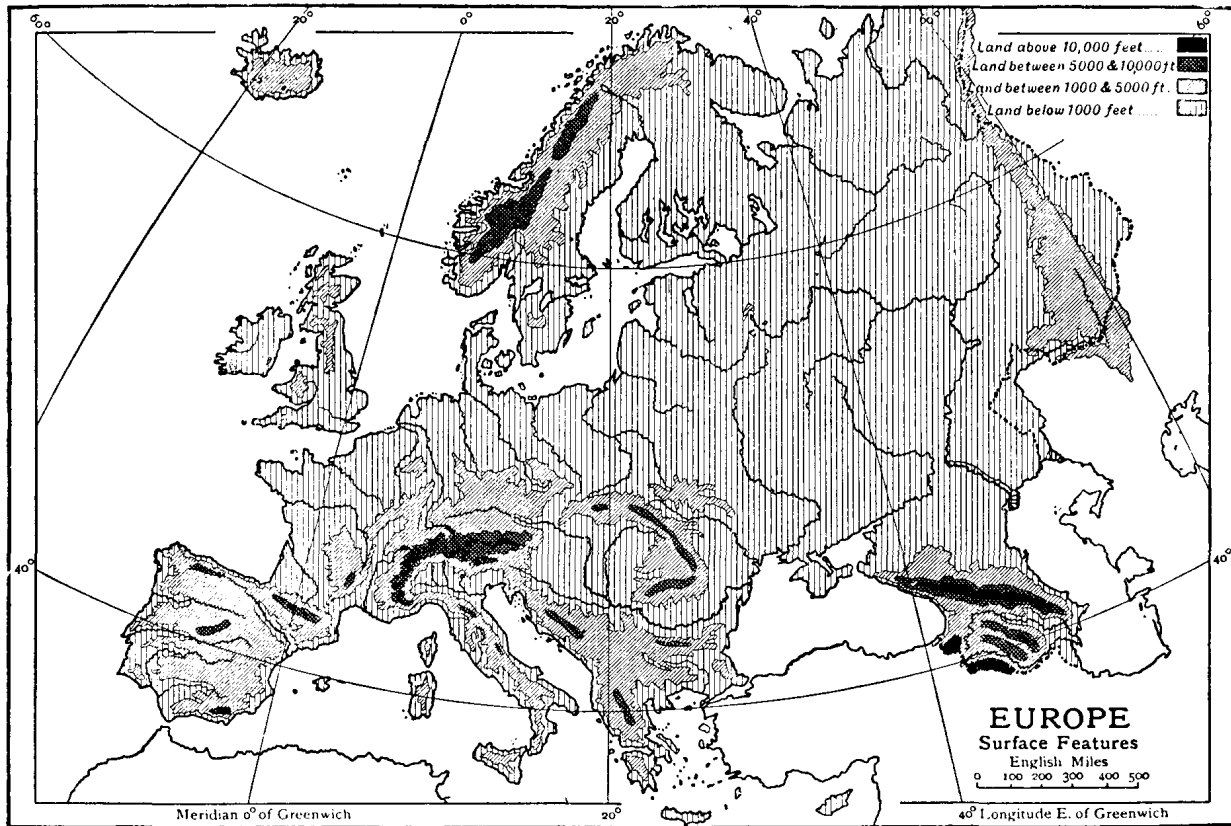
[back](#)



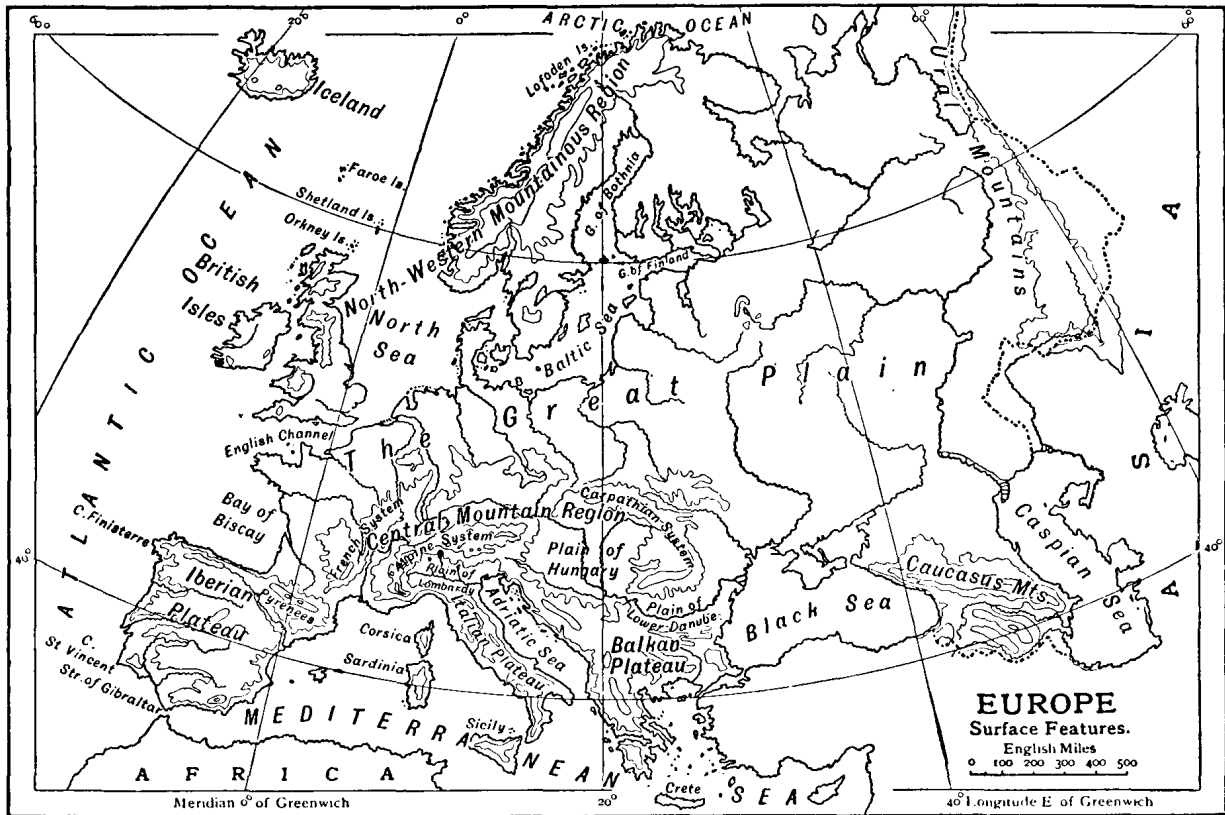
[back](#)

U

[back](#)



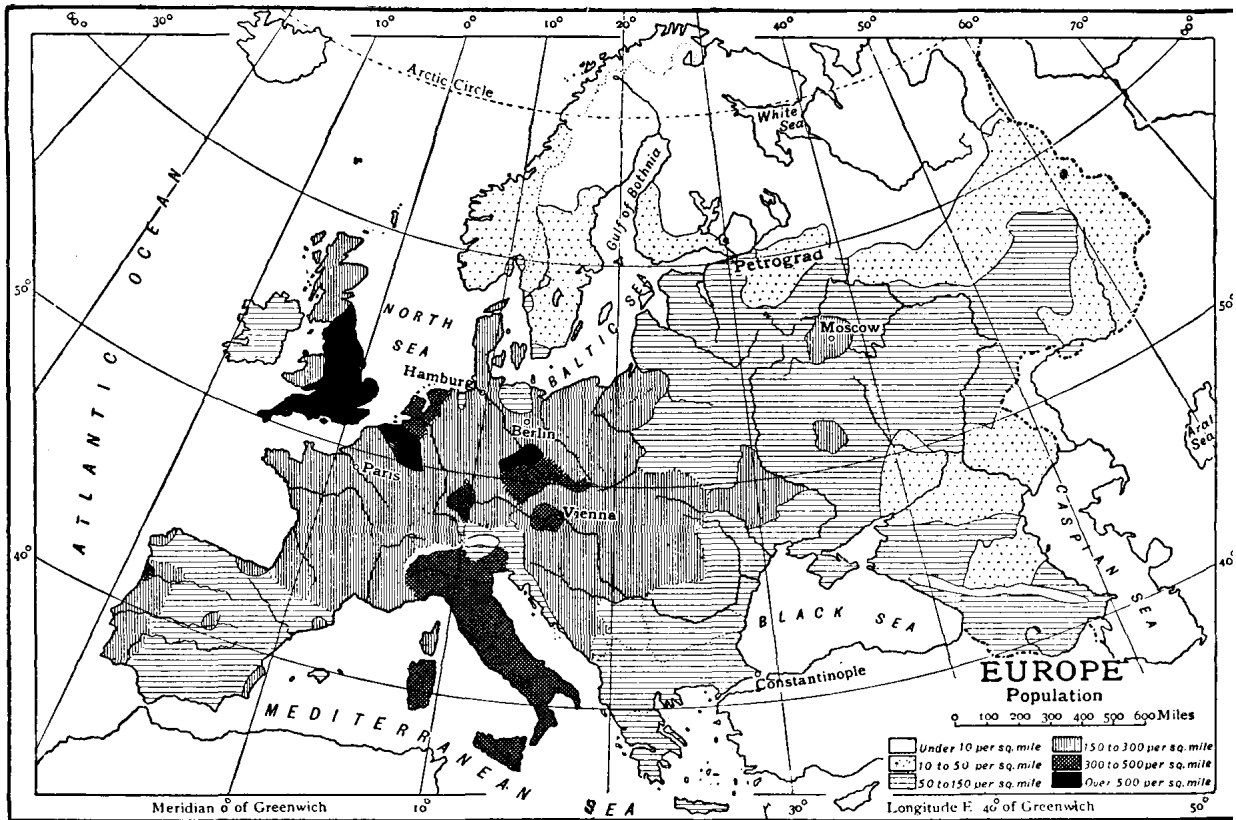
[back](#)



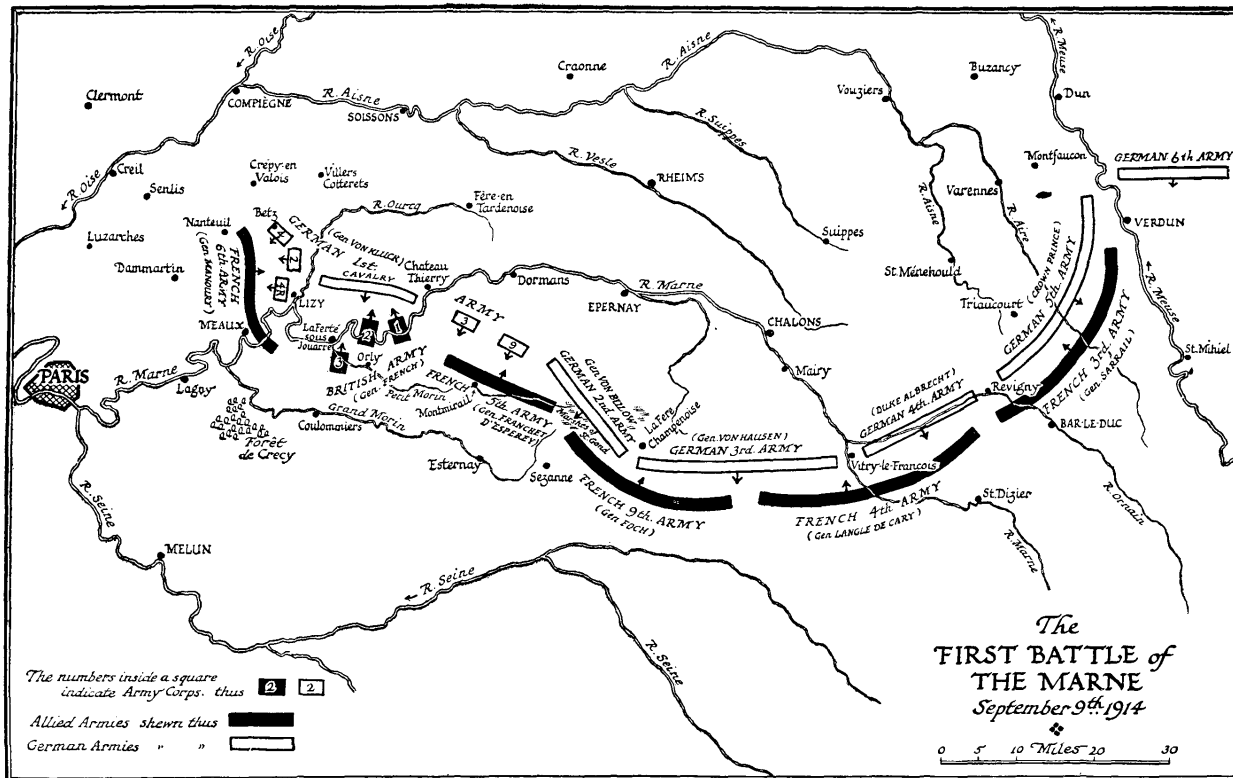
[back](#)



[back](#)

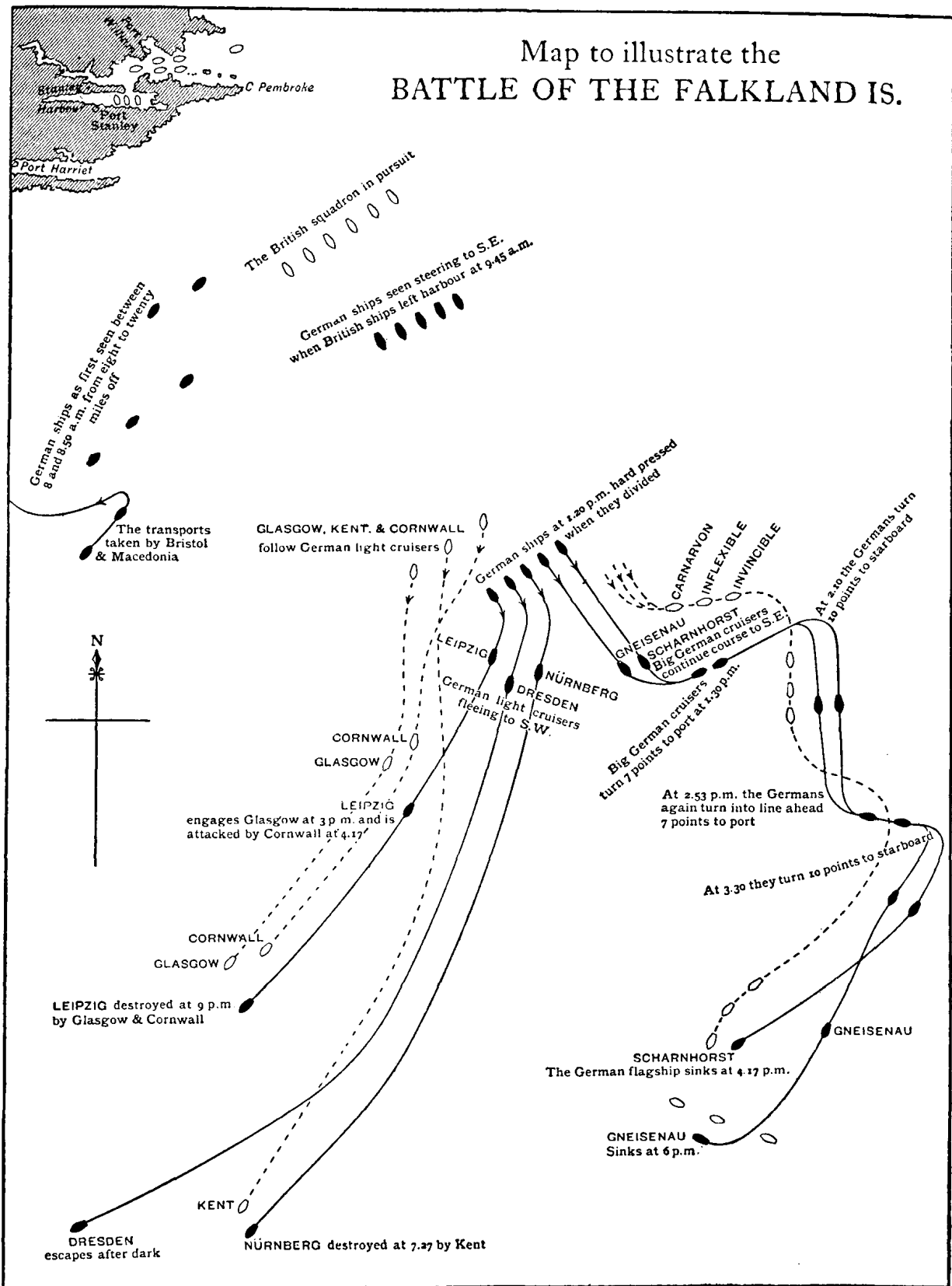


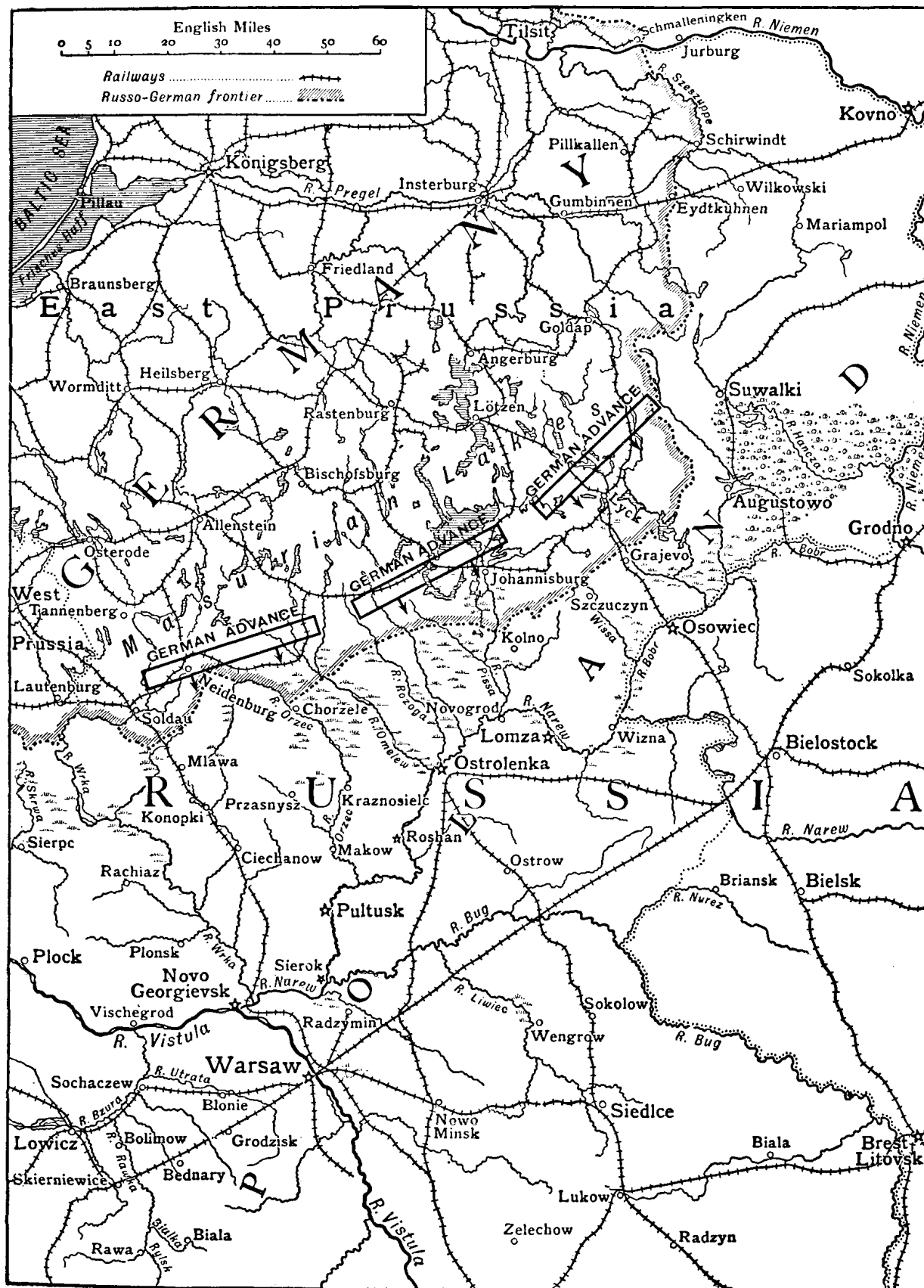
[back](#)



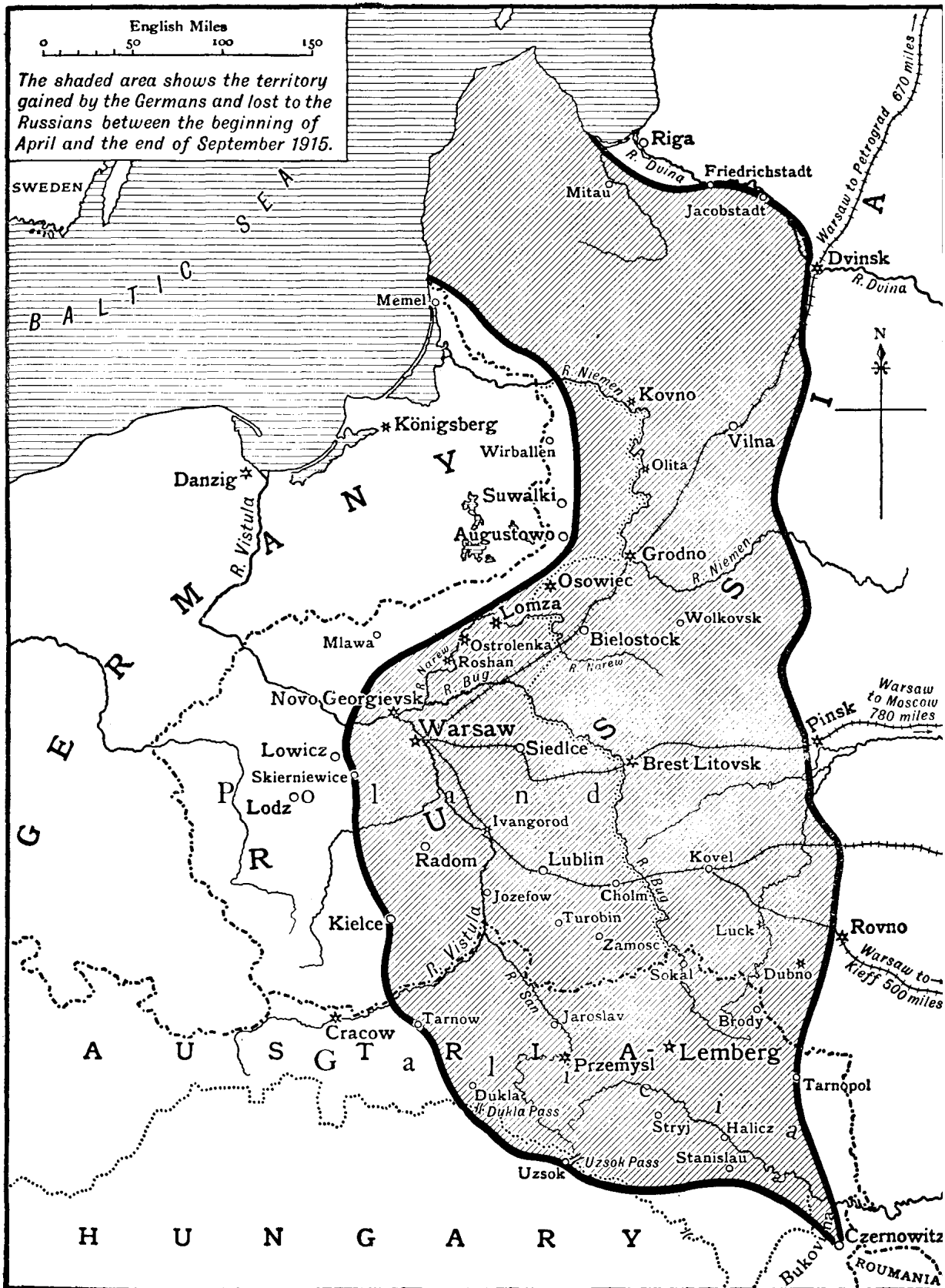
[back](#)



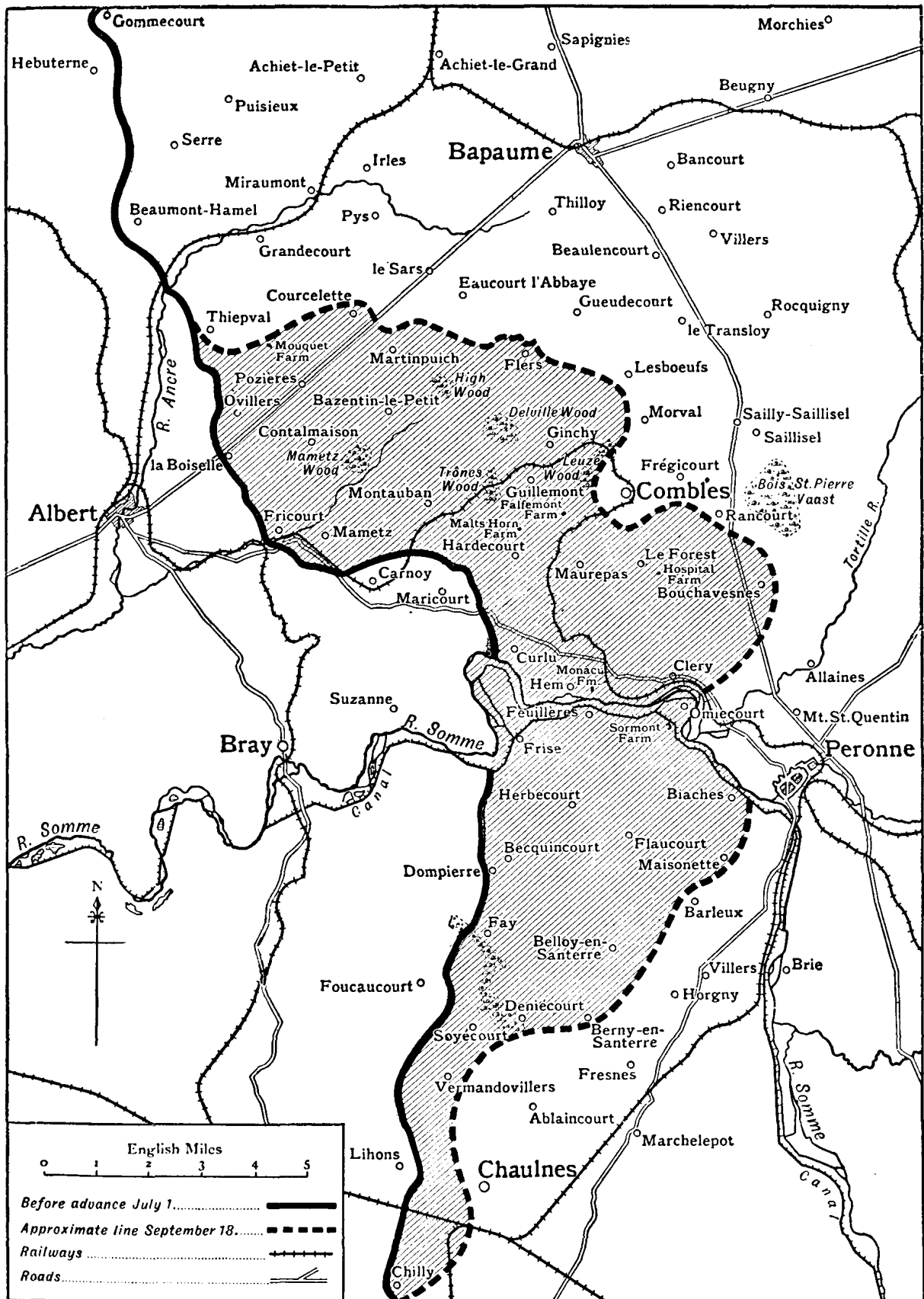




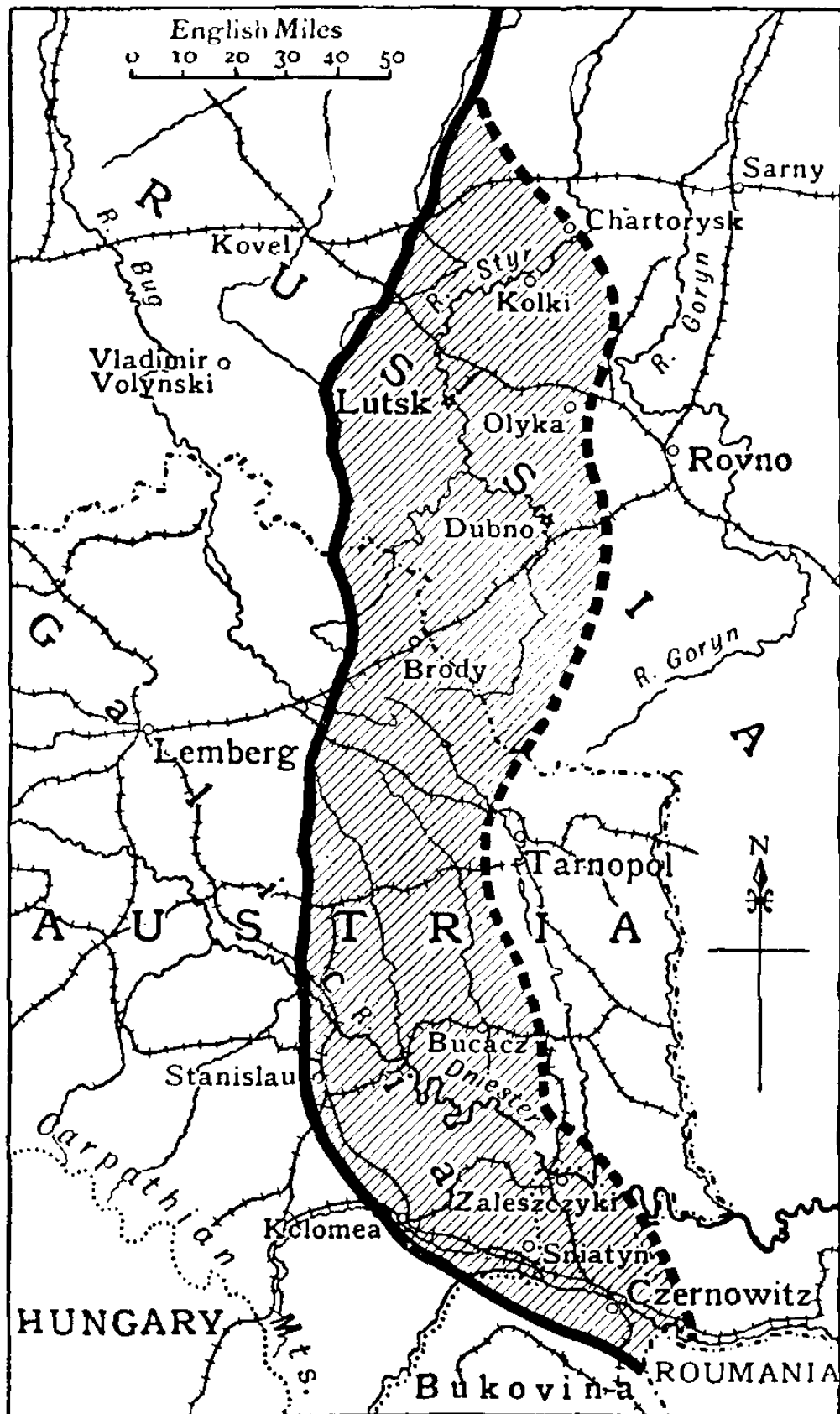
[back](#)



[back](#)



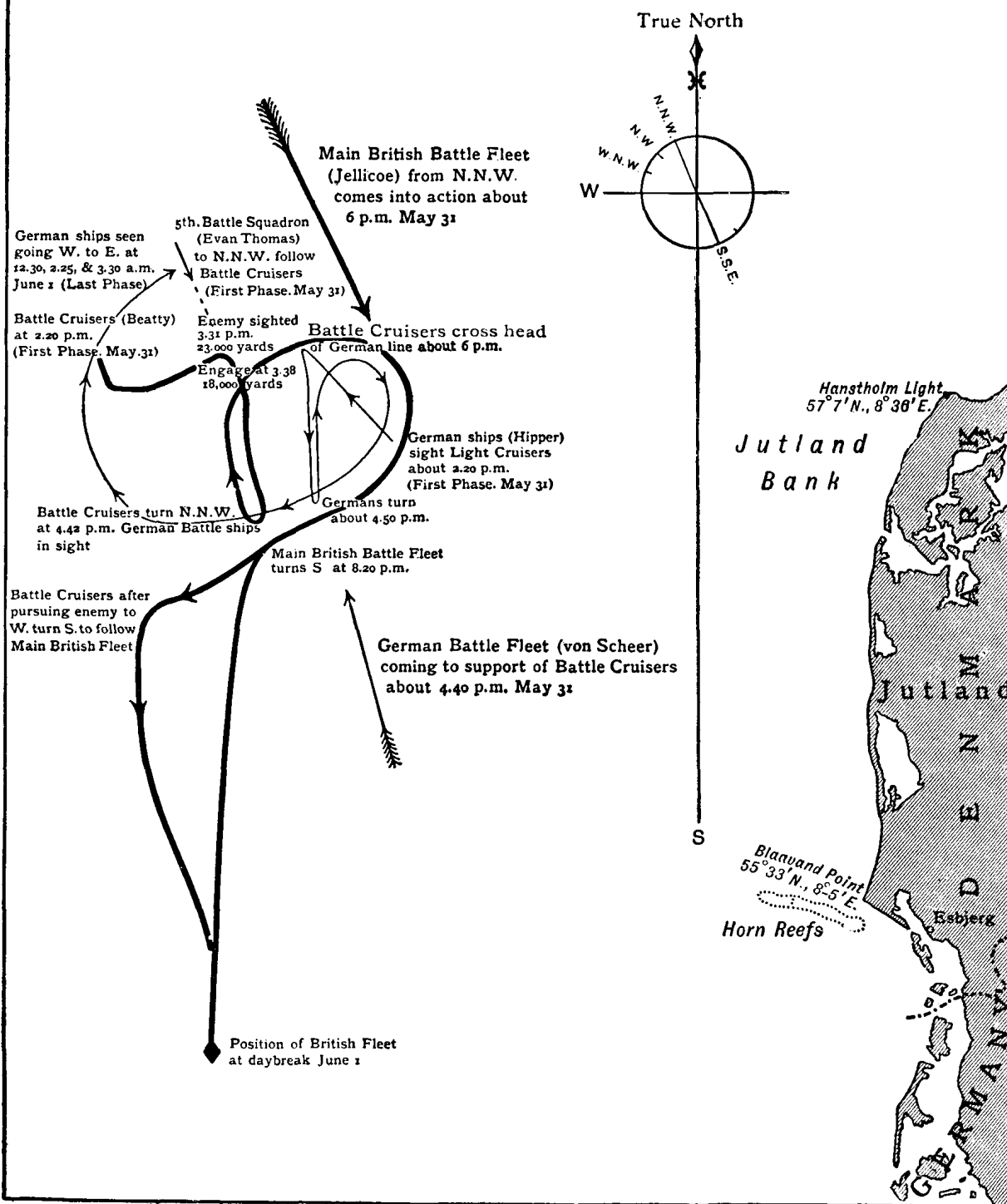
[back](#)



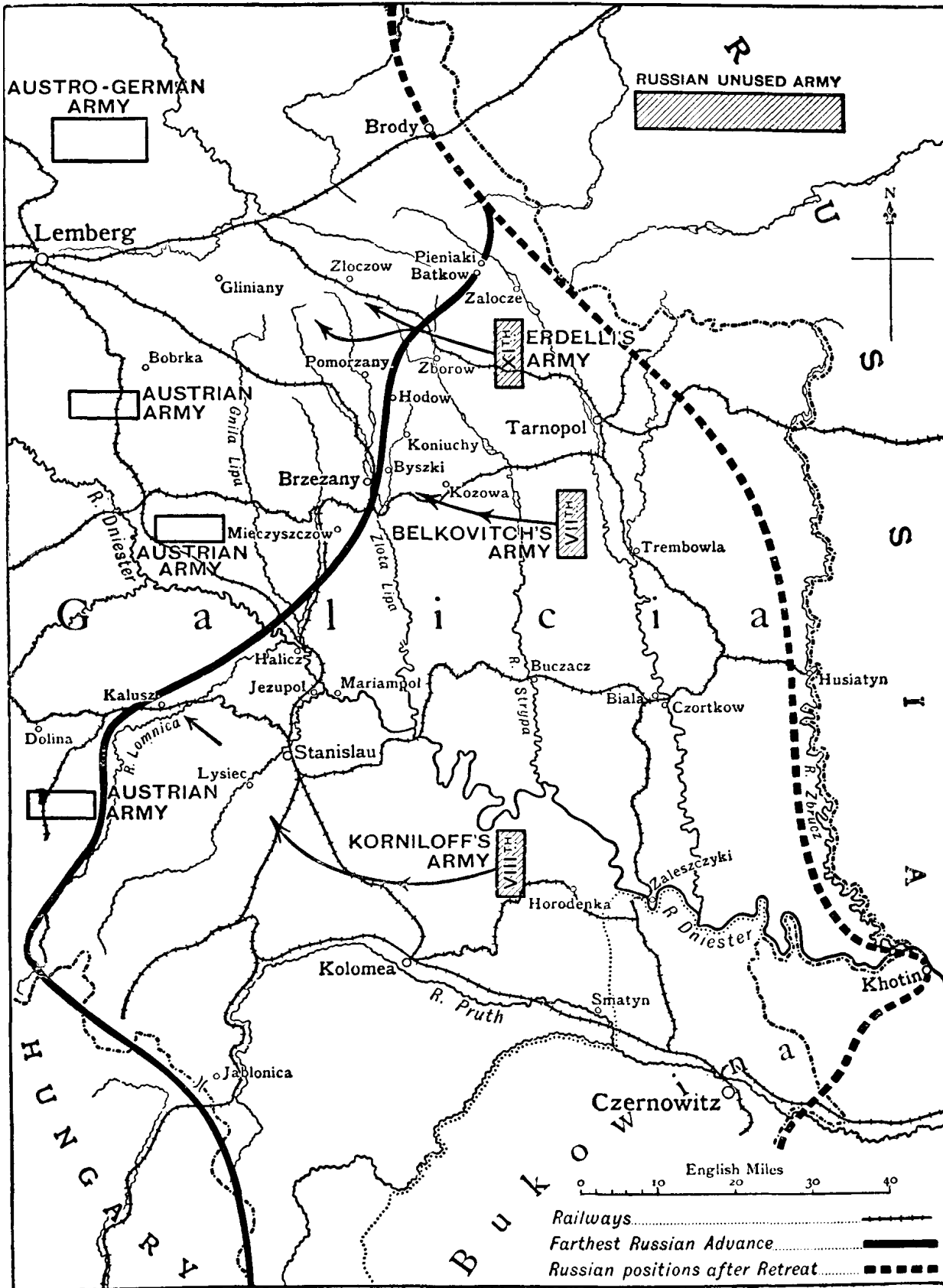
[back](#)

The Battle of JUTLAND BANK

May 31-June 1, 1916



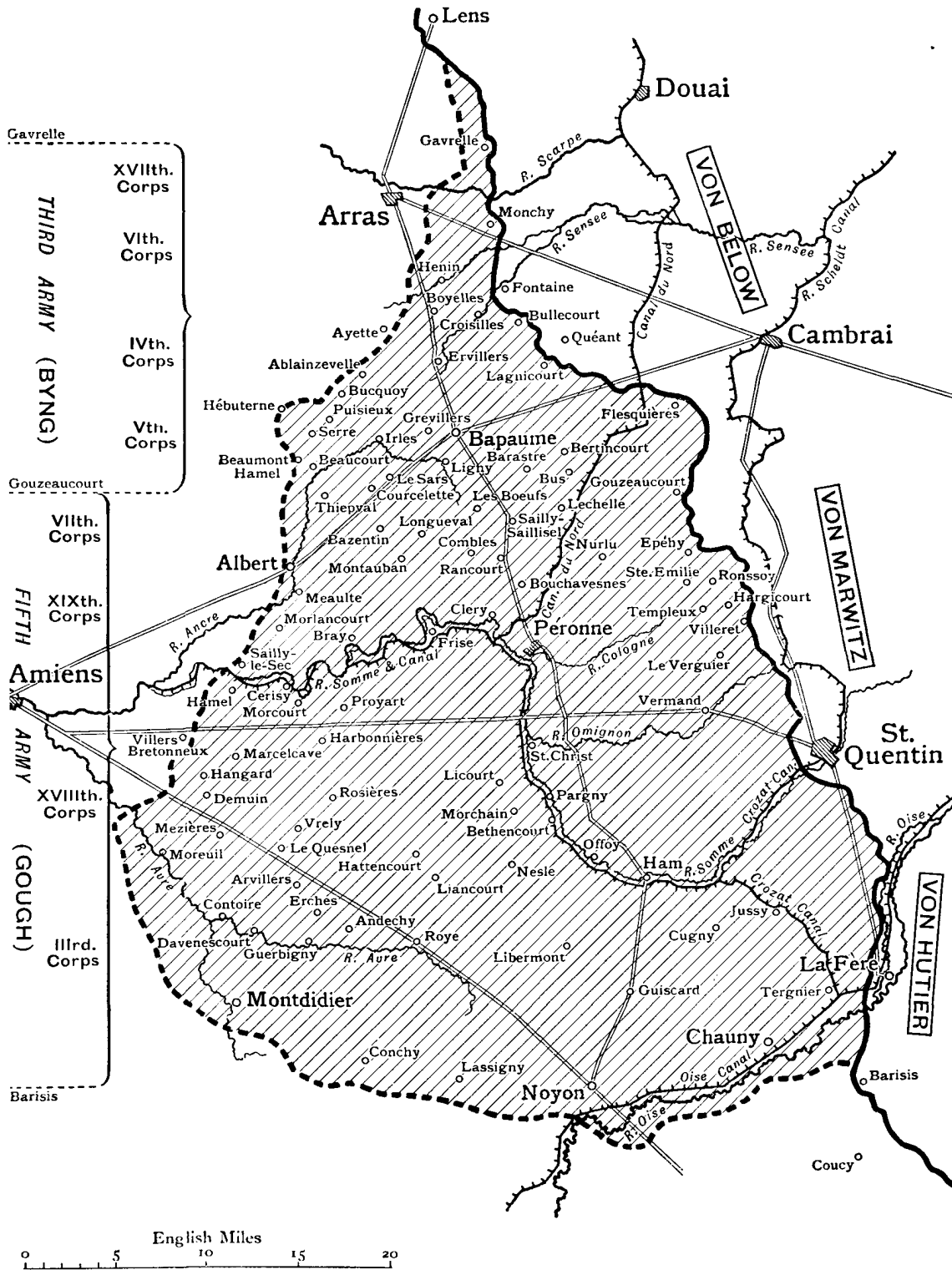
[back](#)



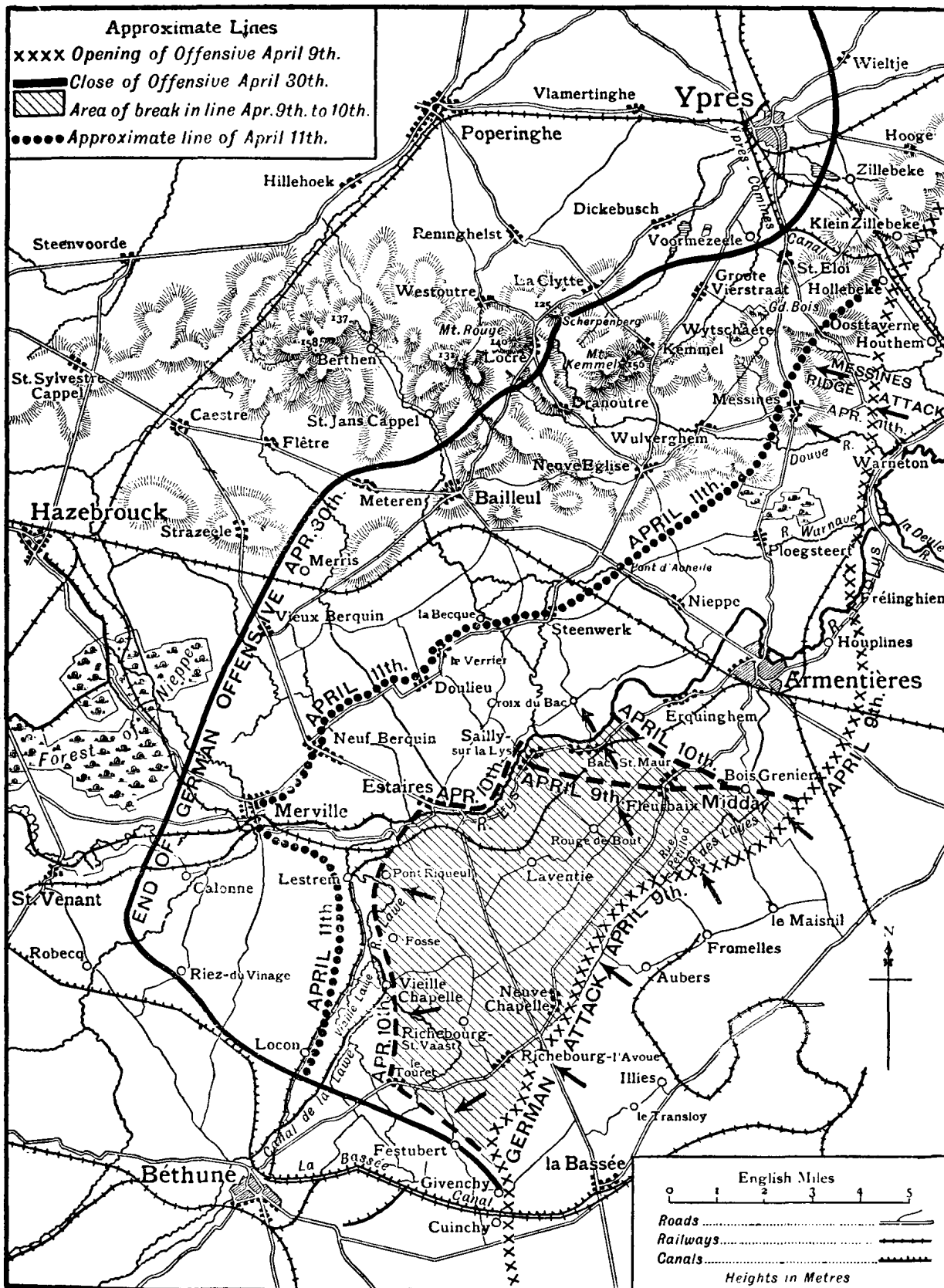
[back](#)

**Map showing the
GERMAN ADVANCE and the BRITISH RETREAT
between March 21st. & March 28th. and the line finally re-established on April 4th. & 5th. 1918**

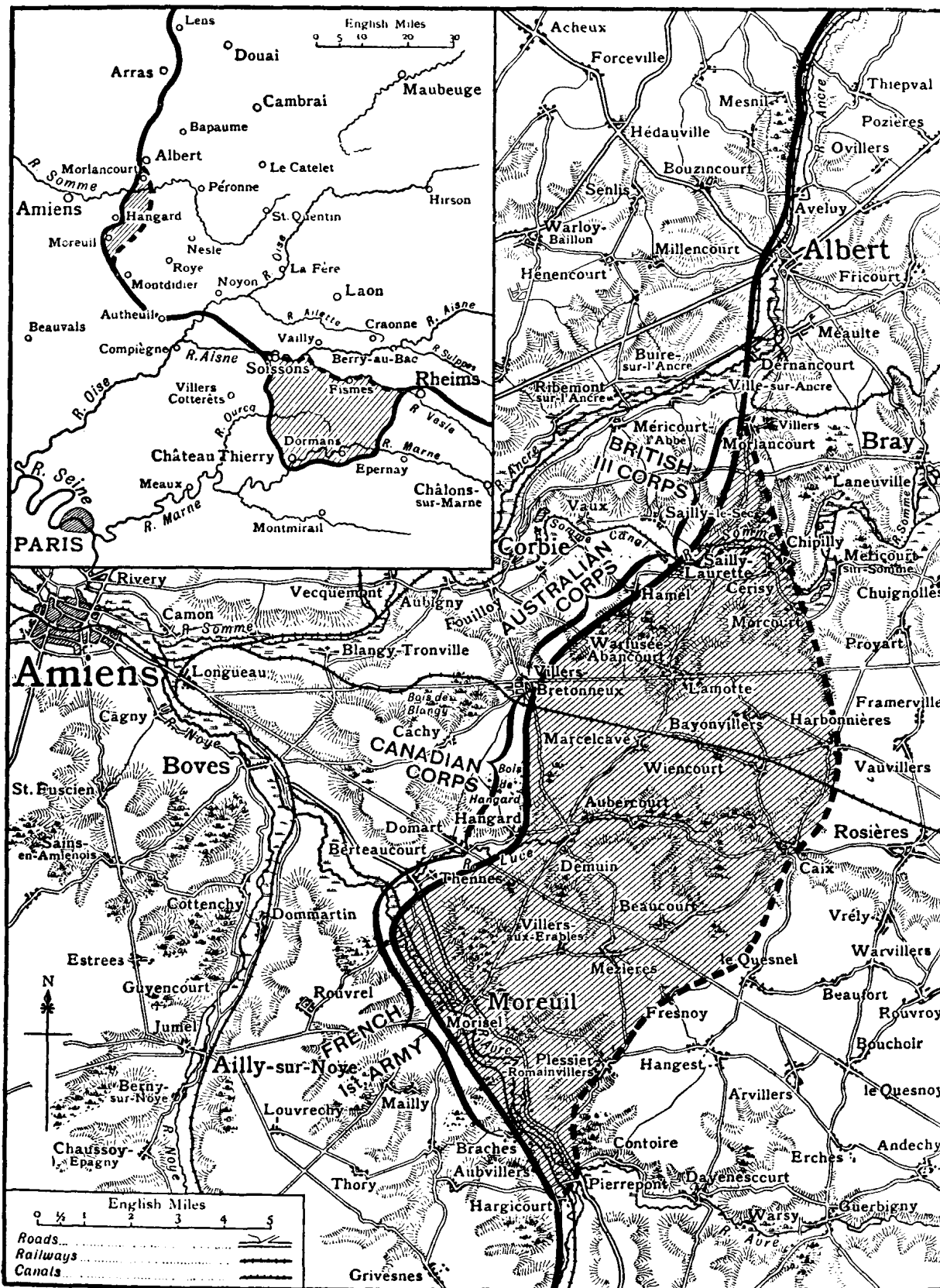
Line of March 21st. Line of April 5th.



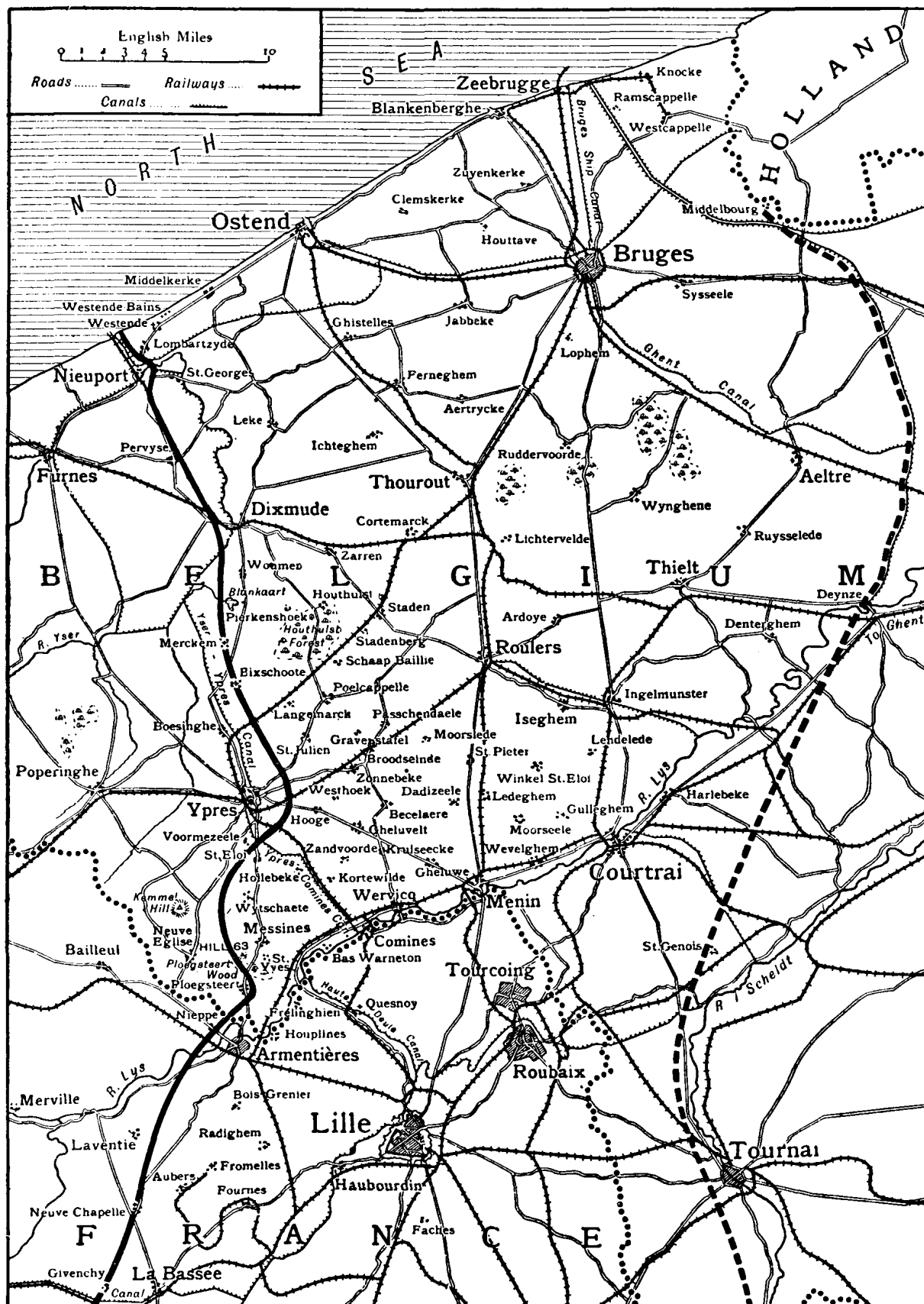
[back](#)



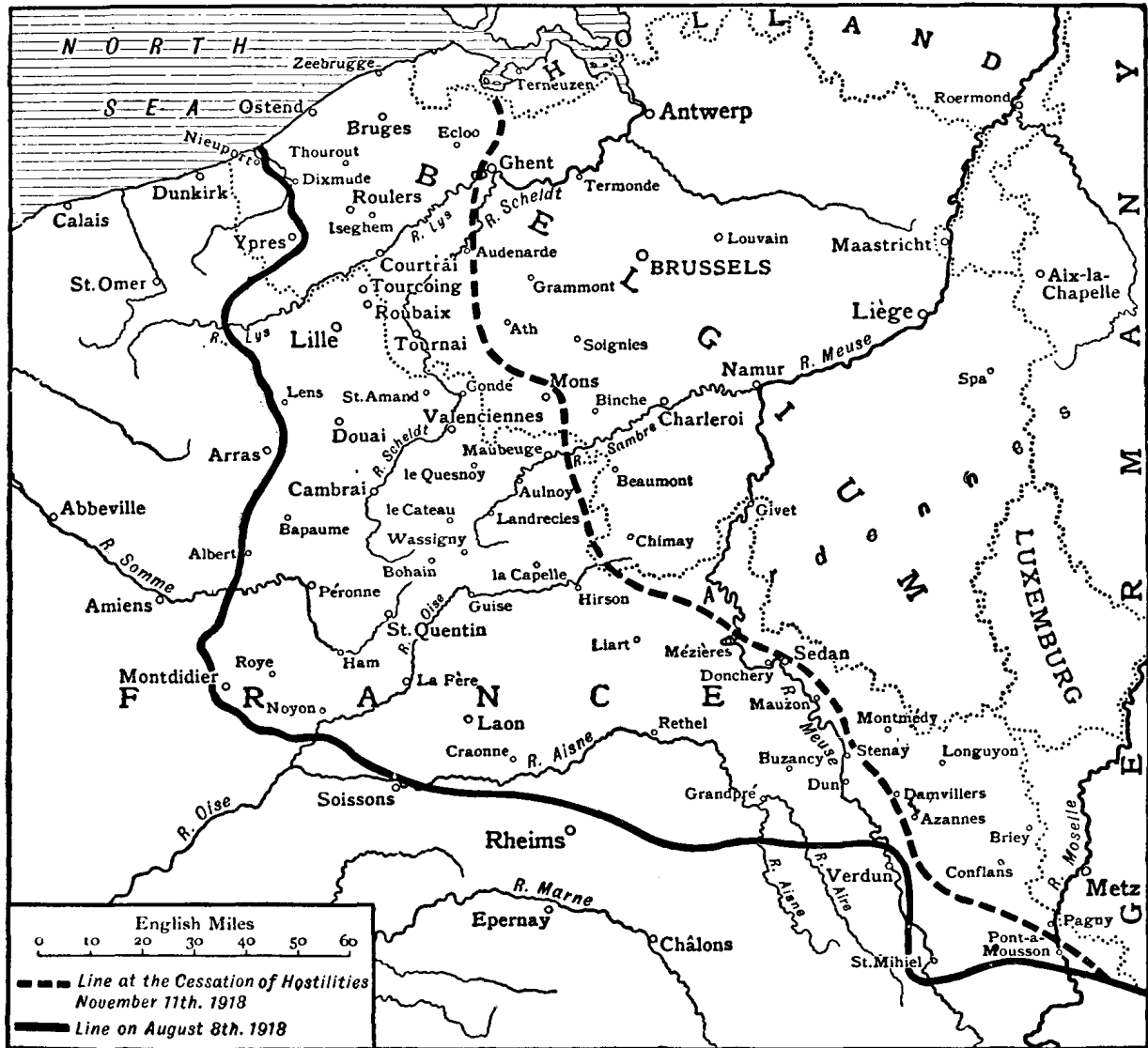
[back](#)



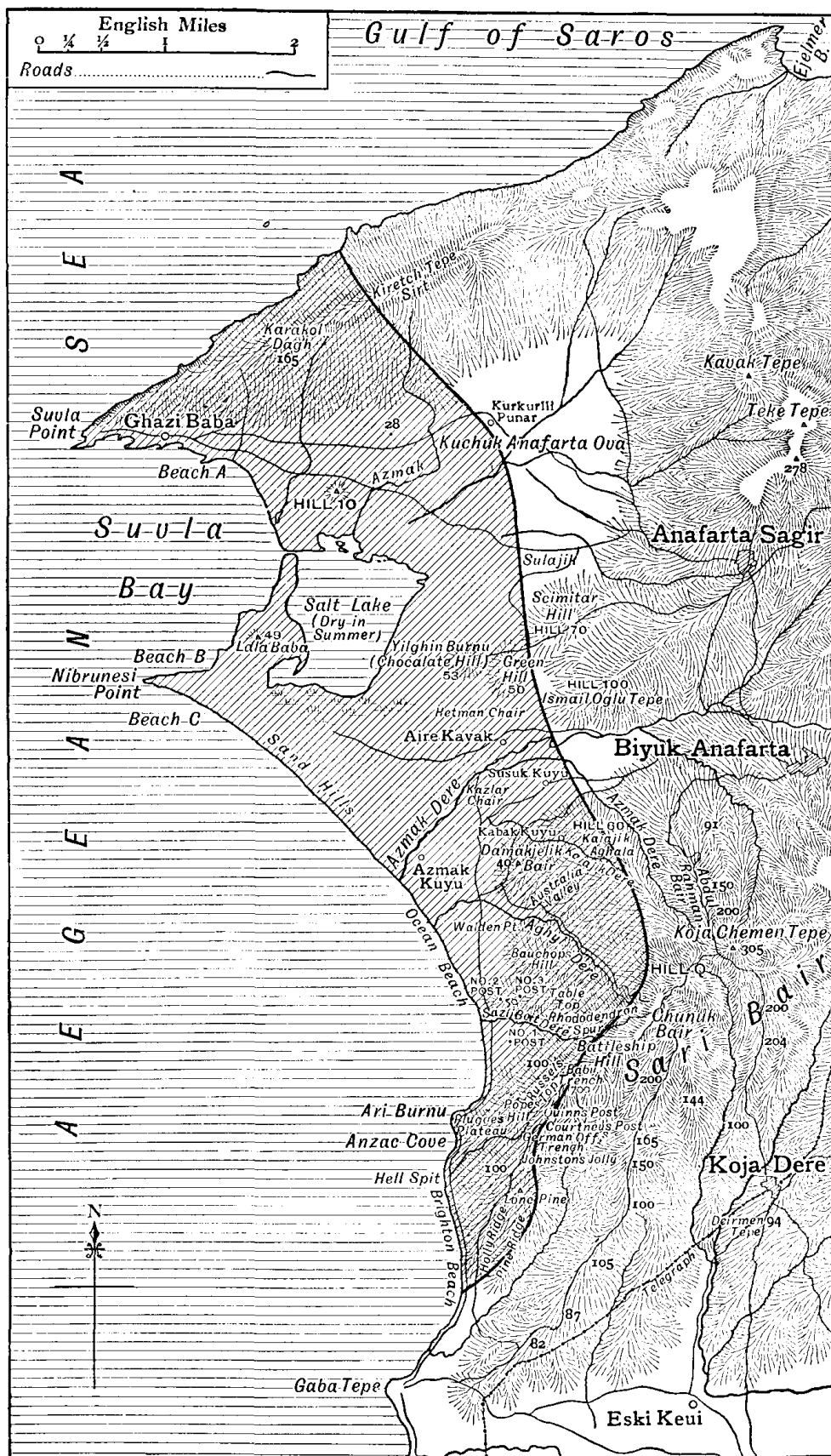
[back](#)



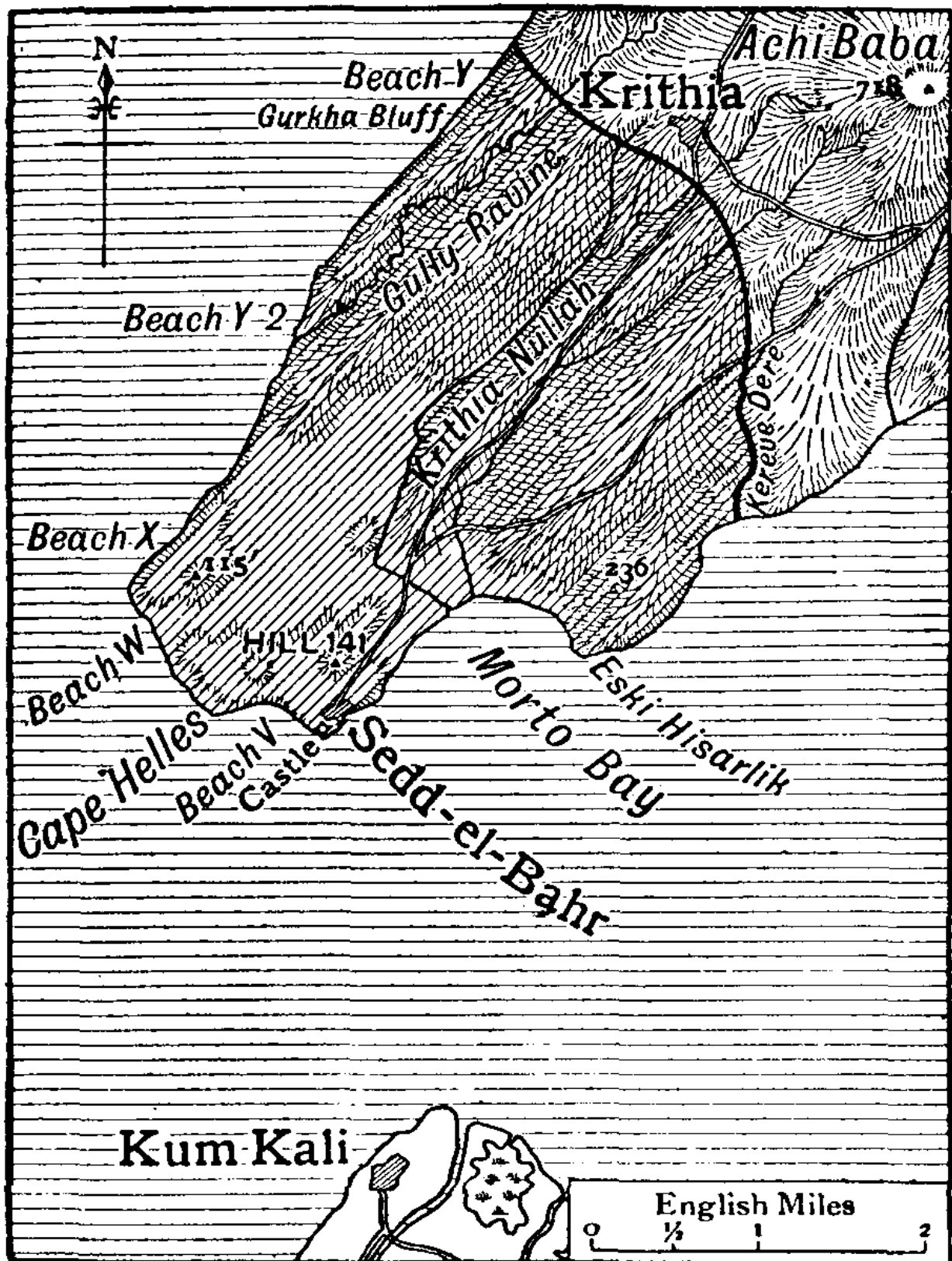
[back](#)



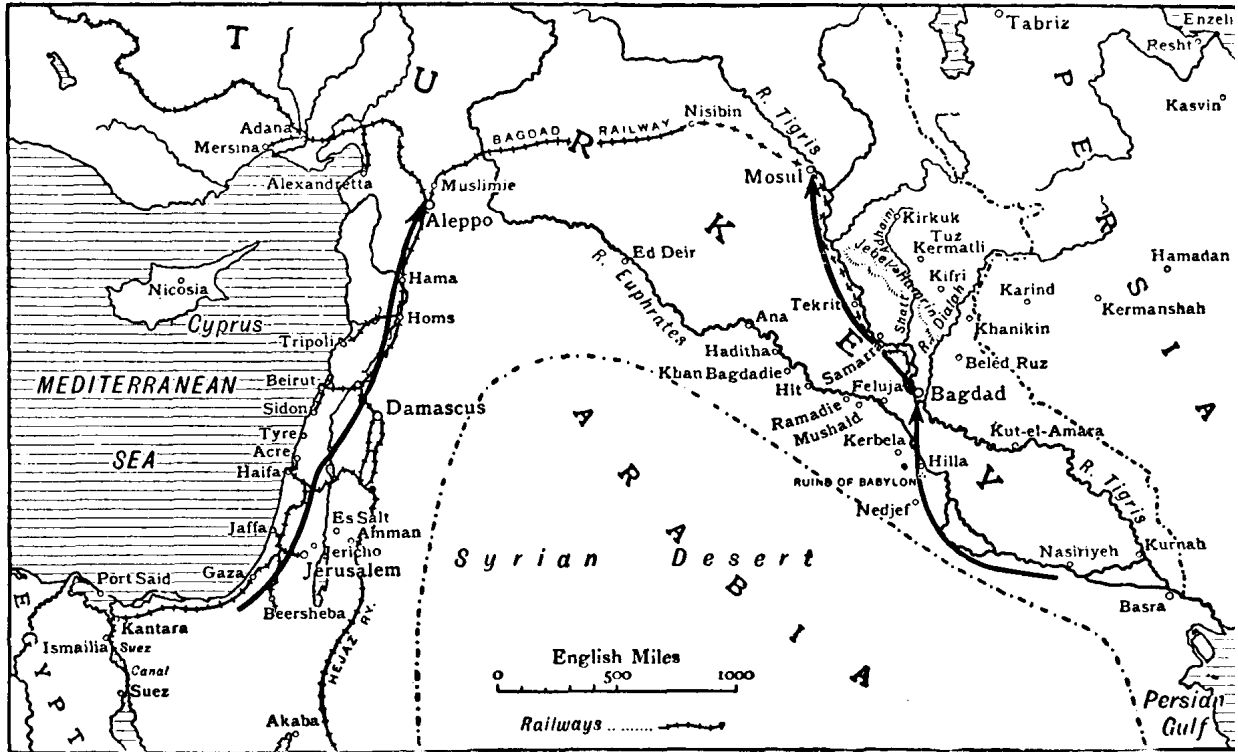
[back](#)



[back](#)



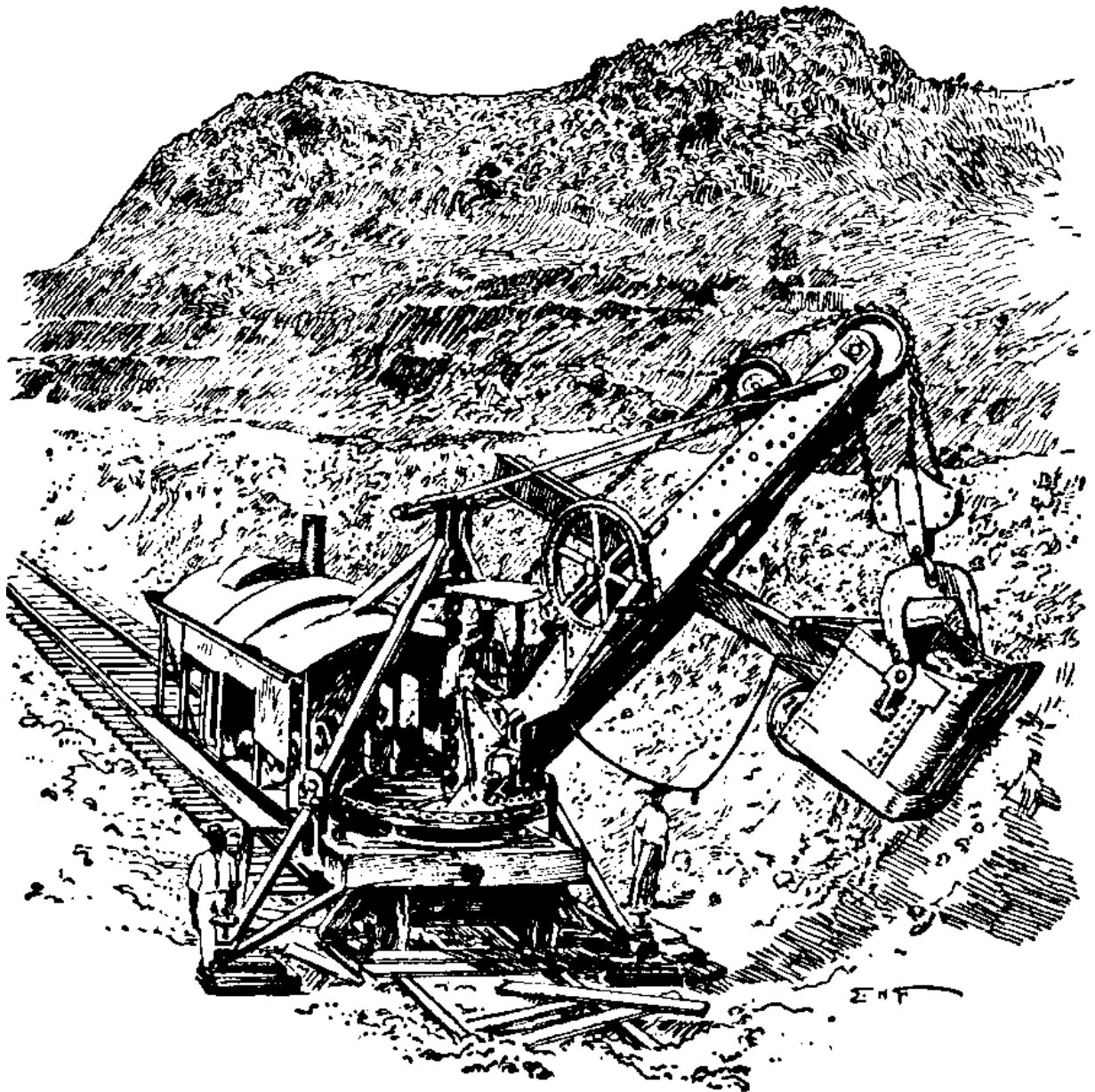
[back](#)



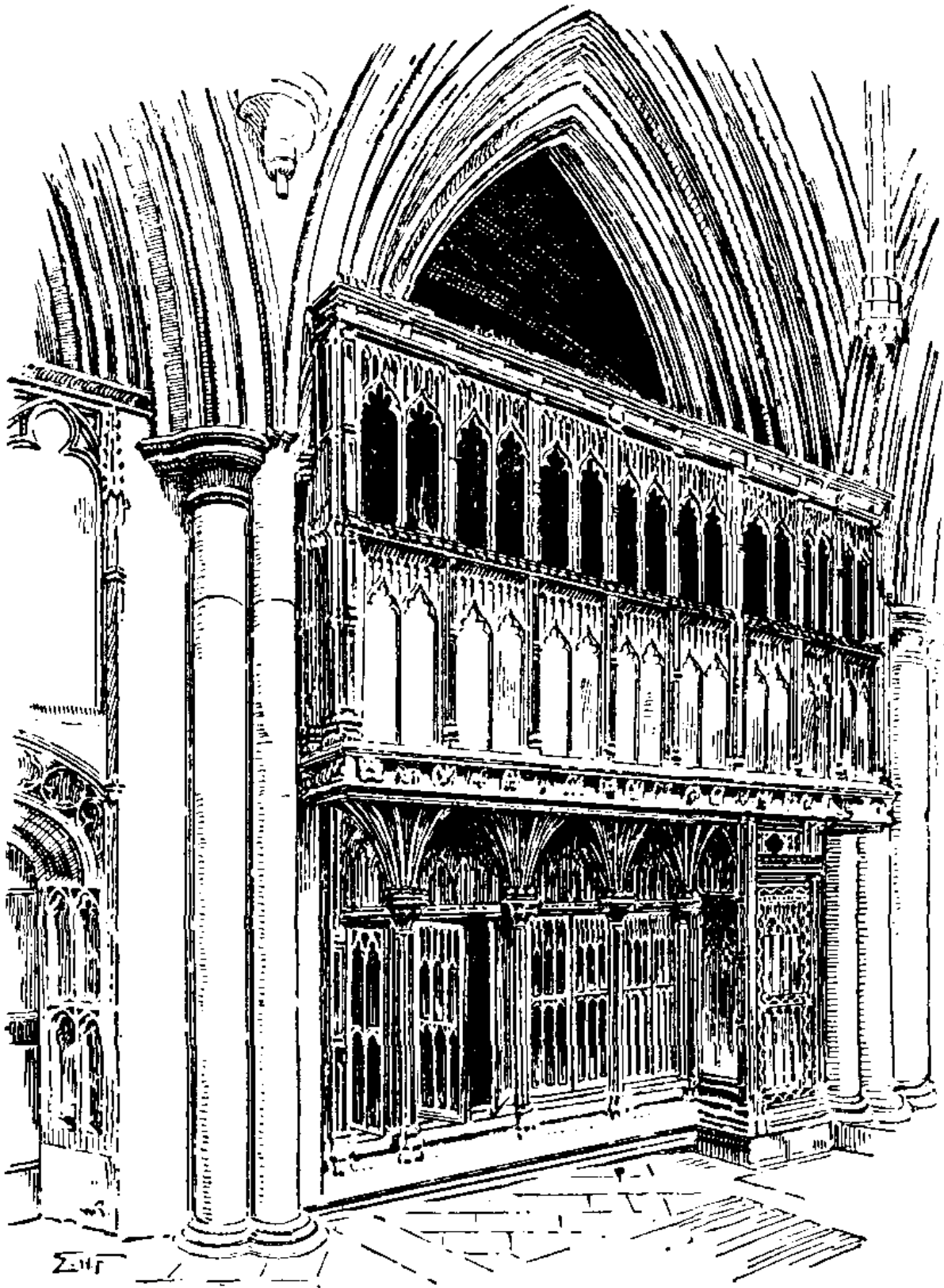
[back](#)



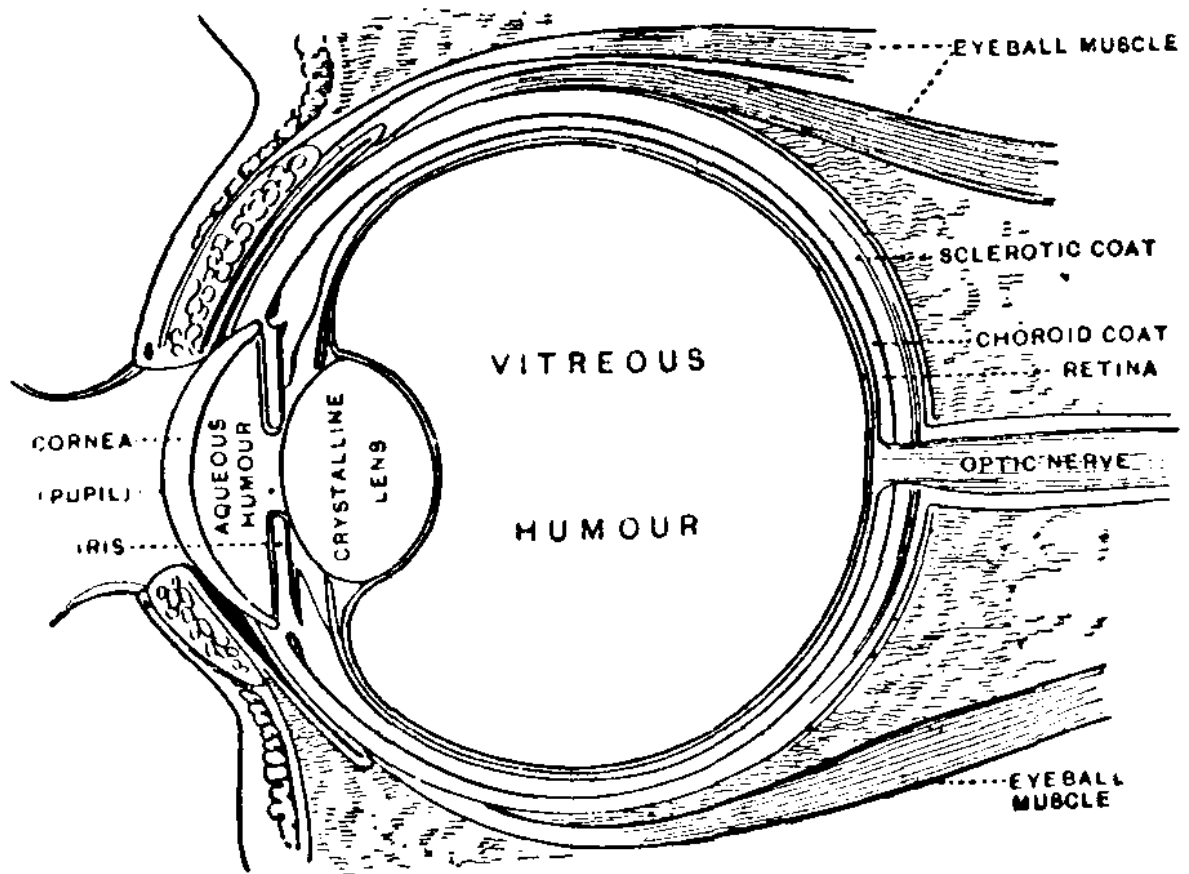
[back](#)



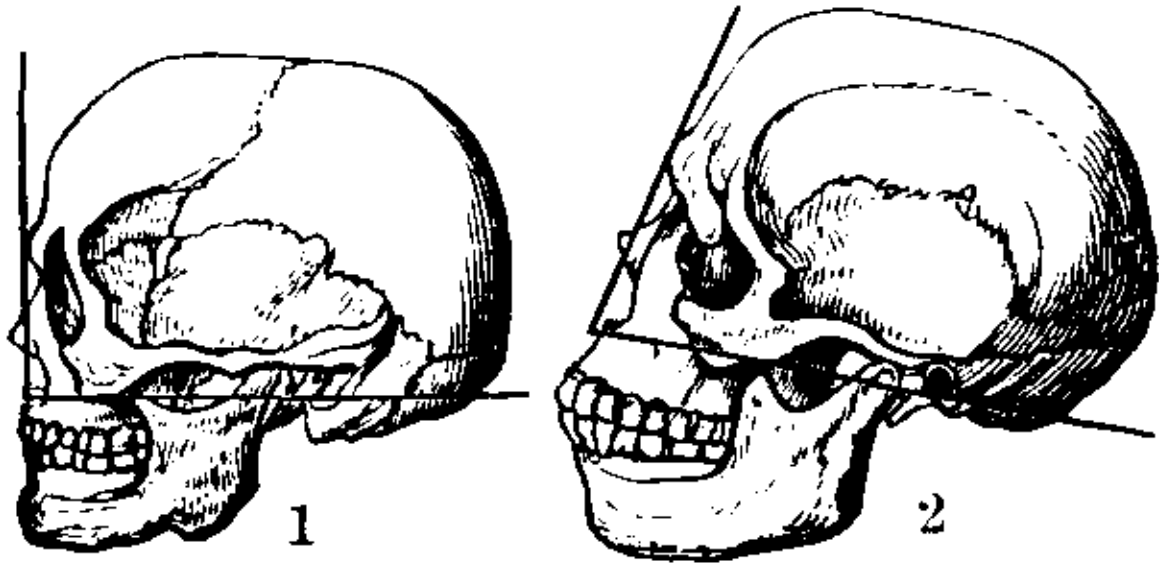
[back](#)



[back](#)



[back](#)



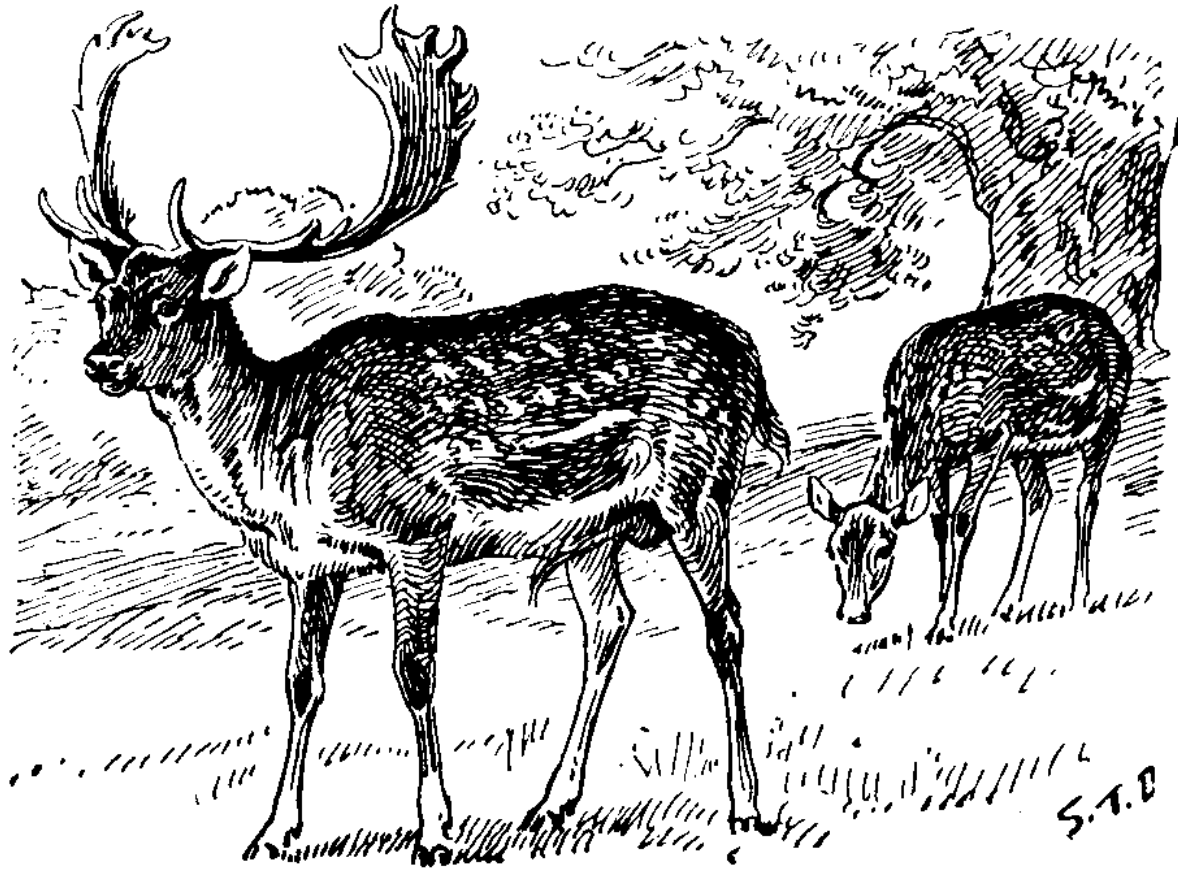
[back](#)



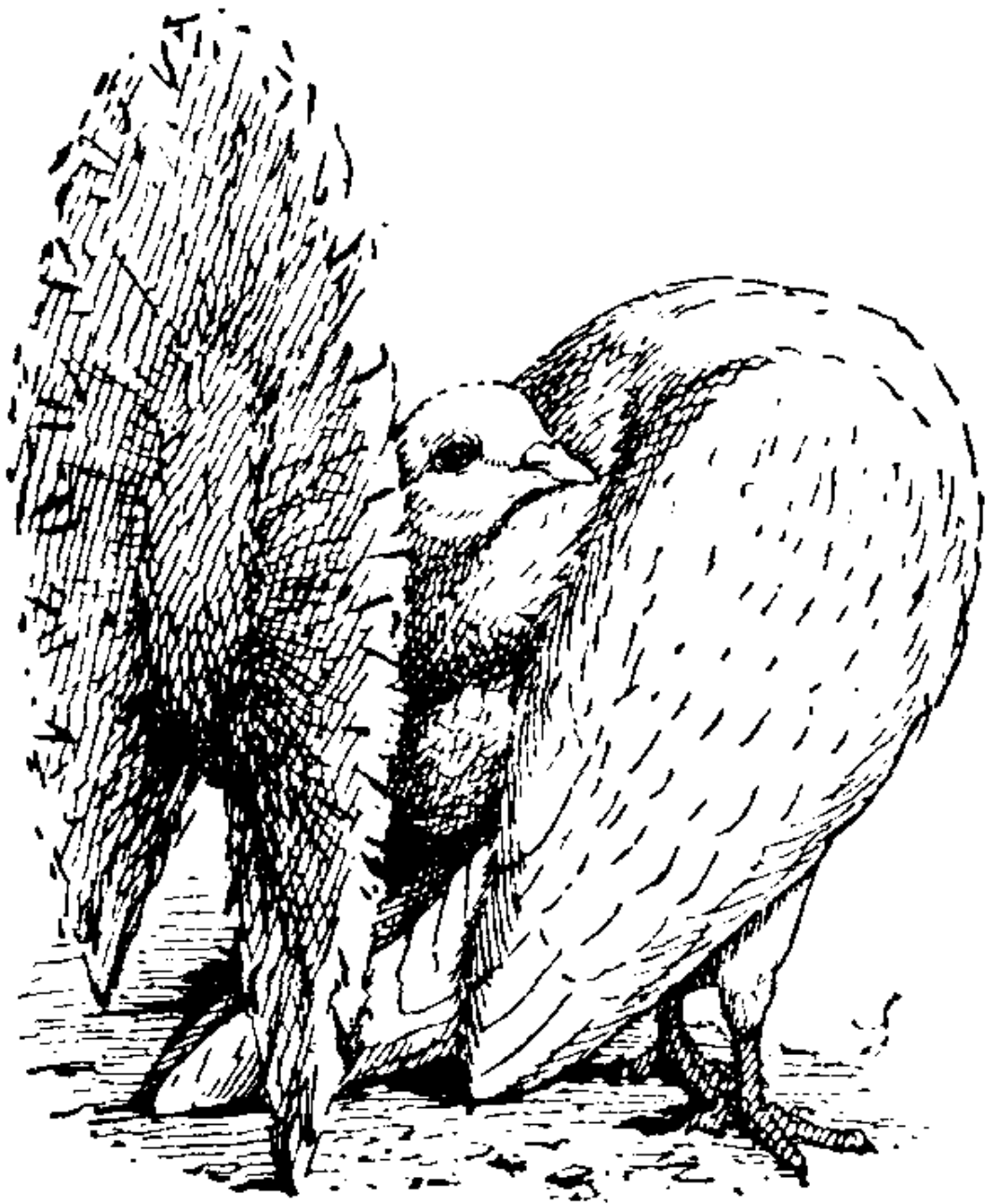
[back](#)



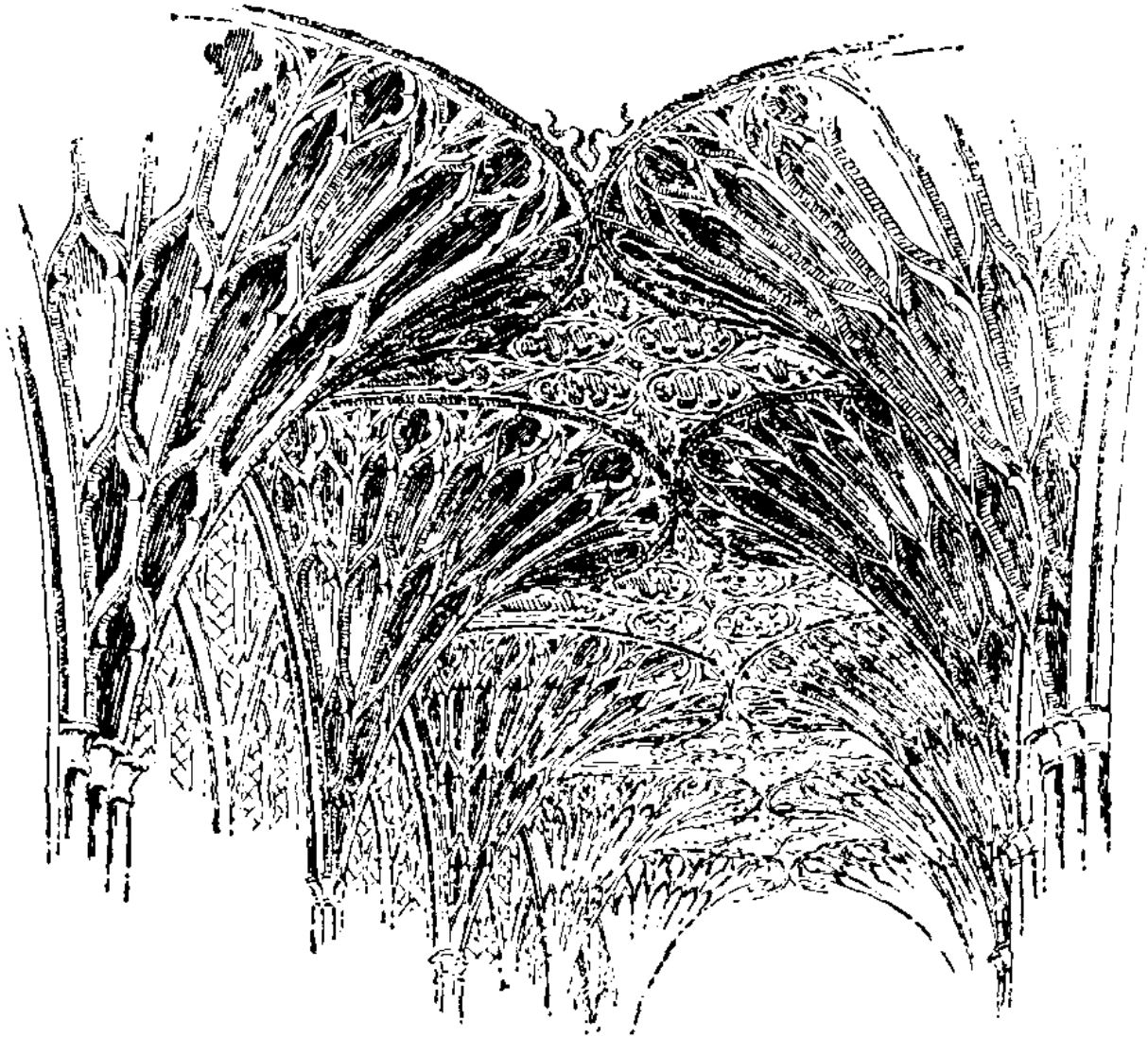
[back](#)



[back](#)



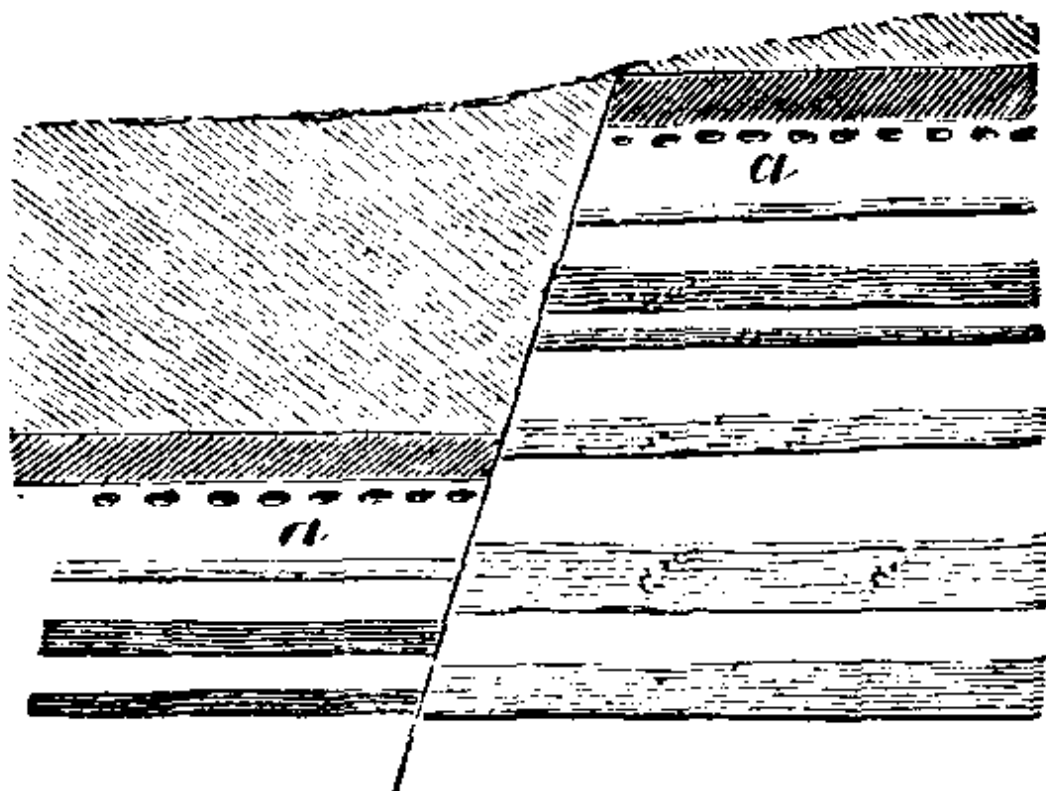
[back](#)



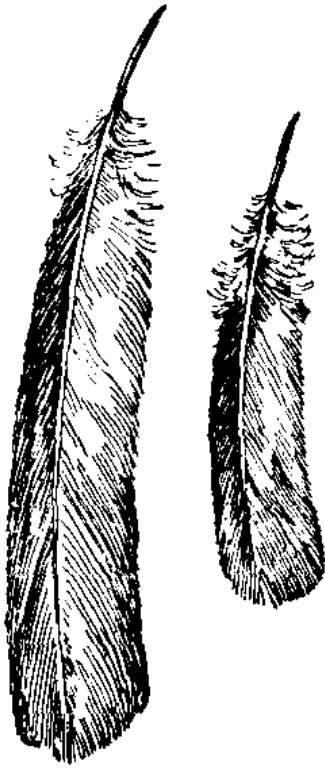
[back](#)



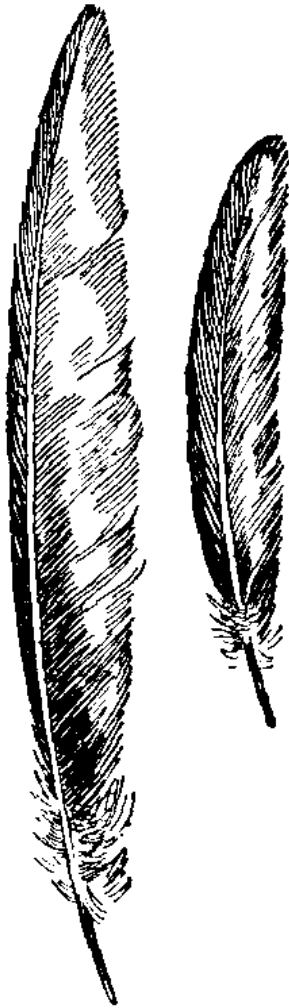
[back](#)



[back](#)



Tail feathers

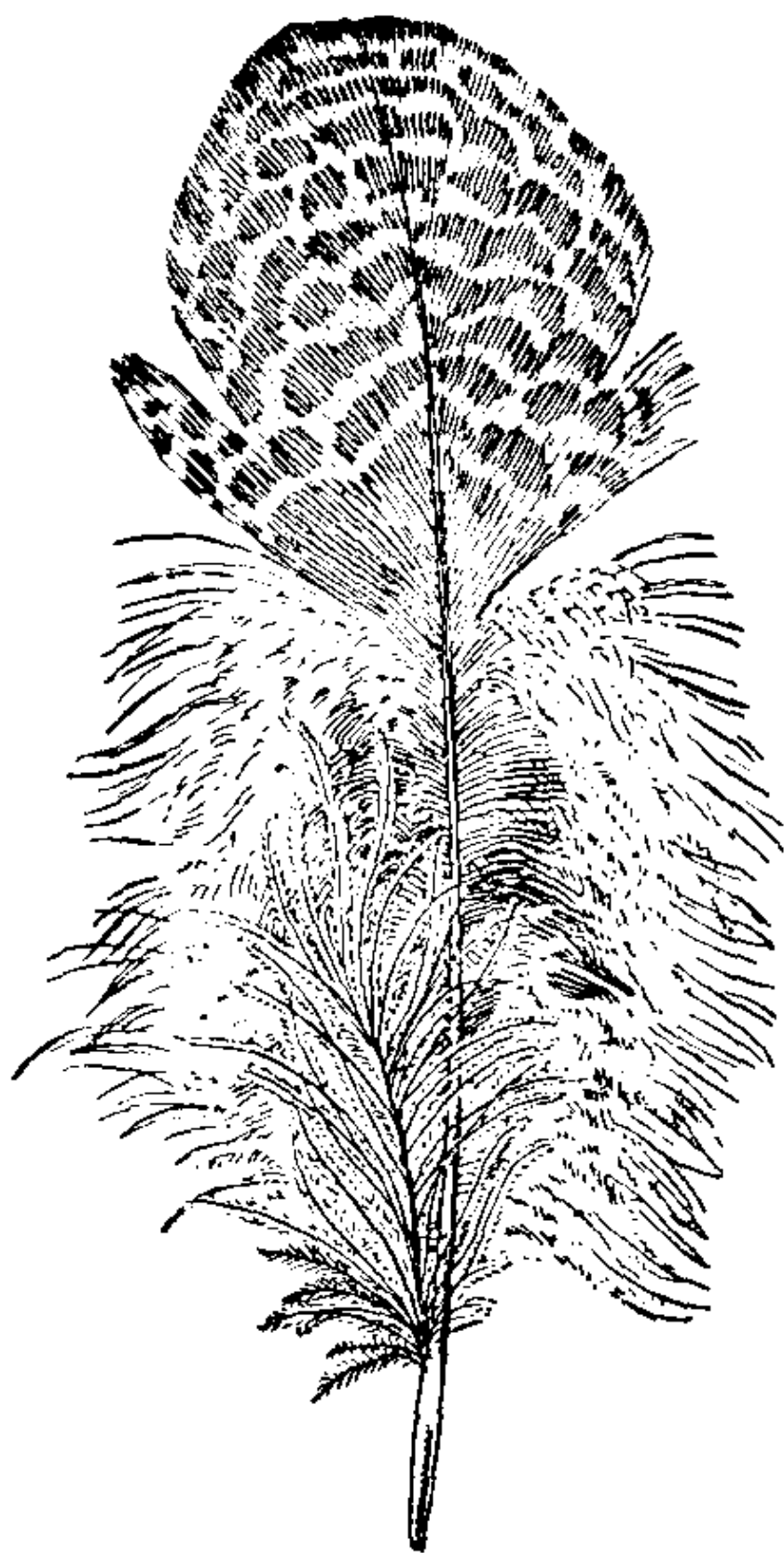


Wing feathers

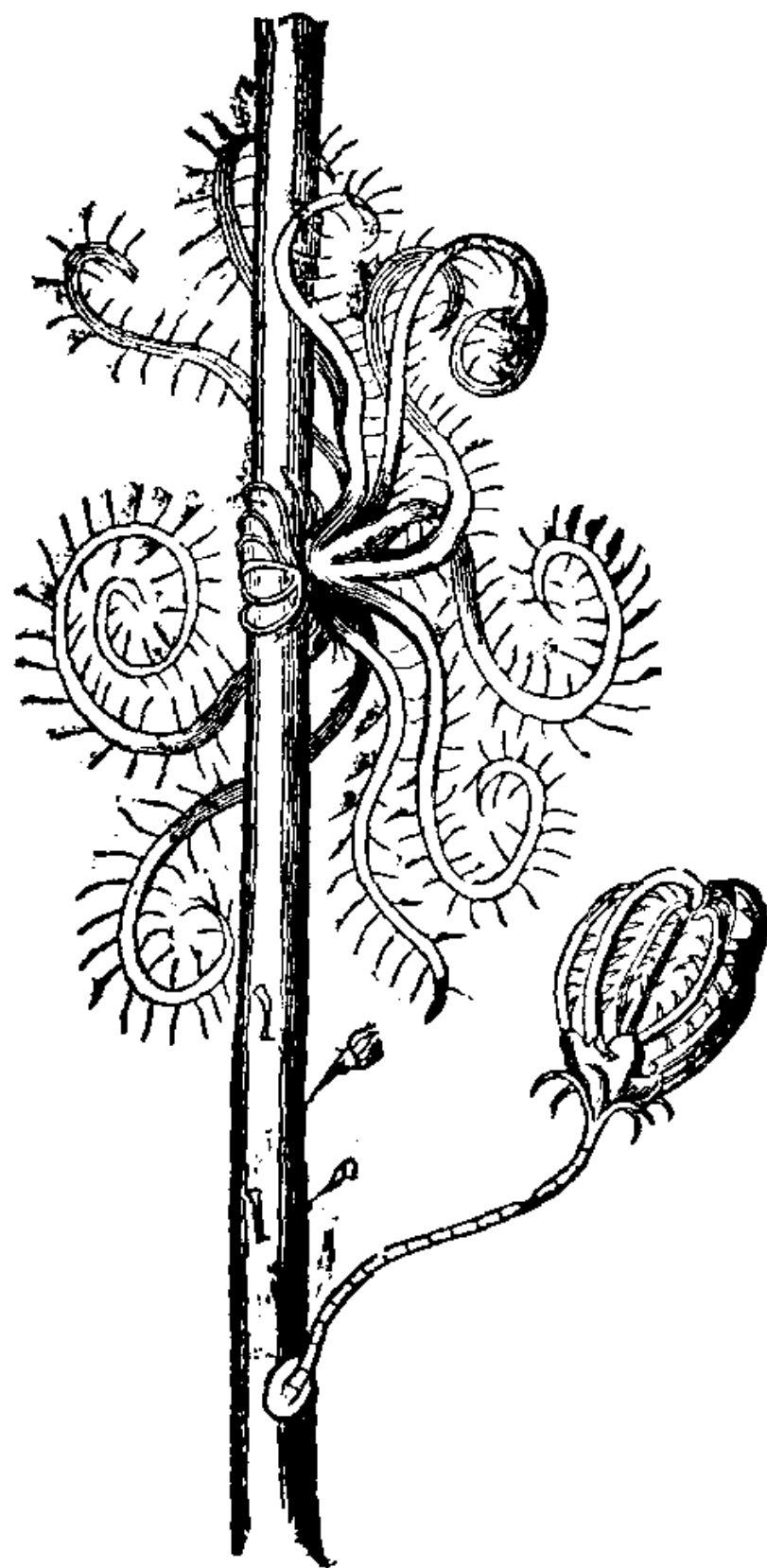


Covert feathers

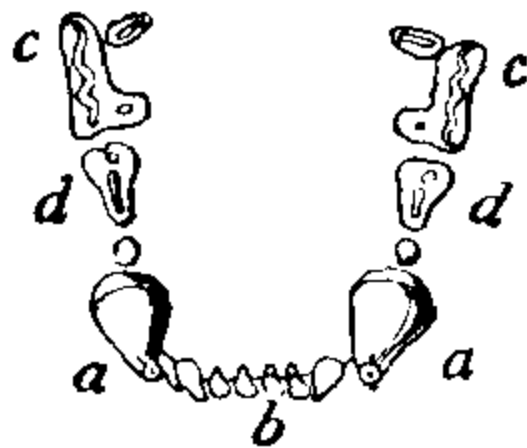
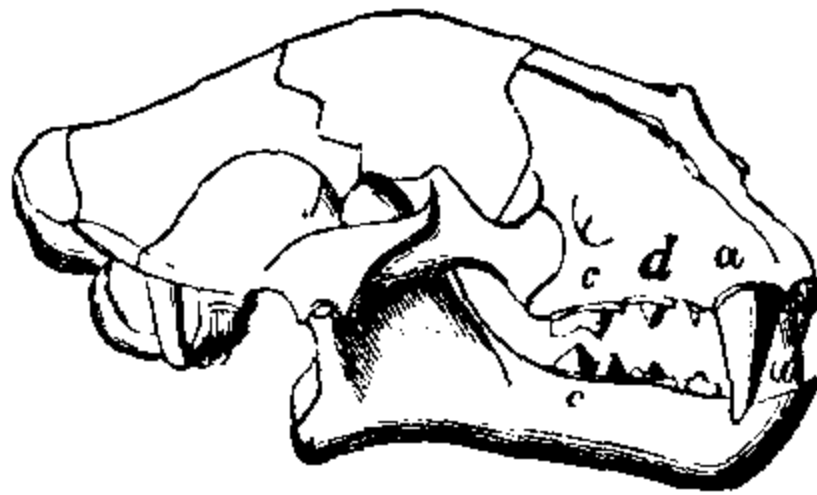
[back](#)



[back](#)



[back](#)



[back](#)